CRITICAL THINKING ABOUT RESEARCH

PSYCHOLOGY AND RELATED FIELDS

SECOND EDITION



JULIAN MELTZOFF AND HARRIS COOPER

CRITICAL THINKING
ABOUT RESEARCH



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JULIAN MELTZOFF AND HARRIS COOPER

AMERICAN PSYCHOLOGICAL ASSOCIATION • Washington, DC

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Foreword

My father, Julian Meltzoff, loved research. He enjoyed mulling over research designs to think about what could—and could not—be inferred from any given study. He had a special gift in this regard, a "research eye" that could detect the slightest flaw in any research article. Some of us can look at a wall of pictures and immediately see one that is crooked amidst a dozen that are not. Other people can hear a wrong note played in a sea of music, or feel minute roughness on an otherwise smooth surface. My father could rapidly detect fatal flaws in research design and gaps in critical thinking that eluded peer review. His ability derived from years of training and a highly logical mind.

Julian Meltzoff believed that this ability to detect errors didn't come naturally but could be cultivated and enhanced through education. When he sat down to write the first edition of *Critical Thinking About Research: Psychology and Related Fields*, he said he wanted to help others learn the value and joy of critical thinking. He also knew that critical thinking was becoming ever more important, as rapid communication and 24-hour news brought a tsunami of information to scientists and the public. What can one do to help people sort truth from fiction? His answer was to write a book to help others to learn to think critically about scientific and everyday information.

Research was exciting and fun for my father, and he wanted to make it fun for students. Readers of this book will see this come alive in the practice articles that constitute Part II. The careful reader will not only find the errors he instructively hid inside the articles but also will enjoy the amusing names he gave the authors of these papers. For example, the author of the article on childhood aggression is Junior Fites, an author on the article about the effects of pleasant experiences on time estimation is Bontiempo, and an author of the article about self-talking is Earl Singleton. There are other buried treasures in the reference lists that will make readers smile as they labor to learn about critical thinking. My father's goal was to show that learning about research could be fun, and that every word counted. Let nothing go unexamined.

Julian Meltzoff was a fellow of the American Psychological Association (APA) for 58 years. He published his first peer-reviewed paper on clinical

psychology in an APA journal in 1952 at the age of 31. He published his last peer-reviewed paper in an APA journal in 2014, at the age of 93. This last article, published in *Psychology of Aesthetics, Creativity, and the Arts*, is a unique piece that unites art and science. It focuses on an intriguing error that has been made by artists throughout history in depicting human gait. Sculptors and painters, including masters such as Rodin and Leonardo, often erred in how they placed the arms in relation to the legs when a person moved. Does the left arm swing forward or backward when the right leg moves forward in natural gait? My father showed that there is one correct answer, and that artists from ancient Rome through modern times did not always get this right. It is a breathtaking piece for a 93-year-old.

I am very pleased that APA has chosen to reissue my father's book and that Dr. Harris Cooper has updated it in ways that make it even more helpful to the modern student. Critical thinking in science and everyday life has, if anything, become more critical in this age of social media. We all need to think critically about not only scientific information but also everyday information. My father had long hoped that this book would make a mark by helping students studying behavioral sciences and everyday citizens learn the essential skills of critical thinking. Happily, with Dr. Cooper's contributions and additions, this classic APA book will now reach an even wider audience.

Andrew N. Meltzoff, PhD
Professor of Psychology
University of Washington

Preface to the Second Edition

It is an honor to have been asked to prepare the second edition of *Critical Thinking About Research: Psychology and Related Fields.* The author of the first edition, Julian Meltzoff, passed away at age 94 after years as a clinical psychologist, researcher, and writer. In the latter role, he wrote about psychotherapy and posttraumatic stress disorder, and he authored this book on how to read and critically evaluate psychological research. He also enjoyed a second career as a sculptor whose bronze piece "Body in Motion" graced the cover of the December 2014 issue of *American Psychologist.* Although I never met Dr. Meltzoff, the opportunity to refresh this book for the next generation of psychologists gave me an introduction to his scholarship, insight, and humor. I am most grateful for the opportunity.

I have tried, in the second edition, to remain true to Dr. Meltzoff's voice, though the content now reflects both of our sensibilities. Part I of the book has been extensively revised to reflect the enhancements in methodology and methodological thinking that have appeared since the first edition. I also added a chapter on research synthesis and meta-analysis, techniques that were just emerging when Dr. Meltzoff wrote the first edition. I have, however, largely remained true to Dr. Meltzoff's organizational structure. I revised the Glossary by adding terms, which are now italicized in the text when they first appear or play a prominent role in what is being discussed. That way, you can turn to the glossary if you need reminding of a term's meaning and know the term will be there. I also have added a summary of the main points at the beginning of each chapter. Other than updating the (fictional) dates and some of the language in the practice articles in Part II, these remain largely unchanged. Dr. Meltzoff did an inspirational job with these, and even his choices of topics remain remarkably current. I added one new example of an article reporting a meta-analysis.

I also thank Linda Malnasi McCarter, acquisitions editor at the American Psychological Association (APA), for thinking of me when this opportunity became available. Beth Hatch, my development editor at the APA, orchestrated two excellent reviews and provided her own cogent

comments. My wife, Beth, put up with me brilliantly and sensitively, knowing when to leave me be and when to drag me away from my writing.

One more personal note. As I worked through the revisions, I realized I was spending hours "listening to" Dr. Meltzoff's reaction to my changes. This led me to read some of his other writings, to ensure I was hearing him clearly. Most notable was his self-published book, *Memories of WWII*. I realized he and my father, Samuel Meyer Cooper, a high school graduate, served together in the European Theater of Operation during the war. They never met either.

Harris Cooper

Preface to the First Edition

This book was stimulated by my having spent the last 17 years teaching graduate students how to design and critique research. During this period, I had the responsibility of critically evaluating far more dissertation proposals than I care to admit, for fear of being too closely associated with Baron Munchausen.

Although there are many good texts on research design, they are written from the perspective of the research producer, not the research consumer. My colleagues and I have had to become inventive and innovative when teaching the important skill of critiquing research. From a pedagogical point of view, the need for supplying flawed research articles on which students could practice their critiquing skills was always apparent to me. The idea of creating the materials to order came later. Construction of such reading matter was informed by my experience in discovering errors that I repeatedly encountered when reviewing published articles, as well as research proposals. I am indebted to all of the students and professional psychologists who unknowingly supplied me with the ideas that went into this book, and who so graciously recognized and corrected the problems once they were revealed to them. My thinking about designing and critiquing research was enriched by frequent detailed exchanges of ideas with my academic colleagues. Their cogent observations, and sometimes their alternative ways of looking at things, stimulated me continuously to reexamine research-related issues, to fine-tune my views, and to reappraise my judgments.

I am most grateful to all of those who reviewed parts of this manuscript and offered helpful suggestions to improve its scope, its focus, and its clarity. I am particularly indebted to Mark Sherman, with whom I discussed research issues so frequently over the years. He helped shape my ideas about the subject and the ways it can be taught. He not only read the entire manuscript, but gave me detailed and expert feedback about it. I am truly grateful to Andrew Meltzoff for his frank, careful, and incisive critique and his ongoing interest and support in this effort. Without the constant encouragement and support of Antonia Meltzoff, I probably would not have been so inclined to

begin this project or to have become so engrossed in carrying it through to completion. As each section was converted from thought processor to word processor, it turned into a prolonged show-and-tell in which she participated with a gusto that fueled my own esprit. Nancy Meltzoff was a most helpful resource when I dared to venture into her area of specialized knowledge. If, with so many who share my surname involved, writing this book is beginning to sound like a cottage industry, it is only because all of them are PhDs who are highly accomplished scholars, scientists, writers, and educators in their own right, whose judgment I trust, whose candor I can depend on, and whose input I value.

I have Peggy Schlegel, acquisitions editor at the American Psychological Association (APA), to thank for the guidance and encouragement she provided from the beginning. My special gratitude goes to Andrea Phillippi, development editor at the APA, for her careful reading and most helpful suggestions. I am deeply grateful to Ida Audeh, whose formidable editorial skills have done so much to polish the text. I very much appreciate the work done by all of the reviewers and editors of the manuscript in its incipient and later stages.

Julian Meltzoff

CRITICAL THINKING
ABOUT RESEARCH

Introduction

How should scientific evidence be evaluated? After reading a research article, one person may be fully convinced by the evidence that is offered; a second person may think that the same evidence might be true but regards it, on balance, to be equivocal and feels obliged to suspend judgment; and a third person may disdainfully dismiss it with scatological words of unmistakable meaning. Their different judgments may be informed by different views about what proof is. They may be influenced by their different beliefs about what evidence is needed to establish proof. They may be limited by the amount of knowledge that they have about the methods used to gather evidence in the search for truth. Finally, they may be affected by variation in their ability to think critically.

Regrettably, there is evidence that people reading the same research paper but beginning with opposing beliefs come away with even more polarized opinions than before they were exposed to the new evidence (Lord, Ross, & Lepper, 1979). This can happen because people misinterpret the evidence they have read, they are motivated to think in a certain way, or because of other cognitive malfunctions (McFadden & Lusk, 2015). This is a quality of human nature that is antithetical to the goals of science. It is what we attempt to overcome when we put on our critical thinking caps.

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We use the phrase *ability to think critically* as a positive quality, not as a pejorative reference to the characteristics of a mean-spirited faultfinder. The phrase refers to the skill of thinking about an issue, analyzing it, looking at it from all sides, and weighing whether there is sufficient evidence of good-enough quality to warrant making a reasoned judgment that is as free of personal bias as possible. The Critical Thinking Community (2016) defined it this way:

> Critical thinking is self-directed, self-disciplined, self-monitored, and self-corrective thinking. It presupposes assent to rigorous standards of excellence and mindful command of their use. It entails effective communication and problem-solving abilities, as well as a commitment to overcome our native egocentrism and sociocentrism. (p. 1)

Texts and curricula are available to teach critical thinking in general (e.g., Moore & Parker, 2015). Our purpose here is a bit more restricted. We focus on critical thinking in the context of evaluating scientific research. We hope (a) to exhort you to adopt a critical mental set when reading research literature; (b) to try to increase your understanding of the principles and methods of research that are used to produce evidence; (c) to alert you to booby traps that can compromise the research and cloud the evidence; and (d) to provide you with materials that can be used for practice, with the goal of improving and honing your critical reading skills when you are exposed to research in the sciences, especially psychology and related fields (education, law, economics, health, nursing).

This two-part book is about how to evaluate scientific evidence. Part I provides an overview of the methods and principles that you need to know about when you critique or evaluate the trustworthiness of research. Part II presents fictional journal articles that provide you with practice in applying the principles outlined in Part I. The book is certainly not a statistics text, and it assumes that you have some familiarity with basic statistical methods. It is not intended to be your only text to prepare you to design and to carry out research. The focus is on reading and critiquing research. The two activities, producing and consuming research, are in no way contradictory, but a person who is adept at critiquing research is not necessarily skilled at thinking up research questions or conducting research any more than a drama critic is necessarily a good playwright. Books on research methods aim to give people who are preparing to do research an insider's view of how to go about designing a study and planning and conducting statistical analyses of data.

This is not a stand-alone methods book for aspiring and practicing researchers, although they can certainly use it to hone their skills as they think about how to proceed. It is intended to assist you in reading and assessing research. A reader who may not be familiar with the intricacies of *research design* (and who may not be aware of all the pitfalls to avoid) should know enough to recognize procedures that have gone awry or to know when plans have not worked out optimally. We review the application of research principles and alert you to things that interfere with valid research. Our objective is to help you bring an inquiring and informed mind to the evaluation of research.

Our presentation of the ways an experiment can flounder, and the emphasis we place on scrutinizing and critiquing research, is in no way intended as an indictment of the research process or as an expression of any reservations whatsoever about its value as a method of seeking the truth. In fact, we know that the perfect study has never been done. All research has strengths and flaws. Your task as a critical reader is to identify these and judge the value of the research for making an informed judgment of its utility in drawing the conclusions the authors intended.

We hope to encourage a form of interactive reading that we believe is central to the process of critical evaluation of research literature, as opposed to the almost reverential acceptance (or rejection) of the printed word. You must be able to bring a skeptical, inquiring, searching, probing attitude to the task; to spot things that do not look quite right; and to recognize how a weakness in one phase of the design or execution of a study can severely compromise the study at a later phase. The "tests" take the form of questions that you, the critical reader, ask of the written research report.

Why Psychologists and Professionals in Related Fields Need to be Critical Thinkers

Psychologists may spend their time in one of three ways: (a) doing research to generate knowledge, (b) transmitting knowledge to others by teaching or by directing the research of others, or (c) applying psychological knowledge in the form of clinical or consulting services. These can be done singly or in any combination. A parallel set of options is available to specialists in other behavioral or social sciences, in education, in law, in medicine and the health sciences, and in the service of organizational improvement. There is variation between and within fields in how much time is spent educating and training students for the functions of generating knowledge, disseminating it, or applying it.

One thing shared by people in psychology and all related fields and subspecialties, no matter what career direction they choose, is that they all must be consumers of research. The bedrock of psychology and related fields consists of theories and data from research that support or refute the theories. All who profess to know a field, and those who want to keep abreast of advances, are necessarily consumers of research. It stands to reason that they must be able to evaluate research critically. Having a good grasp of the scientific method and the principles of research design, and knowing what to look for, are invaluable in this pursuit.

Psychologists who graduate from research-oriented doctoral programs have the advantage in this regard because research knowledge is a vital part of their education and the dissertation requirement gives them hands-on experience in research. Nevertheless, virtually all programs that do not require a research dissertation strive to teach students to think critically and to read empirical work intelligently. Nobody who is considered an expert or specialist likes to be the bearer of misinformation. Throughout the history of psychology and related fields, untested fads, half-baked ideas, and outright quackery have led astray theories, policies, and practices. Many

professionals with weak skills for critically evaluating research have become enthusiastic advocates of unsubstantiated claims. Other professionals have had the wherewithal to resist and have retained their skeptical reserve until solid research data were furnished.

What is necessary, then, is first to learn how to critique research and then to make a habit of sifting all incoming offerings through a sieve whose openings are small enough to permit only the trustworthy evidence to fall through. This two-part book is written to further that goal. We present the principles of research design that you need to know about to critique research (Part I), followed by intentionally flawed, but realistic, fictitious materials on which to practice your critiquing skills (Part II).

Overview of the Chapters

Part I consists of 11 chapters. Chapter 1 deals with the methods of seeking knowledge about our world, the rules of scientific evidence, and what constitutes scientific proof. It discusses how to read critically and presents an outline to serve as a guide for critiquing research. Chapter 2 focuses on the questions and *hypotheses*, or predictions, that motivate research and discusses faulty reasoning about causation. Chapter 3 reviews the critical appraisal of research strategies and the selection of variables. The selection and composition of the research participants, or *sample*, methods of sampling, and the consequences of faulty sampling are covered in Chapter 4. Chapter 5 gets to the heart of research design in a discussion of the variables that should be under the control of the researcher and methods of controlling them. Chapter 6 presents the major *experimental* and *quasi-experimental designs* that are available, and discusses the flaws that you should recognize in some of them. It also gives examples of some of the important threats to the trustworthiness of a study. Chapter 7 discusses the selection of *independent variables*[1] and *dependent variables* (or outcomes) to help you appraise whether the selected techniques were appropriate and used properly in a particular research study. Chapter 8 provides an orientation to the assessment of statistical analyses, graphic analysis, and the drawing of inferences and conclusions. Chapter 9 looks at how data are interpreted and statistics translated to information. It also covers issues to consider when evaluating the completeness and transparency of a report of research. Chapter 10 introduces a different kind of research, the *research synthesis*, or *meta-analysis*, in which the goal of the researchers is not to collect new data but to accumulate and interpret the studies on a topic that have already been conducted. Ethical considerations in research, the subject of Chapter 11, are discussed to alert you to deviations from ethical standards. All the chapters in Part I are written from the perspective of readers looking in on the research report from the outside, with

[1]Most often, we use the term *independent variable* to designate a variable that is being manipulated in an experiment, and therefore is viewed as a cause. The term *predictor* is used in a nonexperimental study, when the variable is measured. *Outcome* or *criterion variable* is used in a study of a treatment or intervention. We will try to use these terms in this manner through the text.

the emphasis on critically evaluating research rather than conducting research on their own.

Part II consists of 17 realistic but short and entirely fictitious simulations of journal articles. The articles have flaws built into them that range from the blatant to the subtle. The collection is essentially a workbook that is based on the principles in Part I and is intended to be used by you to practice your critiquing skills. The frontispiece by the great 18th-century artist, Hogarth, was one of the things that instigated the idea to write this book. The legend at the bottom of the print says, "Whoever makes a design without the knowledge of perspective will be liable to such absurdities as are shown in this frontispiece." So it is with research. Lack of understanding of the principles of research design can lead a researcher to the kinds of errors that are illustrated in the fictitious articles and can prevent you from realizing that there is anything wrong.

The articles in Part II range across a variety of content areas (e.g., clinical psychology, general psychology, psychology and the law, education, sociology) and research methods, but the predominant emphasis is on psychological experiments. Each fictitious article is accompanied by a critique of its built-in flaws. Critical issues are cross-referenced to the chapter in Part I that addresses them, so that if you want a fuller explanation you will know where to find it.

The kinds of statistical analyses in the articles range from simple descriptive statistics to univariate *analyses of variance* with *categorical variables*. Studies that use the techniques listed in the fictitious articles are abundant, and the articles mirror those that can be found in the literature. The articles do not feature highly advanced statistical techniques, so that the book will be useful for the greatest number of people.

These critiquing exercises are meant primarily for the use of students in psychology and related fields, whether they be advanced undergraduates or beginning graduate students. Psychologists whose graduate school experience is well behind them, and who have not themselves been engaged in research, can profit from these as well.

Because the principles of research design and the approaches to critiquing research are not restricted to psychologists or psychological research, students and professionals in other behavioral and social sciences also should find these materials and exercises useful. This book may be helpful to medical researchers, consumers of medical research, educators and education administrators, and members of the business and legal profession who deal increasingly with the evaluation of evidence and opinions that are offered by scientific expert witnesses. With so many scientific specialties, cross talk among them is necessary to function intelligently. Your understanding of research in another field may be circumscribed by a lack of technical knowledge, but you should still be able to read and understand research in a related field.

In this age of science, the media continuously expose nonscientists to science reports from all fields. The ordinary person is served a rich diet of science every day. The portions, unfortunately, take the form of sound bites that must be swallowed whole or of summaries that lack all the details. Efforts to facilitate public engagement with science are plentiful and growing more so (e.g., Alan Alda Center for Communicating Science, 2016; Center for Engagement With Science and Technology, 2016). In psychology, the call for practices to be evidence based is taking hold throughout

the profession (e.g., American Psychological Association Presidential Task Force on Evidence-Based Practice, 2006). Educated nonscientists who want to learn more about research design and how to read research critically could profit from reading in Part I, perhaps skipping sections that appear to be too technical and specialized. As information consumers, they could derive benefit from testing their critical reading skills on the exercises in Part II.

Who Will Find This Book Useful?

Readers who are advanced undergraduate or graduate students may read this book as part of a course's materials, or can use it to assess their own level and needs. Those who have already had considerable training in research design should select any sections of Part I that will fill the gaps. You should evaluate the articles in Part II on your own before reading the accompanying critiques. Concepts that are relevant to the article are cross-referenced with Part I; you should consult Part I for further clarification.

Professional psychologists who have not done any research or who have done some research in the past but have been away from it for a long while may choose to look through the contents of Part I and start to read wherever they see a need for a refresher or wherever something strikes their interest. Those who evaluate the articles in Part II independently and find that they have missed many of the central points in the critiques may benefit from reading Part I more thoroughly.

Readers who believe that they already have a good understanding of research questions, hypotheses, and sampling but that they could use a booster on design options and methods of control might want to read Chapters 5 and 6. Those who are comfortable with various aspects of design but have questions about independent variables and dependent measures could start with Chapter 7. In either case, you should try to sharpen your reading skills by critiquing the articles in Part II.

Knowledgeable and experienced researchers who already know far more than is covered in Part I may begin with Part II and read the fictitious articles for fun, for challenge, and for mental calisthenics.

PRINCIPLES OF CRITICAL THINKING

I

Seeking Trustworthy Knowledge

1

A critical reader of research

- is skeptical of scientific claims on the basis of (a) beliefs that have simply been held a long time, (b) appeals to authority, (c) votes, (d) rational arguments alone (especially if the premises have not been tested), and (e) observation and sensory input (because of the possibility of perceptual distortions);
- understands that because evidence is written down does not mean it is correct;
- is aware that publishing in science has corrective devices (e.g., peer review) that protect against errors and false claims, but this system is not perfect—the Internet, in contrast, has weak, if any, mechanisms for preventing error-filled and fake research reports from being published;
- uses an interactive mode of thinking, which involves active and frequent use of explicit rules to judge whether expectations are met;
- has clear and justifiable reasons when an evaluative judgment is made about the strengths and limitations of an article;
- knows the ability to learn from science requires knowledge of research design and the skillful application of that knowledge; and
- maintains an attitude that the researchers are responsible for providing a complete and accurate rendition of their work.

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Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper
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Critiquing research is not something that comes without learning and practice. We spend many years applying other ways of appraising truths before we learn the scientific method. These modes of thought continue to influence us in deciding what evidence to consider valid, even after we have learned the scientific method.

In this chapter, we review (a) the various methods people use for seeking trustworthy knowledge about their world, and (b) differing views on what constitutes sufficient evidence for something to be taken as "proven." If you aspire to be an informed citizen who makes sound decisions, you need to look on truth claims with a critical eye. The crux of the critical appraisal, in turn, depends on the rules of evidence that you hold a truth claim up against. When scientific evidence is being evaluated, the appraisal rests on your knowledge of research design and how carefully you apply that knowledge.

Methods of Seeking Knowledge

People use many different methods to seek knowledge. These methods may be based on rules of evidence that conflict, coexist, or complement each other. The same individual may use different rules of evidence in different spheres of life. Some people can switch effortlessly from one mode of thought to another (e.g., when you read a historical novel, interact socially, go to church, read scientific studies). Other people find switching the criteria for truth more difficult. They may apply the wrong rules to the circumstance in which they find themselves. A trusted friend may be a good source of information about the quality of food in a restaurant but may be less reliable when it comes to imparting historical facts.

Think about a simple statement you could make that you are convinced is true. Got one? Okay, now think about how you know it to be true. Charles Sanders Pierce (1877), a 19th-century American philosopher and scientist, suggested that we know things based on several sources of evidence. We suspect your truth will be based on one of these.

HABIT OR TENACITY

First, Pierce (1877) suggested that we know things because they appeal to us, we repeat them to ourselves often, and we turn away from any information that might disturb our happy state of affairs. Although we might be denying contradictory facts, we find the state or certainty that tenacious beliefs bring us more satisfying than the unsettled state of mind that accompanies doubt. If we do this long enough, we might forget where and when the original belief took hold, but no matter, "I've believed it to be true for so long, it must be so."

Clearly, this method of knowing has its failings. We will meet people with different facts and discover that we have no way of determining whose facts are more accurate. Likewise, over time, the world will change and our beliefs may become dysfunc-

tional, leading us to act in ways that do not have the consequences we anticipated. For example, if your truth is, "The climate of the earth is not growing warmer," you may have forgotten when and how you learned this, but the consequences may be dire of holding on to the belief simply because you have always believed it.

AUTHORITY

Another way we know things to be true is to take the judgment of one or more authorities as sufficient evidence. Pierce (1877) emphasized the point that "truth by authority" often has a coercive component to it. He felt that religious and political institutions can promote a way of thinking about the world. Our parents, friends, and teachers are also sources of evidence whose authority we accept. We also can view experts as authorities, whose trustworthiness we believe because their credentials make them credible sources of information. Experts can be highly reputable scientists whose education and experience in designing and conducting experiments give credibility to their pronouncements. So, if your truth is, "blessed are the meek for they shall inherit the earth," you probably learned this from the Bible or from a religious authority. So too the truth "hydrogen and oxygen combine to make water" was probably learned from an authority, your first chemistry teacher.

But what if two or more experts disagree, or experts change their mind? With the unerring benefit of hindsight, one can see how mistaken highly qualified scientific authorities have been in their pronouncements. In *The Experts Speak: The Definitive Compendium of Authoritative Misinformation*, Cerf and Navasky (1998) provided retrospectively amusing quotes from outstanding authorities in various fields. The physicist Lord Kelvin, who was president of the British Royal Society, announced in 1895 that heavier-than-air flying machines were impossible. Albert Einstein is reported as having said in 1932 that harnessing the energy of the atom would never be feasible. In 1956, the leading British astronomer Richard van der Riet Wooley alleged that space travel was nonsense. Clearly, there were scientists who disagreed, and if we could ask them today, Kelvin, Einstein, and van der Riet Wooley would regret what they said.

DEMOCRATIC JUDGMENT

Trust in the judgment of many, in contrast to the dicta of single authorities, also is a common way of deciding what is right. Pierce (1877) wrote, "Let the actions of natural preferences be unimpeded, then, and under their influence let men, conversing together and regarding the matter in different lights, gradually develop beliefs in harmony with natural causes" (p. 8). We become confident about the truth-value of statements when a judgment is shared by many people. But consensus is not science. While Pierce had a clear affection for this method of knowing, he also recognized that even democratically arrived-at truths had their failings. They are subject to fashion and thus, rather than being eternal truths, could be prone to swings based on the tenor of the times. Groups can also engage in "groupthink," the process whereby

group pressure leads to the suppression of the views of minorities and the reality testing of ideas (Janis, 1972).

REASONING

Another way of arriving at truth is by reason alone. Under certain circumstances pure reason can lead to what is called *necessary truth*. In *a priori reasoning*, conclusions are deduced from known or self-evident principles or definitions. If your truth is "1 + 1 = 2," you have invoked a self-evident truth. The *syllogisms* of formal logic and the proofs of pure mathematics lead to necessary truth.

However, not all questions can be answered by pure reason alone. Many conclusions about the world call for *inductive reasoning* (i.e., reasoning that begins with things taken as known). Final judgments are based on experience, and principles are generalized from facts that are established by experience rather than by the application of reason alone. If the observations are wrong (the premises of a logical argument), then so are the conclusions.

SENSORY INFORMATION

This reliance on experience for finding knowledge leads to another method of knowing: One of the most compelling and unshakable proofs for people is personal experience. There is a certainty to the subjective truth of the statement "I have seen it, I have experienced it. Therefore, I know that it is true." If you answered our request for a simple statement you know to be true with "I have two eyes," it is because you have looked in the mirror day after day and have seen your two eyes staring back at you.

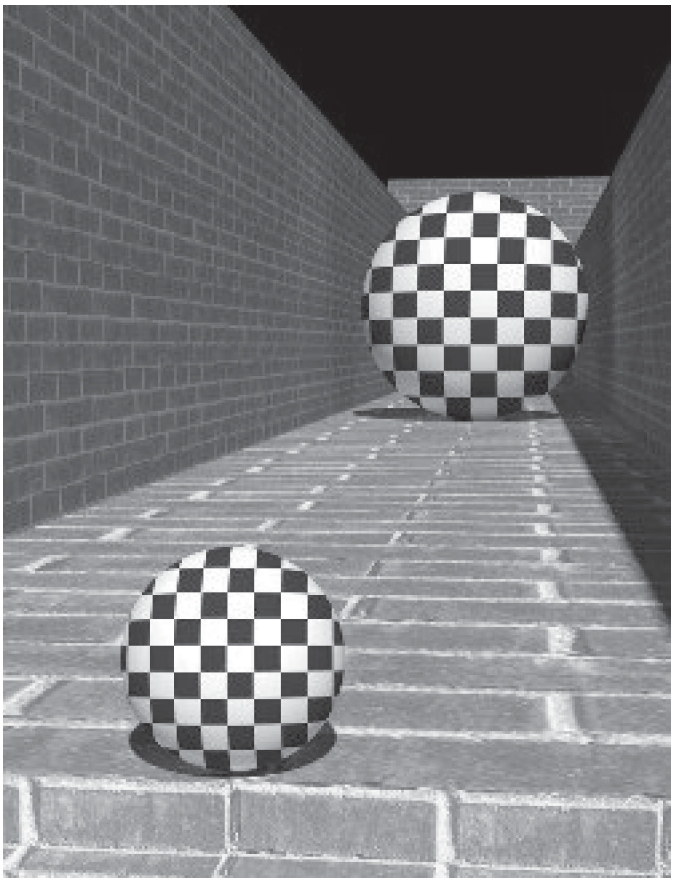
Yet, we know the human senses are fallible. Psychologists have studied optical illusions, and magicians have exploited this fallibility to their advantage. For example, look at the soccer balls in Figure 1.1. Although the ball at the top of the figure clearly appears larger, the two balls are the same size. (Not only does the optical illusion show us why our senses cannot always be counted on to reveal truth, but it can also be used to question the veracity of democratic judgments; if we voted, we would unanimously agree that the ball at the top of the figure looks larger.)

EMPIRICAL AND EXPERIMENTAL METHODS

Far and away, according to Pierce (1877), the best way to know things is through the methods of science:

> It is necessary that a method should be found by which our beliefs may be determined by nothing human, but by some external permanency—by something upon which our thinking has no effect . . . the method must be such that the ultimate conclusion of every man shall be the same. Such is the method of science. (pp. 9–10)

FIGURE 1.1



Illusion of soccer ball size. From http://www.123opticalillusions.com/pages/sphere-size-illusion.php

We can think of the scientific method as one that combines rationality and observation. At its core, scientific knowledge is based on the application of objective and systematic rules to observations or experiments. In experiments, the scientists exercise control over the circumstances under study by careful prearrangement of conditions. In nonexperimental empirical work, the scientists observe, measure, and record but do not control the introduction of the *variables* (when the scientist does control the introduction, these are called *manipulated variables* or *independent variables*).

Let us take a scientific approach to testing the assertion that the top soccer ball in Figure 1.1 is larger than the bottom one. We do not take our observation for granted but instead test it systematically by finding the center of each ball and measuring its circumference with a protractor, a measurement instrument that is objective, unbiased, and not reliant on any person's subjective experience. The measurement

reveals that the balls are of equal size. As good scientists, we then write up our experiment, being careful to fully and accurately describe what we did so that others can repeat our study if they wish, so that the conclusion of every observer will be the same.

The scientific approach to discovering or establishing knowledge reflects a basic skepticism about proof and is characterized by a "show me," "prove it," and "what is the evidence?" attitude about alleged facts. Your truth is just yours until you demonstrate it through means external to yourself and open for scrutiny (and critique) by everyone.

The scientific method also accepts a degree of uncertainty to all its conclusions. Mathematics are used to measure things and statistical tests are applied to results so we can get a sense of how dependable or unreliable the conclusions are. Although any conclusion that scientists reach is never an objective certainty, you can judge how probable it is, through statistical analysis and *replication*, the repeating of the experiment. Even though absolute certainty may be beyond reach, scientists do aspire to attach a level of probabilistic confidence to judgments about what is true. Science encourages an open mind as well as a skeptical mind; implicit in the claim of scientists is the proviso "I may be wrong, but . . ."

The scientific method is the best way to produce knowledge that is unbiased, exists outside any given individual, is open to and testable by others, and has the best chance for permanency and alteration (if the evidence or conditions change). However, it is not the principal way we know things. In fact, if we could count the ways you answered our request for a truth that appeared at the start of this chapter, we would discover that the methods of observation and authority would be the methods used most frequently. Although you may have stated a scientific fact as your truth, in a vast number of instances this fact would be known to you because you read it in a book (or heard it from a science teacher), not because you ran the science experiment yourself. Therefore, the source of your knowledge was an authority, in this case the authority of a science textbook whose contents you trusted. Over time, the written word has grown more and more important in the transmission of knowledge (and falsehoods) and has been joined by video as the way we are most likely to learn about our world. The written word is the principle mechanism for communication of science.

REVERENCE FOR THE WRITTEN WORD

With the advent of the Internet, communicating through writing experienced a revolution unparalleled since the invention of the printing press. The Internet has made it possible for writers to communicate with audiences of enormous size in a matter of seconds. And the cost of communication, the need for writing equipment, and the need for a place to store documents have all diminished greatly.

A few things stand out. First, the ability to write and circulate what is written has been democratized. Anyone with access to a computer and the Internet can do it. But although it is certainly simpler to record ideas and to communicate them to others, this does not mean worthwhile thoughts have become easier to recognize. If anything, the

ability to discern what to believe and what to disregard has grown more difficult and more important. Even videos are open to corruption, through image manipulation and outright lying. On the Internet, you cannot always believe what you read and see.

When almost anyone can disseminate words and thoughts, it becomes critical that you have the skills to separate the wheat from the chaff. This sets the stage for critical reading and makes it a more essential thing for you to do. And because scientific discovery requires an initial skepticism, it is the perfect place to hone your skills as a critical reader.

What Critical Reading Requires

Any critique of what is presented as a scientific study addresses the way the study does or does not meet the scientific standards for evidence and proof. In evaluating such a study, you, as a critical reader, focus on its scientific soundness, not on whether the findings conflict with your preexisting faith, beliefs, or ideas about social acceptability; not on whether the results conflict with expert opinion or clash with other methods of establishing truth and gathering evidence; and not on whether the results are in harmony with your gut feelings. Our soccer ball example in Figure 1.1 shows that one scientist with a protractor can overturn the evidence from our senses or from an otherwise unanimous group.

Even though it cannot provide more than probable truth and must be taken with caution, *empirical* (and particularly experimental) research is still the best way to look for answers to certain types of questions. The emphasis in this book is on the ways an experiment can fail. Scrutinizing and critiquing research is in no way intended as an indictment of the scientific process or as an expression of reservation about its value as a method of seeking the truth. In fact, it is the skepticism of scientists toward their own work that makes their work so valuable.

Doing research and reading research reports critically are arduous processes. It is much easier to say, "This is too hard, let's just brainstorm it and take a vote," or "Let's just ask Dr. Jones, he ought to know." People may become so overwhelmed with all of the difficulties associated with doing research that they detach themselves from it and declare that they are simply not interested in using this method of finding things out. But becoming a skilled reader of research is no different from becoming skilled at anything; it takes time, effort, and practice. Some people dislike the skepticism of science and its assertion that all claims of knowledge are probabilistic; scientists hold assertions with different levels of certainty and might find tomorrow they are wrong. But other people (including you, we hope) would rather hold a true belief with some uncertainty than hold a wrong belief without doubt. Others who have no intention of actually conducting research still feel it is too difficult and too technical for them to even judge by themselves the value of other people's work. They prefer to leave it up to the researchers, who, after all, must know best. They then take, on faith, whatever is printed in the scientific journal. This defeats the whole purpose of presenting research for public scrutiny. It is this attitude we hope to defeat with this book.

The phrase "research shows" is as revered in some circles as pronouncements of authorities are in others. The important thing for the person uttering these words is to have read the research, to have understood it, and to have evaluated it before using its findings to support a position or to inform a practice. When the research is truly sound, however, there is no more powerful way to make a point than to cite research findings.

Critical reading requires a mental set of a particular kind. This mental set can be taught, encouraged, and nurtured. Conversely, it can be discouraged or even forbidden. What is involved is a kind of general open skepticism that enables you to bring a "show me the evidence" attitude toward reading regardless of how authoritative the author may be or how attractively the words are packaged. You, as the critical reader, then engage in an interactive dialogue with the written words. You focus on each assertion that is written and contemplate the meaning behind it. You also think about what was *not* written by the researchers and wonder why it is absent. This cannot be done in haste, and it often requires multiple rereadings and rumination about particularly important or troublesome aspects of the research report.

Critical reading of scientific materials is very different from recreational reading of science fiction or of a novel (although these too can be read critically by literary scholars). When reading scientific documents, every thought that is expressed, every conclusion that is reached, must be screened through your internal skeptical scanner. If everything meets the most stringent cognitive challenges, you come away enriched and gratified. On the other hand, you may accept some of it but have reservations about other parts. A third scenario is that you may reject it in its entirety and resent having spent the time on it. (You can feel gratified, as well, by finding hidden weaknesses in a study, by the way.) None of these scenarios is the same as reading in a passive-receptive mode, with your critical antennae down, and accepting the author's conclusions wholesale.

Uncritical acceptance of conclusions leads to the incorporation of misinformation into your body of knowledge. It is no trivial matter when such material misinforms public policy or becomes a false guide to professional practice. "Research shows" is one of the favorite expressions of experts (a) who are called on by the media to express their professional opinions on a wide range of topics, (b) who are asked to consult with or testify before lawmakers about social issues that affect public welfare, or (c) who are relied on to give counsel to other service providers or to educators.

Research psychologists carry a heavy burden of responsibility for ensuring the accuracy of their claims about their results. In turn, psychologists who cite or apply the research findings of others share that responsibility. They have an obligation to use their critical reading and evaluation skills in reviewing a study before they cite it as evidence that supports a point of view and before they apply the findings in their professional work. Uncritical acceptance of invalid research can impede the development of the field and jeopardize human welfare.

The process of critical reading is far different from that of gaining scientific information from summaries or digests of research results, as busy professionals are often tempted to do. These summaries or digests provide conclusions to be accepted on the reader's faith in the expertise and impartiality of the person who prepared the

summary. But nobody has the time to read all the scientific and professional journals in their field. Digests and abstracts do serve a useful purpose. The best way for you to use them is as a screening device for subject matter that may be of special interest, to be followed by critical reading of the actual articles. There is an exception here. In the relatively recent past there has been an explosion in the amount of scientific research. To accommodate this, researchers have developed a set of rules for accumulating evidence on the same hypothesis (e.g., "What does all the research say on the relative effectiveness of different methods for teaching reading?") in a complete, transparent, and unbiased manner. This approach is referred to as *systematic review, research synthesis*, or *meta-analysis*. We introduce you to these techniques in Chapter 10.

When faced with an original full-length article, experienced critical readers often skim through it quickly so that they can get a general overview of the contents and can decide whether they are sufficiently interested to read it more carefully. If they decide to engage, they then begin to reread the article in a fully-focused and critical mode. Sometimes several readings of crucial sections are required. This is particularly true when the text is fuzzy or lacks detail.

CRITICAL READING

The passive reader is a recipient who takes in what the text says. At the end, the passive reader may have a general impression of the research and some points of agreement or contention. On the other hand, the critical reader is interactive, actively anticipating what is to come. In scientific reading, this anticipation involves the use of rules for conducting valid research. These rules are used like yardsticks that are applied when they are relevant to the narrative about the study. The critical reader then discovers whether these expectancies have been met ("Does the research 'measure up' to this standard?"). Adoption of the critical, interactive mode places you in the best position to evaluate the work critically. Throughout this book, we call attention to the kinds of expectancies that flow naturally from given antecedents. These include the following:

- the research question guides the review of previous literature that sets the context for the scientific report;
- the literature review and the statement of the problem relate clearly to the hypotheses;
- the hypotheses set up research design expectancies and suggest what variables should be manipulated, measured, and/or controlled (so as not to bias the research outcomes);
- the hypotheses, appropriate design, and type of data dictate the method of data analysis; and
- the analysis of the data influences the kinds of conclusions, inferences, and generalizations that can be made.

The critical reader is constantly engaged, checking his or her evolving expectancies against the actual way the study was carried out. Through such interactive reading, the critique, whether negative, praising, or a bit of both, can be focused and justified.

Readers select articles to read because the title or *abstract* suggests it might be of relevance to the answers they seek. Think of a journal article with the title "Effect of Using Phonics and Whole Language Teaching Methods on Reading Ability." You expect to find a cause–effect experiment in which the method of teaching is systematically manipulated and reading ability is compared in groups of children taught by the two different methods. The review of the literature and the statement of the problem reinforce that expectation, and the hypotheses are worded using causal language. Turning to the Method section, you are soon disappointed to find that instead of manipulating teaching methods systematically, the researchers located different elementary school classrooms within a large school system. The method of teaching reading varies from classroom to classroom. You immediately wonder about how similar the various classes of students were when the study began and the equivalence of the skills of the teachers. This means that your expectation that the researchers made sure that the groups were equivalent and that the teachers were comparably expert before the study began was unfulfilled. Proceeding, you anticipate a valid and reliable measure of the dependent variable, reading ability. But instead of finding that standardized reading tests were used, an administrator makes rankings of the various students, and testimonials are given about the accuracy of this ranking procedure. Turning to the analysis of the data, you expect, in view of the *ordinal level of measurement*, that *nonparametric statistics* will be used. Again, your expectancies are not met; the authors use *parametric statistics* instead. Curious about how all these issues are handled in the Discussion section, you find unjustified authoritative pronouncements and unfounded causal statements in support of the hypotheses.

How different is this critical reading process from the passive-receptive mode? In the passive-receptive mode, you might see a scientific research report as excessively detailed and would much prefer to read the abstract and skip the rest. Good "critical" interactive reading does not *require* the uncovering of crucial flaws. A special pleasure and admiration come to you when expectations are met. Indeed, there is little that is more satisfying to the critical reader of science than when expectations are exceeded or when the researchers anticipate and forestall the problems that you have thought about (or even had not considered), when the research procedures not only fit basic requirements (meet the standard on the yardstick) but do so in a particularly clear and original way, where the data are analyzed in an especially convincing manner, where the inferences from the data are unquestionably sound, and where the discussion is impressively profound and thought provoking.

When you critically read sound scientific material, you will say, "I've learned something and will incorporate it into my knowledge base; I'll accept the conclusions and will cite them and apply them when appropriate." When you critically read a flawed study, you cannot accept any of it as true. Exhibit 1.1 summarizes the key characteristics of critical reading.

KNOWLEDGE OF RESEARCH DESIGN

You may sit down in a state of heightened alertness, eager to critique an article of interest. Are you prepared? Do you know what to look for and what to focus your

EXHIBIT 1.1

Key Characteristics of Critical Reading

- Actively anticipate what you expect to read.
- Use explicit rules to judge whether your expectations are met.
- While reading, constantly monitor whether your expectations are being met, not just when you are done reading.
- Have clear and justifiable reasons when you make an evaluative judgement about the strengths and limitations of an article.

attention on? Will you see the defects or be unable to detect them? If you do not have the training to critique the article intelligently, you are bound to have concerns before you begin. You are in a position analogous to that of a person who is contemplating the purchase of a used car. Should faith be placed in the salesperson's appraisal of the car's merits? Do you know how to assess it? Do you know all the potential problems that may be present in the car's mechanical and electrical systems and how to evaluate their presence or absence?

People who lack specialized knowledge can always hire an expert to advise them. For example, you may rely on drama and motion picture reviewers to offer suggestions about what is worth seeing, or you can use websites that accumulate film reviews by experts and untrained movie-goers, a more democratic method of knowing.

PEER REVIEW

When it comes to scientific journal articles, there are usually no reviews to consult, though research that generates attention may have comments written on it that appear with the article later in the same journal or on the blogs of other researchers. Also, articles in some journals have been prescreened by referees and an editor. This is called *peer review*. Peer review can be described as "a formal system whereby a piece of academic work is scrutinized by people who were not involved in its creation but are considered knowledgeable about the subject" (Wager, Godlee, & Jefferson, 2002, p. 3). For master's theses and doctoral dissertations, the committee overseeing the awarding of the degree serves as the peer reviewers.

In a journal that uses peer review, the editor selects reviewers (often two, three, or four) because of their expertise in the substance and methods of the manuscript being evaluated. It is meant to ensure the quality of reports that appear in the scientific record. Peer review is used to evaluate the quality of reports submitted for journal publication, proposals to present at professional meetings, grant applications, reports, and book proposals and manuscripts.

You might think of peer review as the inspection a car dealer says was conducted on a used car before it is offered for sale. Although we do not want to compare science editors with used car dealers, it is important to make the point that such inspections may have occurred but should not lead you to complacency. You need to inspect and test-drive the car yourself.

Also, there are many journals that do not perform peer reviews before they publish articles. The number of these journals is proliferating on the Internet. Be careful here. Your first obligation as a critical reader is to discover whether the journal conducts peer review. Journals also can have more or less stringent peer review criteria before they publish an article. Journals published by the American Psychological Association (APA) have among the strictest requirements for publication. In 2015, the rejection rate of articles submitted for publication in APA (2016) journals averaged 70% and ranged from 45% to 90%. Of course, this rejection rate is based only partly on the quality of the research. It also includes article submissions that were deemed inappropriate for the journal, were not well-written, or do not make enough of a contribution for publication in a prestigious journal. Any rejected article can be revised and submitted to a different journal.

Regardless of how stringent this appears, no article that succeeds in being published is accompanied by a guarantee of excellence. No research is perfect, nor does any research meet all the standards on the yardstick. Flaws and weaknesses appear in all published articles. Critical reading allows you to decide where these flaws are, how serious they are, and whether they lead you to accept or reject the assertions contained in the article. And the existence of flaws should not discourage you regarding the value of conducting and critiquing research. The best-established scientific conclusions occur when evidence accumulates, when multiple studies, none perfect, lead to the same answer.

Any research article has three components: (a) the context and purpose of the research as reflected in the literature review, statement of the problem, and formulation of hypotheses; (b) the research itself; and (c) the interpretation and discussion of the results. A reader who knows much about a topic, but little about research design principles, is not in a good position to evaluate a research article critically. A reader who knows little about the topic, but much about research design, can nonetheless evaluate the design and execution of the research. Such a reader may miss some potentially crucial details regarding the choice and use of specialized apparatus or the selection of the best available measures or most advanced techniques. Generally, however, readers who have knowledge of methods can review material that is within their general sphere of knowledge.

The principles of research design transcend content areas. The effectiveness of a reader is therefore dependent on knowledge of research design and on skillful application of that knowledge. The remaining chapters in Part I discuss those aspects of research design that are essential to know if you are to conduct a comprehensive appraisal of a research article. A systematic approach to the critique of research reports is outlined in the introduction to Part II.

Research Questions and Hypotheses

2

A critical reader of research

- knows the different types of research questions;
- understands that different research questions require different types of research designs to answer them—if the question and design do not correspond, the conclusions of the research are suspect;
- requires that the evidence meet stringent conditions to assert a causal relationship between variables—these conditions are never fully met, just more and less so, and experiments are the best way to test for cause–effect relationships; and
- looks for hypotheses that are (a) clearly and accurately linked to the research problem and its historical context, and (b) stated in operational terms that make clear statements about what the researchers think the is answer to the research question.

In this chapter, we examine the principal types of research questions or problems, give examples of each, and review the design expectations that each type of question establishes. Then we address the complex matter of causation and raise issues that must be resolved whenever the research question involves a causal relationship. Finally, we discuss the importance of defining the terms involved in the research and formulating clear hypotheses.

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Research can be regarded as a process of asking a question (or a related series of questions), and then initiating a systematic investigation to obtain trustworthy answers. When researchers report the research, the research question should be made clear. Research questions are just that: questions. This does not mean that research questions all need to be stated with perfect precision. For example, "What is the relationship between people's religiousness and their satisfaction with their life?" is a fine research question, but so is, "What behaviors are associated with people's sense of well-being?" The first question is subsumed by the second. In both cases, it is clear what question the researchers are trying to answer, more focused in the former example than in the latter.

You will find it disconcerting to read a research report that leaves you feeling the authors may be confused about the question that the research is meant to address. This sometimes leads the researchers to (a) use research procedures that do not seem to be consistent with the question, (b) use procedures that do not test the predictions the researchers have made, and (c) draw conclusions that may not be too closely connected to the original question. For example, researchers interested in behaviors that predict people's sense of well-being might use ethnicity as one predictor, but ethnicity is not a behavior. So, measuring it and drawing conclusions about its relation to people's sense of well-being in a study of behavior and well-being would be problematic, especially if the purpose of the study is to look for things people can change. However, measuring ethnicity would make perfect sense in a study meant to answer the research question "What life circumstances are associated with subjective well-being?" The research question tells you the intent of the study and sets the stage for what is to follow.

Types of Research Questions

There are several types of research questions. You must first understand the characteristics of these types and be able to identify the type of question that is being asked in the study at hand. Different types of questions call for different approaches, or research designs, to seeking answers. To a large measure, the type of question dictates the formal characteristics that are required of the research design. When you read information about the type of question that is being addressed, it sets up your expectations about how the research must be conducted. Having identified the type of question, you can then judge whether the research design is appropriate to a question of that type. You can also tell when a design is simply incapable of providing answers to the type of question that is asked in the research. An example of this kind of disparity would be a study whose question involves a causal relationship but whose design is organized to obtain an estimate of the simple association between the variables of interest. If you, as a critical reader, recognize a discrepancy between the research question and the design, you are on the alert for any erroneous conclusions the researchers might make about causes and effects.

Next, we provide you with a categorization of types of research questions. Other than complexity, you should attach no particular meaning to our order of presentation. All of the questions have a place in science research.

EXISTENCE QUESTIONS

Does *x* exist?

The first question we can ask about something is simply whether it appears in nature. "It" refers here to anything that can be detected by the senses, be it an object or entity, an attribute of an object, an event, a behavior, an ability, a condition, or a state of affairs. Answers to existence questions are important when the existence or nonexistence of something is controversial. These answers become particularly important when some theory rests on the assertion that something exists. The following are existence questions that at one time or another have intrigued psychologists:

- Is there such a thing as the unconscious?
- Can animals use tools to solve problems?
- Can chimpanzees communicate by means of symbols?

It is not necessary to show that the existence of something is widespread. Merely to prove that it occurs, even just once, would be sufficient. Therefore, we would have to produce only a single horse who is clever enough to do arithmetic, for example, to force a major revision of current beliefs. Just this sort of an instance occurred with the famous case of Clever Hans (Pfungst, 1911), who occupies a hallowed niche in the history of psychology. Several illustrious witnesses were deceived into believing a horse had arithmetical prowess. They observed his ability to tap out correct answers to addition problems with his hoof. However, the introduction of simple control procedures (blocking Clever Hans's ability to see nonverbal cues from his handler) shattered the illusion and forced Clever Hans to revert in status from a mathematical celebrity to a normal beast of burden. A horse's ability to do math was not proved to exist, but his ability to read nonverbal cues was. All those who believe that this loss of status affected Clever Hans's self-esteem also need to provide an existence proof.

Answers to existence questions usually require careful scientific work and the application of scientific methods to the study of the evidence, whether that evidence consists of a *single-case design* ($N = 1$, as was the case with the inquiry into the abilities of Clever Hans) or a large *sample*. You should look to see whether the study was designed in a way that systematically rules out rival interpretations of the evidence. Evaluation of the research rests on how comprehensively and effectively this has been accomplished.

In some of the sciences other than psychology, acts of discovery, rather than controlled experiments, serve to demonstrate the existence of things that were not known to exist before. Sometimes such discoveries are accidental, but usually they are the result of planned and informed searches by scientists who have a good idea of what they are looking for and where to look. As an example, astronomers discover that the pattern of known movements of objects in the sky suggests that an astronomical body should exist in a particular place. They then scan the sky for hitherto unseen astronomical bodies (typically with telescopes more powerful than where previously available). Archeologists and paleontologists dig in likely places for specific artifacts that they hope to find. Biologists posit the existence of things (e.g., germs) before they are ever seen.

DESCRIPTION AND CLASSIFICATION QUESTIONS

What are the characteristics of *x*?
To what extent does *x* exist?
Does *x* take on multiple values or is it invariant?
What are the limits of *x*?
Is *x* unique or does it belong to a known class (taxonomy) of things?

When scientists accept an affirmative answer to the question about whether *x* exists, the ensuing questions ask about the description and classification of *x*. Before the expeditions to the moon, the composition of its surface was actively debated. Samples of rocks that were brought back proved that rocks, among other things, existed there. The rocks were then extensively studied in an effort to describe and classify them. This answered questions about whether they shared characteristics with earth rocks and whether they possessed any unique characteristics. Some examples of questions of description and classification in psychology would be the following:

- What are the personality characteristics of salespeople who are successful?
- What are the child-rearing practices of mothers who are addicted to drugs?

Survey research primarily describes, counts, and classifies. Therefore, a survey might find that "85% of successful salespeople score high in sociability." When surveys are designed to yield conclusive and generalizable knowledge, you have reason to expect a research design has been used that ensures the description it produces can legitimately extend beyond the people in the study (the sample) to the *population* or *subpopulation* of people who are of interest (here, all successful salespeople).

COMPOSITION QUESTIONS

What are the components that make up *x*?

Answers to questions of composition call for an analysis or breakdown of a whole into its component parts. In psychology, some examples of research questions involving analyses of components include the following:

- What are the principal components of personality?
- What are the main factors that make up self-esteem?
- Is intelligence a single construct, or does it have multiple independent factors?

Conversely, researchers may begin with several small components, determine which ones hang together to make up an identifiable larger whole, called a *factor*, and ascertain whether the various factors combine to form a larger construct. This type of work is epitomized by pioneers of intelligence such as Spearman (1927), Thurstone (1931), and Cattell (1952). They used a sophisticated analytic technique called *factor analysis* to help them discern which cognitive abilities, such as verbal and quantitative, were associated with one another and which were distinct. In that way, they

could isolate the different components of intelligence. Because factor analysis is a mathematical procedure, the critical reader of research on the composition of intelligence expects that (a) care has been exercised to assure the accuracy of an large number of computations; (b) the sample of people to which the factor analysis is applied will be large and representative enough for the procedures and conclusions to be valid; (c) *experimenter bias*, the preconceptions of the experimenters, played no part in the identification and naming of factors; and (d) the individual components (in this case, the items in the pool of questions and activities meant to measure intelligence) are well-constructed and are comprehensive in their representation of the various aspects of the construct under investigation.

DESCRIPTIVE–COMPARATIVE QUESTIONS

Is Group *x* different from Group *y*?

Answers to many research questions call for more than simple descriptions. They require (a) statements about the general applicability of the description to the sample of people in the study (e.g., of all the successful salespersons, how many had these characteristics?), and (b) statements about the uniqueness of the description to the people (the population or subpopulation) that interest researchers (e.g., did unsuccessful salespeople also have these characteristics?). These statements turn descriptive questions into descriptive-comparative questions. Examples of descriptive–comparative questions in psychology include the following:

- Are men more aggressive than women?
- Are younger people more liberal than older people?
- Are wealthy people happier than poor people?

The descriptive–comparative question is an elaboration of the simple descriptive question. For these questions the researchers intend to compare two or more preexisting groups. The defining characteristic of the groups may be some *organismic variable* (e.g., sex, age, weight) or an *attribute variable* (e.g., socioeconomic status, educational level). These are characteristics that the researchers can identify and describe but cannot influence the values people take on of the characteristic. Clearly, the researchers cannot influence a person's age. Surveys can ask not only, "How liberal are people?" (a descriptive question) but also, "Who is more liberal, young or old people?" Political surveys often ask these comparative questions.

When examining the effort of researchers to determine whether the young are more liberal than those long-of-tooth and to assure that their age accounts for the difference, you must anticipate that the groups of young and old being compared are equivalent in as many other ways as possible (e.g., having equal numbers of men and women, wealthy and poor). You must also expect that if any conclusions are drawn about age and political orientation, the sample of people studied is taken from the population of interest. Your critical assessments about whether the measures measure what they claim (*construct validity*) and whether they do some consistently (*reliability*)

also must be considered before you decide whether the conclusion is to be deemed trustworthy.

When descriptive–comparative questions are posed, researchers must also say whether the description is unique to the subpopulation under study. Therefore, the critical reader of a study designed to answer a question about the child-rearing practices of mothers who are addicted to drugs would expect researchers to show that (a) the sample from which the description is generated is truly representative of mothers who are addicted to drugs, and (b) the observed methods of child-rearing are unique or distinctive to mothers who are addicted to drugs and does not also describe the child-rearing practices of mothers with other kinds of disorders, or, indeed, of mothers in general. Lacking statements about commonality and distinctiveness, the description would be incomplete from a critical assessment point of view and potentially misleading.

RELATIONSHIP QUESTIONS

Is there a relationship or association between *x* and *y*?

Relationship questions are a special type of comparative question. Relationship questions can compare two groups (e.g., the previous child-rearing example comparing mothers who are addicted to drugs with those who are not addicted to drugs), but they address variables that are continuous in nature as well. They are also typically answered with statistical calculations, most simply with *correlation coefficients*. Examples from psychology include the following:

- Is happiness related to income?
- Is there an association between time spent studying in college and college grades?

If we change young versus old to "age" in the previous example of whether young people are more liberal than old people, we have changed this into a *continuous variable*, and a correlation coefficient can be used to answer the question about its association with political leanings (rather than a comparison of the mean "liberalness" for young and old).

More complex questions of relationships among several variables also can be asked. So, we could add other variables to the question about study time in college and grades, say, class attendance and peer relationships, and broaden this question to, "What variables are associated with grades in college?" Using *multiple regression* techniques, we can ask questions about whether several variables individually and collectively predict some dependent variable, and what is the relative contribution of each. The research question may concern the practical association of the group of predictor variables (studying, attendance, peers) with the criterion (grades) or may ask whether the patterns of intercorrelations fit a proposed theoretical model, say about the importance of peers for success in college. In these cases, again, you expect that (a) researchers have used valid and reliable measures, (b) the sample is representative and of sufficient size for the number of variables under investigation, (c) the computations are accurate, and (d) interpretations do not go beyond the data by making insupportable statements about causality.

CAUSAL QUESTIONS

Does *x* produce, lead to, or prevent changes in *y*?

Questions about *cause* are different from questions about relationships. Causality implies production (i.e., one variable plays a role in generating the other), which is more than just association. Two things can be associated (e.g., housing prices in Seattle and crime in Detroit), though they are not causal of one another (both sharing the cause of United States economic conditions). Causal questions do not have to specify a single cause and effect. The questions "What is/are the consequence(s) of *x*?" and "What is/are the cause(s) of *y*?" are legitimate questions. Sometimes research of this type is called *exploratory research* because of its imprecise nature. But usually the researchers may make informed guesses, which focus the investigation and become the research hypotheses. Examples of causal questions in psychology include the following:

- Does smoking marijuana reduce epileptic seizures?
- Does playing violent video games make children more aggressive?
- Does psychotherapy change behavior (an exploratory question because the behaviors are not specified)?

Causal questions call for experimental research in which the experimenters manipulate (control the introduction of) the independent variable, treatment, or *intervention*.[1] Here, the researchers create a set of conditions under which the research subjects will be asked to behave. Or the researchers find and use a set of conditions that have been fortuitously manipulated by nature or circumstances. For example, the introduction of a new reading curriculum at a school would be a *naturally occurring experimental manipulation*.

Regardless of how the experimental condition came about, the experimenters then contrast the consequences of the manipulation with observations taken on a *control group* condition not exposed to the manipulation. Seeing that a causal question has been asked, you should anticipate that the experimenters (a) meticulously assigned individuals (in the best case, using *random assignment* or chance) to the treatment or no-treatment condition, and (b) controlled as many extraneous variables as possible from influencing the experimental and control conditions in different ways. These conditions help you rule out anything else besides the manipulation that might affect the results. Without these two criteria being met, there is little difference between research meant to be experimental, with the controlled introduction of the independent variable, and research in which the variables being related to one another have simply been measured without control by the researchers (relationship research). For example, if experimenters manipulate two video game conditions with more and less violent games, but children get to choose which video condition they

[1]The term *independent variable* is typically used in laboratory studies and ones that explore basic processes, whereas *treatment variable* or *treatment* is used in clinical studies and *intervention* is used in other applied settings. Because our discussion is relevant to all these situations, we use them interchangeably here. More on this in Chapter 3.

are in, you cannot tell if the games made them more aggressive or if the children who choose violent games were more aggressive at the beginning of the study. If the researchers do not control where the games are played or how loud their volume is turned up, these extraneous variables might cause their effect on aggression. In both cases, we might know that the games are associated with more aggression but we cannot assign causality. Finally, as with other research questions, causal research also must demonstrate that it (a) manipulated what the researchers intended to manipulate, (b) removed possible experimenter bias, (c) used valid and reliable measures of the effect (e.g., aggressiveness by the children after exposure to the games), and (d) analyzed the data accurately.

When causal studies are done in the form of single-case designs (e.g., one child is measured for aggressiveness before and after playing violent video games), you should expect to find that the researchers (a) used care in the application and timing of the experimental conditions, (b) accurately measured or judged the child's behavior before and after experiencing the experimental treatment, and (c) protected against experimenter bias when analyzing and interpreting the results. Also, you hope enough replications exist to warrant generalizations or that the researchers caution against any claims about generality.

CAUSAL–COMPARATIVE QUESTIONS

Does *x* cause more change in *y* than does *z*?

Causal questions are comparative in the sense that the effects of *x* must be compared with non-*x* (the absence of *x*). In causal–comparative research, the effects of *x* are compared with a rival manipulation or treatment and not simply with the absence of the experimental manipulation. Examples of causal–comparative questions in psychology include the following:

- Is studying for a test alone more effective for improving grades than studying in groups?
- Are antidepressant drugs more effective than psychotherapy in decreasing depression?
- Are face-to-face job interviews better than online interviews for making successful hires?

All the things that are expected of research on causal questions apply to causal–comparative questions, with the additional provision that the second experimental manipulation must also be valid and must be introduced in an unbiased manner.

CAUSAL–COMPARATIVE INTERACTION QUESTIONS

Does *x* cause more change in *y* than does *z* under certain conditions but not under other conditions?

Causal–comparative *interaction* questions are just elaborations of causal–comparative questions. When two or more independent variables are manipulated in a study, the

researchers can determine whether the independent variables have different causal effects depending on the levels of the other independent variables. The variables can be preexisting characteristics of the experiment's *subjects* or participants, environmental conditions, time or order factors, and so on. Examples from psychology include the following:

- Is studying alone more effective for humanities classes but studying in groups more effective for science classes?
- Is a certain medication more effective than psychotherapy in treating depression among men than women?
- Are face-to-face job interviews better than online interviews for making successful hiring decisions depending on the type of position being filled?

You should expect the same standards be met for causal–comparative interaction research as for causal–comparative research. However, you should pay additional attention to the careful application of the added independent variables.

Hypotheses

Experimental hypotheses are predictive statements about the expected outcome of research. We provided many examples of research questions previously. Hypotheses are predictions about the answers to the questions. So, the research question "Does playing violent video games make children more aggressive?" can be stated as the hypothesis "Violent video games will make children who play them more aggressive than children who do not play them." Hypotheses call for a test and they embed a conclusion. They dictate the method and design of the research, and they should give you a good idea about what the research design of the study will have to look like to test it. For example, in the video game hypothesis, an experiment with random assignment is called for because the hypothesis implies a causal relationship.

Explicit statements of experimental hypotheses are a must in master's theses and doctoral dissertations, but they are often omitted in more succinctly written journal articles. When this is the case, the summary of the theoretical basis for the study, the overview of research literature that preceded it, the synthesis of these materials, the statement of the problem, and the reason for the research to follow should leave little doubt in your mind about what the author's predictions are concerning the results of the study (assuming there are predictions).

When comparisons are predicted, they must be clearly explicated. For example, in a study that has boys and girls doing mental tasks under conditions that include events meant to distract them, the hypothesis might state, "Girls will score higher under the distraction condition." This hypothesis is imprecise. Does it mean that girls score higher than boys when they are both distracted? Or does it mean that girls score higher when distracted than when not distracted? The prediction "higher than" requires that a comparison of two conditions be clearly identified for it to acquire meaning.

NULL AND EXPERIMENTAL HYPOTHESES

A *null hypothesis* predicts that no relation exists; for example, "Boys and girls do not differ in how distractible they are." In contrast to the null hypothesis (often labeled H0), experimental hypotheses (H1, H2, H3, etc.) take a stand. They predict, for example, that groups assigned to different treatments will perform differently (two-tailed prediction), or they may predict what direction the expected differences will take (*one-tailed test* or prediction). When researchers do statistical tests, what they do is ask how likely it is that the null hypothesis is true. That is why these tests are referred to as *null hypothesis significance tests*. If the null hypothesis is rejected (typically because its likelihood of being true is less than 1 in 20, or $p < .05$), the direction of the group means must be in the predicted direction (if this was specified) for the experimental hypotheses to gain support.

What Is Causation?

Having phrased some of the research questions in terms of cause and effect, we need to clarify how this term is used here and throughout the text.

In common usage, *Merriam-Webster* (n.d.) defines cause as "something or someone that produces an effect, result, or condition; something or someone that makes something happen or exist." An effect is defined as "a change that results when something is done or happens; an event, condition, or state of affairs that is produced by a cause." The definitions are circular; each definition uses the other term to define it.[2]

Given the circularity in definitions of cause and effect, it should not be surprising that there is lots of debate among philosophers over the nature of causality. The history of the debate about the meaning of causality can be traced back to Aristotle's (384 BCE–322 BCE) concepts of formal, material, efficient, and final cause, through the views of David Hume (1711–1776), Emanuel Kant (1724–1804), and John Stuart Mill (1806–1897), and up to the philosophers of the modern era, such as Bertrand Russell (1872–1970). Brief treatments of all these philosophers can be found in the *Stanford Encyclopedia of Philosophy* (Zalta, Nodelman, & Allen, 2016).

Philosophers have argued for millennia over the meaning and, indeed, the value, of the notion of causality. The Scottish philosopher David Hume (see *A Treatise on Human Nature*, 1739–1740/1978) stated the dilemma nicely (and, some would say, led social scientists astray; see Maxwell, 2004). He argued that for something to be considered a cause it had to meet three conditions: (a) the cause had to occur before the effect (even if by only a nanosecond), (b) the cause and the effect had to happen

[2]*Merriam-Webster* also differentiates between a cause and a reason. A *reason* (Merriam-Webster, n.d.) is defined as "a statement or fact that explains why something is the way it is, why someone does, thinks, or says something, or why someone behaves a certain way." For example, the cause of a car accident might be the failure of the brakes, whereas the reason is negligence on the part of the owner for not having had the brakes inspected. Thus, a cause for one person may be considered a reason by someone else.

together (though the interval between them could be long), and (c) there must be a "necessary connection" between the two events.

Hume's first two conditions are relatively easy to meet. First, the temporal sequence of events is typically observable with a high degree of reliability, regardless of how short the interval is and even if sometimes events of interest to psychologists are caused by the anticipation of other events (in this case, anticipation is the cause).

Second, researchers need to show that events *x* and *y* co-occur more often than would be expected by chance. They need to show this because sometimes the role of chance makes things look like they are appearing together when this is purely fortuitous. Many superstitious behaviors are based on chance relationships between events. This implies that we can never make the determination of causal relationships between things that happen together with absolute certainty, hence our need for the use of statistical probabilities in research.

Hume's third condition presents the greatest challenge to researchers. He argued that people can never see one event produce another, so we can never know with perfect certainty that the event we are calling the cause was a sufficient or necessary condition for the effect to come about. A thought experiment will help clarify his claim. Suppose we place a pair of spectacles near the edge of a desk and one of our elbows slides into a book that then knocks the eyeglasses to the floor and they break. What caused the glasses to break? If we were to ask a group of people who all watched the same event to independently identify the cause of the fall, we would be confronted with multiple nominations. Most observers would say "the book" or "the elbow" but the more playful in the group might nominate "a gust of air just prior to when the book hit the glasses" or even perhaps "a poltergeist." Are they wrong? Probably, but can you prove these explanations' falsity conclusively?

Hume asserted that events happen in an unending flow. Designating where one event ends and the next begins is subjective, that is, it requires an a prior theory or perspective that is supplied by the observer. Therefore, he claimed, whenever we identify a cause it remains possible to argue for other causes in two ways.

First, we can identify another event that takes place between the asserted cause and effect. So, if we claim the elbow caused the spectacles to fall, you can counterclaim that it was the book that provided the necessary connection. Yet another observer, perhaps a physicist could claim gravity was the cause. After all, without gravity, the eyeglasses would have remained suspended in air. Events preceding the errant elbow might also be viable alternate nominations—it was a writing deadline that caused the elbow to be at the desk.

Second, Hume argued that causal systems are open to outside influences. That is, an outside event can enter the claimed causal sequence. The "gust of air" or "poltergeist" explanation for the spectacles' spectacle would be outside an elbow-book-gravity-crash causal system. Could an invisible mischievous spirit have knocked the eyeglasses just a nanosecond before the book hit them?

Let us look at a more complex example to illustrate some of the problems in pinning causation down. The following allegory is intentionally as anthropomorphic as it can be. During a heated argument, John takes out a pistol and fatally shoots Jack. The

coroner's report states the cause of death as "Gunshot wound caused by 8-mm bullet that tore into the carotid artery causing massive bleeding that deprived the heart and brain of oxygen until they ceased to function." The heart says, "That artery was the cause of it. It stopped feeding me so that I could no longer do my job." The artery says, "It wasn't my fault, that bullet tore into me so that I could no longer function." "Don't put the blame on me," says the bullet, "I was resting peacefully in my chamber when the powder behind me exploded and sent me headlong into the artery." "I'm not the cause," says the powder, "I was quietly sitting there when that firing pin slammed into the casing and caused me to explode." "I wouldn't have moved," says the firing pin, "if that finger hadn't pulled the trigger which released the spring that drove me forward into the shell casing. John is the cause. His finger pulled the trigger, and he is the only one in this episode who acted with intention." John admits that he pulled the trigger, but he says, "I'm not to blame because, as my therapist explained to me, I am a very angry person and have difficulty controlling my impulses. The cause is that my father, who was an alcoholic, left home when I was 3 years old, and my mother had to bring me up all alone. She had to work hard for a living and was rarely at home to raise me properly."

Parts of this causal chain may sound familiar; it embodies the social policy debate over preventing gun violence. At the extremes, it includes the arguments of those who want to make guns harder to obtain because they believe the availability of firearms is the cause of gun violence. It also includes the argument of those who say misuse of guns is more a result of the user's mental health. Although the debate is generally more complicated than this example, for our purposes it illustrates Hume's third point by showing that debaters are focusing on different points in the casual chain at which to intervene. (You could even argue that guns do not kill people, bullets do.)

Necessary and Sufficient Causes

Another unresolved issue concerns the necessity and sufficiency of causes. The necessity argument states that the cause must be present for the connection to hold; the effect cannot happen unless the cause is there, but the effect might not happen even if the cause is there because other conditions must be present as well. For example, you must press the power button on your cell phone to get it to come on. But, it will not come on if your battery is dead. Pushing the button and having a live battery are necessary conditions to cause the phone to come on.

Sufficiency of cause is even more restrictive. It argues that when the cause is present, the effect will always happen. In our example of the cell phone, neither the button push nor the charged battery are sufficient of themselves to make your phone come on. In the social and behavioral sciences, we hold, quite reasonably, the view generally regarded as the plurality of causes: Psychological and social events are the result of multiple necessary causes being present, none of which is sufficient to make the effect happen without other conditions being met. We rarely talk about *the*

cause of an event; instead, we talk about *a* cause, perhaps one of many. John Stuart Mill maintained that events happened because of a whole set of conditions, some of which are contextual variables. We might refer to these as *enabling conditions*. They all have to be present for an event to occur. Social scientists have long recognized the plurality of causes.

A Working Definition of Causation

If we accept Hume's argument that we never actually see one event cause another, and we never know causes with complete certainty, then how are we to proceed to answer the question "What events cause other events to happen?" Shadish and Sullivan (2012) presented the strategies that three influential thinkers—drawn from social science, statistics, and computer science—have offered for how evidence can be mapped onto causal claims so as to make the claim more or less plausible. The most influential of these strategies was first proposed by Campbell and Stanley (1963), then modified by Cook and Campbell (1979) and Shadish, Cook, and Campbell (2002). In the following discussion, we also draw on the work of Simon (1970) to put together our working definition.

When social scientists talk of one event causing another, they typically mean that six conditions have been met:

1. *The cause precedes the effect:* This condition is straightforward. The length of time between the cause and effect can be very short, and the anticipation of an event can be viewed as a legitimate cause (especially in psychology).
2. *Many alternate explanations have been ruled out:* Alternate explanations are sometimes called *third factors* or *plausible rival hypotheses*. Suppose someone claims, "My elbow knocked my eyeglasses to the floor." You say, "But a gust of wind might have knocked the eyeglasses off the table just before your elbow hit it." The person counters with, "All of the windows in the room were shut." This would have ruled out your alternate explanation. The more of these alternative explanations that are anticipated and ruled out, the more credible is the claim to have identified the cause. Campbell and Stanley (1963) referred to this second criterion in our working definition as the *internal validity* of causal claims.
3. *The relation must hold with few side conditions:* This third consideration means that the cause–effect relationship must hold over a wide range of circumstances, it must have *generalizability*. Suppose we made the claim "Warm weather causes a decline in attendance in college classes." You might agree at first, reasoning that warm weather creates other attractive alternatives for how students might spend their time. But, what if we then said, "However, this relationship only appears in colleges in relatively colder climates, during fall semester (when warmer weather and finals do not co-occur) and only in classes of certain subjects." These side conditions would undermine your certainty that our causal statement was correct often enough to warrant the claim. Campbell and Stanley (1963) referred to this as the *external validity* of the claim.

4. *The variables being measured are good reflections of, or capture adequately, the underlying cause and effect:* This condition raises the issue of needing measurements of the cause and effect variables that merit our trust. Suppose we make the claim that "an impending snow storm causes grocery sales to increase." We measure the former, the proposed cause, by selecting some northern cities, recording whether the weather service's December forecasts predicted snow for the next day, and defining a snow storm as impending if four or more inches of snow are forecast to fall. We also gather sales tax receipts from the cities' grocery stores for each day in December. If we find the proposed relationship (i.e., higher grocery store sales on the days snow is forecast for the next day), you are likely to buy our causal statement (pun intended) because our indicator of cause (i.e., the forecast) seems to be a good reflection of "impending snow" and "grocery sales" seems to be a good reflection of the effect (in fact, a very good reflection). Cook and Campbell (1979) referred to this as the *construct validity* of causes and effects.
5. *The statistical association between cause and effect is credible and substantial*: To support a causal claim, the statistical relationship between the cause and effect variables must be calculated properly. This means that the correct measure or test of association is used and its *assumptions* are met. It also means that the relationship is strong enough to be of interest. If we say, "Talking on a cell phone causes car accidents" but then say, "The correlation is $r = .0001$," you are less likely to accept our causal claim than if we say, "The correlation is $r = .5$." We use null hypothesis inference testing to show that an $r = .00$ relationship between variables is unlikely, the first step in establishing what Cook and Campbell (1979) called *statistical conclusion validity*.
6. *The cause and effect relationship is deduced from or compatible with a general framework or theory*: Simon (1970) pointed out that causal claims are more credible if they fit into broader explanations. These can be theories. John's explanation of why he shot Jack is more plausible because psychological theories exist that suggest a lack of positive role models during a child's formative years causes antisocial behavior. There are lots of other instances, not just murder, in which this theory has proven predictive.

In sum, as a good critical thinker, you should ask if the evidence about a proposed causal relationship answers six questions in the affirmative:

- Did the cause precede the effect?
- Were other plausible alternative explanations ruled out?
- Is there evidence that the relationship holds under many varying conditions?
- Were the measured variables closely tied to the variables they were meant to measure?
- Was the statistical evidence of a relationship properly calculated and strong enough to be of interest to us?
- Is the relationship consistent with a theory or theories that already has shown an ability to make causal predictions?

Causality and Experiments

Experiments in which conditions are manipulated by the researchers are the best way to establish causal precedence. When designing an experiment, there is no doubt about what the causal agent is that is being tested; it is the conditions manipulated in the experiment. Researchers control the experimental conditions and introduce them in advance of the presumed effects, meeting the requirement of temporal sequence (Condition 1).

If random assignment of participants to conditions is used, the researchers have gone a long way toward establishing that alternative explanations cannot explain the result (Condition 2), because the groups are as similar as possible to one another, on average. For example, when randomly assigned, there should be roughly equal numbers of men and women in the *experimental group* and *control group* and the two groups should average roughly similar ages. This rules out these two individual differences as third variables that might explain any differences found between the groups on the dependent variable. All other individual differences, ones the researchers might not have considered, are similarly controlled through random assignment (see Chapter 5).

Also, researchers must treat the participants in the experimental and control conditions as similarly as possible in all respects except for the manipulation. If not, any other way that the groups are treated differently (e.g., different characteristics of research assistants administering the different conditions) might be the cause (Condition 2, as well).

That said, researchers also must concern themselves with contextual variables (or *moderator variables*) that might limit generalizability (Condition 3). For example, if a study of impression formation is conducted in a psychology laboratory, the results might not be generalizable to a barroom. Researchers also recognize that *intervening variables* (or *mediator variables*) are affected by the treatment that, in turn, bring about changes in the effect. In our optical debacle example, the elbow immediately affects the book (the mediator) which in turn affects the eyeglasses.

The researchers also must measure the dependent variable in a way that gauges reactions to the manipulation with fidelity and sensitivity (Condition 4). If you are suspicious that the measurement may be measuring something different from what is claimed, the construct validity of the claim about what is being effected is cast in doubt.

Researchers in psychology also accept the idea of multiple, contributory causes rather than single, unitary causes. They do not expect the manipulation or treatment to influence every participant in the same way. Some will not react to the manipulation at all. Some will react more strongly than others. And in the control condition some participants will have scores on the dependent variable (the effect) that fall within the range of scores in the experimental condition. For example, in a study of the effect of violent video games on aggressiveness, there will be some children who are highly aggressive who get randomly assigned to the control condition, and who are still more aggressive than some children even after those children have played the violent games. A basic truth of the human condition is that behavior is multidetermined

and people react differently to the same stimulus. Therefore, we must rely on statistics in research (Condition 5) to help us distinguish the signal from the noise.

Unsubstantiated Claims of Causation

Somewhat easier than making assertive claims of causation is the ability to dismiss false claims of causation from studies that do not provide evidence for it. In critically reading research, your emphasis is not so much on proof of cause as it is on rejecting false or shaky claims of causation.

The reason for this is simple. Research is used to help us understand our world better and to inform public decision making. It is used to help people form attitudes and opinions, and to influence social action. Researchers bear a professional and social responsibility to do research that supports their claims and to report it accurately. Journalists and science writers who reconstruct and transmit the findings to the public have the responsibility to understand well how research is designed, to summarize it competently, and to interpret it accurately.

You have only to read the nearly daily news reports of scientific studies to see evidence of confusion between finding evidence of a relationship and evidence of a causal connection. As a critical reader, you must guard against accepting the false notion that if one event preceded another, it must have caused it to happen (our friends studying logical errors call this *post hoc ergo propter hoc*, which translates as "after this, therefore because of this").

Reports of nonexperimental medical studies have become routine. Many of these are *epidemiological studies*. They are based exclusively on population statistics, not on demonstrations of the full complement of conditions needed to make claims of causality. They are correlational in nature and find associations of interest, but they can give the public a false idea about causality. For example, one day the public is informed that the rate of arteriosclerosis is comparatively low in countries where there is high wine consumption and that therefore wine prevents arteriosclerosis. The next day an article explains that rice consumption prevents heart attacks in countries where rice consumption is high. Then, a correlation is presented between broccoli consumption and low colon cancer rates. Thousands of people are influenced to protect themselves from all three ailments by gorging on diets of rice and broccoli in wine sauce.

The frequent logical error of post hoc ergo propter hoc is based on the correct antecedent–consequence time sequence (Condition 1), but the "rule out third variables" condition for causality (Condition 2) is not demonstrated. It is a fallacy to think that a happening that follows another must be the result of it, and you need to be on the lookout for authors who fall into this trap.

Multiple Correlation and Causation

Correlations between two variables show the relationship or association between them and do not imply that one is the cause of the other. Multiple regressions show the associations between two or more independent or predictor variables and a

dependent or outcome variable. You cannot say that the former cause the latter to happen. With the development of *structural equation modeling*—an application of multiple regression and factor analysis combined—you can construct models of multiple hypothesized causal relationships. The inferences can be of great theoretical interest, but as Tabachnick and Fidell (2001) cautioned, the demonstration of causality is a logical and experimental, rather than statistical, problem:

> One of the few considerations not relevant to the choice of statistical technique is whether the data are experimental or correlational. The statistical methods "work" whether the researcher manipulated the IV. But attribution of causality to results is crucially affected by the experimental-nonexperimental distinction. (p. 3)

STATING THE HYPOTHESIS FOR DIFFERENT RESEARCH QUESTIONS

The generic hypothesis about concomitant variation ($y = fx$) states, "Dependent variable y is a direct (or inverse) function of independent variable x." For example, as gun ownership in a community increases, the number of shootings increases. As x increases, y increases; as x decreases, y decreases. The inverse refers to y increasing when x decreases to yield a negative correlation. For example, as the number of gun owners in a community increases, the number of home invasions decreases. You should check carefully that, in noncausal studies in which only relationships or associations are predicted, the hypotheses should be stated in these relationship terms and not promise more than the design can deliver.

Comparative questions call for hypotheses such as, "Other things being equal, Group A will score higher on y (the dependent variable) than will Group B." Usually the phrase "other things being equal" is assumed rather than stated. In studies in which preexisting groups are compared, the assumption of group similarity is often more of a hope than a reality.

For cause–effect experiments, the hypothesis for the simplest cases is, "(Other things being equal) the mean score of the Experimental Group A will be higher (or lower) on the dependent variable measure y than will be the mean score of the untreated group." For example, a community-based gun buyback program caused the number of shootings to decrease. You should evaluate whether the experimental challenge to make "other things" as equal as possible has been met.

The hypothesis would become even clearer if the participants, treatments, time, and setting (the who, what, when, and where) were specified (e.g., "At posttest, middle-age participants randomly assigned to a condition in which they attempted unsolvable problems with a gun visible will express less aggression than similar participants assigned similar tasks but with no gun present"). Expectations have been established. Armed with your knowledge of research design and critical reading skills, you can now look forward to seeing how well the house is built. At the end, when viewing the conclusions, you can look backward to see how well they match the predictions that were made at the beginning.

HYPOTHESES AND THEORIES

Diffen (2016) defined a *theory* as "a well-substantiated, unifying explanation for a set of verified, proven hypotheses" (p. 1). Therefore, theories are broader than hypotheses and can overarch many of them. Hypotheses and theories are testable (and therefore falsifiable). However, a hypothesis is typically "based on a very specific observation and is limited to that instance" (Diffen, 2016, p. 1), whereas a theory "is the establishment of a general principle through multiple tests and experiments, and this principle may apply to various specific instances" (p. 1).

Some hypotheses spring from experiential observation; they are not offspring of any formal theory. Therefore, we might observe that children on a playground appear more likely to get into fights after playing violent video games. You cannot help wondering where hypotheses came from and where they lead. Will the results of a test of such a hypothesis merely be added to a collection of isolated and homeless facts, each of which has yet to be placed in a small chamber within the home of some theory? More fortunate hypotheses are deduced from theories. They benefit when their broader and more general origins are explained to the reader. So, we might predict greater aggression after playing video games based on a more general theory that rests on the notions of "desensitization" or "heighten arousal." Or we might predict less aggression on the basis of a theory that rests on "catharsis." If we want to test the theories against one another, we would propose a two-tailed hypothesis (violent video games will affect aggression), and use the rejection of the null hypothesis and direction of the finding to decide which theory is supported (Condition 6).

As a critical reader, you are well-served when authors situate hypotheses in these more general, theoretical frameworks. These theoretically derived hypotheses, the summary of the literature on which they are based, and the relevant findings from earlier research form the foundation on which the new study is to be secured. The experimental hypothesis can then be stated in its more highly specific way within this broader context.

Consistency Between the Parts of a Scientific Investigation

Consistency between the research question, the hypotheses, the manipulations (if any), the measurements, the design, the analysis, and the conclusions is something that a critical reader expects to find in good research. You should evaluate whether the variables you encounter in the description of the problem are clearly reflected in the statement of the hypothesis and in the variables that were measured or manipulated. You decide whether the research design allows the researchers to make the inferences they want to make, whether the statistical procedures were appropriate for the design and measurements, and whether the final conclusions were properly drawn from the evidence. If a group of researchers of depression use only outpatients who were treated in the study (there is no control group) and then correlate Depression

Scale scores with time in treatment, the design is not consistent with the causal hypothesis and does not test it. Instead, the study as it was conducted addresses the question of whether there is an association between length of time in treatment and depression. This may be a worthwhile question, but it is not the one you were expecting to be tested.

If there is more than one independent variable, additional hypotheses (H1, H2, H3, etc.), including hypotheses predicting interactions between independent variables, need to be stated, if they are anticipated. A research report is always suspect when the authors conclude that the hypothesis was supported even though the hypothesis was not stated and cannot easily be inferred from the material in the report's introduction. Your suspicions are compounded if the conclusions are based on a one-tailed test of significance, which is predicated on the preexistence of a directional hypothesis.

Summary

In this chapter, we examined the different types of research questions: existence, description and classification, composition, relationships, descriptive–comparative, causal, causal–comparative, and causal–comparative interaction questions. We paid particular attention to how we establish causal connections in scientific research and how to spot errors in logic that lead to false claims of cause and effect.

Clear definitions of terms and unambiguous hypotheses give you an understanding of precisely what the researchers aim to accomplish. They also set up expectations about how the study should be organized to test the predictions and answer the research questions. Consistency between the questions, the hypotheses, the design, the analyses, and the conclusions is essential if a scientific investigation is to contribute to knowledge.

Research Variables and Strategies

3

A critical reader of research

- knows the definitions of variables used in research, especially the difference between experimental (manipulated) and measured variables;
- understands that causal claims in research can only be made in studies that manipulate the variable presumed to be the cause;
- can identify the different ways measurement levels can be defined and their implications for the kinds of inferences researchers draw;
- knows how studies vary in time sequencing (prospective or retrospective, longitudinal or cross-sectional), actuality (simulations, real-life), and setting (laboratory, field); and
- can evaluate the likely generality of research results across people, researchers, settings, times, and procedures.

Once you know what the research questions are going to be and what the predictions (hypotheses) are, you next examine the researchers' choices of a number of important data collection strategies. These strategies will determine whether the study can and does answer the research questions. This chapter provides a discussion of the first set of these choices and strategies. The issue of generalizability, which is partly dependent on the decisions that are described here, also is discussed.

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The initial set of decisions that researchers make involve the following considerations:

1. The research variables: Does the research involve (a) a variable (or variables) that is manipulated by the experimenters or (b) naturally occurring measures (i.e., measures simply collected by the researchers)? If so, what levels of the variable were chosen or measured by the researcher? How many different values can the variables take on?
2. The timing sequence: Is it a *prospective study* (i.e., researchers predict the value of a variable in the future from variables measured now or in the past) or a *retrospective study* (i.e., researchers infer the value of a variable in the past from variables measured now)?
3. The setting of the study: Does the study take place in an actual situation or is it simulated? Is the study done in the laboratory or in the field?

Variable Definitions in Research

To understand the research question and the details of the study, you must clearly comprehend the meaning of the terms that are used. Scientists are forever coming up with new concepts that do not yet have a name. Vocabularies must be invented so that ideas can be communicated. The terms used in research can be new, or they can be old terms used in new or distinctive ways, or general terms used in specific or restricted ways. As a critical reader, you expect the authors to define terms at the outset to increase the intelligibility of the report and to avoid any ambiguity or misunderstanding.

A *variable* is a quantity attached to some person, event, or situation that can take on different values. This can happen because the experimenters deliberately manipulated the values of the variable, or the experimenters simply measured variables that occur naturally without influencing their values. Now it is time to be more precise about the different types of variables you will encounter in a research report, and the different dimensions and labels that will be applied to them. We have summarized these types of variables in Exhibit 3.1, and they can also be found in the Glossary.

Conceptual and Operational Definitions

The variables involved in psychology research must be clearly defined in two ways. First, the variables must be given abstract or *conceptual definitions*. Conceptual definitions describe qualities of the variable that are independent of time and space but can be used to distinguish observable events that are and are not relevant to the concept. For example, a conceptual definition of *intelligence* might be "the ability to learn or understand or to deal with new or trying situations" ("Intelligence," n.d.). You will typically find the conceptual definition of terms in the introduction to the research report. That said, there are instances in which researchers may be interested

EXHIBIT 3.1

Labels for Different Types of Variables Used in Research

If the study involves a manipulation, you might see this variable referred to as the following:

- the *independent variable*—most often in a study of basic processes or one conducted in the laboratory;
- the *treatment*—most often when the manipulated variable is meant to improve the health or well-being of participants; or
- the *intervention*—most often when the manipulation is meant to change participant's behavior but not necessarily remedy a health problem.

The variable being tested for the effect of the independent variable might be called the following:

- the *dependent variable*—the variable the manipulation is meant to effect;
- the *outcome variable*—used more frequently when the manipulated variable is a treatment or intervention; or
- the *criterion variable*—an outcome variable that is central to the evaluation of a treatment or intervention and might even have a cutoff for when the treatment or intervention would be deemed successful and unsuccessful.

If the study does not include a manipulation, you might see the measured variables referred to as the following:

- the *static-groups variable*—characteristics of people that can be used to sort them into groups (e.g., organismic, status, and psychological individual differences);
- the *predictor variable*—any variable used to explain *variance* in another variable (this designation is most often used in correlational or multiple regression designs); or
- the *criterion variable*—the variable the predictor variable(s) are attempting to predict.

in variables that are very concrete, leaving little need for conceptual definitions, for example, in a study of whether adolescents finish high school.

Sometimes concepts are named after a person (e.g., Hawthorne effect), sometimes a name is formed from a Latin or Greek root (e.g., dementia praecox, schizophrenia), sometimes words are combinations of smaller descriptive words (often used in German; e.g., *Schadenfreude*), and sometimes names are just apt descriptions of meaning (e.g., self-esteem). When the term is new, as all of them were at one time, it must be defined to be understood. It should also be defined the first time it is used in a research report. If a study is about psychotherapy, for instance, the authors should say, "*Psychotherapy* in this study refers to short-term cognitive treatment." The terms *short-term* and *cognitive* would also have to be defined. Terms that have standard meanings and are used in standard ways need not be defined. If the researchers do not provide clear conceptual definitions in the introduction, you, as a critical reader, should note this as a weakness of the study.

Conceptual variables, by definition, cannot be directly measured. We cannot open a person's brain, remove the section labeled *intelligence* and weigh it. Therefore, to relate concepts to events or to the things researchers can measure, a variable must also be operationally defined. An *operational definition* is a description of the observable characteristics that determine if the event represents an occurrence of the conceptual variable.

For example, the operational definition used in a study of intelligence might be "the score received for doing the tasks on the Wechsler Adult Intelligence Scale administered by a qualified and experienced examiner." As Stern and Kalof (1996) pointed out, "The first requirement for observations to have scientific value is that abstractions be concretized" (p. 12). The notion of the correspondence between concepts and operations in research is another way of defining construct validity or measurement validity (Condition 5 for claiming a causal relationship; see Chapter 2).

Experimental and Measured Variables

Conceptual and operational variables get their meaning from their relations to abstract ideas and empirical events, respectively. The new variable's definitions get their meaning from how they came to take on their values and what those values might be.

EXPERIMENTAL VARIABLES

As noted in Chapter 2, to critically evaluate a study, you must understand the theoretical and logical bases for the research, the choice of methods, and how correspondent are these considerations. If the statement of the problem that accompanies the research question and the hypotheses are clear to you, then it should be easy to identify whether manipulated or unmanipulated variables are involved in the study. If the research takes the form of an experiment, you expect the experimenters to have manipulated (i.e., to have intentionally and systematically created the values of) at least one variable, so that the effects of this manipulation on a subsequent variable can be observed. The manipulated (presumed causal) variable might also be referred to as the *independent variable*, *treatment variable*, or *intervention*. The affected variable might be referred to as the *dependent variable*, *outcome variable*, or *criterion variable*. In this arrangement, the manipulated independent variable is an experimental treatment that is clearly the antecedent of the effect. For example, in an experimental study of the effects of stress, the experimenters introduce and vary the stressful event experienced by the subjects, and then observe the consequences.

Naturally Occurring Experimental Variables

Researchers may use an independent variable that is being manipulated not by them but by some real-life occurrence. For example, researchers may be aware that some teachers at a school are about to adopt a new way to teach reading. After it is introduced, the researchers compare students who received the new approach with students who did not. Even though the researchers have not manipulated the type of instruction, the same labels for variables may be used to describe them (e.g., treatment, outcome), but this should be done with caution. You must remember that the causal status of conclusions is still in question. Because random assignment to types of instruction has not occurred, the type of instruction (the naturally occur-

ring variable) may have caused higher reading scores, or higher reading scores may have caused the teacher to adopt the new approach (i.e., risk a new technique). So, the causal variable remains at issue, and it does not matter when adoption of the approach and reading level were measured (Condition 1 for claiming a causal relationship; see Chapter 2).

In other circumstances, alternate temporal sequences are pretty implausible. For instance, if a natural disaster (e.g., an earthquake in a community—a naturally occurring variable, if there ever was one) is followed by an increase in the divorce rate, it is hard to argue that the divorce rate caused the earthquake. However, it could be argued that some other event happening at the time of the earthquake (e.g., an economic downturn) might be the cause of higher divorce (Condition 2 for claiming a causal relationship; see Chapter 2).

MEASURED VARIABLES

Measured variables are variables that are not manipulated by experimenters or naturally occurring events. They are measured with minimal interference on the part of the researchers. The variables playing the role of presumed causes mentioned previously are measured variables. These variables are often labeled *predictor variables* when they are related to another measured variable. So, if researchers correlated school attendance with grades, they might label school attendance the predictor and grades the criterion, but to suggest causality would be wrong. The researchers are finding the association between two continuous variables; that's it. As an astute reader, you must bear in mind that even though the researchers are using attendance to predict grades, no causal connection is being tested. There is no evidence in the study about whether poor attendance causes poor grades or poor grades causes poor attendance, even if one variable does strongly associate with the other.

Static-Group (Nonexperimental) Variables

A *static-group variable* is a special type of measured variable. Here, researchers may select participants from appropriate preexisting groups, whose identifying characteristics constitute the predictor variable. Static-group variables are characteristics of people that can be used to identify their assorted group memberships, including (a) *organismic variables* that are part of the individual's physical being (e.g., sex, race/ethnicity, age), (b) *status variables* (e.g., education, occupation, socioeconomic status, marital status), and (c) *attribute variables* (e.g., diagnosis, personality traits, political affiliations). Most static-group variables cannot be manipulated by the researchers. For example, researchers cannot manipulate subjects' age, and although they might try to manipulate their personality, this would be very difficult and unethical.

In the case of static-group variables, we eschew the use of the term *independent variable* because it connotes a causal relation. It is more precise to refer to static-group membership as a predictor variable because no manipulation of it has occurred; group membership may have simply occurred before the variable it is presumed to affect.

Static groups can be used in research by selecting proper contrast groups. The static-group variable becomes analogous to an experimental treatment (but without the causal connotation), with the contrasting group serving as a rival treatment, or a control group receiving no treatment at all. Men and women constitute two static groups that can be contrasted on other variables. People with Alzheimer's disease (a static-group identity) can be contrasted on other variables with a group not exhibiting the disease.

The analogy between experiments using random assignment of subjects to conditions and static-group comparisons holds only if the people in the static groups are truly equivalent in all other ways. Are the men and women being compared to examine sex differences roughly the same age? Do those subjects who have Alzheimer's disease differ in wealth from those who do not? When the organismic, attribute, or status variable in a static-groups design is the predictor variable, comparison with an appropriate *contrast group* allows you to determine whether there is anything else that distinguishes the groups.

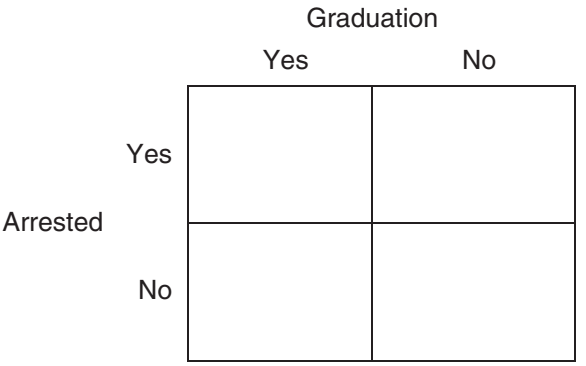
Sometimes the organismic, attribute, or status variable is set up as the outcome variable. This is done so that researchers can gain some clues about how the groups acquired their characteristics. For example, if we discover that people with Alzheimer's disease were more likely to have lower wealth, this might provide clues to why they contracted the disease (e.g., exposure to environmental contaminants). But again, similar to naturally occurring experiments, static-group comparisons cannot rule out all rival explanations for a relationship, regardless of how implausible one temporal sequence might be or how similar the two groups compare on other variables.

Two Static-Group Variables

Some studies have two or more organismic–attribute–status variables predicting a different *continuous variable*. For example, suppose a study is conducted by researchers who are interested in the association between family income during the first 12 years of children's lives and problems in adolescence. The researchers choose the variables (a) completing high school and (b) having been arrested as a juvenile as the two criterion variables. The thinking is that low family income in childhood predicts these two difficulties in the adolescent years that follow. The researchers set up "graduation" and "arrest" as the static-group variables in the 2×2 design shown in Figure 3.1. They then calculate the means of family income in childhood in each of the four cells, and conduct an analysis of variance to determine whether each predictor is associated with early income—either one static-group variable is and the other is not, or the two interact such that an association occurs only when both are present.

The outcome variable is family income during childhood, even though this is an antecedent of behavior in high school and not a consequence of it. The possibility that "high school graduation" and "whether or not the juvenile was arrested" influence past family income is patently absurd. Here, too, the problem could be reframed: Researchers may ask whether past family income is a variable that discriminates between groups of high school students who graduate and display criminal problems and those who do not.

FIGURE 3.1



A static-group table for grouping students by their experience of graduation from high school and whether they had been arrested.

Risks of Causal Inferences

It is worth reiterating that when preexisting static groups or continuous predictor variables are used, causal inferences, though tempting, are risky. Dependency of one variable on another cannot be verified. For example, suppose two researchers have opposing hypotheses about the relation between obesity and self-esteem. One proposes that obesity causes low self-esteem, and the other proposes that low self-esteem causes obesity. Now suppose that a third researcher concerned about causal direction in the obesity–self-esteem problem decides to make it into an experiment instead of a static-group design. The first plan is to enlist a group of participants of average weight and, over time, to increase the weight of a random half of them by means of a high-calorie diet. The design calls for measuring the self-esteem of both groups before and after this treatment. On further reflection, and the influence of collegial counsel, the researcher realizes that there is an ethical problem in doing something that could have negative health consequences for some of the participants. The situation is therefore reversed, and a study is designed in which one group of overweight participants is placed on a low-calorie diet to lose weight. Participants in a randomly assigned control group of equal starting weight continue with their regular eating habits. Self-esteem is measured before and after treatment (as in the original plan). The hypothesis is that the self-esteem of the diet group will increase from pretest to posttest, whereas the self-esteem of the untreated group will not.[1] Another researcher decides to approach the problem by making self-esteem into the independent variable and manipulating it. An

[1]Generally speaking, given the widely accepted notion that obesity has negative consequences for health, this study is likely to raise fewer ethical concerns, especially if all participants are desirous of losing weight and the control group is offered the diet plan after the study is concluded, or the study continues with them in the experimental group after a period of time.

experiment is designed in which the self-esteem of half of a sample of obese participants is increased by giving them a series of success experiences. The remainder of the participants have no such experiences. The hypothesis is that the group whose self-esteem is raised will lose weight, whereas the other group will not.

These last two researchers each started out with the proposition that obesity and self-esteem were causally related. They both succeeded in designing cause and effect experiments, but were these experiments consistent with their thesis? The first researcher wanted to show that increase in weight (independent variable) lowers self-esteem (dependent variable). Demonstrating that losing weight raises self-esteem does not prove the converse (i.e., that gaining weight lowers self-esteem). The second researcher reversed the independent and dependent variables. Demonstrating that raising self-esteem lowers weight does not prove that gaining weight lowers self-esteem. As a critical reader, you must look for logical errors of this sort when examining the chain of reasoning used as the research proceeds from thesis to hypothesis to independent and dependent variable selection to design to conclusions.

Unidirectional and Bidirectional Paths

When there is a logical two-way association between the predictor and criterion variable, the thesis of the study may tell which one is to be considered as the causal variable. For example, the researcher studying the relationship between obesity and self-esteem, mentioned previously, may offer the thesis that being overweight lowers self-esteem. The predictor variable would consist of an overweight group to be contrasted with a group of individuals of average weight. The criterion variable would be self-esteem. The second researcher may posit that people who have low self-esteem eat excessively and gain weight. For this study, weight is conceived to depend on self-esteem. In both instances, the researchers have chosen a causal direction. But as long as weight and self-esteem are both measured variables, the causal thesis remains just that, a thesis. In this instance, however, it is not only possible to posit a *unidirectional causal relationship* but also to suggest that either causal direction exists in nature (i.e., a *bidirectional relationship*). Being overweight can cause low self-esteem and low self-esteem can cause overeating.

Contrast a thesis about obesity and self-esteem with one about height and self-esteem. The researcher proposes that height is associated with self-esteem, and selects height as the independent variable and self-esteem as the dependent variable. The self-esteem of a group of tall people is contrasted with that of a group of short people. In this example, switching the independent and dependent variables, making height into the dependent variable, would be illogical. Few would believe that increasing people's self-esteem would actually make them grow significantly taller (as distinct, perhaps, from just standing taller). Of course, conducting the study as an experiment does not raise the ethical concerns the obesity study would (wearing shoes with high heels might be the manipulation).

One-way unidirectional paths are fixed by the logic of antecedents and consequences. It is reasonable to think that early childhood experiences could have a bearing

on adult adjustment. It would be conceptually backward to begin with a group of well-adjusted adults and a group of poorly adjusted adults as two levels of the independent variable and to make early childhood experiences into the dependent variable. If you encountered this, you would be puzzled unless the author clearly acknowledged that it is a retrospective study featuring *postdictive reasoning* instead of predictive reasoning. If so, you would be anticipating a *discriminant analysis* or *logistic regression* to test the hypothesis that early childhood experiences discriminate between well-adjusted and poorly adjusted adults.

As you critically read research, remember that because logic rules out one causal direction in a study that uses only measured variables, it does not mean that causality has been established. The research still must meet the five other conditions before it can be used to make a good case for causality (see Chapter 2). If a cause–effect problem were to be approached by using static groups, you would be automatically on guard against insupportable statements about causality.

Establishing Levels of the Independent or Predictor Variable

One early decision facing researchers is whether to make the independent or predictor variable continuous or categorical. If the variable is a conceptually continuous variable and that is the way it was measured (e.g., if IQ was used), the researchers must decide whether to (a) leave it that way and treat the full range of numerical values, (b) make it a *dichotomous variable* (e.g., low IQ vs. high IQ), (c) arrange it in graduated multiple levels (e.g., low IQ vs. average IQ vs. high IQ), (d) make the levels into discontinuous groups (e.g., IQ between 70–80, 95–105, 120–130), or (e) use predefined cutoff scores (e.g., use the labels of *intellectually challenged*, *average intelligence*, and *genius* on the basis of IQ scores).

CATEGORIZING CONTINUOUS VARIABLES

Researchers are rightly reluctant to tamper with a continuous variable by transforming it into dichotomies or into categorical levels because information is lost in the process. At issue is whether the researchers really have any interest in the information to be sacrificed, and whether it is worth compromising the central focus of the research to preserve that information. The researchers may have a theoretical, empirical, or pragmatic reason for formulating hypotheses in categorical terms, such as studying the differences between genius IQs and all others, or first-born children versus all later-born children (rather than preserving all birth orders). Either of these dichotomies may represent the central focus of the study. The interest may not extend to intermediate levels, and, in fact, their introduction might cloud the picture.

If the researchers state the hypothesis in categorical terms, it is appropriate to keep the independent variable consistent with its phrasing. On the other hand, if the

hypothesis is stated as a relationship that varies across a graduated range, then it is not appropriate to break the predictor variable into dichotomies, or *nominal levels of measurement*. The choice is driven by the theory and the rationale of the study, not by statistical considerations. The researchers cannot be faulted for setting up the predictor variable in a way that permits a direct test of the hypothesis. If you disagree with the way the hypothesis is stated, you should find fault with it, not with a design that is appropriate to test that hypothesis.

All of this applies to variables that are inherently continuous. There is no choice with variables that are inherently categorical (e.g., man vs. woman, blue eyes vs. brown eyes). Also, when variables are experimental in nature, their level is controlled by researchers, and you should not expect them to manipulate every possible level of it. For example, if researchers are studying the effects of stress on performance, you expect to find a stress and no stress condition, or maybe three levels of stress, but introducing all possible levels of stress in the study would be unrealistic.

As a critical reader, you should examine how the levels of the measured variables were established to see whether they are consistent with the hypotheses. Consider the hypothesis "Older people have more difficulty learning new things than do younger people." The age groups that the researchers selects to represent older people and younger people may be crucial in this study. There are several options:

1. *Extreme groups:* Researchers select a group that is 80 years of age or older for the *older* group and 20 years of age or younger for the *younger* group. This would maximize the effect, if there is any. But it would also minimize the amount of information about the relationship of age and learning because it is restricted to two extremes with no attention to anyone in between. What if researchers find no difference between the two extreme age groups, but the relationship is curvilinear (e.g., people between 20 and 80 learn more easily than the young or old)?
2. *Range of categories:* A continuous range of age categories could consist of the following groupings—younger than 20, 20 to 39, 40 to 59, 60 to 79, and 80 or older. A discontinuous range could consist of the following groupings—20 to 25, 40 to 45, 60 to 65, and 80 to 85. Continuous and discontinuous age ranges give differential information across the adult lifespan. Neither gets at exactly what this hypothesis predicts, because the hypothesis calls for a comparison of "younger" people and "older" people. If the researchers define the term *younger* as ages 20 to 25 and *older* as ages 80 to 85, the hypothesis predicts nothing about ages in between these groups. On the other hand, if the hypothesis predicts a progressive decline throughout the lifespan, the use of the intermediate levels would be necessary.
3. *Median split:* The strategy of dividing groups along the median, though frequently used, presents two problems. Falling above or below the median does not necessarily place a person in a category that is consistent with the theory. Being more than 50 years old is not necessarily "older" for the purposes of this sample hypothesis. A better example of the problem would be a sample split in the middle of the IQ range into an above-average IQ group (> 100) and a below-average IQ group (< 100). The expectation that this classification yields an above-average

group containing highly intelligent people is vitiated by the presence of so many people in the higher group whose scores are only slightly higher than the median. The same holds for the classification of those who fall slightly below the average. An individual with an IQ of 101 is labeled above-average IQ, whereas one with an IQ of 99 is labeled below-average IQ. The 2-point difference is smaller than the standard error of the measure, and the two individuals are essentially indistinguishable on this variable. The placement of them in contrasting groups that are expected to score differently on some other variable would work against the study's hypothesis, if it is true (i.e., it would make it more difficult to reject the null hypothesis). A more rational way to divide the groups would be to separate them by some *standard deviation* unit, like $+0.5\sigma$ or $+1\sigma$. Researchers who wish to maximize the contrast by using extreme groups could use an interval of 2σ or 3σ.

LEAVING VARIABLES CONTINUOUS

With a continuous variable like age, as opposed to a categorical variable like sex, researchers my find an advantage in studying the question at all points rather than breaking the continuous variable into arbitrary categories. The hypothesis informs the decision about the best way to proceed. Returning to the age and learning hypothesis, researchers could treat age as a continuous variable and would obtain a progression of learning scores across the whole range of ages. This, however, is not how the hypothesis is worded. If, in fact, it is not until an advanced age is reached that there is a relationship between age and learning new things, linear correlation across the lifespan would not be very revealing. Suppose, for example, that in a study of visual acuity, farsighted changes begin to show around the age of 40, after which there is steady decline. Correlation across lifespan is not linear. The graph of visual acuity is flat for the years before age 40, and then begins to drop off. *Curvilinear correlation*, or the use of categories like ages 30 to 35, 36 to 39, 40 to 43, 44 to 47, and 48 to 51, might show the picture more accurately.

THEORY-DRIVEN LEVELS FOR VARIABLES

Instead of selecting levels arbitrarily, researchers are advised to use the theory that underlies the study as the guide for selection. Predictions would come from the literature on age and learning. The age group for the onset of learning deficits would be identified and then bracketed with other age-level categories for comparison. The hypothesis would be stated in a way that was consistent with the theory.

STRENGTH OF THE RELATIONSHIP OR MAGNITUDE OF EFFECT

How strongly two variables might be related or how strongly one might cause the other also needs to be considered when researchers consider how to level their variables. In the previous example of extreme groups, the potential for effect was maximized. In a study of the effects of stress, use of a powerful stressor increases the magnitude of

effect. However, with both choices of extreme levels comes a decrease in generality because the findings of the study are limited to extreme conditions. Researchers who obtained predicted results with weaker stress conditions could probably safely assume that, in most cases, results would apply for greater stress as well. The results of such a study could therefore have greater generality.

Examination of the levels of the measured and manipulated variables give you an idea about whether they match the hypothesis and whether the choice was meaningful and appropriate. You might ask whether the results would have been the same if other levels had been selected.

Time Sequencing of Measurements

PROSPECTIVE STUDIES

In prospective studies, the researchers predict consequences or effects from known antecedents or causes. This can be done with or without manipulating the variable assumed to be causal. To illustrate the former condition, consider medical researchers who place a random half of a group of people who have moderately high cholesterol on a medication presumed to lower cholesterol. The other half are given a *placebo*. Five years later, the coronary illness rates of the two groups are compared. In a prospective study that does not involve manipulating the assumed causal variable, the future coronary illness rates of a group of people who have high cholesterol is compared with the rate for a group whose cholesterol is well within the normal range. In examining this study, which uses preexisting groups, you focus your attention on how the researchers attempted to control all the other variables that could contribute to coronary illness. You know that the researchers, who were not in control of the predictor variable, must be exceedingly cautious about claiming causality.

RETROSPECTIVE STUDIES

In retrospective (*ex post facto*) studies, researchers use postdictive reasoning (i.e., infer backward) to predict antecedents or causes from known consequences or effects. Obviously, the predictor variable cannot be manipulated because what is under investigation has already happened. In the cholesterol example, the best that the researchers can do is to go back in the records and look at the cholesterol content of the premorbid diets of people who have coronary disease as compared with those who are free of this ailment.

Because the researchers have (a) no control over the amount, duration, or timing of the levels of the antecedent; (b) no control over the selection or assignment of participants to the antecedent conditions; and (c) no control over other events, situations, and circumstances that could have a bearing on the criterion variable, you should be cautious when weighing the credibility of causal statements and causal inferences from retrospective studies.

Longitudinal Versus Cross-Sectional Studies

Another aspect of time sequencing applies to studies in which the passage of time is a factor. In a *longitudinal study* of human development over time, for instance, the predictor variable (age) is a marker of levels of the passage of time. The same individuals are reassessed at specified age intervals. Contextual and experiential events are intervening variables that operate during the passage of time to bring about observed changes. The study is prospective, but instead of manipulating the predictor variable, the researchers just wait to see what happens.

If the strategy is to make the study *cross-sectional*, the researchers select representative samples of people who have already attained different age levels but are equivalent in other respects. The researchers assume that participants have all passed through the same earlier stages of development and have had similar kinds of relevant life experiences before reaching their present age. The cross-sectional strategy is quicker and easier to carry out than a longitudinal study, but there is greater chance for making erroneous inferences. Individuals at different age levels are not the same individuals as those at other age levels. They may not be as equivalent in important respects that the researchers would like. Assumptions about their intervening growth and experience may not be entirely justified. For example, assume researchers are doing a cross-sectional study of the physical development of children from ages 1 to 4. They should not choose a cohort of 1- and 2-year-olds who differ from older children in the way they are recruited (e.g., by drawing younger children from birth records and older children though preschool enrollment). Also, the children might differ because of historical events that influence one group of children and not the other, for example, the introduction of new dietary guidelines affecting 1- and 2-year-olds but not older children.

Real-Life Versus Analogue-Simulated Situations

In some studies, a real-life experience, with or without independent variable manipulation, is introduced or is found for the study. When possible, such studies can be powerful, but ethical considerations can prevent this strategy if any harm can come from it. For example, it would be unethical to study the effect of alcohol or marijuana on the in-flight performance of commercial airline pilots. In other instances, the in vitro situation is abandoned because the time or cost is prohibitive or because real facilities are not available to the researchers.

Experimental arrangements can be set up to be analogous to real-life situations. They are "as-if" experiments, with simulation of times, places, persons, or situations. The study with the airline pilots could be done in a flight simulator, which realistically duplicates the genuine experience. But, of course, the simulated setting does not exactly duplicate the real-life one; there are no consequences for crashing a

simulated flight, and it might be sobering to be piloting a real airplane full of real passengers while "under the influence." Still, these caveats do not override the necessity of an analogue approach in this situation. Some analogue studies are of considerable interest and great ingenuity, but as a critical reader, you must scrutinize the kinds of claims and generalizations that are made.

Settings

Research can take place in the field or in the laboratory. Strictly speaking, the setting merely describes the venue of the study. In practice, some correlates of location may be important, because people may behave differently in different settings. The setting of a study often overlaps with its actuality (real life = field, in vitro = laboratory), but this is not always the case.

FIELD STUDIES

Field studies are meant to take place in a naturally occurring environment. In field studies, chimpanzees are studied in the jungle as opposed to the zoo, educational studies are conducted in the classroom, and psychotherapy clients are studied in the therapist's office. These are frequently mislabeled *in vivo studies*, a term that refers to occurrences within the living organism as opposed to being isolated from the living organism and artificially maintained in a test tube (*in vitro* means "under glass"). As psychological experiments generally deal with living organisms, it is more appropriate to describe the setting as in situ (i.e., in its original place).

LABORATORY STUDIES

In laboratory experiments, participants are removed from their natural habitat and brought into a special setting that is used for research purposes. The setting does not necessarily make any difference, but it might under some circumstances. The setting, the realism of the situation, and the use of manipulation of the independent variable are all relatively independent decisions made by the researchers. Let us use a study of psychological therapy as an example. It is possible to have a therapy field study using real therapists in a simulated (analogue) therapy situation with simulated written or spoken vignettes or films of patients or actors playing the part of patients. The film or reading can be stopped at strategic points and the therapist asked to respond to the client's remarks. This could be done equally well in the therapist's offices, or clients could come to a laboratory to participate. Eells et al. (2005) conducted a study using vignettes that were prerecorded and played over the telephone. You can also manipulate a condition (e.g., type of therapy) in a study of the effect of these interventions on clients' impressions of the value of the session. This could be done in the regular therapy context and setting with real therapist–client dyads. The real dyads may be brought into the laboratory, simulating the place and circumstances. An analogue study could be set up with participants who are not real clients being interviewed by people who are not their therapists.

It is also possible to simulate real life in the laboratory. For example, as subjects arrive for a study, they can spend time in a waiting room with another participant who is a confederate of the experimenters. The friendliness of the confederate might be manipulated and the effect of the exchange on the subjects' subsequent behavior could be the outcome variable.

From the point of view of critical evaluation, you should recognize that it is as possible to do analogue studies in the field as it is to make the field into a laboratory. You should also understand that it is possible to do careless and imprecise studies in the laboratory or in the field, and it is possible to do precise cause–effect studies in either setting, as well.

EXTERNAL VALIDITY AND GENERALIZABILITY

Questions can always be raised about the generalizability of a study, whether it takes place in the laboratory or the field. Dialogues about the comparative merits of field and laboratory studies in psychology have been going on intermittently throughout the past century. For example, in discussing the difference between reaction time measurements outside and inside of the laboratory, Woodworth (1938) stated,

> The automobile driver cannot hope to equal the short R.T. [reaction time] of the laboratory because his preparation is not so good, he does not get a Ready signal two seconds before the emergency. He has to shift his own internal transmission when the stimulus arrives. A second or two must be allowed him for shifting his set and adjusting himself to the new situation which has arisen. If he is really startled several seconds may be needed. (p. 339)

Egon Brunswik (1951/1943) introduced the term *ecological validity* in advocating the use of study participants and settings that are representative of the real world. Laboratory studies were subsequently challenged by many critics as lacking in ecological validity. In counterargument, Berkowitz and Donnerstein (1982) deplored the "widespread equation of experimental value with ecological validity" (p. 2). They asserted that laboratory experiments are designed to test causal hypotheses, not to determine how probable it is that they happen in particular situations. Berkowitz and Donnerstein insisted that laboratory experiments "give us a truer image of human complexity than do uncontrolled, naturalistic investigations" (p. 247).

But Mook (1982) counterattacked. He stated that when one "isolate[s] a single factor from that complexity [the natural environment] and var[ies] it independently of the rest of nature . . . the complexity of nature is taken directly in hand and discarded [in the laboratory]" (p. 126). Mook recognized this was not necessarily a liability; some experiments are designed to test generalizations instead of make them. Conducting studies in "unnatural" settings such as laboratories can be a virtue if you are trying to find out whether something can happen. This is sometimes referred to as a *proof of concept*. Whatever is being tested does not have to happen all the time, it just has to happen once to demonstrate it exists. This is a valid point: If experiments are viewed as attempts to falsify hypotheses by putting them to the sternest test in the laboratory, credence is awarded to a hypothesis that holds up under conditions that are far more rigorous than in real life.

In a detailed treatment of the issues, Kerlinger and Lee (2000) listed the virtues of laboratory experiments as follows:

- it is possible to attain complete (or near complete) control of extraneous influences on behavior,
- the independent variables can be manipulated most easily,
- random assignment of subjects to conditions can be accomplished most easily,
- measurements can be taken with great precision, and
- internal validity is high.

The weaknesses of laboratory studies include the following:

- there can be a lack of strength to independent variable manipulations (they lack the impact of real-world events),
- the context in which behavior is occurring is artificial, and
- external validity may be poor.

Kerlinger and Lee listed the virtues of field experiments as follows:

- they are most suitable to study social and educational problems where external validity is important,
- they are subject to independent variable manipulation, and
- they are subject to random assignment.

The principal weakness of field studies is that controls may not be as tight as you might like.

Ray (2012) added another dimension to the discussion: the difference between the scientist as observer (typical in field research that involves the use of naturalistic observations) and the scientist as participant (typical in laboratory experiments where the scientist controls what is happening). He suggested that in field studies there is less control over environmental factors but more ecological validity and fewer *demand characteristics* (i.e., qualities of the experimenters or study context that place a demand, or expectation, on research participants to respond in a particular way).

As a critical reader, you will judge the appropriateness of the study setting on the basis of what the researchers are attempting to demonstrate. If they place a high premium on their hypothesis holding in real-world situations, the field setting is most appropriate. If they are simply determining whether a phenomenon can happen (i.e., proof of concept), the lab may be just fine. These are extreme examples, so you must judge more nuanced variations in claims as you consider how appropriate a setting is for testing a hypothesis.

Efficacy and Effectiveness Research

Fowler, Seligman, and Koocher (1999) differentiated between efficacy and effectiveness research in psychotherapy to reflect the difference between two types of field study. Efficacy has tight controls that narrow the scope of the study, whereas effectiveness portrays things the way they actually are. According to Kazdin (2016), *efficacy*

research occurs when a treatment is conducted under highly controlled conditions, often with nonclinical samples. Clients are screened to maximize the homogeneity of participants and to provide a strong experimental test. In contrast, *effectiveness research* is designed "to evaluate treatment in clinical settings, with real patients, and under conditions more routinely seen in clinical practice" (p. 42).

An efficacy study in clinical psychology, for example, will ensure that the treatment is administered correctly, to a sample from the appropriate population, and the subjects will adhere to the regimen. An effectiveness study will be looser on these controls, revealing an estimate of the treatment's impact more like that practicing psychologists might expect among their patients. In a sense, you can think of an efficacy study as a broad proof of concept (can it work under ideal conditions?) and an effectiveness study as a real-world test (will it work in a clinician's practice?).

On seeing that an efficacy study is being reported, you should be attuned to the issue of whether the same results would hold outside of this experimental situation (external validity). When encountering an effectiveness study, you should be more concerned with whether the results are truly attributable to the experimental treatment (internal validity). Fortunately, the choice available to researchers is not necessarily dichotomous. Studies can be designed that close the major potential loopholes to demonstrations of efficacy while still preserving the real-life components. Efficacy studies typically occur when a new and unproven intervention is being tested. Effectiveness studies often occur after efficacy studies have shown an intervention has potential.

Generalizability in the Context of Variables, Timing, Realism, and Settings

The statement of the problem gives you an idea about the purpose of the research and the intent of the researchers. It tells you whether the intent is to describe how things are in the real world; to determine whether a phenomenon or a relationship can exist under any circumstance; or to predict, to explain, or to test a theory. Knowledge of the purpose of the research enables you to judge whether the researchers intend to generalize from the circumstances of the study to other circumstances not included therein. Then you can decide whether it is legitimate for the researchers to make generalizations, and what generalizations are justified. On the other side of the coin, researchers cannot be faulted for not doing something that you wish had been done. If there is no intent to generalize, the researchers do the right thing by not generalizing. Criticism is justified only when unfounded claims of generality are made.

Consider a study of the effect of sugar intake on children's hyperactivity. The researchers do not care if the findings do not generalize to other childhood behavioral disorders. Generalizability of the data to other foods, say food colorings, is also not the intent. External validity is bounded by intent and by claims. Because there is no universal study that generalizes to everything, everywhere, and every time, the observation that the study has questionable external validity is itself questionable

unless the researchers claim generality without furnishing the grounds for making such an assertion.

If a claim of generality is made and you disagree, it is your job to explain why the generalizability of the study may be limited. This will involve comparing the participants, treatments, measurements, and settings used in the study with those to which generalizability is desired. Are there differences between the subjects used in the study and those to which you wish to generalize that suggest a treatment might lose its effectiveness? Is there something unique to the situation created in the study? Of course, this is conjecture on your part (more or less convincing), and only additional studies can determine whether the generalizability of results is or is not restricted.

The importance of generalizability, and whether it makes a difference, depends on the nature of the research question and on the intent of the study. Generalizability is of no import if the study addresses a proof of concept or existence question. For example, researchers would have to produce only a single chimpanzee who could act appropriately in response to printed symbols to show that it is possible for chimpanzees to read.

Applied research that does not aspire to make generalizations cannot be faulted for not doing so. Studies can be designed to answer a question about a single setting, for example, the case of outcomes in a particular mental health clinic, with no interest in the generality of the findings to other clinics and no claims that the findings apply anywhere else. That said, most research that is designed to establish a psychological or behavioral principle is expected to be generalizable. No one would have much interest in such a study if its findings did not extend beyond the circumstances of that study (except, perhaps, to explain why). As a critical reader, you can view generalizability as the demonstrated correspondence between (a) the generalizations that the researchers intended the research to make at the outset and (b) the generalized inferences that the researchers offered in the Discussion.

In summary, an automatic criticism about the generalizability of the study is not especially fruitful. It does not matter whether the study lacks some unreachable kind of collective generalizability. Instead, the spotlight is on what, to whom, under what conditions, and how far you can extend the results beyond this single study. Some of the main aspects of generalizability are as follows:

- *Persons:* Do the results apply to people who were not research participants but who share the same subclass memberships as the participants (e.g., diagnosis, age, sex, socioeconomic status, IQ, education)? Do they apply to people who belong to other subclasses as well?
- *Researchers:* Would the same results be obtained with different researchers, data collectors, judges, or raters?
- *Places, environments, settings:* Would the same results be obtained if the study were conducted in a different environment, place, or setting? If the study were done in a laboratory, would the results generalize to the natural environment (the issue of ecological validity)? Would it generalize to other geographical locations? If the study was a simulation, would the results generalize to real-world situations?

- *Time:* Is there temporal generalizability? Would findings obtained at one time apply as well at other times, or are the results time bound? This includes other days and hours of the day, as well as other years or historical eras.
- *Levels of treatment:* Do the findings pertaining to the levels of treatment used in this study apply to other levels of the treatment? Or are the conclusions restricted to the kind, amount, intensity, and frequency of the treatment given in this research? For example, can general conclusions about the benefits of psychotherapy for the treatment of severe anxiety be made on the basis of a study in which ten sessions of behavior therapy were used generalize to treatments of shorter or longer duration?
- *Procedures, conditions, and measurements:* Would you get the same answer to the question under different conditions and with different procedures and apparatus or with different methods of measurement? In other words, are the results generalizable beyond the specific operations used in this study?

It should be evident by now that at least some of these questions might be answered in the negative for almost any research. Others might be dismissed a priori in a given study. For example, researchers studying memory loss in patients with advanced Alzheimer's disease might see no reason why a study that was done in a New York retirement home should not hold as well in a New Orleans retirement home. The effects of the disease are unlikely to change on the basis of geography or setting. Of course, each of the other aspects of generalizability would have to be addressed in turn by you, if not by the author(s) of the study. And you should be sensitive to claims made by researchers that go beyond what appears to be reasonable. Sometimes generalized conclusions are grossly overblown. For example, a generalized claim that exercise elevates mood and self-esteem, made on the basis of the study using a small sample of females in an aerobics class, could lead a discriminating reader to ask questions about the kinds and amount of exercise (e.g., does weightlifting have the same effects on mood as aerobics?), and the characteristics of the females who are susceptible (e.g., did the females in the class volunteer, come from the same geographic area, social class?) before accepting the conclusion as a generalizable finding.

REPLICATION

In addition to the correspondence between the conditions in a study and the conditions to which we want to generalize, another key to generalizability is whether the results of a study can be reproduced or were a one-time occurrence. Could the study's results have come about by an accidental confluence of participants and conditions that uniquely accounts for the obtained results? Replication of findings is central to all science; unless findings can be replicated, science is not really moving toward the discovery of those universal truths it seeks.

A *direct replication* is an exact-as-possible repeat of the same procedures but usually with different participants at a different time and setting, though the attempt is also to make these as similar as possible. A systematic or *conceptual replication* uses

different manipulations or measures but is meant to test the same conceptual variables. For example, assume researchers are interested in testing how fear-inducing messages influence intentions to talk on a cell phone while driving. A first study manipulates fear by showing a video of a violent car crash. A direct replication would use the same video, instructions to subjects, and measures of intention. Subjects and settings would also be as similar as possible. A conceptual replication would use a different manipulation (perhaps a magazine article or a discussion group) and different instructions and measures.

A successful replication demonstrates that the results of the first study were not a chance or situation-specific happening. Direct replication using similar procedures but different experimenters and different participants would extend generality and would show that the results were not unique for the original researchers. Changing the setting as well as the participants and the experimenters (conceptual or systematic replication) would add further generality.

Successful replications on a broader sample of participants and under additional or altered conditions would further extend the generality of a study's results. For example, suppose an original study demonstrates the successful use of a drug on a sample of people with tension headaches. It is then replicated successfully by other researchers on a different sample of tension headache sufferers. Still other researchers obtain similar results on a sample of people given a different dose of the drug. Each replication makes claims of generality more credible.

Summary

This chapter focused on the definitions of variables and the establishment of variable levels. We considered the advantages and disadvantages of using extreme groups, a range of continuous or discontinuous categories, median splits, or a full-range *distribution*. You, as critical readers, must evaluate the correspondence between the intent of the study, what it is meant to demonstrate, and the characteristics of the variables' definition and leveling. We also discussed the implications of adopting strategic choices about time sequencing (prospective or retrospective), the realism of the situation, and setting (laboratory or field) for how researchers can interpret their study, and how you can evaluate the legitimacy of that interpretation. Finally, we examined how you can assesses the generalizability of the findings of a study. Inferring from a single study to other persons, researchers, settings, times, levels of treatment, or other procedures, conditions, or measurements must be done with caution.

The Sample 4

A critical reader of research

- determines how participants in a study have been sampled and recruited;
- assesses how the sampling procedures might influence the generalizability of the study's results;
- considers how the source of the data (e.g., directly from participants, from archival records) might influence the representativeness and reliability of the data;
- considers how (a) the volunteer nature of participation, (b) participant attrition, and (c) removal of participants' data after the study has begun might influence results;
- determines whether the sample size is large enough to allow for a fair test of the predictions; and
- knows the difference between random sampling and random assignment.

In this chapter, we explore how the characteristics and size of the sample from which data are collected can affect inferences made from a study's results. Who participates in research affects the generalizability or external validity of research findings. We review a variety of commonly used approaches to probability and nonprobability sampling. We describe some of

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the ways that samples can become biased (i.e., not representative of the population they are meant to represent). Alternative methods of assigning participants to groups are elaborated and appraised.

Sample Characteristics

A description of the sample is usually the first thing that is presented in the Method section of a research report. For you to fully comprehend what was done, enough information must be given about the number of participants and their *demographic characteristics* (e.g., age, sex, marital status, socioeconomic status, education, race/ethnicity).

But how much information is "enough"? From your point of view, as the critical reader, the reported information must be sufficient to enable you to judge how well the sample represents the population, and how well the sample's characteristics mirror those of the *target population*. For readers who wish to compare the results of a study with other studies on the same topic, enough information would include details about participant characteristics that might influence results if they differed from one sample to another. For readers who wish to repeat the study, enough information would include sufficient detail about participants to serve as a recipe for recruiting a similar new sample. Of course, the specific characteristics involved will vary depending on the topic under study.

Target and Accessible Populations

The target population of a study includes all those individuals that the researchers hope to represent in the study. Before a critical reader can appraise a sample for its representativeness, the target population must be given a clear definition. For example, if the researchers are studying the attitudes of college students, the target population would be "all college students." This is the population to which the researchers want to make inferences.

But the target population may be different from the *accessible population*. The sample itself includes those individuals who actually take part in the study. The *sampling frame* includes those individuals the researchers can actually access for the sample. For example, if the researchers must draw their sample from one specific college, the accessible population might differ from the target population of all college students because of that school's geographic location, entrance requirements, and any other special characteristics of the school that might attract one type of student more than others. These are just a few ways that target and accessible populations can differ. The sampling frame might be "all students at College X duly enrolled on January 1, 2018."

Critically evaluating the inferences researchers draw about to whom their results apply will depend on (a) how the target population, accessible population, and sampling frame differ from one another and (b) whether these differences matter, considering the phenomenon under study. For example, differences between target and

accessible populations can be a big issue if researchers use a college subject pool to recruit participants. College subject pools typically draw participants who are enrolled in psychology classes. Taking part in studies might be a requirement of their class. If the researchers want to draw inferences about "all human beings" the participants drawn from subject pools will be younger and more intelligent than the target population (not to mention more interested in psychology). Subject pools even differ from "all college students." As a critical reader, you must ask, "How might the target and accessible population differ?" and "If they do, could these differences be related to the results of the study?"

Representativeness of the Sample

Studies that are meant to lead to the establishment of general principles about human behavior are expected to obtain data from samples that stand for the population under study. The representativeness of samples in experimental studies is, in many instances, simply assumed. This assumption is not unreasonable if the participants are truly members of the population and fit into its major distinguishing categories. Therefore, if researchers are doing a study on the effects of an antipsychotic drug on hallucinations in patients with schizophrenia, the participants would have to have schizophrenia, and they would have to represent the various types, durations, and levels of severity of the illness. If the sample does not mirror the target population in these regards, you might worry that the effectiveness of the drug has not been tested across the full range of possible patients. On the other hand, it probably could be assumed that a sample selected in Chicago would respond in similar ways to a sample selected from New York City or London. It is hard to think of ways that the city in which the study was conducted would be related to the effectiveness of the drug (unless, say, the admission of patients to clinics relates to the availability of services in different cities).

When the research question is being approached using a survey, however, the representativeness of the sample can be even more crucial. Hardly anyone writes about the folly of nonrepresentative sampling in surveys without mentioning the *Literary Digest* poll of 1936 (see Opinion Poll, n.d.), which occupies a unique niche in the history of sampling. More than 2.5 million people were asked their preference during the presidential election that pitted Franklin Roosevelt versus Alf Landon. The poll predicted a rousing victory for Landon. For this huge sample, which makes contemporary polls look puny by contrast, the pollsters selected respondents from automobile registration and telephone listings (the sampling frames). Because the country was in the midst of the Great Depression, however, people who owned cars and telephones were hardly representative of the voting public (the target population), and they were less likely to be in the Roosevelt camp. When President Roosevelt was reelected in a landslide, the *Literary Digest* magazine became a metaphor, lost its social authority, and never recovered. Pollsters today spend enormous time, effort, and resources attempting to ensure that their sampling frame accurately reflects their target population, though they are far from infallible.

Knowing that a sample should be representative of the target population and armed with descriptive information about the study sample, you should be able to judge the representativeness of the sample and to assess the consequences of deviations from representativeness. Many approaches to achieving *representative samples* (i.e., choosing elements from the sampling frame so the sample mirrors the characteristics of the target population) are available to researchers; some are more likely to get there than others. The two major classes of sampling strategies are called *probability* and *nonprobability sampling*. A brief but excellent introduction to sampling methods can be found in Tourangeau and Yan (2012).

Different Methods for Drawing Samples

PROBABILITY SAMPLING

In *probability sampling*, as you might have surmised, each member of the population has a calculable, nonzero, chance of being chosen to be in the study. The probability has to not be zero because if it is, that member has no chance of being in the sample. The sample is no longer representative of the target population. Most often researchers hope the chance of being drawn into the sample is equal for all population members, But, in some cases, which we discuss next, this is not so. There are several methods for drawing probability samples.

Random Sampling

Random sampling requires that each member of the population have an equal chance of being selected into the sample. The ability to select participants at random from among a large sampling frame is well beyond the resources of most researchers. For example, it might be feasible to select a random sample from among "all registered students at College X," but even with this relatively simple sampling frame there might still be the possibility of some bias. The researchers could get a list of registered students from the registrar. But is it perfectly up-to-date? Does it include students who recently dropped out or enrolled? How will students be contacted? By phone? What if some students have two phones? In some cases, these minor variations in students' probability of being in the random sample will be of little consequence. In others, such as polling for political elections, pollsters go to great lengths to even out the odds of being drawn into the sample. In other cases, random sampling is very difficult. For example, it would be near impossible to select a random sample of individuals with schizophrenia from the entire population of people with schizophrenia. The critical reader should be sensitized to unsubstantiated claims of random sampling.

Stratified Random Sampling

Stratified sampling occurs when the researchers first divide the sampling frame into categories, called *subpopulations* or *strata*, and then sample participants from within

these groups of possible participants. Stratified random samples are used most often when the target population includes a rare subgroup that the researchers want to make sure is represented in the sample. For example, if the researchers are studying the tolerance of racial/ethnic differences on a college campus, they might want to ensure all racial/ethnic groups, even those on campus who are few in number, are represented in the sample. They would then stratify students by race/ethnicity and sample from each, then weight their results on the basis of the population proportion of each stratum.

It is important that the strata are defined so that each element in the population can appear in only one category. If persons with multiple racial/ethnic identities appear once in each of their racial/ethnic groups, these persons have a greater chance of being sampled than other people listed with a single race/ethnicity.

Here is another example. If researchers want to learn the attitudes of psychologists who are American Psychological Association (APA) members toward managed health care, you should recognize that the results might vary with psychological specialty (e.g., clinical, experimental, organizational), region of the country, age, and clinical experience, to name a few variables. The attitudes of experimental psychologists would not stand for all psychologists any more than would the attitudes of a sample of clinical psychologists in independent practice stand for all clinical psychologists. To get a stratified sample of psychologists, the researchers would get a count of all psychologists in the APA by sex, specialty, age, geographic region, and so on and would obtain a sample that matched the entire membership on these subcategories as closely as possible. In actual practice this is feasible only if the researchers limit the number of strata to the ones that are considered most important.

Other than in well-funded large-scale research, opportunities for obtaining truly stratified samples are limited. Most small-sample researchers do not find it feasible to obtain samples of participants who match the target population on all demographic and other variables. A physical scientist who is studying the properties of silver can assume that 1 gram of pure silver is the same as any other gram of pure silver worldwide. In contrast, a psychologist who is studying depression in women and wants to draw conclusions that are generalizable has to be concerned with a host of demographic and personal variables. Information obtained from a sample of sorority members from a small college in New England who are depressed cannot safely be generalized to middle-age women working in a factory in Hong Kong who are depressed.

Cluster Sampling

In *cluster sampling*, the researchers first randomly sample clusters of units (e.g., schools within a city). Then, all students within the school are sampled, or the researchers use random sampling again to draw students from within these schools. In this form of random sampling, the larger group is randomly broken into smaller units to simplify the task of random sampling from within these units. It is often used to reduce the cost of large-scale sampling (e.g., instead of visiting every school, only a random sample must be visited). It is rarely used in experimental studies.

Systematic Sampling

Sometimes systematic methods are adopted in the hope that they help produce unbiased samples that are equivalent to random sampling. *Systematic sampling* involves choosing every *k*th element from a list of the sampling frame members. In this case, you must be alert to determine whether the particular form of systematic sampling is unbiased or whether it introduces some unanticipated inequalities. For example, if the researchers pick every third person from an alphabetical list of all students who attend a college, it would be hard to think of ways students might differ systematically on the basis of the first letter of their last name (and the proportion of first letters in the sample will reflect their proportions in the population). However, if the researchers pick participants from a list organized by age and they started with the first person listed, they are guaranteed to have a sample with a younger average age than if the participants were picked at random.

Nonprobability Sampling

Nonprobability sampling techniques do not allow the researchers to estimate the probability each member of the target population has of being included in the sample. In fact, the members of a sample chosen using nonprobability sampling might have differing chances (some zero) of being tapped to be in the study. There are several different techniques for drawing a nonprobability sample.

Self-Selected Sampling

When conducting surveys on all kinds of topics, the media (radio, TV, newspapers, magazines, websites) issue open invitations to listeners, viewers, or readers to call or to submit their responses online. This sample, known as a *self-selected sample*, is limited to people who are tuned into that program and to the subsample among them who have the desire and the means to respond. Surveys conducted on computer websites are restricted to people who have computers, who are online, who are aware of the survey, and who have the inclination to respond. It is unclear whom these self-selected surveys represent, though we can make some educated guesses (e.g., people with computers will be better off than people without, people who respond to surveys on websites with a known political slant will overrepresent that stance).

A somewhat different form of self-selected sample is one that is exemplified by surveys taken by *Consumer Reports* on commercial products or TripAdvisor or Yelp on restaurants, travel destinations, the arts, and so on. Survey forms may go only to subscribers from within that group who then self-select again by whether or not they respond.

Haphazard Sampling

Much research draws on people who are selected with *haphazard sampling*. Enterprising researchers often recruit participants in public places where people congregate

and where they may have a little time on their hands. Laundromats, airports, shopping centers, beaches, and parks have all been used as a source of research participants. There is a long tradition of researchers going to places where people might volunteer to participate.

Haphazard samples are particularly difficult to replicate. They also run the danger of being biased. People who travel on airlines or people who go to laundromats may differ in systematic ways from people who do not (e.g., socioeconomic status). For some topics (e.g., the study of the perception of optical illusions), this may make no difference. These human abilities are pretty much hardwired into everyone. For example, the optimal illusion in Chapter 1 of this volume is shared by almost all humans. For other topics (e.g., social attitudes), these differences could be pivotal. Differences are influenced more by upbringing and context. Some researchers may think that they are randomly selecting participants and may report their method as such, when in fact their selection of participants has been haphazard. As a critical reader, you need to identify the sampling method for yourself and consider any biases it may introduce into who participated in the study.

Convenience Sampling

In *convenience samples*, also known as *samples of opportunity*, researchers typically study a pool of participants who are available to them (e.g., sophomores at the college where researchers teach, clients with obsessive–compulsive disorder who come to a clinic researchers work at or is near to them, criminals in the prison where the researchers work). The researchers cannot be faulted if these samples are drawn from the target populations (but not with calculable probabilities) and if they make no sweeping generalizations. If the behavior of the sample with obsessive–compulsive disorder is much like what would be expected from clients with obsessive–compulsive disorder anywhere, regardless of how they differ on demographic and other personal variables, representativeness of all participant characteristics is not crucial. However, the more that the dependent variable is associated with variables other than the variable under study, the more important representativeness becomes.

In a variation of convenience sampling, researchers may obtain participants from an organization such as a school, a hospital, or a clinic that has more than enough people who meet the requirements of the research. Instead of picking and choosing, the researchers randomly select as many as are needed. The randomly selected sample permits the researchers to generalize to the entire population of the institution. Further generalizations can be made in accordance with how representative that institution is of all institutions of that type. If the institution is a state mental hospital, for example, the issue is how representative patients in that hospital are of patients in state mental hospitals in general.

Homogeneous, Restricted, Purposive Sampling

Researchers might target one specific subgroup of a convenience sample. *Purposive samples* can be restricted to a group that is *homogeneous* by selecting a subset of people

who share some feature in common. For example, when the point of the research is to study the personal adjustment patterns of college freshmen, the sample should be kept homogeneous by restricting it to college freshmen. Nevertheless, the sample would be expected to represent all college freshmen, not just those from one university, or only one sex, or only one ethnic group.

To illustrate the biasing effects of too restricted a sample, consider the state-by-state differences in average SAT scores for 2015 (SAT Scores 2015, 2015). The global average of the combined verbal and math scores on the SAT was reported to be 1,490 (out of a possible 2,400). Students from Mississippi received an average score of 1,713, even though schools in that state are reputed to be below the standards of most other states. New Jersey, with an average SAT score of 1,520, was nearly 200 points below Mississippi, even though New Jersey is regarded as having a far more advanced school system and spends much more per capita on education. The explanation for the paradox lies in the restricted sample selection in Mississippi. Only 10% or less of Mississippi high school seniors ($n = 858$) took the SAT in contrast to 81% to 90% of high school seniors in New Jersey ($n = 85{,}021$). The New Jersey sample represented a much broader range of ability than the select few from Mississippi. This might occur because more high school students in New Jersey than Mississippi aspire to attend college. For a true test of comparative scholastic aptitude, representative samples of high school seniors would have to be selected, if the total population could not be used, and students were required to take the test.

Network Sampling

For some studies, researchers may not have a pool of readily available participants. They begin with a few people and ask them to refer friends and acquaintances. Participants who are recruited in this way are asked to refer others, creating a *network sample*. Networks of people are used to generate a sample of the desired size (e.g., mothers of infant children refer their friends with young children). Such samples can suffer from the narrowness resulting from people picking people known to them—the sample might be more homogeneous and uniform in composition than would allow for broad generalization. The sample is also vulnerable to contamination of the results as a consequence of participants' talking with each other about their research experiences.

Systematic Nonprobability Sampling

Researchers sometimes confuse nonprobability samples that are systematically drawn with random sampling. For example, a report may state that random selection was achieved by asking the first 50 people who walked through the door at a counseling center to take part. Because the researchers do not know who will come through the door first, they assume this is a random sample. However, the participant's time of arrival is not a random event. Early birds may be different from latecomers in some systematic

ways. Perhaps they are more likely to work jobs from 9:00 a.m. to 5:00 p.m. and are less likely to be college students. Because the sampling procedure uses a system that does not contain an explicit bias that is easy to identify does not mean it produces a random sample.

DIRECT AND ARCHIVAL SAMPLING

In *direct sampling*, researchers obtain data directly from people who are selected to participate in the study. All of the examples in this chapter involved direct sampling. Experimental studies in which some variable is manipulated typically require direct sampling, and nonexperimental and quasi-experimental studies also commonly obtain data through direct contact with participants.

But this is not always the case. Researchers who do *archival sampling* do not obtain their data by direct contact with a sample of participants. Archival studies use data that have already been gathered and are a matter of record. Actuarial records (e.g., vital statistics, births, deaths, divorces) and census data are types of archival data. Data from archives such as medical records, psychological test scores from patients' school records, therapy summaries from case files, and so on are also sources of archival data for research.

Because archival data have already been collected, the researchers' biases can have no influence over how participants respond. The disadvantages are that the researchers are forced to rely on the accuracy and the timeliness of the recorded materials. Income statistics, for example, that were collected 5 or 10 years ago may not be very reflective of a person's income today. With psychological test data from files, there is no control over the qualifications of the examiner, the accuracy of the scoring, and other such crucial aspects of test administration and interpretation. The advantage that the data are not contaminated by the researchers because they were collected for other purposes, might not be balanced by the disadvantage of lack of quality control and the inability on the part of the researchers to fill in what is missing.

Other disadvantages are that archival data are subject to *selective deposit* and *selective survival* (Webb, Campbell, Schwartz, & Sechrest, 2000). In selective deposit, the impression that can be gleaned from some records may be distorted because selective factors governed people's entry into the data pool in the first place. For example, an archival study might report the relation between the college people attended and their subsequent place of residence (e.g., rural, urban, suburban). If the researchers used alumni association entries as data, the result is limited to the individuals who chose to join the association. It is unlikely that a graduate who became a homeless alcoholic would have furnished such information.

With regard to selective survival of data, the concern is that even if records did exist, some records are more likely to be lost or destroyed than others. If so, as a critical reader, you must consider whether the survival of the record is related to its content and whether this might be related to the variable of interest to the researchers and you. For example, are the records of students in poorer school districts more likely to

be lost (as these children may move around more) than records in wealthier school districts (where children's addresses are more stable)?

On balance, the risks of using data-of-record from archival samples probably outweigh their convenience. But some research problems call for use of archival data. For example, if the researchers are interested in the effects of a natural disaster on the divorce rates in a community, this is a good candidate for use of archival records. Tracking down and interviewing all members of the community directly, or even sampling them, might be considerably more expensive and introduce its own problems. Also, archival materials may have to be used because the events under inquiry took place long ago and cannot be reproduced (as in retrospective research). As a critical reader, you should be aware of the limitations of archival records.

Sample Bias

BIASED SAMPLING BECAUSE OF USE OF VOLUNTEERS

The sample in a study may become unrepresentative because some people volunteered or agreed to participate in research, whereas others declined. If the research depends on volunteers, at the very least, the generalizability of the data could be threatened. We have already pointed out that self-selected samples can be biased. In fact, the issue is much broader than that; in one sense, all participants in research are volunteers to some extent. The APA's (2017) *Ethical Principles of Psychologists and Code of Conduct* states that nobody can be coerced, deceived, or duped into participating in research (Standard 8: Research and Publication).

Even college subject pool participation is voluntary in that students in the pool must be given an alternative way to fulfill this course requirement, perhaps reading some research articles and answering questions on them. The APA Ethics Code requires researchers to obtain signatures from participants (or the parents of minors) attesting to the fact that they have been told enough about the nature of the research to enable them to make an informed decision about whether they wish to participate, and the participants have to give their consent to take part in the study.

Although almost all participants nowadays are volunteers, there are different levels of volunteering. For example, there is a difference in commitment among people who (a) volunteer to give an anonymous opinion about a social issue while strolling through a mall; (b) agree to a professor's invitation to spend a half-hour (for course credit) participating in an interesting experiment that has no noxious effects; or (c) answer a newspaper ad to participate in a project that requires them to show up at a specific location at a specific time, takes a good deal of time to complete, and may cause some discomfort and entail some risk.

As a critical reader, the best that you can usually do is to speculate about what effect the means of participant recruitment could have on a given study. In judging this effect, it could be helpful to consider the level of volunteering, because all volunteers are not equal. The volunteer spirit can range from people who are weakly

inclined to consent to people who are eager to be involved and who are prepared to go well out of their way to participate. Ascending levels of volunteering are as follows:

1. Agreeing when asked to participate in a research project that consists of a small addition to a regular routine at work or school. The procedure is not stressful, and almost everybody in the group participates.
2. Agreeing upon personal solicitation to participate in a procedure that is slightly inconvenient.
3. Agreeing when asked to participate in an anonymous, non–ego-involved research procedure that is at most a minor inconvenience.
4. Agreeing to participate upon personal solicitation. The procedure requires considerable time and inconvenience.
5. Going out of one's way to respond to a public advertisement for participation in a study that requires some time and inconvenience.
6. Going out of one's way to respond to a public advertisement in a newspaper, website, or bulletin board. The study requires considerable time, inconvenience, and (perhaps) some risk.

Another factor in your judgment of how representative a volunteer sample is of the target population is the compensation or tangible gain that is or is not being offered to participants. Clearly, if there are incentives to participate, such as monetary payment, volunteering is more likely. Course credit for subject pool participation is also an incentive, but subject pool participants are still free to choose between studies that might differ in the time and effort they are committing to.

We would hypothesize that the greater the sacrifice or inconvenience required of the participants, the fewer the volunteers and the greater the chance of a nonrepresentative sample. In other words, as more and more people decline to participate, the usable sample narrows and generalizability suffers. Unfortunately, the number of nonvolunteers cannot be known in situations where participants are not personally solicited (e.g., where participants are those who responded to advertisements and announcements). For these studies, all that the researchers see are the volunteers. From your point of view, questions can be raised about generalizability when it is evident that the recruitment of participants has yielded a highly select group, and especially a group whose characteristics might in some way affect the evidence about the study's problem.

The most comprehensive presentation of volunteer bias effects was offered by Rosenthal and Rosnow (2009). On the basis of an extensive review of the literature, they concluded that there is reason to have "most confidence" in the finding that volunteers, in contrast to nonvolunteers, are better educated, have higher occupational status, score higher on intelligence tests (especially among men), are more approval motivated, less authoritarian, and better adjusted. Rosenthal and Rosnow also had "some confidence" that volunteers were more likely than nonvolunteers to be sociable, seek arousal, and to be unconventional, first-born, and women when the task was standard but men when the task was unusual.

You may reasonably expect the researchers to provide information about how the participants were recruited, the implications this might have for the results, and the reasons that may have been given by people who declined to participate.

BIASED SAMPLING BECAUSE OF ATTRITION

At the beginning of a study, a sample may be representative but later become unrepresentative. This can happen because of *attrition* (i.e., participants agree to take part in the study but leave the study before it is over). Different from the refusal of people to take part in a study before it even begins, attrition involves the loss of participants while a study is in progress.

Researchers distinguish between two types of attrition: *overall attrition* and *differential attrition* across conditions. A large amount of overall attrition throughout all the conditions of the study can suggest that the study results may lack external validity (i.e., it does not produce results that generalize to certain subpopulations within the target population). For example, suppose a study is examining the effects of a new nutrition program on the development of infants from poor families. When the study begins, there are 200 total participating families, but when the final measurement is taken 3 months later, only 140 families are available. What happened to the other 60 families? Did they change residence? Are more mobile families (maybe being the poorest of the poor) different from more stable ones? Might this difference relate to the effectiveness of the nutrition program?

Differential attrition occurs when more people choose to leave a study from one condition than another. When this happens, you must consider whether the participants in each condition remain comparable on all variables and whether the differential attrition might be related to the treatment. In this case, differential attrition might pose a threat to internal validity, or the experimenters' ability to draw strong causal inferences about the intervention. For example, suppose researchers are testing the effectiveness of an anger management therapy. One hundred subjects are randomly assigned to the experimental condition, and 100 subjects are randomly assigned to the traditional-treatment control condition. Three months after the treatments are completed, the researchers collect a final measurement and find that while nearly all the clients in the control condition return for the follow-up measurement, only 60 return from the new therapy condition. If you compare the follow-up scores on the available clients, is any difference the researchers find because of the treatment, or might it be due to differences between clients, because they have self-selected into the follow-up? What might this tell you about the effectiveness of the treatment?

Attrition is more apt to happen in longitudinal or multisession studies than in those in which data are collected in a single session. The more times that participants are required to return, the more opportunities there are for them to drop out. Attrition can happen in a single-session study as well, however, if participants refuse to answer particular measures.

In sum, then, it is easy to see how a sample could no longer represent the population if the dropouts are more likely to be from one subgroup than another (e.g., women,

middle-age adults, people of low socioeconomic status). Even greater damage is done when more people drop out of one of the treatment conditions than from the others, and in so doing the group samples becomes biased for or against one of the treatments.

When you read an article, you should look to see how many participants began in each group, how many were in the analyzed data set, whether there is a difference in attrition rates between the groups, whether an account is given of any characteristics of the dropouts that differentiate them from the completers, and whether the researchers elaborate on the reasons for all dropouts (e.g., illness, moved, no longer willing to participate).

BIASED SAMPLING DUE TO DROPPING SUBJECTS

Biased sampling can also occur because the researchers decided to drop or remove subjects from the data after they have responded to the measures. The decision to remove any participants from the sample after the study begins should be completely explained by the researchers. Most important, the reasons for dropping subjects should be unaffected by knowledge of whether subjects performed in a way that is contrary to the hypothesis. For example, suppose in a hypothetical learning experiment, the time to solve a problem is the criterion measure. One member of a group that is expected to do poorly cannot solve the problem and gives up after 20 minutes. Most people solve the problem in less than a minute. Should this subject be dropped from the study? Casting out the single individual would have eliminated the score of the "worst" case in this study. It is a procedure that could make considerable difference in research with small samples. If the decision is made not to drop the subject, what time should be recorded for solving a problem that was never solved? Assignment of an arbitrary score is a possibility. There are several ways of doing this, each with a different consequence. You should look at how the issue of missing data was handled and what effect the solution might have had on the data and on the results. You must judge whether the researchers' course of action biased the results in any way.

When subjects are lost, either through attrition or removal by the researchers, simply replacing them by resampling will not restore random sampling. Despite the truth of this, researchers are more prone to discuss the loss of participants and the reasons for it, rather than abandon the research entirely. The researchers are more likely to cite it as a limitation of the study. Loss of participants is inevitable because of real-life necessities (e.g., moving, illness, loss of interest). The critical reader can tolerate these losses if they are small in number and not selective, even if this creates a limitation to the study's conclusions. The severest problem arises if more people are lost from one group than from the other(s) and when the differential loss upsets the equivalence of the groups on some important characteristics, or when the loss is not only selective but appears to be a consequence of one of the treatments.

Opinions vary about the practice of dropping participants when a *manipulation check* reveals that they did not receive the treatment or when they did not perceive the experimental condition as intended. Kazdin (2016) and others have regarded the

dropping of such participants as inappropriate because it violates random assignment. There are instances, however, in which it might be a worse error to retain them. For example, in a study evaluating the effects of a drug, to include people who reveal that they did not take the drug as prescribed makes little sense. It presents a major threat to the internal validity of the study; for example, you cannot study stress effects in people who are not stressed by the manipulation. Researchers cannot study the effects of noise on people who are deaf or use people who are unable to see in a visual perception study. Violation of randomization might make a difference, but retention of those who do not receive the experimental treatment as intended inevitably makes a difference.

The rationale used by the researchers for dropping participants should be made clear. To judge the credibility of the study, you must determine whether there was a problem with attrition, whether the loss was associated with the experimental treatment, and whether the problem was handled effectively.

Sample Size

One of the most frequent criticisms of studies that did not work out the way the researchers had hoped is that there were not enough participants. The sample size must be sufficient to provide a fair test of the predictions (i.e., a reasonable chance that a false null hypothesis will be rejected or that the strength of the relationships being studied can be estimated accurately). There are no immutable answers to the question "What sample size is large enough?"—but there are good ways of making informed choices.

The soundest way for researchers to go about deciding on the size of the sample is to do a statistical *power* analysis (Cohen, 1988; Kelley & Maxwell, 2012) in advance, and to use this as a guide to making a wise decision. The probability of accepting a null hypothesis that is false (i.e., making a *Type II error* or *beta* [β] *error*) is something that researchers obviously want to keep to a minimum. The power of a study is $1 - \beta$. For example, if the probability of a Type II error is 10%, the power would be 90%. In contrast, if β is 70%, the power would only be 30%.

Power is a function of the sample size, the criterion that the researchers adopt for statistical significance (called *alpha* [α]), and the *effect size*, the magnitude of the relationship or treatment effect that the researchers wish to detect. Increases in power can be achieved by (a) raising the level of significance required (which also means increasing the chance that a true null hypothesis will be falsely rejected), (b) increasing the magnitude of the relationship or experimental effect (e.g., by using measures with less error or by using stronger treatments), and (c) increasing the size of the sample. In planning the study, prudent researchers decide what level of power is desired, what level of effect size to anticipate, what *significance level* to set, and what statistical method is to be applied. Armed with this information, the size of the sample that is needed can be determined. Researchers can simplify their task by consulting Cohen's (1988) power tables or using one of the tables that is available online.

On seeing that a study could not reject the null hypothesis because the alpha level was $p < .05$ in a study with only 10 people in each of two groups, you could surmise that there was not enough power to detect a significant difference. A post hoc analysis might show that if the sample had been increased to 20 per group and if the same level of differences between groups (effect size) had been sustained, the results could have been significant at $p < .05$. If so, it would be reasonable for you to surmise that there was insufficient power in the study. You could conclude that the researchers should have done a power analysis before deciding on the size of the sample. This issue is discussed further in Chapter 8.

Using power analysis has become de rigueur for publication in the best psychology journals and is required by most agencies that fund research before it begins. If the researchers have no clue about the size of the effect they seek, the answer to the question "How many subjects do I need?" is usually "As many as feasible." A variety of different sample sizes have been suggested as guides for determining sample size.

We think that if significant results are obtained (a) with 20 to 30 participants per group, (b) the distributions of data within conditions are reasonably normal, and (c) the assumptions of the statistical technique are met, there usually is no good reason for you to dismiss the results on the grounds that the sample is too small (see Chapter 9 for a more elaborate discussion of this issue). If inconclusive results are obtained, however, the size of the sample could be the culprit. At the other extreme, when the sample is extremely large, you must look carefully to see whether the obtained difference has theoretical or practical meaning (see Chapter 9). Trivial differences are statistically significant if the sample is very large.

Assignment to Treatment Groups

After learning how participants were selected to participate in an experiment, you must consider the crucial issue of how they were assigned to treatment groups.

SYSTEMATIC ASSIGNMENT

The potential for bias is always present when participants are systematically assigned to treatment groups. For example, suppose researchers assign the first 25 allergy patients to a control condition when the study begins during the winter season and the next 25 to the experimental treatment during the spring. Clearly, the treatment and the presence of pollens in the air are confounded. Other researchers assign one class of students who have high scholastic aptitude to the favored experimental teaching method and a second class who have lesser scholastic aptitude to the control group. When criticized for the biased assignments, the researchers insist that the assignment was random because it was decided by the flip of a coin. Whichever way the coin landed, the procedure was ill-advised because of the presence of an obvious *confounded variable*. The teaching method is confounded with the achievement level of the class, as well as any other variables, such as the experience of the teachers,

that distinguish the two classes. When it is necessary to assign a whole group as a unit, the researchers must convince you that the two groups being compared were equivalent when the experiment started and that the timing or treatment makes no difference.

RANDOM ASSIGNMENT

Many people are confused by the difference between random sampling and *random assignment* (sometimes called *random allocation*, especially in studies from medicine). Sampling brings participants into the study; assignment places them into treatment groups once they are in the study. Random sampling does not resolve the problem of nonequivalent groups within the study. Random assignment should take care of this. Random assignment means each sampled participant has an equal chance of being in any of the conditions within the study. Some researchers state this as "each condition has a random sample drawn from the same population." Can you see how these two statements are equivalent?

It is important to ensure that the assignment procedure is truly a random one. Assignment made by a table of random numbers or a random numbers generator (easily found online) is the simplest and surest way to do so. Sometimes clues in the text of a research report reveal that the claimed random assignment could not have been as carried out. For example, suppose one group in a study consists of 13 men and 12 women participants, and a second group is made up entirely of 25 women participants. What are the chances that this assignment occurred at random? Less than one in 10,000 times. Either subjects were not randomly assigned to conditions or the researchers were very, very, very unlucky! Either way, the conditions and the sex of subjects are confounded. At other times it becomes evident that some systematic method of assignment was used, and that method could not have been random. This might occur, for example, if you discover that the pretreatment weight of subjects in a study of a new diet suggests that those receiving the new treatment weighed noticeably more (or less) when the study began.

Summary

Our discussion in this chapter focused on the importance of sample representativeness and the two main issues with research samples: selection and assignment. In selecting participants, experimenters must consider the advantages and pitfalls for nonprobability methods. Samples can be biased by using volunteers, by overall attrition, by differential attrition from conditions, and by the experimenters removing participants from the sample in a way that alters the samples representativeness or the equivalence of subjects within conditions. We highlighted the advantages of random as opposed to systematic methods of assignment of the participants to treatments. Power analysis can serve as a valuable guide to selecting the size of the sample.

Confounding Variables and Their Control

5

A critical reader of research

- understands the potential impact extraneous variables may have if they are confounded with the independent or predictor variable of interest;
- can identify the different methods (e.g., random assignment, homogeneous samples, blocking and matching of participants, statistical control, within-subject designs) that the experimenter can use to promote group equivalence;
- considers other processes (e.g., maturation, history) that may have caused change in participants over the course of a study;
- thinks carefully about how the participants' motivation, role perceptions, response sets, expectations, and communications with other participants might influence their responses to the study's conditions;
- considers how researchers might treat participants differently, whether inadvertently or on purpose, to elicit different responses within different conditions; and
- looks for influences on participants' responses from situational variables, apparatus, procedures, and the data collectors that might cause confounds or increase variance in participants' responses.

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Extraneous or *confounded variables*, ones that are irrelevant to the independent or predictor variable but that also vary systematically from one group to another in a study, create a problem for researchers. When they are present, researchers cannot be certain whether the variable of interest or the extraneous variable with which it is confounded is responsible for the observed relationship or difference between the groups. This chapter presents the principal sources of confounding: participant variables; placebo effects; experimenter variables; stimulus, procedural, and situational variables; apparatus, judge, rater, and scorer variables; and nuisance variables. For each of these sources, methods are presented that, during a critical reading, you may counter that researchers use to control extraneous variables.

It is of utmost importance for researchers who are seeking causal inferences to keep variables other than the independent variable equal, on average, across conditions. As a critical reader, you must consider what factors ought to have been kept equal and search the research report to find out which ones actually were kept equal. You should also be interested in learning how this equating was accomplished by the researchers.

Confounded variables could include anything besides the variable of interest that could account for or influence the dependent variable. Any of these could lead to a rival explanation of the results. You must search for confounded variables because there may be some that the researchers did not consider and try to rule out. For example, researchers who want to test for a relationship between geographic residence (e.g., rural, suburban, urban) and happiness might have thought to control for the respondents' age and sex. Considering the problem, you might think that income, a factor certainly related to where someone lives, also needed to be taken into consideration. This would be a confounded variable the researchers missed.

In critiquing a study, you should (a) concentrate on the major categories of confounding variables, (b) note the various ways that these confounds or *artifacts* were (or were not) controlled for in the study, and (c) reflect on what else might have been done to control or remove them from consideration. Developing an understanding of what variables need to be controlled is an important step in the critical evaluation of research. This part of the task is not complete until you evaluate the appropriateness of the methods that were used to control the potential confounding variables. The reason for this is that researchers sometimes correctly identify confounds but do not satisfactorily control them. There are a number of choice points along the way where researchers can select a control method to use from among a variety of options.

Participant Variables

Characteristics of the participants other than those under study as predictor or stratifying variables might contribute in some way to the results. There are an enormous number of ways that people can differ from each other. It is doubtful that all are ever accounted for, but a systematic search through categories of such variables is helpful. Once these have been identified, you should ask, "Might these confounded partici-

pant characteristics relate to the dependent variable in a way that would augment or diminish the apparent effect of the independent variable?"

DEMOGRAPHIC AND PERSONAL CHARACTERISTICS

The demographic characteristics of the participants make up a large category containing subcategories that can easily be associated with the results. You should examine the research report to find out whether the groupings of participants in the study are comparable on such characteristics as age, sex, race/ethnicity, education, socioeconomic status, marital status, living arrangements, and employment status. Depending on the topic under study, some of these might be important for the researchers to control and others not so much.

Personal characteristics such as habits and traits form another large category of considerable potential importance. For example, if IQ is a variable that could influence the scores that participants obtain on the dependent variable measure, are the groups the researchers have formed comparable in IQ? Are there any characteristic behaviors of the participants that could have a bearing on the results? If the research groups are from clinical sources, are they equivalent in diagnosis, length and severity of illness, medication, history of previous treatment, history of previous disorders, and presence of concurrent disorders? Are the members of one group more suggestible than those in another group, and is their performance more likely to be influenced by their suggestibility? These variables might be especially important to consider if one treatment condition occurred in one clinic and another in a different clinic. Here, the treatment is confounded with any differences that might exist between the clinics' clientele or other features of their settings.

Methods for Equating Groups of Participants

In studies that call for the use of more than one group (e.g., an experimental group, a control group), the experimenters searching for a causal relationship aim to establish equivalence between the groups on all but the independent variable. You should know the various options available to the researchers and judge whether the one used was satisfactory.

Random Assignment

Random assignment (as distinct from random selection; see Chapter 4) is the preferred method for placing people in treatment groups. It is the easiest and surest way of scrambling all possible confounding variables across groups. You should be aware that sometimes assignment procedures to which authors refer as random are in fact not random at all. For example, a procedure that assigns the first 50 participants to Group A and the next 50 to Group B (i.e., systematic sampling) is not random. The first 50 people who respond to an advertisement might have characteristics related to their "eagerness" to participate in the study that also relate to the independent

variable, perhaps their motivation to adhere to a treatment's demands on them. In random assignment, each participant has to have an equal chance of being assigned to any of the treatments. Then, you, as the critical reader, would know the motivations of participants was roughly equally distributed across conditions.

Regrettably, random assignment is the best way to promote group equivalence, but it does not guarantee it. For example, if the size of the sample in each condition is small, random assignment may fail to meet its purpose. Think about randomly assigning 16 people to two conditions. If it leads to six men and two women in one group and the opposite sex makeup in the other group, there are 3 times as many people of different sexes in each group. We would be much less likely to get a confound with sex this large if 48 people were available to randomly assign to each condition.

Also, because chance is involved in random assignment there is always the possibility that the researchers have simply been "unlucky." Think of 20 ways that people might differ. If participants are assigned to two conditions at random, chances are good that the two groups will differ (at $p < .05$, 1 in 20) on one of these variables, even if the random assignment was done properly. This part is not luck; it is a fact of life. The researchers are unlucky if the difference that happens by chance also happens to be related to the dependent variable.

Randomized Block Designs

Instead of ruling out the effects of a static-group variable (e.g., sex) using random assignment, researchers may be interested in studying the effect of such a variable and trying to determine whether it interacts with the independent variable. In effect, the researchers welcome the extraneous variable into the experiment and make it into another predictor variable. The researchers divide participants into two groups, a *block* or *blocking variable*, on the basis of their sex and then randomly assign an equal number of participants of each sex to each of the treatments. This then becomes a *randomized block design*. For example, in a study of the effects of meditation on headache pain relief, the researchers may create groups of only men and only women and then assign equal numbers of each to each meditation condition at random. This procedure assures an equal number of men and women in the treated and an untreated group, thus neutralizing the effects of sex as a source of confounding. The researchers can then determine whether men and women react differentially to the meditation condition.

The blocking option is not limited to one static group variable. The experimenters can also block for age with several levels (e.g., young and old men, young and old women). Of course, as more static-group variables are added, the participant pool must increase because more and more cells must be filled. Because it is very possible that the various cells will have unequal numbers of participants, the experimenters must be careful to handle this problem.

The blocking option can also be used to prevent possible critical confounds when the sample is small. So, if the researchers only have 16 participants available to study and think men and women may differ on headache pain without the meditation

treatment, they can block on sex and randomly choose four men and four women for each condition. This would be an indication that the researchers have thought carefully about possible confounds and the problem that a small sample creates for them.

Homogeneous (Restricted) Sampling

Another way to avoid nonequivalence of groups is to keep the sample characteristic narrow by restricting it to a homogeneous group of participants. For example, consider a study of a treatment to reduce the effects of rumination (repeated thoughts) in depression. The researchers decide to assure equivalence of the experimental and control group participants by restricting the sample to highly motivated, bright, male, White, affluent college graduates, age 30 to 35, with mild to moderate reactive depression of short duration. With the pool so restricted by this process, members of the experimental and control groups resemble each other on these individual variables, and the experimenters succeed in controlling all of these potential confounds. The price to be paid for this success is that the generalizability of the study can be challenged. This is particularly so if the researchers use the data to make assertions about the treatment of depression in the general population.

Matching Participants (Person-for-Person Matching)

The procedure of *matching* is related to the blocking procedure mentioned previously, but here the matches are done on a case-by-case basis. Also, we mentioned blocking in situations in which the experimenters can conduct random assignment. In fact, blocking or matching can also be used when the researchers do not control the introduction of the independent variable (in naturally occurring experiments) or when they wish to make static groups that differ on the variable of interest as comparable as possible on other confounded variables.

The procedure used for matching people who are assigned to two different groups is to go into the pool of potential participants and identify pairs of individuals who match each other on one or more key variables. Assume the rumination and depression study did not involve an experimental manipulation, but the researchers went out and found one clinician who used a treatment for rumination and another who did not. The two groups of clients might differ in lots of ways other than which treatment they were receiving because different clinicians might attract clients of different age, sex, race/ethnicity, and education. To rule out these confounds, the researchers may pair two male, White, college graduates with depression of short duration from the different clinicians. The next pair might be two female, high-school graduates with moderately severe depression of long duration.

This kind of person-for-person matching would continue until a sample of the desired size and diversity is attained. Because the groups would be nearly identical on the matched variables, none of the between-group differences at the end of treatment can be attributed to differences in these particular initial characteristics. In studies with three groups, demographic "triplets" would be located.

Dropping Participants Without Matches

Another way of promoting group equivalence is to drop individuals from the study for the sole purpose of equating static groups. If a group of clients that received a rumination treatment are older than control group members, the researchers can drop the oldest members of the experimental group and/or the youngest members of the control group. If this can be done, it succeeds in equating the age means, and no further adjustment is necessary. If there is a surfeit of available participants (e.g., the clinicians have many clients), the researchers simply trim the sample by excluding a few of the oldest from one group and the youngest from the other group. Unfortunately, the process could very well change the means of other variables that distinguish participants and throw the groups out of balance in other ways.

Problems With Matching Designs

You can see the problem with any of the equating procedures that involve a matching strategy: It is exceptionally difficult to find enough people who have a match in the other group on more than a limited number of variables. Finding matches becomes less likely as the researchers identify one, then two, then more participant characteristics they want to control between groups. Because most researchers do not have access to the large samples that would be required for matching on numerous variables, they prefer to match on a restricted number of variables and to find other ways to control the remaining ones that might present problems. You, as the critical reader, must weigh the researchers' claim of group equivalence when it is based exclusively on having matched participants on a couple of variables and when there is reason to think that other personal characteristics might still be confounded with the independent variable.

A study that has blocked, matched, or paired participants in groups can do so before or after the experimental manipulation is administered and the dependent variable is measured. Clearly, it is preferable to do so before beginning the rest of the study. This is because (a) the pool for finding matches might be larger, and (b) the cost of administering the manipulation to participants who do not have matches in the other group might be avoided. However, sometimes post hoc matching cannot be avoided because the manipulation has already occurred (e.g., the participants in one clinic have already learned to meditate).

Whatever equating procedure is used, it should be described by the researchers so that you can decide whether the procedure truly equates the groups, is free of bias, and was done in a manner uninfluenced by the values of the dependent variable. Dropping participants to achieve equivalence after the data have been collected could increase your skepticism about the procedure's freedom from bias. By whatever procedure, when equating occurs first there is no chance that the decisions about what constitutes a good match (e.g., how close in age participants need to be to be considered matches) or which match to choose when there is more than one good match for a participant can be influenced by how the participants responded to the

dependent variable. You should be careful to read what was actually done to equate the groups in a study, not just rely on the label the researchers used.

Statistically Equated Groups

In addition to the equating procedures discussed so far, group equivalence can be established by statistically adjusting the data so that the means, medians, or percentages of the groups on important participant variables are equivalent across groups. For example, for a continuous variable like age, the researchers may report that the mean age of participants in the experimental group was 12.5 years as compared with a mean age of 14.1 for the control group participants. These means are found not to be significantly different from each other. Ideally, the two groups should not only be not significantly different ($p > .05$) but should be significantly alike ($p > .95$). Studies that do not reach a strict standard of equivalence are not faulted as long as the difference between the groups is clearly marginal and has no discernible practical significance. For example, the difference between these ages might be irrelevant for a study of a new middle school English curriculum but might be considered a confound if a curriculum on sex education is being studied (the two groups might contain different numbers of students who have and have not reached puberty).

Differences between groups are assessed by statistical tests (e.g., the chi-square test, *t*-test). The critical thing you should look for in a report of a study that did not use random assignment is a presentation of such data. At a minimum, the researchers should provide statements that support and document group equivalence on important variables and an acceptable procedure for dealing with nonequivalence. It is not unusual for reports to include tables that show the group differences on important extraneous variables before and/or after they have been equated.

If concern remains about group equivalence, instead of prearranging group equivalence by blocking or matching, the researchers can inspect the frequencies of subjects who fall into the different combinations of values of the confounded variables and apply statistical controls to those that are not equal in subsample size. *Covariates* are another name for confounded variables, other than the independent variable, that correlate with the dependent variable. It is important that they be controlled lest the effects found in the study be attributed erroneously and exclusively to the independent variable. Such sources of confounded variation can be dealt with statistically rather than experimentally. Before the researchers can do so, however, the covariates must be identified. This identification can be done by making informed judgments about any variables that might be confounded with the dependent variable and then looking at correlations among such variables. For example, in a study in which people have been preassigned to conditions and the mean ages of the groups are different, age can be treated as a covariate in an *analysis of covariance*. The technique adjusts (i.e., removes any relationship between) the scores on the dependent, outcome, or criterion measure for any age effects.

In some cases, the researchers may have overlooked covariates that you, as a critical reader, can specify as possibilities, can wonder about, and can wish that the

researchers had taken into account. Such doubts can justifiably affect your readiness to accept the conclusions of the study. Also, if an analysis of covariance was used, you should be satisfied that the data meet the assumptions of the statistical method.

The important thing you should look for in a study that used statistical equating is a presentation of how this was accomplished. At a minimum, the researchers should provide statements that support and document group equivalence on important variables and an acceptable procedure for dealing with nonequivalence.

Within-Subject Design

Sampling error is the largest error that is built into a design that has different people in each group. The different people within the same group are going to show variability in their initial level of the dependent variable (e.g., the severity of their headaches) and how they respond to the way they are treated (e.g., people will have different levels of acceptance of meditation as a treatment and this will impact its effectiveness). Indeed, the error term in inferential statistics (e.g., the *t*-test, analysis of variance) provides the estimate of these between-subject differences on the dependent variable, free of the effect of group membership.

In some situations, researchers may choose to avoid sampling error by using the same participants under two or more different conditions, known as a *within-subject design*. In addition to controlling for between-subject sampling variance, within-subject designs can be adopted to control some of the confounding variables that materialize when different groups are used. For instance, if members of one group are experiencing greater headache pain than members of the other group, this difference could be confounded with the vigor with which clients seek out an alternative treatment. Such problems are ruled out when participants serve as their own controls. Instead of assigning clients to a meditation or no-meditation group, participants pain level is measured before the study begins (control condition) and again after they have learned to meditate and have been doing it for a while (experimental condition). This design minimizes differences in participants between conditions because the researchers can examine the change from before to after treatment as the outcome variable (and the initial level of the headache pain is controlled). However, some kinds of studies do not lend themselves to within-subject designs. For example, in a comparative study of two long-term psychotherapy approaches, the researchers could not very well give the two different kinds of treatment to the same person.

This simple design has other problems as well. As a critical reader, you should look for answers to a new set of questions that apply to this approach. Although the same participants are involved on multiple occasions, are they truly the same on the second occasion as they were on the first, or have they changed over time? Would the headache have faded anyway with time even if it was not treated? Has exposure to the initial experience (e.g., talking with a concerned professional) changed them in a way (e.g., relaxed them) that would alter their pain on a second occasion even if no meditation had occurred?

Imagine a study on the effects of noise versus soothing music on test performance. A test is taken by the same participants under these two different conditions. The

researchers have controlled for between-subject achievement differences by using a design in which each subject serves as her or his own control, but could this one group have done better the second time because they had taken the same test before? If alternate forms of the test were used to rule out this possibility, could the second performance have been enhanced because of familiarity with the test situation and instructions? Could the two tests be sufficiently different in difficulty to affect the results? We will delve deeper into the weaknesses of this design in Chapter 6.

CHANGES IN PARTICIPANTS OUTSIDE THE EXPERIMENT

Regardless of whether a multiple group or within-subject design has been used, people may undergo changes that have little or nothing to do with the experimental treatment. This should especially concern you if considerable time elapses between the pretest and posttest measures. Some of these changes can be due to alteration in the internal states of the individual. People grow, develop, age, or become more depressed, anxious, or happy for reasons unrelated to the treatment. Changes also may result from external events other than the experimental treatment. Shadish, Cook, and Campbell (2002) referred to the former category as *maturation* and to the latter as *history*. Both can cause a confound in a study and lead to misattribution of the extra-experimental influences to the treatment effect.

Methods for Control

Extra-experimental change cannot be prevented over the long run: people change. But, it can be controlled by random assignment in between-group designs. People may change, but if the assignment has been truly random, the odds are greatly against anything (e.g., growth, historical events) happening systematically to one group that does not happen to the other.

PARTICIPANTS' MOTIVATION AND ROLE PERCEPTION

The motivation to participate in a study might be different for participants in the study's different groups. Is one group more ego-involved than the other? Does being a part of the study count more for some participants so that they try harder or respond differently, or are they more threatened and made more anxious by it? Are the members of one group more motivated to cooperate and to perform as well as they can because the benefits that they anticipate are greater than they are for the other group? Have the two groups been promised unequal rewards for participating? Might one group have volunteered in the hopes of obtaining highly desired symptom relief, whereas the other group did not share this motive?

Participants' perception of their role in the research also might differ systematically in the study's groups. *Role selection* is a set of behavioral prescriptions adopted by individuals to create a particular impression. Do the participants in one group have a greater need to be "good participants," "bad participants," or "negativistic or resistant

participants"? Might one group have a greater tendency to "scope out" or second-guess the researchers? Does one group contain more people with a negative *response set* (i.e., who tend to disagree) or more people whose tendency it is to agree? Any of these kinds of variables, either singly or in combination and when confounded with group membership, may affect the outcome of the study.

Methods for Control

Again, you can see how powerful random assignment can be in ruling out these possible confounds. It ensures that people with different role selection and response sets are evenly distributed across the study's conditions.

Where possible, you should judge whether all participants received the same benefits and rewards for participating and whether every effort was made to hold the level of motivation constant across groups of participants. *Unobtrusive measures* and *nonreactive measures* (Webb, Campbell, Schwartz, & Sechrest, 2000), measures taken that do not intrude on the natural flow of behavior, are constructed with an eye to controlling role selection and response sets. They can help to minimize the influence of response sets and to foil dissembling and role selection. They are described further in Chapter 7, in a discussion of issues in measurement.

Finally, researchers use *deception* in research. They can do this (a) actively, by deliberately misleading a subject by giving then false information about the purpose of a study, or (b) passively, by omitting important details of the study (Christensen, Johnson, & Turner, 2014). The intent of the deception is to lead the subject's thinking away from the study's actual purpose, thus making it less likely that they will act based on what they think is the "correct" response rather than react naturally and spontaneously. Of course, deception violates informed consent, and must be used minimally and only with the approval of a human subjects review committee. Deception will be covered again in Chapter 11, but we wanted to point out here its role in reducing confounds.

COMMUNICATION AMONG PARTICIPANTS

The possibility that participants who have already been in the experiment communicate their experiences with those who are waiting their turn is a potential source of concern that must be controlled. Any study in which (a) the participants are drawn from within the same hospital, clinic, school, club, church, or social group and (b) a task is used on which performance could be affected by foreknowledge, is potentially vulnerable to contamination by communication among participants.

Communication among participants is not a problem when participants are to undergo some straightforward procedure (e.g., measuring auditory acuity or motor performance). However, where there are right and wrong answers that can be communicated, problems to solve, deceptions that require debriefing, judgments that are to be made—in short, where equality of naiveté is a requirement—communication can contaminate a study.

For example, if children in a class come into a room one at a time to see the researchers, you can expect that the word will soon spread about what happens in there. A prepared and informed (or misinformed) child could behave or respond differently than one who is uninformed. For some children, a request by the researchers to keep a secret is almost tantamount to a hard-to-resist invitation to divulge it. If the setting lends itself to communication, you must search for an explanation of how intersubject talk was minimized.

Communication can become a concern with adults when different experimental groups are seen over a period of time and participants happen to know each other and to exchange information. Suppose there is an experimental-control or experimental-alternative treatment-control design on stress reduction, weight reduction, or parenting. Because the subjects are all recruited from the same clinic, some of the members of different groups know each other and share acquired knowledge of techniques learned during the course of the study. This could contaminate the study by destroying the integrity of the independence of the treatment conditions.

The danger of this happening increases in situations in which participants are sampled from within the same organizations or when *network sampling* is used. Under these circumstances, the critical reader should look for an acknowledgment of the problem by the researchers in their report, an assurance that the issue was examined, a description of the precautions that were taken, and a discussion of how any problems that were uncovered were resolved.

Methods for Control

As each participant begins the study, the experimenters ask in a friendly, conversational tone what his or her expectations are and what others might have said about it. If the response appears to be a candid expression of lack of information, it enhances the experimenters' confidence that there has been no communication in that case. Those who have been told anything of importance can be thanked and excused, or they can be put through the experimental procedure, although it is predetermined that their data will be discarded.

At the end of the experiment, a control method that works fairly well, at least with adults, is for the experimenters to explain to participants the reasons why they should not talk about the session. They will explain that the validity of the results will be ruined if participants know the study's procedure or purpose before they take part, and sincerely request cooperation in keeping silent. This request is made during *debriefing* at the end of the session with the participant. Also during debriefing, participants can be told that they can receive a copy of the research report when the study is complete. This "reward" for silence can also help gain the participants' cooperation. Researchers who communicate that such care was taken in debriefing can raise your confidence in the entire research procedure.

Sometimes the problem of participants communicating with one another can be avoided by obtaining participants from different places. The hope is that they are not likely to know each other or that they do not have an opportunity to talk

about the study. Another method is to have a team of data collectors work quickly and finish before communication can become much of a factor. When possible, some researchers collect data all at once in an assembled group setting that can be monitored for communication.

Experimenter Variables

It has long been known that experimenters can have unwitting effects on the outcome of a study. Rosenthal and Rosnow (2009) presented a comprehensive discussion of these artifacts and the history of the artifact problem in the behavioral sciences. They divided experimenter effects into noninteractional and interactional effects.

Noninteractional experimenter effects include observer and interpreter effects. What has been called the "personal equation" has a long history in the sciences; it came into prominence when it was realized that astronomers differed from each other in their recordings of their observations. Psychologists, too, are subject to systematic differences in observations and vary in the recording of time and other measurements.

Interactional experimenter effects include selective ways that experimenters and participants influence each other's behavior during the research. Rosenthal and Rosnow (2009) mentioned biosocial experimenter effects, psychosocial effects, situational effects, and modeling effects. The word *selective*, as used here, refers to confounding influences that are different for one group than another. Variables that differentially affect the outcome in different conditions are confounded with the experimental manipulation. Other experimenter variables that affect all groups alike do not confound the group differences but could influence external validity (e.g., the effect appears only when the experimenters are a man or a woman).

Biosocial experimenter effects are parallel to what we have referred to as the demographic characteristics of the participants. In addition to what the experimenters do, personal characteristics of data collectors who come into direct contact with participants can have an unplanned effect. As a critical reader, you must ask (a) whether characteristics (e.g., age, sex, race/ethnicity) of the person who interacted with the participants could have had any bearing on how the participants responded, and (b) whether the same results would have been obtained had the person been from a different group. For example, in research on sexuality female participants may react differently from male participants to a data collector of the same or opposite sex. These differences can confound the experimental effect. So, if all participants (men and women) in one condition met with a data collector who was a woman and in the other condition all participants met with a man, it would be a confound. If all the data collectors were of the same sex, it could increase within-group variability, perhaps by making same-sex participants more open in their responses than opposite-sex participants. It could also affect external validity by making the results generalizable only to experimenters who were the same sex as those used in this study.

In many instances, it will be hard for you to discern from a research report whether these types of experimenter effects were present. However, in some instances the researchers should recognize that their efforts to minimize these effects will be

appreciated by readers, as when they describe the sex of the experimenters in a study of sexuality.

Psychosocial experimenter effects parallel the participant's personal characteristics, habits, and traits. Was there anything about the experimenter's personality, demeanor, behavior, manner, or appearance that could have affected the way the participants responded? Characteristics of the data collectors (e.g., regional accent, dress, appearance) can contribute to differential response of groups of different compositions.

The *self-fulfilling prophecy* of researchers refers to the effect of the researchers' expectancies on the participants' responses. Did the experimenters, wittingly or not, exert any unplanned influence over the participants' behavior? Were any of what are commonly called *demand characteristics* (Orne, 1962) present—words, acts, or expressions of the data collector or aspects of the setting (e.g., a lab coat, a diploma on the wall) that serve to inform the participant about what is expected and lead some participants to oblige? Experimenters can grimace, smile, look anxious, look relieved, look pleased, sigh, frown, praise, show warmth, be curt, be distanced, appear angry, act annoyed, and look disdainful. All these things can influence the way the participant behaves or performs even though the experimenters may not be consciously aware of having done them. These experimenter behaviors are called demand characteristics because the experimenter is perceived to be making demands (no matter how subtle) on the participant to respond in a particular way (e.g., a lab coat suggests authority and seriousness of purpose).

Sometimes the experimenters are plainly encouraging certain kinds of responses, but in other instances the methods and procedures themselves provide the participants with response cues. As a critical reader, you are interested in knowing whether the experimenters expressed any expectation that the participants would respond in a particular way, through vocal inflection in giving instructions, facial expression, or demeanor, or if the experimenters gave unintended extra-procedural positive or negative reinforcement of the behavior or responses of the participants. Because you were not there, this will be difficult, if not impossible, to determine from the written report. You can, however, look for the methods that the researchers might have used to control for these biases. If they are present, the research should gain credibility in your eyes.

You must be aware of the role that experimenter bias may play in influencing research outcomes. The ideal of the completely neutral and dispassionate experimenters, who are interested only in the truth regardless of the outcome, would be hard to find in real life. Researchers have a vested interest in studies coming out as predicted because they want support for their positions. Slight but consistent tipping of the scales in favor of a hypothesis can occur through time-recording errors, errors of measurement, leanings in making judgments or doing ratings, errors in entering or transcribing data, slanted interpretations of events and of data, tilted reporting of the results, and drawing of unsubstantiated conclusions, either singly or in combination. The good researcher is aware of these possibilities and takes actions to do the research as fairly as possible. As a critical reader, you must look for what these actions might have been.

METHODS FOR CONTROL

Principled and self-disciplined scientists do their best to control their biases. The institutionalized process of critical peer review also helps their efforts to keep themselves from becoming part of the experimental treatment. Nevertheless, hoping, wishing, and expecting can have an influence on the results. Intentional self-discipline, although helpful, is no guarantee that experimenters will not say things, do things, or reveal things that can affect a participant or jeopardize the objectivity of the research report.

To the extent that it is possible, researchers try to bring experience and self-discipline to bear in a conscious effort to control these influences and to hold conditions as constant as possible for all participants except on the experimental variables of interest. You must ask yourself, "Could the behavior, or even the presence, of the experimenters reasonably have had any differential effect on the various participants in ways that were not intended?" If the answer is yes, you should take a careful look at what procedures were introduced and what precautions were taken to eliminate or minimize these effects.

In the attempt to control biosocial effects researchers can (a) anticipate them, (b) rule them out or minimize them by design, (c) analyze them, and (d) report them to readers. If, for example, it was anticipated that the experimenter's sex might make a difference, it would be wise for the researchers to *counterbalance* sex by using male and female research personnel and having them administer the manipulations in all conditions. Astute researchers might block same-sex and opposite-sex pairing within each condition and look to see if the sex pairings interacted with the effect of the conditions (and whether sex had a main effect). The data would be analyzed to determine whether different results were obtained on the basis of the sexes in the pairing. If the results were comparable, the data could be pooled. If not, the experimenter's sex would be used to qualify and interpret the results.

Psychosocial effects that relate to the experimenters' group identity can be dealt with in the same way as biosocial effects. Those that have to do with the experimenters' behavior or demeanor can be kept to a minimum by doing pilot studies in which the experimenters have the opportunity for practice and rehearsal. It helps if the experimenters' behavior is monitored by impartial observers and feedback is given. Whenever it is important to gauge how respondents perceived the session and how they felt about it, an inquiry to participants during the postsession debriefing can be helpful. Printed, recorded, or computerized instructions to participants are applicable in some studies. Such instructions are uniform, and there is reduced risk of experimenter influence. You should be reassured when you read that steps have been taken to anticipate and control experimenter variables.

Finally, the preferred way to ensure that experimenter bias does not influence participants' data is by using *masking* techniques that keep the people who interact with participants unaware of which condition each participant is in. Using the example of the effect violent video games have on aggression (see Chapter 2), the experiment can be devised so that the researchers who meet the subjects have no idea which type

of game each subject will play. They read the instructions, answer questions, sit the player at the console, then leave. The computer then randomly places the subject in a condition. The subjects' measures could be taken directly by the computer (a good way to ensure data collection is unbiased), or when the game-playing session is over, a second experimenter, again unaware of conditions, can come in and administer the postintervention measures.

Stimulus, Procedural, and Situational Variables

The variables that must be controlled include the nature, amount, and duration of the experimental treatment and the conditions of its application. Was there anything about the procedure, the wording of the instructions, or the way that they were given or comprehended that could have affected the results? Good experimenters try to keep these factors as similar as possible across conditions, sometimes, for example, by using scripted instructions in each condition that vary only in intended ways. Other questions for you to ask include the following:

- Were there any unplanned events or malfunctions of equipment or apparatus that could have influenced results?
- Could the order in which experimental treatments were applied have made a difference by favoring one group (e.g., could listening to cacophonous noise first influence a subject's reaction to the soothing music that would not happen if soothing music came first)?
- Might the timing of the introduction of procedures or the length of the intervals between procedures (too short or too long) have disadvantaged one of the groups?
- Was one group seen at the end of the day when tired and hungry?
- Was the magnitude or intensity of the stimulus situation sufficient to have any effect, or was it so small and weak that it was unlikely to produce any measurable result?
- Was there equality of the experimental setting for all participants (e.g., was one group seen in quieter or more comfortable surroundings)?
- Was the experimental manipulation delivered as intended, and was it perceived as intended?

Answers to some of these questions come from a manipulation check. This takes the form of (a) ongoing checks during the experiment to assure the integrity of the manipulation delivery or (b) a postsession interview or questionnaire that is given just before debriefing. A manipulation check is used to try to determine how participants perceived the experiment. If the treatment was a drug, you want assurance that it was actually ingested, not merely prescribed. If the treatment was a particular modality of psychotherapy or a new method of teaching, you want assurance that the therapists or teachers were competent to deliver what was intended and actually did so to the exclusion of other methods. You should look for spot-checks or evaluation of recordings of sessions for assurance. If the independent variable was a condition

that was meant to create a state of mind or an affect in the participants, you need to know whether they actually perceived it that way.

METHODS FOR CONTROL

The obvious way to keep such "other things" as stimulus, procedural, and situational factors from differentially affecting participants in different conditions is to hold everything constant except the intended manipulation or at least to keep them as invariant as possible. As a critical reader, you must ask the following questions:

- Were instructions given and data collected the same way by the same people, in the same place, at comparable times, under identical conditions? Or was there any systematic variation that could have biased the results?
- Were the place, the temperature, the comfort level, the ambiance, the noise, and other interferences that are features of the research setting held constant, or equally variable, for all conditions, unless the setting variables themselves were the independent variables?
- Were the nature, intensity, frequency, amount, and duration of the experimental treatment held constant by doing the same thing to the same degree and in the same way with all people assigned to the same condition? If the nature of the treatment is the independent variable, was the nature of treatment varied while the frequency, intensity, amount, and duration of different treatments held as similar as possible across groups? If the independent variable is frequency, were the nature, amount, intensity, and duration held constant?
- Were convincing efforts made to keep the treatment consistent with what was intended? The description of the details of the research should reassure you that procedures that were meant to be invariant were truly held constant.

Instrumentation (Apparatus and Raters)

APPARATUS

Changes in apparatus (or *instrumentation*) during the course of a study can be a confound with the independent variable. Experimenters may not necessarily inform you that this happened, so you can only look for indications that the researchers succeeded in holding things constant across conditions. You should be assured by the researchers that the apparatus was of good quality, that it was reliable, and that it was frequently checked and recalibrated during the research.

JUDGES, RATERS, AND SCORERS

Judges, raters, and scorers can differ in their observations, interpretations, and recording of events. Were there sufficient safeguards in the procedures to protect against the systematic influence of rater prejudgments stemming from their knowledge of the hypotheses and their awareness of the expected outcomes? Were necessary precautions taken to protect the research from bias and unreliability on the part

of data collectors, such as incompetence (e.g., were they trained well?), inattention or boredom (e.g., were the sessions too long?), drift (e.g., could definitions of measured variables, like the occurrence of particular behaviors, have changed over the course of data collection?), and ambiguity in task definitions (e.g., were tasks clearly defined?)? Issues involved in judging, rating, and scoring and a detailed discussion of methods that can be used to reduce their influence as confounding variables, are presented in Chapter 7.

Placebo Effects

Placebo effects can be quite powerful. Christensen et al. (2014) described a placebo effect as occurring when participants "experience improvement in symptoms as a result of thinking that they have received a useful treatment, even when the treatment condition has no value" (p. 13). Placebo effects are of importance in any study in which participants expect to experience some change. This can hold for many kinds of experimental conditions that have no clinical or practical application and are designed to have immediate but transitory effects. A simple example would be a study of whether ibuprofen reduces headache pain in which one group gets a salt pill but is told it is a headache reliever. A placebo effect occurs if this group's headache pain quickly diminishes.

Placebo effects also operate when under one of the experimental conditions some participants expect that they will perform particularly well. It is the expectation, not the treatment, that affects behavior. The placebo effect is of greatest concern when the experimental treatment is a clinical intervention that the participants hope will benefit them. In situations in which expectancies of participants can affect results (e.g., therapy and drug studies, interventions meant to enhance performance), the accepted procedure is to keep the participants unaware of which manipulation they have received by administering an inert placebo to a random half of the sample. However, this procedure is not always appropriate. In a psychotherapy outcome study, for example, the participants obviously know which treatment they are receiving. The placebo effect can be assessed here by giving an "attention placebo" to another group. This group receives equal attention over an equal period, but because nothing intentionally therapeutic is done during this time, the intervention is considered to be inert. If the only active ingredient in the psychotherapy group was "added attention," then there should be no difference between the experimental group and the control (placebo) group. If the study you are reading tests an independent variable that might result in a placebo effect, the researchers should address the issue in the way they treat the control group.

Nuisance Variables

Accidental, inadvertent influences on participants that have an unintended effect on the outcome are another source of uncontrolled variance. For example, during a task requiring great concentration, an airplane flies closely overhead and distracts the participants in one of the groups. This effect is confounded with the condition

manipulation. For individual participants, maybe a computer locks up at a crucial moment in a timed exercise. Perhaps one participant has a fender-denting accident on the way in, and another has a family squabble just before leaving home. Some of these events may not even be known to the researchers. Any kind of unplanned event of this sort could become an issue unless it happens randomly throughout the sample or procedure, and is balanced out. Still, these events will create within-group variance that will make it more difficult for researchers to find significant effects.

METHODS FOR CONTROL

The best that researchers can do is to anticipate as many things going wrong as possible and to be prepared for them should they occur: selecting a quiet room where disturbances will be minimal, being prepared with an ample supply of computers so that a rapid substitution can be made, or having rehearsed quick ways of restoring things to working order. Doing a *pilot study* and trying out all procedures, equipment, and instructions until they work smoothly should help but can never guarantee a trouble-free experience. The researchers should be prepared with decision rules that tell when to eliminate a participant's data because of contamination by some extraneous variable, and they should report any such cases to readers. A critical reader who sees that these types of precautions were made will gain confidence that many potential confounds were anticipated and minimized.

Summary

In this chapter, we examined the effects of extraneous confounding variables and the need for their control. Methods for controlling participants' demographic and personal characteristics were described, including using random assignment, homogeneous samples, matched groups, equated groups, blocking variables, statistical control, and within-subject designs.

The ways to control extra-experimental changes in participants, participant motivation, role perception, and communication among participants were reviewed. The chapter addressed the importance of controlling experimenter variables, including biosocial, psychosocial, and experimenter bias.

We examined the effects and control of stimulus, procedural, and situational variables, changes in apparatus, judge, rater, and scorer variables. We placed an emphasis on sensitizing you to all of these sources of confounding and alerting you to effective methods of control. Furthermore, the ways that researchers can handle the issues of concomitant variation and error variance have been introduced as elements in your critique of the research.

Research Designs and Threats to Internal Validity

6

A critical reader of research

- can identify the kinds of threats to the internal validity of an experiment that might or might not remain plausible, given the design of the research;
- knows that quasi-experimental designs call for a manipulation but do not use random assignment;
- is aware that quasi-experimental designs differ in the kind and number of threats to validity they rule out, but cannot be used to make strong causal inferences;
- understands that true experiments use random assignment of participants to conditions and are best for demonstrating causality; and
- knows that single-case designs can have good internal validity, depending on the number and arrangement of treatments, withdraws, and measurements, but the generalizability of results may be questionable.

In the preceding chapters we (a) defined research questions and hypotheses, (b) described the principal research strategies typically used in the social and behavioral sciences, (c) outlined how the levels of an independent or manipulated variable are established, (d) cataloged methods for sampling

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participants, (e) assessed the potential use of different strategies for making groups of participants who are placed in different conditions in a study as similar as possible, and (f) offered ways to control situational and experimenter variations that can be confounded with the variable of interest. Your attention can now shift to the type of research design that was adopted by the researchers and how these designs compare in their ability to rule out threats to internal validity of a study.

The material in this chapter catalogs the major threats to the internal validity of experiments meant to draw causal conclusions about the effect of an independent variable on a dependent variable. This chapter also introduces and critiques the major preexperimental, quasi-experimental, and true experimental research designs. It provides examples of commonly encountered circumstances that jeopardize or protect internal validity, the trustworthiness of the design for drawing causal inferences. These standard designs are well-known to all who have been introduced to research design. They are illustrated in this chapter from the point of view of a reader's critique of the most salient flaws of the weaker designs and the design alternatives that experimenters can use to answer these critiques.

Let us begin with a brief description of the major classes of threats to internal validity. This is not a complete list, just the ones most likely to be encountered. Unless we discussed an issue in a previous chapter, we will illustrate the design and show how each eliminates threats or leave them as plausible alternative explanations for results the researchers might mistakenly attribute to the effect of the independent variable.

Classification of Threats to Internal Validity

Campbell and Stanley (1963) categorized the principal factors that jeopardize internal validity—factors that create doubts that the independent variable was responsible for the results and that raise the specter that the credit should go to confounding variables. Shadish, Cook, and Campbell (2002) modified this list. Once alerted to these threats, you should be able to identify them when they occur. A few of the major kinds of threats that you might look for in a study are defined and illustrated next. In each case, a brief summary of a study is followed by a critique that covers the key threat to internal validity being addressed.

SELECTION

Selection refers to systematic differences in the characteristics of participants in the different conditions of an experiment. These differences typically exist before the experimental manipulation occurs, and they arise because a mechanism other than random assignment is used to place people into groups. We discussed how static group differences can be confounded with experimental variables and how random assignment prevents these differences from happening in Chapters 4 and 5, this volume, so we will not reiterate that here.

HISTORY

History refers to extraneous events that take place during an experiment and that have an effect on the participants. If these events are concurrent with the manipulation of the independent variable or occur between two measurements, historical events can work to the detriment of the internal validity of the study. These events can be general historical events (events that happen to all participants in a study that has only one condition) or local history events (events that happen to one group in a study with multiple groups).

Local History Example 1

Researchers who are studying the benefits of group therapy for anxiety in wives of service personnel select three groups of wives at Camp Pendleton, California, and three groups of wives at Camp Lejeune, North Carolina. Wives at both installations are pretested for anxiety on a valid and reliable test and are found to be equivalent on mean scores and standard deviations. Comparisons on demographic characteristics (e.g., age, length of marriage, number of children, living arrangements, rank and years of service of husband, nearness to relatives) show no differences.

The wives at Camp Pendleton attend group therapy sessions twice a week for 12 weeks. The wives at Camp Lejeune meet a comparable number of times for social interaction but receive no group therapy. Posttest shows a reduction in anxiety for the wives treated at Camp Pendleton. The results are interpreted as showing clear benefits of group psychotherapy. A sentence in the Discussion section of the research report indicates that at the time of the group therapy, Camp Pendleton also hired its first woman liaison for military families.

Critique

That the participants in the two settings had different experiences in addition to the therapy during the study period could easily account for the different results. This is considered a local history effect because the therapy and new liaison are confounded at one of the sites.

Local History Example 2

The effectiveness of a new drug for the treatment of depression is researched in Nome, Alaska. The control group (n = 50) is seen in January, February, and March and receives a placebo. In April, May, and June the treatment group (n = 50) receives the experimental drug. The treatment group shows a decrease in depressive symptoms (as measured by a valid and reliable symptom checklist) over the 3-month period. The placebo group, by contrast, shows an increase in depression. The researchers report that the hypothesis has been supported and recommend the use of the new drug in practice.

Critique

The first group was seen during the winter months in the far north where the days are long, dark, and cold, and there is little sunlight. The second group was seen at the beginning of spring and during the summer, after the end of the long, dark winter. The weather may have had as much or more effect on the participants' depression as the drug. This illustration of the history-by-assignment interaction contrasts with the effect of the historical event that could not be anticipated in the earlier example of the Marine wives. The researchers should have realized that assignment to experimental and control groups at different times of the year could have had predictable consequences in this particular environment. As a discerning reader, you should pick this up. Can you think of some of the ways the researchers might have rearranged their treatments and measurements to result in a more valid conclusion?

General History Example

A group of wives at Camp Pendleton are used in a study about anxiety, for which they all receive group therapy. Between the first and second assessment of the wives' level of anxiety, many of the husbands are shipped overseas to a hot spot where fighting is occurring. On the second assessment, the researchers find an increase in anxiety and conclude that the therapy had a negative effect.

Critique

Because you do not know if the increase in anxiety was caused by the treatment or the added stress of spouses being deployed overseas, it is not legitimate to conclude the therapy was a failure. Anxiety in the group might have been even higher had the deployment not occurred. This is considered a general history effect because everyone in the study experiences the same extraneous event.

MATURATION

Maturation refers to changes within the participants during the study that would happen anyway with the passage of time. These can be long-term changes (e.g., reaching puberty) or short-term changes (e.g., growing hungry or tired).

Example

A group of school counselors perform group counseling with high school sophomores, for whom tests indicate problems adjusting to school. The counselors do a follow-up test during the students' senior year, which indicates 40% of them are now well-adjusted. This success is attributed to the counseling intervention.

Critique

Adolescents of this age mature and undergo internal changes, some that might improve adjustment others not, with or without intervention. There is no justification for

crediting the better adjustment to counseling that occurred 2 years earlier. You must consider whether it was possible that 40% of the students would have "outgrown" their adjustment problem (facing graduation can itself be a maturing experience) had they not been in counseling. Indeed, 60% of students might have been well-adjusted by their senior year, suggesting the counseling was counterproductive.

TESTING (PRACTICE EFFECTS)

Simply taking a test can have an effect of its own. People react to psychological (and physiological) tests. Some learn from the initial experience and do better the next time. Others do worse because difficulty with the test increases their anxiety the second time they take it. Some people adapt to physiological tests, whereas others do not.

Example

Researchers give two groups an achievement test in mathematics. One group is then given a particular type of instructive tutoring for 2 weeks. Another group receives a rival type of instruction. A posttest shows that both groups have improved over pretest. The researchers conclude that the two methods of tutoring are equally effective.

Critique

A practice effect from the initial experience may have enhanced the performance of both groups and overwhelmed any increase because of the instruction itself. If the tests are identical, this is the second exposure to the test in 2 weeks. If the tests are different but similar in format and material, the first exposure might have lessened anxiety and allowed the students to practice answering strategies before they took the second test. Because most or all the improvement might be attributable to the practice effect, you would find the conclusion that both treatments were "effective" is not justified.

MULTIPLE TREATMENT INTERFERENCE

Some experiments require participants to receive more than one experimental treatment. The first treatment can influence the second treatment to an extent that makes it difficult to draw any conclusions about its effects.

Example

For a study on the effect of distraction on memory, participants memorize a list of 10 nonsense syllables while listening to music. The next day, the same participants are asked to memorize an alternate set of nonsense syllables in a no-music condition. When performance with music turns out to be superior to no music, the researchers conclude that music facilitates memory.

Critique

The nonsense syllables learned the first day could be interfering with the memorization of the new list on the second day. Had the interval between trials been extended, multiple treatment interference would not have been a problem. You must consider the possibility that what was learned on the first day interfered with learning on the second day, and music had no effect at all.

INSTRUMENTATION (RATERS)

When the dependent measure comes from raters (not the subjects themselves), it is essential for the raters to be trained, motivated, conscientious, and able to give ratings that are consistently reliable and valid. If the researchers do not report evidence on rater reliability, it is a shortcoming of their study.

Example

For a study on social interaction among children, researchers use a group of undergraduates to code the frequency of positive exchanges between children. The raters view and count positive interactions from a 2-hour videotape taken at an afterschool program. The raters are trained to high *interrater reliability*. The ratings of behaviors are carried out in an evening session. The frequency of coded positive behaviors reveals that they diminish as the afterschool day grows longer.

Critique

The long session that required 2 hours of careful attention and the fact that the session took place in the evening possibly contributed as the session progressed to a wavering of motivation, increased fatigue and boredom, and even a change in what raters viewed as positive interactions (called *coder drift*). These may have led the raters to miss instances of positive behavior later in the session (corresponding to later in the afterschool day), as well as contributed to the decline of interrater reliability.

REGRESSION TO THE MEAN

Sometimes subjects are selected into an experiment on the basis of the extremity of their scores (i.e., because their initial scores are far from the mean of people who are sampled and tested). You need to be most carefully on the lookout for possible *regression toward the mean* when a treatment under study is meant to improve participants' scores (e.g., achievement level, mental health index). The experimenters might be prone to choose participants who have the lowest or highest scores because these are the people who need the most help. In such a case, however, the initial extreme scores will move toward the overall sample mean on a second testing. This will happen simply because some subjects picked for the experiment are extreme because of bad or good luck guessing on the pretest, illness on the day the pretest was given, and so

on. From the experimenters' point of view, these chance influences are unlikely to be repeated for the same subjects the next time the test is taken.

Example 1

For a study on extrasensory perception (ESP), the experimenters prepare a deck of 100 3 × 5 cards; 50 are marked with an X, and 50 are marked with an O. The order of the cards is randomized by means of a random numbers generator. The experimenters tell a group of 100 college students that they will look intently at the back of each card in turn for 10 seconds. The students are instructed to concentrate on what they are doing and to mark down whether they think the card is an X or O, even though they cannot see it. At the end of the exercise, the scores are tallied, and 10 students with the lowest scores are culled out. The mean score of this subgroup is 25 correct out of 100. These 10 people are then given an intensive 1-week training course designed to teach them ESP skills. When the instruction has been completed, they are retested. Their mean score has doubled from 25 to 50. In a triumphant discussion of the results the researchers say, in effect, "With training, people can be taught ESP skill."

Critique

We would expect that subjects would get 50 ESP guesses correct by chance alone. The lowest 10 scorers were likely to be the lowest by chance (or mostly chance, depending on how strongly you feel ESP is real). The mean of the second trial for this group, with or without training, would be expected to regress toward the mean of 50. Had the experimenters trained the 10 students with the highest scores on the first run, the chances are that their group mean after training would have gone down, toward 50, instead of up.

Example 2

A school is instituting a new reading curriculum to help struggling readers. Because the teachers have most concern for the students who are the poorest readers, the teachers choose the students in the bottom 20% of readers (based on recently completed reading tests) to learn using the new curriculum. After a semester using the new curriculum, these students show significant improvement on reading test scores.

Critique

In this example, chance (i.e., bad luck) might have put some of the students in the bottom 20%. Other transient factors (e.g., illness, a fight with a friend) before the pretest was taken could have reduced their reading scores as well. It is also still possible that the new curriculum did have a positive effect, but the researchers will not be able to isolate this from the plausible occurrence of regression to the mean. As a critical reader, you should always look out for studies that chose participants because

of the extremity of their scores and consider regression to the mean as a plausible threat to internal validity.

ATTRITION

Attrition involves the loss of subjects from an experiment after it begins. In Chapter 4 we discussed how differential attrition (i.e., the unequal loss of subjects from the different conditions of an experiment) can threaten the internal validity of an experiment.

Now that we have illustrated the major types of threats to internal validity, let's look at how these threats relate to individual research designs.

Preexperimental Designs

Preexperimental designs are ones that have no control group. This makes it impossible for the researchers to even think about drawing causal inferences from their study. However, for limited purposes not involving causal inferences, preexperimental research has an important place in the social sciences.

DESCRIPTIVE RESEARCH

Preexperimental studies that are designed to describe the characteristics of objects, subclasses of people or living things, or natural or contrived events or phenomena all have a special challenge. These descriptions often include assertions about how and why things got the way they are, though no testing of these assertions has been made. You need to be on the lookout for these types of assertions. For example, researchers might want to describe what happened when a major earthquake hit a community. If they find the divorce rate increased, they might be very tempted to write that "the earthquake caused stresses on family life and economics that weakened the bond between married couples." This statement might be reasonable, but it is not part of the description—is it conjecture. As a critical reader, you would need to check whether it was clearly labeled as such. Even if the researchers had a measure of family stress (e.g., incidences of domestic violence), the assumption of the causal link remains a hypothesis and as such, is not supportable by the design of the study.

Even if researchers confine themselves to description, a complete description must not only (a) fully describe what is being observed, but it must also (b) describe what the object has in common with all other members of its classification, and (c) what distinguishes it from members of other classifications. Describing the Pacific bonito provides a clear example. A description would first identify them as fish, as they are cold-blooded vertebrate animals that live in the water and have gills, fins, and scales. This description is obviously incomplete because it does not differentiate them from any other kind of fish (e.g., flounder, striped bass). The description must indicate what distinctive characteristics they have that identify them specifically as a Pacific bonito, including information about their size, shape, coloration, number and placement of dorsal fins, number of spines in each fin, and all other identifiable

characteristics. Not until these descriptive elements are put together can one definitely conclude that the fish is a Pacific bonito and not, for example, an Atlantic bonito, a mackerel, or a tuna.

The same process also must be used to describe any subclass of people (e.g., people addicted to heroin, girls who are anorexic, individuals with borderline personality disorders, successful CEOs). In each case, the description would have to tell us what the group members have in common and what makes them distinctive from other groups. This holds also for descriptions of the way people react or behave (e.g., behavior under stress, behavior when in crowds).

Descriptions or profiles that accurately depict commonalities but do not describe distinctiveness are not very illuminating. For example, after extensive study of the characteristics of successful executives—on the basis of interviews, observation, and questionnaires—researchers might describe them as being assertive, being alert, working long hours, and being tired after a long day's work. These descriptors may indeed apply to successful executives, but they are not useful because they also could apply to unsuccessful executives, as well as taxicab drivers or bartenders. The description does not address the crucial issue of distinctiveness. Does the ability to make quick judgments about people, to be tactful with others they do not like, and to listen carefully distinguish successful and unsuccessful executives? (Remember, the researchers may be tempted to call these differences the "cause" of success, but they remain only a hypothesis.)

When reading a descriptive study, you want (a) to be assured that the sample or phenomenon selected is truly representative of a group; (b) to know how the descriptive information was obtained (interview, observation, survey questionnaire); (c) to be convinced of the description's objectivity and validity; (d) evidence that the description is shared by all members of the group; and (e) evidence that this particular group is different, unique, or distinctive in specified ways.

Description occupies a very important place in the social sciences. It is where the study of many phenomena begin. Description is especially important when it is used to set standards for clinical diagnosis. Standards for delineating groups with recognized psychological problems stem from descriptions that appear in the American Psychiatric Association's (2013) *Diagnostic and Statistical Manual of Mental Disorders*. These descriptions have been agreed on by committees of expert clinicians and physicians. For example, what set of symptoms must a person experience, and how often must the symptoms be experienced, to be categorized as experiencing posttraumatic stress disorder (PTSD)? Symptoms of PTSD include flashbacks, bad dreams, and frightening thoughts, as well as avoidance symptoms (e.g., staying away from situations that evoke stress). Experts not only list these symptoms but also consider if there are any other disorders that look like this set of symptoms. These descriptive standards dominate and regulate the practice of diagnosis in mental health. Like all descriptions, they should meet the requirements of commonality and distinctiveness.

When describing an event rather than an object, researchers still must talk about distinctive, as well as shared characteristics. Take this brief description of an event: "A black-robed dignitary is conducting a ceremony. Some members of the audience

have tears in their eyes, others are openly weeping." This could be a description of a scene in which a clergyman is officiating at either a wedding or a funeral, a judge is pronouncing a sentence on a defendant, or high school seniors are graduating. Or consider this descriptive statement: "A group of men with sticks in their hands rush up, crowd around one man and knock him down; they all fall on him and pummel him repeatedly." This could describe a fearsome gang assault or the joyous end of a hockey game after one member of a team has scored the winning goal.

For the value of the description to be established, the researchers must offer support for the assumption that a qualitative description—no matter how accurate and comprehensive it may be—is applicable to the particular group of people, the particular event, or the particular phenomenon that is being described. That support comes in the form of complete information about how members of the group or the event category are similar to each other but different from others.

ONE GROUP, POSTTEST ONLY

Suppose researchers conduct a study on the effects of long-term psychotherapy on a group of clients who entered therapy with a primary complaint of PTSD. The sample consists of 40 clients who had completed therapy for a year or longer. "Completion of the therapy" is defined as termination mutually agreed on by client and therapist. Assessment is accomplished by administration of the Taylor Manifest Anxiety Scale (Taylor, 1953) at the time of termination. The mean score of the group is reported to be within the normal range of anxiety scores, and the researchers claim that positive effects have been demonstrated. This one-group, posttest-only design can be diagrammed as follows:

$$G_1 \; X \; O_1$$

In this notations, G_1 stands for a group that exists before the study began, X for treatment, and O_1 for the observation or measurement.

The research on the effects of long-term psychotherapy on PTSD is also a preexperimental design (Shadish et al., 2002). The clients entered therapy with a presenting condition of PTSD, but there was no measure of the severity of their anxiety at the outset. With only a posttest to go by, there is no way of knowing whether the clients experienced any change at all. In our description, the researchers introduced the notion that the "mean score of the group is reported to be within the normal range." This was their way of setting up a comparator for you to gauge the effect of the treatment. Do not be fooled—the clients may have been within the normal range when the treatment began. In the absence of data from an untreated group for comparison, it is not possible to interpret what changes, if any, took place as a function of the treatment.

This design also excludes from the data analysis those clients who discontinued therapy because they felt that they were not improving or who believed that they were getting worse. Therefore, the clients remaining to take the anxiety test at the

end of the study might have self-selected into a group that the therapy would help. Although this limitation is not specific to a one-group, posttest-only design, the restriction of the sample to people who have completed therapy introduces a potential bias in the results.

You will rarely see the one-group, posttest-only design in psychology. It leaves too much to be desired regarding its ability to establish relationships or causality. If you see it used in this context, look on the results with great skepticism. Where you may see it more often is when the researchers are after existence proofs or deep descriptions of a group. Here, a comparison group may not be needed.

Quasi-Experimental Designs

In *quasi-experimental designs*, there is an identifiable manipulation or treatment, but the subjects have not been randomly assigned to conditions. As Shadish et al. (2002) described,

> Assignment to conditions is by self-selection . . . or others decide which person should get which treatment. However, researchers who use quasi-experimental designs may still have considerable control over selecting and scheduling measures, over how nonrandom assignment is executed, over the kinds of comparison groups with which treatment groups are compared, and over some aspects of how treatment is scheduled. (p. 14)

ONE GROUP, PRETEST AND POSTTEST

Heeding part of the critique, the researchers of the PTSD study perform another study, this time with an anxiety pretest obtained before treatment begins. All else remains the same. A within-subject comparison of the means of the pretest and posttest anxiety is statistically significant, and the researchers claim that this is evidence of the positive effect of treatment.

This one-group, pretest–posttest design is diagrammed as follows:

$$G_1 \; O_{A1} \; X \; O_{A2}$$

The notations O_{A1} and O_{A2} distinguish between the same observation or measurement (A) taken at two times.

The one-group, pretest–posttest design is an improvement for making causal inferences, but the problem is that subjects serve as their own controls (see the discussion of within-subject designs in Chapter 5). There still is no untreated control group measured at the same time as the treated group. Therefore, any change found by the researchers may be caused by the subjects maturing over time, spontaneous remission, changes in tests or raters, other general historical events that happened between observations, regression to the mean (if subjects were chosen because of their extreme scores), or other factors related to the passage of time. It remains a quasi-experimental design with many threats to internal validity.

TWO GROUPS, POSTTEST ONLY

Next, the PTSD researchers add an untreated contrast group consisting of clients who have not yet been in therapy for PTSD. This is a static-group design without random assignment (see Chapter 3). Lower anxiety scores of the treated group are interpreted to demonstrate the benefits of treatment. This design is diagrammed as follows:

$$G_1 \; X \; O_{A1}$$

$$G_2 \; \not{X} \; O_{A1}$$

G_1 and G_2 stand for two groups that existed before the study began (no random assignment) and $\not{X}$ stands for no manipulation or treatment.

Because random assignment was not used, the untreated group is not necessarily equivalent to the treated group. The unwarranted assumption is that the two groups would have been equivalent had scores been taken before treatment. However, without a pretest there is no evidence that they were equivalent to start with. Therefore, it is not a perfectly suited group against which to compare the treated group. Was the PTSD treatment given to clients who responded to an ad or on a first-come-first-served basis (meaning there may be motivational differences between the groups)? Were clients chosen by an administrator (perhaps based on the severity of disorder)? Any of these explanations remain plausible rival hypotheses on the basis of the characteristics of this design. Maturation, testing effects, instrumentation and multiple treatment interference are ruled out. Can you explain why?[1]

TWO GROUPS, PRETEST AND POSTTEST

In this design, the researchers give a pretest to two static groups, one of which then receives the treatment or manipulation, and wait an equivalent period of time before giving both groups a posttest. In our PTSD example, the difference between the two groups is that the treated group receives PTSD therapy and the untreated group does not. On finding that the treated group improves more than the untreated group, the difference is attributed to the treatment. The research diagram is as follows:

$$G_1 \; O_{A1} \; X \; O_{A2}$$

$$G_2 \; O_{A1} \; \not{X} \; O_{A2}$$

In a research article, you might see the results of the study presented like this:

	Pretest	Posttest
Treated Group		
Untreated Group		

[1]Because the two groups were measured at the same time.

Although the pretest anxiety levels of the two groups might have been comparable, this is a still a nonequivalent control group design because the two groups may have differed in other ways that could have been associated with change in symptoms. One group may have been brighter, better educated, or more advantaged, on average; subject to different kinds of life stresses; or at a different stage of life. Local history may still be an issue. So, this is still not a true experiment.

By now, our persistent researchers are getting discouraged. Each of these studies has taken several months, and with each, critical readers point out that the researchers' conclusion that the treatment is effective is still not warranted.

OTHER QUASI-EXPERIMENTAL DESIGNS

There are numerous other variations on the quasi-experimental designs (Shadish et al., 2002). One way these vary is in the number of observations that are taken. For example, the one-group, pretest–posttest design using a double pretest might be diagrammed as follows:

$$G_1 \; O_{A1} \; O_{A2} \; X \; O_{A3}$$

The use of two pretests helps rule out the effects of maturation (which should happen gradually over time) and testing (which should be most pronounced after the first testing) as plausible rival hypotheses. So, if our intrepid researchers used this design and there was relatively little change in clients' anxiety scores between the first and second testing but a large drop between the second and third, it is less plausible that maturation, testing, or regression to the mean, as opposed to the treatment itself, caused the change. Of course, general history remains a plausible alternative explanation.

Finally, the PTSD researchers might have collected a second dependent variable at the same time they collected data on anxiety. This variable might have measured a similar construct to anxiety but one that should not respond to the treatment. The one-group, pretest–posttest design using a nonequivalent dependent variable might be diagrammed as follows:

$$G_1 \; \{O_{A1} \; O_{B1}\} \; X \; \{O_{A2} \; O_{B2}\}$$

Suppose the PTSD researchers collected a *self-report* measure of self-regulation at the same time they measured anxiety before and after the treatment. They expect their therapy for PTSD will reduce anxiety, but it has no component to increase self-regulation. If they find a change in anxiety but not in self-regulation, this would make maturation and testing explanations for their finding less plausible and even help rule out general history.

We hope the logic of how a quasi-experimental study's design and results can be used to help rule out rival hypotheses is becoming clear. As more measures and more testings are added, the study becomes more effective at ruling out rival hypotheses.

If you find it an interesting exercise to puzzle through the different designs and what can and cannot be inferred from their different results, you can find numerous other examples in May (2012), Reichardt and Henry (2012), and Shadish et al. (2002). Many of these designs require considerable statistical sophistication, however.

True Experimental Designs

RANDOM ASSIGNMENT WITH TREATMENT AND CONTROL

Now, our PTSD researchers start with a sample of participants showing up at a clinic and randomly assign half to receive a treatment and the other half to a waitlist control. They measure anxiety in both groups after the first group completes the treatment and at the same time. The design diagram looks like the following:

$$R_1 \; X \; O_1$$

$$R_2 \; \not{X} \; O_1$$

Here, R_1 and R_2 refer to the fact that clients were randomly assigned to the two conditions.

If the results show that anxiety is significantly lower in the treated group, this difference can be attributed to the effect of the treatment. It cannot be due to client differences between the groups because they should be statistically similar, on average (see Chapter 5). It cannot be due to historical or maturation differences because the two measurements were taken at the same time. Testing is ruled out because there is only one testing. Even if the clients were chosen because of the extremity of their PTSD, the effect of regression to the mean should be equal in both groups.

RANDOM ASSIGNMENT, TWO EXPERIMENTAL GROUPS

But what if the researchers are reluctant to let some clients go untreated until the experiment is over? In the case of clients with PTSD, this might raise ethical concerns. So, our PTSD researchers start with a sample of participants showing up at a clinic and randomly assign half to Treatment A, the experimental treatment, and the other half to a rival Treatment B, perhaps treatment as it is given currently at the clinic. The diagram would be as follows:

$$R_1 \; X_A \; O_1$$

$$R_2 \; X_B \; O_1$$

If anxiety is less in the Treatment A condition, the study has succeeded in demonstrating the superiority of Treatment A over Treatment B. Finally! But, unfortunately, the design does not enable the researchers to know whether either of the

two methods was better than no treatment at all. Perhaps, Treatment A had no effect and Treatment B made clients even more stressed than when they arrived at the clinic.

RANDOM ASSIGNMENT, THREE GROUPS

Here, the researchers randomly assign participants to three groups: two treatment groups and one untreated control group. All groups are tested at the same time after the experiment. Treatment A is found to be significantly better than both Treatment B and no treatment. Perhaps the Treatment B group and the untreated control group do not differ from each other.

This design succeeded in demonstrating that Treatment A was superior to rival Treatment B and to no treatment. The researchers have also succeeded in testing the effectiveness of Treatment B. Adoption of Treatment A is advocated as the most effective approach.

RANDOM ASSIGNMENT WITH PRETEST AND POSTTEST

In this design, both groups, or all three groups, receive a pretest and a posttest. This design allows the researchers to examine whether the anxiety scores would have increased or decreased over time without treatment. Equally important, it allows the researchers to use the pretest score as a covariate in their assessment of treatment effectiveness (see Chapter 5). This will allow them to control for much within-group variation in client scores and make their study a more sensitive test of treatment effectiveness.

SOLOMON FOUR-GROUP DESIGN

The *Solomon four-group design* is a method of assessing the pretest effects. This design has two experimental groups and two control groups. In each case, one of these respective groups receives a pretest and the other does not; participants are randomly assigned. Its design would be diagramed as follows:

$$R_1 \quad O_1 \quad X_A \quad O_2$$

$$R_2 \quad\quad X_A \quad O_2$$

$$R_3 \quad O_1 \quad X_B \quad O_2$$

$$R_4 \quad\quad X_B \quad O_2$$

This design enables the researchers to look at the effects of the pretest independent of the treatment effect.

With success in hand, let us leave our PTSD researchers and take up a new example. Suppose researchers report a study on the effects of a training course to improve the SAT scores of high school juniors. Participants are randomly assigned to

an experimental group that receive instruction over a 3-week period or to a control group whose members are advised to study on their own. All students are given the SAT before and after the 3-week interval. Results reveal that both groups improve, but the difference between the group that received special instruction and the study-only group is not as great as expected.

The experience of taking the pretest may have been beneficial regardless of whether any special training was given. The pretest may have given students in both groups experience in the test-taking situation and may have provided them with familiarity with the kinds of questions asked. Therefore, the pretest itself could be an active independent variable. The effect of the course can be gauged by comparing the mean of groups $R_1 + R_2$ with groups $R_3 + R_4$. The pretest is now a second independent variable whose effect can be gauged by comparing the mean of groups $R_1 + R_3$ with groups $R_2 + R_4$. And the interaction of the two variables (e.g., did the course only work when a pretest was given?) can also be tested.

RANDOM ASSIGNMENT, TWO GROUPS, EXTENDED REPEATED MEASURES

The designs that have been presented thus far are prototypes that have used pretest and posttest assessments. Effective designs can include subsequent tests at more than one time. These follow-up assessments later are important for treatments that are expected to last. In experiments in which the "treatment" is an experimental condition that is hypothesized to have an immediate and transient effect, there is no need for any follow-up. For example, in a study of the time it takes to memorize nonsense syllables, there is no point to a follow-up unless the researchers are expressly interested in how long the memories are retained. Treatments that are designed to change people and not merely to produce a transitory effect can benefit from one or more follow-up measures.

Progressive measures can be analyzed by trend analysis, a *repeated-measures analysis* that appraises the significance of linear and curvilinear effects over time. For example, many weight reduction programs, smoking cessation programs, and brief psychotherapies all seem to have a short-term effect. However, it is important to determine whether their effects are transient and soon wear off. After the first flush of enthusiasm, people have been known to backslide. In some studies, a single intensive workshop experience is assessed by the participants on the spot. Participants may be temporarily imbued with the spirit of the experience, or they may desire to please the charismatic leader who was so eager to help them. You cannot be sure that any real change has taken place. As a critical reader, you should be more impressed by intervention outcome studies when follow-up measures are included in the design. A two-group design with a single follow-up would be diagrammed as follows:

$$R_1 \; X \; O_1 \; O_2$$

$$R_2 \; \not{X} \; O_1 \; O_2$$

CROSSOVER DESIGNS AND COUNTERBALANCING

Crossover designs are those in which the researchers give one manipulation to one group and then switch that group to another manipulation partway through the study. Assessments are made before and after each manipulation. A second group is given the second manipulation first, followed by the first manipulation. The design diagram might look like this, although the design can be used in studies that use random assignment or static groups:

$$G_1\ X_A\ O_1\ X_B\ O_2$$

$$G_2\ X_B\ O_1\ X_A\ O_2$$

For obvious reasons, this design is most useful for the appraisal of manipulations that take effect quickly and have transient effects. The effect of the first manipulations on each group needs to wear off between the first measurement and the institution of the second manipulation. Even then, the *multiple treatment interference* that comes from exposure to the different conditions can pose a threat to the internal validity of the study. You must assess this possibility on a study-by-study basis. There may be an order (sequence) effect or a manipulation-by-order interaction. Here, the experience of exposure to X_A may have more effect on the participants' subsequent response to X_B than the experience of having X_B first has on responses to X_A. For example, consider an experiment on visual perception in which subjects are exposed to very bright light (X_A) and very dim light (X_B). Performance on a perceptual task after X_B may be affected by the time it takes to get dark adapted after coming out of bright light. There may be less interference going from dark to light.

DESIGNS WITH MULTIPLE INDEPENDENT VARIABLES

All the previous designs are limited to one independent or treatment variable. In one design—the random assignment design that used two treatments and a no-treatment control group—there is one independent variable with more than two levels. There are also designs with more than one measurement, so time of measurement (pretest and posttest or posttest and follow-up) is another controlled variable in the experiment.

Many studies will either manipulate or measure more than one variable simultaneously. For example, researchers could manipulate the effect of playing violent videos games (yes or no) and whether players are exposed to loud or soft sounds while playing the game. Perhaps they want to isolate the effects of playing in the game from the effects of exposure to strong sensory stimuli. This is called a 2×2 factorial design with two manipulated variables. Alternatively, the researchers could manipulate playing violent video games and want to investigate its effects separately for men and women. This is a 2×2 factorial design with one manipulated and one measured variable. Just as the Solomon four-group design allows the researchers to separately estimate the effect of the pretest, these multifactored designs allow separate estimates of playing violent video games from the level of noise in the first instance and violent

video games from the sex of the subject in the second. In both instances, the interaction of the two variables can also be estimated (e.g., Does the effect of violent video games differ depending on the level of sound? Do men react differently to violent video games than women?).

Single-Case Designs

Single-case designs are experiments in that the researchers exercise control over the experimental conditions, especially their introduction and withdrawal, and attempt to control all the artifacts that could explain the results. If they are successful, the researchers are able to make causal inferences about this one case or this single group. Single-case designs are within-subject designs that feature a series of repeated measures for a single individual or a single group.

The basic idea of the design is to obtain a series of successive observations or measures of a target behavior or physical measure until a stable *baseline* (a starting point for measurement, typically notated as A) has been established. Then, a planned experimental condition (B) is introduced, and additional measurements are taken. This is followed by a cessation of Condition B and a return to Baseline A accompanied by still another series of measures. The baseline measurements are then used for comparison with measurements taken under the experimental condition. This design would be diagramed as follows:

A B A

From the continuous record of assessments, the experimenters can determine whether changes in the measurements co-occur with the introduction and the removal of the intervention. They can examine the trend lines, the slope, and the variability of the data (Kazdin, 2011).

VARIATIONS OF THE SINGLE-CASE DESIGN

You might come across numerous variations on the simple single-case designs (Barlow, Nock, & Hersen, 2009). For example, an extension of the A B A design is the additional step of reintroducing the experimental condition after the second baseline series of observations. This design may be diagramed as follows:

A B A B

Support for a causal hypothesis in A B A B designs is garnered if the researchers can successfully switch the target behavior on, then off, then on again.

The single-case design can also be applied to a group instead of to an individual. For example, in a study of the effect of music on assembly-line productivity, the Baseline A production rate for the assembly line could be measured with no back-

ground music, whereas in Condition B the production rate would be counted when background music was played. A longer string of alternations could be used, say, A B A B A B. The length of the baseline and manipulated conditions could also be varied. And, a third condition, C, could be introduced, perhaps varying the kind of music that plays in the background so that comparative effects of no music, fast music, and slow music could be examined. This design is also referred to as a *time series design* when it is used with intact groups (McCleary & McDowall, 2012).

Another useful arrangement for research and for clinical application is the *changing criterion design*. Here, the performance criterion is repeatedly changed as subgoals are reached and new subgoals are specified. For example, some smoking cessation programs use this technique by setting an attainable target of number of cigarettes smoked in a day. Once this target is attained, a new and lower target is set. By the end, the changing criterion reaches the final goal of no cigarette smoking.

Finally, *multiple baseline designs* are designs in which baselines are established for several different target behaviors that the treatment is aiming to change. If there are three such behaviors (X, Y, and Z), a specific intervention designed to affect X is first introduced. Successive observations aim to show changes in X, but not in Y or Z. When that has happened, an intervention aimed at Y is introduced. A further series of observations show changes in Y but no change in Z and/or no relapse in X. Finally, an intervention to change Z is initiated and is followed by a further series of observations.

STRENGTHS AND WEAKNESSES

The obvious strength of the single-case design is its ability to rule out important threats to validity. Because the experimenters control the introduction and removal of the manipulated condition(s), this makes the design good for ruling out many threats to internal validity. For example, maturation and testing effects are associated with specific patterns of data (the former should look like gradual change, the latter should occur as a large change from the first to the second observation) that would look nothing like the pattern of observed changes you would see from an effective treatment in an A B A B design (a change from A to B, a return to the level at A, and a final return to the B result). History also becomes less plausible. What are the chances that another event coincides with the introduction and withdrawal of B each time it occurs that would be associated with the same change in data? As phases are added to the design these threats to internal validity become less and less plausible.

One of the concerns raised about single-case designs is their inability to establish the generalizability of a finding beyond the single case. The way researchers using single-case designs establish generality is to replicate the study on more than one subject or group. Exact replications are not possible because each single-case study is an individualized experiment with an individualized baseline. If generalizations are attempted, the study must be repeated as closely as possible on as many different people as is feasible. As a critical reader, you should be more impressed with articles that contain multiple replications.

When researchers repeat the study on an additional number of people, another problem emerges if they obtain mixed results. This can easily happen because each of the participants is different and there is variation from one of the mini-experiments to another. Researchers should address this issue in a convincing discussion of their findings. Why did it occur? Are there discernible characteristics of subjects associated with whether the manipulation was effective?

Another possible problem is the need for a stable baseline, or at least one with a clear ascending or descending trend from which to judge a change in observations. Sometimes the baseline does not settle down to the point where it can rightfully be called stable, so identifying a change becomes difficult.

Still another issue is that of finding ways to analyze the kind of continued repeated measures that single-case studies generate. Some single-case experimenters rely chiefly on visual inspection. Sometimes their graphic presentations are unmistakably clear, but at other times the graphs can be open to various interpretations. The graphic presentations are there for the reader to see. For this to be convincing, the change needs to be dramatic. Other researchers use sophisticated statistical analyses. Making a complex issue overly simple, these techniques gauge the probability of obtaining the data observed in Condition B if the observed pattern of data in Condition A would have continued.

Summary

We began this chapter with definitions and examples of studies that exemplified threats to internal validity, such as general history, local history, maturation, testing (practice) effects, multiple treatment interferences, instrumentation (rater) effects, and regression to the mean by extreme scores. Then we described a panoply of research designs ranging from the nonexperimental descriptive approach to preexperimental, quasi-experimental, and true experimental methods. *Descriptive research* studies must provide supportive evidence of commonality and distinctiveness of the categories of persons, events, or phenomena under scrutiny. We also offered illustrations and critiques for preexperimental research designs, quasi-experiments, true experiments, and single-case designs.

Manipulation and Measurement Validity

7

A critical reader of research

- knows that all variables in a study, whether manipulated or measured, must have demonstrated construct validity;
- looks for evidence that a manipulated independent variable has construct validity established by experts, pilot testing, and/or checks on the manipulation;
- appraises dependent variables for their validity, reliability, and potential for self-serving biases (whether they are completed by the participant or an observer);
- looks favorably on studies that use multiple operations, which likely do not share potential operational biases, to measure dependent variables; and
- understands the different levels of measurement, types of measurement data, and how these relate to the validity of measurements and the statistical techniques used to study them.

This chapter focuses on the variety of measures that can be used in psychological, behavioral, and social research and their construct validity and reliability. We survey the types of tests and measures that can be used (or misused) in research. We also address the effective use of raters and judges in research and the use of reactive and nonreactive (unobtrusive)

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measures. As a critical reader, you need to know how trustworthy are a study's measures. If a measure does not measure what it is meant to (has poor construct validity) or does not do so in a reliable manner, it cannot be trusted to yield believable results, no matter how well-thought-out and well-conducted a study is in other respects.

The Independent or Predictor Variable

Unless the variable being manipulated in an experimental study represents what it is intended to represent, the study lacks construct validity of the cause. This is also true when the variable being used to predict another variable's values is not an experimental manipulation but rather a preexisting static group variable (e.g., men and women as categories of an organismic variable).

When you read a research report, it should be the case that by reviewing the procedures you can quickly learn what the definition is of the independent or predictor variable. For example, if the study is on the effects of stress, you should be attentive to the conceptualization of stress that is used. Further, you want to know whether stress is going to be manipulated or measured. Does the stress being studied originate from the conditions that are arranged for the study by the experimenters, or are people who are experiencing real-life stresses enlisted as participants? If the stress situation is arranged by the experimenters, how realistic and how severe is the stress? For example, are difficult or unsolvable tasks given to participants who are led to believe that how they perform has important consequences? Are they shown distressing film clips or photos? Are they "stressed" by merely asking them to subtract serial 7s from 100? If real-life stress is measured, do the experimenters use people who are waiting anxiously to take an important examination or people undergoing divorce or a job change?

As a critical reader, you must assess the experimenters' operationalization of the independent or predictor variables, to ask whether this situation is indeed stressful, and to ask whether it provides the kind of stress and an intensity level (i.e., the impact) that would be necessary to cause the hypothesized reaction. This is especially important when independent variables are manipulated, considering the necessity for protecting the welfare of human participants. Ethical considerations will dictate to researchers that they keep any experimenter-imposed stress as low as possible. You should be interested in knowing whether the kind of stress (e.g., physical or performance based) and its intensity level are sufficient to produce any effects. For example, stress from unsolvable puzzles is likely to be great, and therefore more likely to show relationships with other variables related to stress, rather than stress from subtracting numbers backward. If the impact of the independent variable is too low, a promising experimental hypothesis may be rejected even though it is true.

MANIPULATED VARIABLES, TREATMENTS, AND INTERVENTIONS

In a research report of an experiment, the independent variable must be explained and operationally defined so that you can understand (a) what is the construct that

the researchers intend to study and (b) what is operationally manipulated to create varying conditions of the construct. Without this information, you cannot judge whether the manipulation appears valid. A board of experts agreeing that its contents match the intent is helpful. The establishment of construct validity (detailed next) make it even more convincing. Most important, the participants themselves should experience it in the way that is intended.

To illustrate these issues, assume that the researchers conceptually define stress as "any condition or situation that is perceived by an individual as a severe threat to that person's physical or psychological integrity." Assuming the definition is reasonable, you discover that the researchers intend to induce stress by having participants engage in parachute jumping. Therefore, the researchers operationally define stress as "a condition of physiological arousal and uncertainty that exists when a person is about to jump out of an airplane for the first time and is equipped with a parachute." This fanciful example (and one that is likely not to pass muster with an internal review board) is being used only for the purpose of illustrating an operational definition that most people would be likely to accept. Still, stress in actual subjects is being assumed because of the compelling condition that the experimenters have arranged. This is called *face validity*. You could, nevertheless, justifiably wonder whether all the participants perceived the situation as stressful. Just because a situation is meant to accomplish something and appears to be capable of doing it does not guarantee that it succeeds in reaching its objective. Going for a drive through a verdant countryside on a Sunday afternoon might be intended to be pleasant, and it might be perceived as pleasant by most participants. On the other hand, people who are phobic about riding in automobiles or who are allergic to insect bites could experience it as extremely stressful. Likewise, jumping from an airplane for the first time might produce little or no stress in a high risk-taker, who views it as exhilarating but experiences no fear.

Pilot Testing and Manipulation Checks

Aware of the need to accommodate the perception of the participants, researchers might conduct a check of the construct validity of the stimulus situation before beginning the study. Accordingly, a preliminary study, called a *pilot study*, is carried out in which people are placed in the stressful situation, and such physiological measures as heart rate, blood pressure, sweating, respiration rate, and muscle tension are recorded before the experience and again at the time that they are exposed to the stressor. Then, to determine that the observed changes were due to stress and not to anticipatory arousal, it would be worthwhile to get verbal reports from the participants describing their feelings or to have them fill out a rating scale about the stress that they experienced (or both). Your confidence in the validity of the condition would be high if all these checks are consistently confirmatory.

Researchers might also introduce a manipulation check into the main investigation. As a check on how the actual participants reacted to jumping from an airplane, they could interview participants after the jump (and nonjumpers in the control condition) and ask them to fill out a rating scale about how they felt during the jump. If the stress

reported by jumpers is significantly higher than nonjumpers, the construct validity of the manipulation gains additional credence.

What if some participants were found not to perceive the situation as stressful? After all, there probably are people who would be thrilled to jump out into the void from an airplane while bathed in utter confidence that the chute would open and would lower them gently to earth, just as there are people who enjoy hurtling down incredibly steep roller coaster slopes. Because this is possible, as a critical reader, you also must be interested in what was done with participants who were found by the manipulation check not to experience the situation as intended. To include them would reduce the possibility that the study finds an effect even if it exists; the effects of stress on people who have not been stressed cannot be studied. That would be conceptually equivalent to studying the effects of aspirin on headaches in people who did not have headaches. You should look to see how the researchers handled this; in this example, it would probably be best to exclude such individuals from the study.

Ethical Constraints on Manipulation Impact

The scenario in the hypothetical study on stress is greatly exaggerated to make a point. The study could be justly criticized on ethical grounds (see Chapter 11), because inducing this much stress could be very threatening to some people and cause adverse reactions. Because of ethical constraints, researchers must temper experimental situations, and thereby reduce the magnitude of experimental effects. This makes manipulation checks even more important. Compare our example with another study of stress in which participants are shown photos of parachutists as a way of inducing stress and the success of the stress induction is simply assumed, no pilot test or manipulation check. No ethical issues here. But also, as a critical reader, you should be inclined to believe that stress had been induced more certainly in the first study than in the second one.

The general set of procedures for demonstrating construct validity is essentially the same for most studies in which conditions are devised and arranged by the experimenters. As additional examples, suppose you are reading a study of the effects of success and failure experiences. What is success and what is failure must be defined. Situations that are thought to be capable of inducing success and failure must be concocted and checked to determine that they really do what they are supposed to do. There would need to be a manipulation check in the final experiment to verify that the participants perceived the situations as intended. Or, in comparing two types of psychotherapy, the therapies must be defined, experts or clear proponents of the types of therapy must be used as therapists, therapists must be instructed to restrict themselves to applying the type of therapy requested of them, and procedures (e.g., time sampling of observations, ratings of tapes or transcripts) must be introduced to make sure the therapists comply. Only then can you be comfortable in judging that the types of therapy in question were appropriately, validly, consistently, and differentially applied. Failure to define the variables of interest, to arrange valid operationalizations, and to check the effectiveness of the manipulation may jeopardize the construct validity of the study and reduce your receptivity to the claimed results.

If the independent variable is something more tangible (e.g., a drug used in a study on alleviating anxiety), the researchers must assure you that the drugs used are representative samples of the intended medication, that the dosage is sufficient to bring about the hypothesized effects, that the subjects in the experimental condition actually take the drug, and that those in the control group do not take either the drug or any rival substitute.

STATIC GROUP VARIABLES

The construct validity problem is generally not as complicated when the predictor variables in the study are organismic, psychological, or status characteristics of participants. Here, their group membership preexists. *Static group variables* (e.g., age, sex, socioeconomic status, race/ethnicity, education, marital status) are matters of record (and we hope there is no reason to distrust self-reports). If the predictor variable is some measurable psychological quality or aspect of people (e.g., IQ score, placement in a diagnostic category), then test scores, expert judgments, or a combination of these can be used to place people into groups.

Data about the reliability and calibration of apparatus are needed for an appraisal of physical and physiological measures. The validity of these types of measures is not something that you can simply assume is up to standards—some physiological tests and measures are notoriously unreliable. The researchers are expected to furnish evidence of the reliability of the apparatus, tests, and measures. The researchers' presentation of the reliability and validity of the measures should guide you in making a judgment about their validity.

Psychological Tests

When the predictor variable is obtained from a psychological test, evidence should be presented about the test's reliability and validity. The test must be valid for this specific variable, not merely valid in some broader sense. Therefore, an intelligence test like the Wechsler Adult Intelligence Scale (Wechsler, 2008) might be a valid test of intelligence, but it would not be appropriate to use it as a measure for a variable such as self-esteem.

When test scores are used to classify people into categories, it is important to determine whether well-established cutoff scores have been used. Participants who have been placed in a group because they are anxious, depressed, or have a gifted IQ should have test scores that fit the cutoff level to ensure valid classification. To a critical reader, the construct validity of the study would have to be suspect unless all of the participants truly belonged in their assigned groups.

Classification by Judges

Classification by qualified (preferably expert) judges is a favored mode for some static group variables. For example, the two levels of a predictor variable in a study of individuals with depression might be set as "borderline" and "normal." For this classification, the

researchers decide to use the judgment of a panel of experts. Working independently, each of the judges examines potential participants and studies their mental records. As a critical reader, you must be convinced of the expertise of the judges and look for the standards of agreement required of them. Were enough judges used, and were the standards of agreement among them sufficiently rigorous (did all have to agree?) for you to be confident that the participants were correctly classified?

The rule governing selection is that each participant must fit unequivocally into one and only one of the available categories. Anyone who does not meet this requirement cannot be used for this study. You must be confident that participants who have been selected to represent the borderline group fit neatly into that category and that those selected for the normal group also are correctly classified.

The Dependent Variable

The dependent variables in a study also must be relevant, valid, clearly defined, and stated in advance. For example, if a manipulation of stress is meant to cause changes in a participant's desire to affiliate with others, the measure of the affiliation variable must be subjected to the same scrutiny for construct validity and reliability as the independent variable. Because they are measured and not manipulated, the procedures used to establish these qualities of a dependent measure are similar to those discussed previously for static group variables.

BIAS IN DEPENDENT VARIABLE MEASUREMENTS

Measurements on participants for a study can be derived from various sources and may be of several different types. Each of these bears a greater or lesser degree of potential bias, of being systematically influenced by something other than the variable the researchers wish to measure (e.g., the respondent's desire to make a good impression on the experimenter). As you think about the measures used in a study, it is important to consider the possibility of bias influencing the responses.

The potential for bias depends on what is being measured, who does the measuring, and how the measurements are made. For example, in a study of hyperactivity in children, trained independent observers of the children's behavior would be the optimal collectors of data because they have no stake in the outcome of the study. In a less optimal situation, the observers could be the experimenters and/or the research assistants. They may be well-trained and motivated, but they also know the condition of the participant, and they have a stake in finding results that confirm the hypothesis. Parents, supervisors, teachers, or therapists may be harder to train, less motivated, and may have biases caused by their strong desire to look like effective caregivers of children with hyperactivity.

Self-reports (e.g., asking children how often they get "antsy") are not the best choice because of the obvious potential for self-serving bias or for distortions of self-perception. If used at all, reading critically, you should prefer to see self-report data

accompanied by data from the vantage points of other observers or raters and/or data assessed by physiological criteria as well. However, when conducting studies of subjective experiences (e.g., physical pain, joy, sadness), it is hard to find out how people feel without asking them for a self-report. Even such an objective-looking personality inventory as the Minnesota Multiphasic Personality Inventory (Friedman, Levak, Nichols, & Webb, 2014) is, after all, a collection of self-report items.

In other instances, the measurements can be made by instruments or by physical tests and measures. For example, an activity monitor can be placed on the wrists of children with hyperactivity. The general principle is that there is less potential for bias when there is less involvement of the data collector in the participant's performance or behavior, less ambiguity of the behavior, and less judgment required of the collector. An activity monitor scores high on all these criteria. Its failing is that it cannot distinguish between activity that would be viewed as hyperactive (e.g., fidgeting in class during lessons) and those that would not (e.g., playing at recess).

Measures that involve ratings that are entirely judgmental or evaluative (e.g., *bad–good*, *improved–unchanged–worse*) are more prone to bias than are decisions about whether a specified observable behavior did or did not occur. When evaluative data are used, it is even more important that the persons making the judgment be free of bias. This is also true of descriptive data or other data of a qualitative nature because they are dependent on the perspective of the beholder.

Status data also can be used as dependent measures. Many of these are not as bias-prone as other types of measures. For example, status data (e.g., whether someone is employed or in a hospital) can be used to measure the effectiveness of therapies. You must identify the mechanism being used to measure the dependent variable and assess how well the threats of bias accompanying that type of measure have been eliminated.

REACTIVE AND NONREACTIVE (UNOBTRUSIVE) MEASURES

In Chapter 5, we alerted you to ways in which participants' perception of and reactions to the test situation can influence their behavior and bias the results. Webb, Campbell, Schwartz, and Sechrest (2000) presented a comprehensive review of the development of nonreactive or unobtrusive measures in research in the social sciences. They discussed four types of nonreactive measures:

1. *Physical traces:* These include measures of physical erosion, such as (a) the carpet or floor tile wear in front of various exhibits in a museum to determine their comparative popularity and (b) the tears and dirt smudges on pages of books to ascertain selective reader interest. Webb et al. also discussed measures of attrition or buildup, such as the study of inscriptions in public restrooms or the number of liquor bottles that were thrown out with the trash in selected residential areas. An activity monitor on the wrist of a child is also an example of a physical trace measure.
2. *Archives:* Included here are actuarial records; political (e.g., to measure attitudes), judicial (e.g., arrests for a study of delinquency), hospital (e.g., effectiveness of

health and mental health interventions), and other governmental records; material found in the mass media; sales records; industrial and institutional records; and private written documents.

3. *Observation:* The observations referred to by Webb et al. are all unobtrusive in nature, meaning the actors are unaware that they are being observed. This includes observations of physical locations (where things happen) and clusters of people (when and where people might congregate), expressive movements, language behavior in the media or as overheard in public places, the amount of time people spend looking at things in public displays, or time sampling of observations to determine whether occurrences are temporally linked.
4. *Contrived observation:* Webb et al. discussed the use of hidden hardware (e.g., video monitors) and the use of confederates in contrived situations (e.g., hidden camera television shows), where the observed individuals are unaware that they are in a study. People's behavior when asked to sign petitions or to volunteer has also been studied. At the outer edge of contrived situations are those in which unsuspecting people are entrapped into doing things and are studied in the process.

Ethical issues that are raised in some studies that use nonreactive measures are discussed in Chapter 11. From your point of view as the critical reader, the importance of unobtrusive measures is that they offer a way of studying certain issues without running the risk that the results are in part due to the reaction of the participants to the measurement. No matter how ingenious the measure, you must still evaluate its validity and its accuracy.

USING MULTIPLE MEASUREMENTS

Realizing that measures may have biases, many researchers use multiple measures of different types coming from different sources. Under certain conditions, this is sometimes called *multiple operationism*. Here, it is hoped that the different measures will not contain the same source of bias due to how the measures were constructed and collected. So, lots of self-report measures might qualify as multiple measures but would not qualify as multiple operations because they all are prone to the same sources of bias (positive self-presentation). In studying hyperactivity, why not train observers to watch the children, ask parents and teachers to rate the children, and put an activity monitor on them? Seeing this in a research report, you cannot help but be impressed if all of the measures give the same results.

But a problem arises when the findings are not consistent. Multiple measures increase the probability of obtaining some significant differences by chance, and the danger of this happening increases as the number of measures increases. Therefore, you should note whether an appropriate statistical correction for this has been applied. If inconsistencies remain after such corrections have been made, you should look for a fair and balanced presentation of the conflicting findings, in contrast to an undue emphasis on the favorable results and a discounting of the unfavorable ones.

Reliability

When you read about a study's procedures, you should look for documentation of the *reliability* and construct validity of the static group or dependent variable measures. Establishment of reliability can take any of three forms, and each tells you something different about the measure. First, a test can be given on more than one occasion. The *test–retest reliability* correlation can be used to establish whether a measure that is meant to measure a stable characteristic of people shows the desired consistency over time. If a claim is made that a test is measuring IQ but finds little correlation between test scores when it is taken twice 2 weeks apart, you have every right to question whether IQ is really being measured.

Reliability also can take the form of a measure of *internal consistency*. This measure indicates whether the individual items on a multi-item test correlate with one another and therefore are likely measuring the same underlying construct. If the items on a scale do not show some relationship to one another, how can they be measuring the same thing? Internal consistency is typically measured by statistics such as the split-half procedure or coefficient alpha (Cronbach, 1951). In *split-half reliability*, the items are randomly divided into two groups and the halves are correlated with one another. *Coefficient alpha* uses the correlations between the individual items to estimate what correlation you would expect if the sum of these items (the test score) were correlated with the sum of an equal number of items drawn from a pool of items meant to measure the same construct.

Finally, *interrater reliability* gauges the similarity of observations or ratings by different judges about the same people performing the same behaviors at the same time. High interrater reliability means that the raters are interchangeable; therefore, the possibility of idiosyncratic bias in their observations or ratings is diminished. Agreement among raters will be influenced by the complexity of the behavior they must judge, the clarity of the judgement definitions, and the quality of the training.

Validity

For critical readers, a satisfactory demonstration of validity is as essential for the dependent variable as it is for the independent variable. Validity can be established in several different ways. The most appropriate way depends on the kind of measure. Face validity, the appearance that a measure is measuring what it is meant to measure, is usually not enough.

If the dependent variable is meant to assess mastery of subject matter, a test of *content validity* is valuable. The content is judged by a panel of experts and it is considered to be valid if the experts agree that the content of each item is appropriate and that all important areas in the domain of knowledge have been covered adequately by the entire test. For example, measures of achievement must demonstrate content validity. The content of standardized tests is the result of input from many experts.

Construct validity can be determined by an appraisal of the correlation of the test with other measures of the same trait or ability (called *convergent validity*) and the lack of correlation with measures of different traits or abilities (called *discriminant validity*). So, a measure of self-esteem should show a strong correlation with other measures of self-esteem but little, if any, correlation with measures of intelligence. These two constructs are believed to be largely unrelated to one another. Therefore, if they are highly correlated (and the measure of IQ is well-validated), the validity of the self-esteem measure should be called into question.

The construct validity of a measure can also be demonstrated by showing that it has been successfully used to make a theoretically derived prediction in past research. If a self-report measure of risk-taking has shown in the past that high scorers are more likely to agree to jump out of an airplane than low scorers, this speaks well for the construct validity of the measure. It can be used to distinguish risk-takers in future studies.

Finally, measures can be assessed for *criterion-related validity*. Here, the measure is meant as an indicator of the person's status on some other variable. Three types of criterion-related validity are defined, depending on the temporal relationship between the test and the criterion. For *concurrent validity*, the test is meant to correlate with another variable at the same time the test is taken. When police officers ask car drivers to walk a straight line, they expect that this will indicate whether the driver can safely operate the vehicle. For *predictive validity*, the test is meant to correlate with the person's status in the future. So, employers might give applicants for an insurance sales job a test that they hope predicts the amount of insurance they will sell in 5 years. *Postdictive validity* is used most often in forensic psychology, when psychologists are asked to give tests today to people accused of crimes that reveal their mental state at the time the crime was committed. It is used most often in cases where the defendants have claimed they were temporarily insane when they committed the crime.

SITUATIONAL INFLUENCES ON VALIDITY

Above and beyond these standard forms of validity, you must also consider whether a measure is valid when used in a particular research situation with a certain type of participant for the specific purpose that is designated. A psychological test can come bearing impressive validity coefficients, yet be of questionable validity for use in a particular study. A highly valid verbal IQ test might have poor validity if it were used with children who mostly spoke a different language at home than the one used in the test. The validity of a widely used and well-validated paper-and-pencil personality inventory could be challenged if the respondents had difficulty understanding the vocabulary because of age or educational level. The test must measure what it purports to measure with this sample of people and under the special conditions of this research. That is why researchers are asked to report reliability and validity evidence obtained using their sample, as well as the evidence gathered when the measure was first validated. Researchers who report this information have taken an important step toward gaining your trust in the study's results.

Also, using a valid test of anxiety to measure depression is an example of the misuse of a test that is valid for measuring something else but not the construct of interest. No test has omnibus validity (i.e., no one test does it all). Further, the test or measure must be sufficiently sensitive to yield a valid assessment of what the researchers are trying to measure. Therefore, it would not do to use a bathroom scale to measure grams or a grandfather clock to measure fractions of a second, as they are not sensitive enough.

Influence of Measurement on the Dependent Variable

As a critical reader, you should also attempt to discern any effect that the act of measurement might have on the dependent variable. For example, participants in research on smoking cessation programs are frequently asked to keep a running record of the number of cigarettes they have smoked. However, this process of "charting" has been used clinically as one of the methods used to help people gain control of smoking. Just measuring a behavior that is sometimes done automatically and without much awareness brings it into conscious focus. The charting of food consumed and of daily weight can have a similar effect on the outcome of a weight reduction study. Not all measurement effects are positive effects. For example, the act of measurement in sleep studies, where participants are wired to electronic apparatus and are asked to fall asleep in laboratory surroundings, could make it harder to achieve sleep and negatively affect the behavior under investigation. Even a monitor worn on the wrist can make children aware of their activity level and influence how much they move around. In all these instances, the act of measuring is itself an intervention.

Types of Measurement Data

Measurements can take many forms. To be able to read a research report critically, you should have a basic understanding of the various types of data that can arise from different approaches to measurement, when it is appropriate to use them, and their advantages and limitations.

BEHAVIOR COUNTS

Observable physical and psychological events can be counted. Counts are direct recordings of the frequency of occurrences coming from yes–no decisions. Sometimes the decisions are clear. "How many times did the U.S. Marine subjects touch their left ears with their right hands during an interview?" To avoid ambiguity in this example, Marines were used so that observers would not have to judge whether the hair or the ear under it had been touched. Contralateral touching was picked to avoid ambiguity over inadvertent ear touching while resting one's chin on one's hand. With

these ambiguities ruled out, and guided by good definitions of *touching* and *hand*, any competent and conscientious observer could make a reliable count of the frequency of the behavior. Other behaviors, such as "How many times did the person sigh during the interview?" are more ambiguous. After having satisfactorily defined the word *sigh*, unbiased judges would have to be trained to recognize them reliably before being asked to make the determination for the research.

Sometimes counts can be made by using instruments. Heart rate and respiration rate counts can be recorded directly with proper apparatus. Counts of the number of words used in the Thematic Apperception Test or other storytelling assessments (Teglasi, 2010) can be made by a computer. By contrast, more ambiguity would be introduced if the dependent variable were the number of affective responses. The researchers would have to define *affective*, establish standards, train judges to identify affective responses, set and determine interrater reliability, decide how many raters to use, settle on what level of agreement to require, and determine what to do about disagreements.

Behavioral observations, either within the context of naturally occurring situations or in situations that are devised and arranged by the researchers specifically for the study, are an important type of measure that involves counting. For example, suppose you are reading a study designed to measure aggressiveness in children as a consequence of experimentally induced frustration. Children are placed in a room that contains a life-size plastic doll. The number of times that the children punch the doll, the operational definition of aggression toward people, is taken as the dependent variable measure. In another study, a more direct measure may be used. Several children are observed together, and the number of overt aggressive acts toward each other are counted. Sometimes the behavior in experimentally arranged situations or in naturally occurring situations can be counted. At other times trained judges and raters must be brought in (this is discussed in a subsequent section).

Behavioral observations can take the form of dichotomous yes–no counts. These indicate whether the behavior was or was not enacted, or whether a criterion was met by the participant. A paradigmatic example of meeting a criterion in clinical psychology is Lazarus's (1961) study on the treatment of phobias. One part of the study involved a pretest and posttest of acrophobia before and after treatment. Behavior was called successful (i.e., the criterion was met) if the patients could climb 50 feet to the third-story landing of an external fire escape of a high-rise building, take an elevator to the eighth-floor roof, and count the cars passing for 2 minutes while looking over the parapet. Needless to say, the acrophobic participants could not do this at pretest. The outcome criterion was an all-or-none, dichotomous measure.

Some people tend to be more impressed by readings from instruments than they are by psychological test scores. But even instruments are not perfectly reliable. An apparatus that has been designed for use in stable laboratory conditions may have to be moved from place to place and used under conditions that are less than ideal. A portable and inexpensive apparatus may not be as reliable as one would like and may even be less reliable than good psychological tests. Experimenters who use an apparatus need to provide its name, assure you that the apparatus is reliable, and that its accuracy has been calibrated against a known standard.

PSYCHOLOGICAL SCALES

Measures of psychological variables (e.g., love, hostility, friendliness, anxiety) are less tangible and are difficult to observe and count reliably. Although researchers can count the number of times two people kiss and use it as a measure of love, they cannot count "loves." However, they can use scales on which people rate their love for one another or a third party observes interactions and makes a "love" judgment.

Measures of psychological states that are as reliable as yardsticks and thermometers have not been developed. Standard units of hostility are not housed in the Bureau of Weights and Measures.

Standardized Psychological Tests

Standardized psychological tests that measure the variables under study are extremely useful when available. Norms for these tests are already published, and reliability and validity data should be presented so critical readers can review them. Well-established tests generally have multiple studies to support them. Most test scores yield good distributions (variability among people taking them that is normally distributed) and lend themselves to the use of powerful parametric statistics.

Instead of using individuals' test scores, some studies find it more meaningful to use as data the number of participants who fall beyond some normative cutoff score. These categorical data are analyzed by nonparametric statistics (e.g., chi-square). In some situations, this is more appropriate than comparing group means with parametric statistics. Two group means could be significantly different, for example, even though neither group mean reached an established cutoff point. In this case, you might read that the researchers claim the treatment to be effective, although the amount of change is trivial from a psychological point of view. For example, in a study of a depression treatment, the mean depression score in the treated group might be significantly lower than the untreated group, but it still may be the case that there is no difference between the groups in the number of participants who would be categorized as clinically depressed, defined as a cutoff value on the scale being used.

Test scores and behavioral measures sometimes give results that appear to be inconsistent. For example, in a study of a treatment for delinquency, the mean disruptiveness score given to adolescents by teachers could go down significantly more for a treated group than for a control group, but the incidence (count) of arrests could remain comparable for the two groups. This does not necessarily mean that the disruptiveness scores are invalid; differences in arrests might not manifest themselves until some critical cutoff score value in disruptiveness is reached. Reading critically, you should search for the psychological meaning of significant results and not be guided exclusively by *p values* (for more on this, see Chapter 9).

Tailor-Made Measures

After searching for a standardized test (e.g., using PsycTESTS, http://www.apa.org/pubs/databases/psyctests/), psychological researchers may find themselves unable to

locate a valid and reliable measure that can be used as a dependent variable. In this case, inventing a measure and confirming its reliability and validity becomes part of the research endeavor, and the success of the entire project depends on it. You can admire when it is done well, or criticize when it is not, but unless it is properly done, you can have little faith in the conclusions of a study using tailor-made measures. Therefore, this dictates that you look with special care at the validity of measures that have been tailor-made for a study. Some researchers adapt preexisting measures by altering them to fit the special needs of the study. You want the researchers to convince you that the alterations have not affected the validity and reliability of the measure.

Measurement Scales

To understand psychological research and to be able to evaluate dependent measures, you will profit from a familiarity with the development, application, and limitations of the rating, scaling, and ranking methods that are used in research. Most measures are offshoots of one or more of these ways of quantifying qualitative data. As a critical reader, you need to recognize when they have been poorly selected or badly applied or misused. The techniques that are in use today have developed over time, and classic texts on tests and scaling still make for a good read (e.g., Anastasi & Urbina, 1997; Nunnally, 1978).

LEVELS OF MEASUREMENT

A brief review of the different levels of measurement will assist you in understanding the different characteristics of the data collection methods used in the social and behavioral sciences. At the most rudimentary level, measurements can be taken at the *nominal level of measurement*. The nominal level of measurement simply divides the units into categories. Sex, race/ethnicity, and religious affiliation are variables used frequently in social research that are measured at the nominal level. The important statistical characteristic of nominal-level data is that the categories are qualitatively different and cannot be compared on the basis of possessing different degrees of the variable. Because people are simply placed into groups, the appropriate statistical procedures will be nonparametric. Nonparametric statistics make no assumptions about the distribution of the variable (for more on this, see Chapter 8).

The next level is the *ordinal level of measurement*. This measurement level describes data in which participants are rank ordered. It tells which participant has more or less of the variable, but not how much more or less. High school class rank would be an example of ordinal data. This tells you where each student sits in the distribution of students' cumulative grade point averages—how many students achieved higher and lower—but it does not tell you by how much. Because of this, nonparametric statistical procedures are also used with ordinal data.

For interval-level data, the size of the difference between numbers corresponds to the size of the differences on the variable. For example, IQ scores are measured

at the *interval level of measurement*. This means that an IQ score difference of 10, say between 90 and 100 and between 100 and 110, represents an equal difference in intelligence. The equal differences in scores allows researchers to use parametric statistics on interval-level data. Parametric statistics assume that the distribution of the data approximates a normal curve. Because of this added assumption, parametric tests are more sensitive to differences in score than nonparametric ones, meaning they are more likely to find statistically significant differences when they exist.

Measures taken at the interval level can be added and subtracted, but they cannot be divided. This is because they do not have an absolute zero value, so dividing would be meaningless. For IQ scores, no one has an IQ of 0 (the average IQ score is set at 100), and it would be ludicrous to say that someone with an IQ of 120 was twice as smart as someone with an IQ of 60.

Data collected at the *ratio level of measurement* do have an absolute zero. The behavior counts discussed previously are at the ratio level (e.g., two people who have kissed eight times have done so twice as often as two people who have kissed four times). In psychology, ratio data will most often be expressed as frequencies, duration, latency (how long it takes for something to happen or response time), or amplitude (how strong a response is). Parametric statistics are used with ratio level measurements.

THURSTONE SCALES

Thurstone (1931) and Thurstone and Chave (1929) applied the method of equal-appearing intervals to the measurement of attitudes. To summarize the method briefly, the scale developers compose a large number of statements reflecting attitudes that range across the spectrum of attitudes about the issue under investigation. These items are submitted to a large number of judges who, working independently, place the statements in piles that range from one extreme to the other in attitude. The median value for each item is obtained from the judges' placements. Selection is then made of items whose median values are spread out as evenly as possible along the scale. Because the intervals between items have been established by judges to appear equal, Thurstone considered the final scale to be an interval scale that was amenable to analysis by statistical methods that require interval data.

The *Thurstone scale* is a type of *normative rating scale*. These are scales in which an individual's responses are compared with those of other respondents sampled from the same population.

LIKERT SCALES

The method that Likert (1932) devised for the measurement of attitudes was much simpler than that of Thurstone in that it did not involve laborious prejudging to establish *equal-appearing intervals*. Instead, in a *Likert scale*, stepwise adjective choices that have the semblance of equal intervals are offered to the respondents for a series of statements about the issue under investigation. Because of its simplicity, it is used much more frequently in psychological research.

Likert originally presented a 5-point scale with the adjective choices *strongly disagree*, *disagree*, *undecided*, *agree*, and *strongly agree*. For example, a participant might be asked which choice best expresses his or her attitude toward the legalization of marijuana. These choices are scored 1, 2, 3, 4, and 5 or −2, −1, 0, +1, and +2, and a total score is arrived at by adding across all items. If an item is worded to reflect a negative or unfavorable attitude, these scores are reversed to keep the meaning of the score consistent. Therefore, a *strongly agree* (+2) response to a negative item would be scored as −2 when the item scores were added together.

For a Likert scale that includes numerous items, an item analysis is performed. Each item is correlated with the scale's total score, and items that do not correlate satisfactorily with the total score are discarded. A factor analysis of responses can also be conducted to see how individual items perform or whether the scale might be measuring more than one thing. Scales that are devised ad hoc and that retain some of the elements of Likert scales are commonly called Likert-type scales even though all the recommended procedures have not been followed.

Adjectives are used to label levels of agreement on Likert-type scales. Arranging them in sequential fashion is an attempt to create a scale that has intervals that appear to be equal. To be a true interval scale, the difference between *strongly agree* and *agree* would have to be the same as the distance between *agree* and *undecided*. Because this is not necessarily the case, the resulting scale, strictly speaking, yields ordinal data that are limited in the type of statistical treatments that can be applied. This has become less and less of a concern over the more than eight decades in which Likert scales have been widely applied.

Research methodologists and statisticians have taken positions that run the gamut from strong disapproval to benign acceptance of the application of parametric statistics to Likert-scale data. Strictly speaking, Likert scales yield ordinal-level data. Those disapproving, say, Likert scales can be used to rank respondents but cannot be used to express how much more favorable one respondent is than another. Others argue that Likert-scale data are close enough to interval levels of measurement that the parametric tests can still yield legitimate results.

Midpoints

True Likert scales have a midpoint. Some respondents tend to gravitate toward the center when faced with an obligation to make rating scale decisions. Some researchers are sufficiently concerned with the centrist tendency of respondents that they eliminate the midpoint and force the respondent to make a choice. The previously mentioned 5-point scale of agreement becomes a 4-point scale (*strongly disagree*, *disagree*, *agree*, *strongly agree*), with score values of 1, 2, 3, and 4, respectively. However, the psychological distance between *disagree* and *agree* does not appear to be equivalent to the distance between *agree* and *strongly agree*. The gap in the middle and the inequality weaken the argument for considering this as an interval scale. Also assumed is the notion that a midpoint rating is not a valid representation of a person's judgment. Although some people may use the midpoint to avoid making a choice, there are others whose best and most honest response is in the middle when they feel truly

ambivalent. A 4-point scale also reduces the number of data points and the chances for finer discrimination between groups. On balance, it appears that more is lost than is gained by dropping the midpoint. As a critical reader, you should assess whether a Likert-type scale without a midpoint is appropriate for the attitude being assessed.

Likert-type scale construction has several important characteristics that you should look for:

1. Items should be written in a way that matches the reading level of the respondents.
2. Compounded items should be avoided because they can be hard for respondents to answer. If the item contains two referents (e.g., "I am afraid of heights and open spaces"), the respondent may agree with one part and disagree with the other. They can also be hard for researchers to interpret.
3. The double negative should be avoided. Consider the item "I do not have trouble falling asleep at night." If the response is *Disagree*, the respondent would have to go through mental gymnastics to figure out if this was the answer they wanted to give. The double negative could be avoided by changing the wording of the item to "I have difficulty falling asleep at night." A positive alternative would be, "I fall asleep easily at night."
4. The items should be arranged in the questionnaire so that the response to one item will not affect the response to the next. If responses might affect one another, items should be randomly ordered on each questionnaire so that different respondents get different orders.

Item Weights

Adding up Likert-type item scores means the researchers have assigned equal weight to each item. It is also possible to weight each item. As an illustration, here is a Likert-type item:

My boss listens to my ideas:

strongly disagree	*disagree*	*undecided*	*agree*	*strongly agree*
(1)	(2)	(3)	(4)	(5)

Here is a weighting item to follow it:

How important is it to you that your boss listens to your ideas?

of little importance	*somewhat important*	*very important*

When the weighting is done by multiplication, the possible scores in this illustration range from 1 to 15. However, in multiplicative ratings, prime numbers by definition can never be obtained. Scores of 7, 11, and 13 cannot be reached by multiplying

two numbers that are larger than 1, so in this example there are 12 possible scores and three gaps. This clearly makes the scale ordinal. No such problem exists for additive weights. Instead of multiplying, adding either 1, 2, or 3 could make this 5-point unweighted scale into a 7-point weighted scale. When multiple items are used to make up a scale (e.g., if the question was part of a scale on satisfaction in the workplace), the weighted items would be summed.

Adjectives and Number of Scale Points

Sequential adjectives are not restricted to scales of agreement. They can describe frequency, using such scale anchors as *always*, *usually*, *sometimes*, *rarely*, and *never*. They also could describe magnitude, approval, or any other meaningful dimension.

Also, the number of scale points is not limited to 5; in fact, scales with 7 points, 9 points, and 11 points have all been used. The advantage of a scale with more points is that it allows for finer discriminations and appears to make the interval assumption more tenable. A disadvantage is that it is sometimes difficult to find enough appropriate adjectives. One way that this has been dealt with is to leave blanks in between every other adjective and to give instructions about their use to the respondents. For example,

strongly disagree		*disagree*		*undecided*		*agree*		*strongly agree*

Respondents are encouraged to mark anywhere along what has now been converted into a 9-point scale. When the final score comes from the summation of the scale values of several different items, a wider range of scores is possible than if five or seven responses were possible. When the scale consists of a single item on which respondents are asked to make a global rating (e.g., "I feel better than I did before undergoing psychotherapy"), it is important to have enough points on the scale to provide sufficient sensitivity and to permit statistical analysis. A global scale that has only 5 points would restrict the distribution of scores. Any tendency of respondents to avoid extreme ratings by rating safely in the middle range would reduce the scale to 3-point scores and make statistical analysis hazardous. Use of a 9-point global rating scale would reduce this possibility. This is the kind of thing that you should notice as you read and take into account as you assess the credibility of a study's findings. Was the rating scale sensitive enough to capture subtle differences on the dependent variable?

NUMERICAL (DIGITAL) RATING SCALES

Instead of adjectives, a numerical scale of any length can be presented to respondents with anchors at each end, representing the most extreme scores (e.g., *not at all*, *very much*). People have become quite accustomed to being asked to rate things on a scale from 1 to 10. For example, clients in therapy could be asked to rate their experience on a 10-point scale. Decimals can be added to make for even finer distinctions.

Numerical rating scales are used in a similar way to rate all kinds of psychological variables in research.

VISUAL ANALOGUE SCALES

Visual analogue scales, originally called graphic rating scales, are simply straight lines 10 cm to 20 cm in length, with adjective anchors at either end. Suppose that a scale has two anchors, *strongly disagree* and *strongly agree*, and respondents are asked to make a slash along the line at the point that best represents their level of agreement:

My experience in therapy was excellent:

Disagree ———————————————————————— Agree

The closer to the right-hand side, the stronger the agreement. The item score is obtained by measuring the distance of the slash mark from the left-hand edge. Item scores can be summed to obtain a total score.

The advantage of the visual analogue scale is its simplicity and the unrestricted choices afforded to the respondent. It can be repeated after a short interval for a test–retest reliability check. A numerical rating can be remembered from trial to trial, but the exact position that was marked on a visual analogue scale at the first trial cannot be recalled as easily on the retest.

The disadvantages to using visual analogue scales should be considered carefully. Such scales require a great deal more work on the part of the researchers to measure each individual response.

PICTORIAL ANALOGUE SCALES

Pictorial analogue scales consist of a series of pictures that are graduated along some dimension. They are commonly used with young children. One of the most popular is a series of circular faces with two dots for eyes and a line for the mouth (see Figure 7.1). For example, a kindergarten student might be asked, "Which face best shows how much you like school?" The series might feature a frowning face that gradually morphs into a smiling face. A series of five such faces can be given numerical values for scoring purposes and used as a 5-point rating scale.

FIGURE 7.1



Example of a pictorial analogue scale.

As a critical reader, you should seek assurance from the researchers that all of the children understood the task and were capable of following the instructions and making the necessary distinctions. You should also be aware that some children have difficulty with the idea of having to make fine distinctions represented by intermediate choices and tend to go directly to either end of the scale.

FORCED CHOICE QUESTIONS

Forced choice questions require that participants choose from a set of statements, most frequently two, the one that best expresses their preference. Rotter's (1966) original measure of locus of control had people choose between two statements describing their beliefs about whether the things that happened to them were caused by internal forces or external events. One question looked like this:

25a. Many times, I feel that I have little influence over the things that happen to me.
25b. It is impossible for me to believe that chance or luck plays an important role in my life. (Rotter, 1966, p. 12)

If higher scores reflect an internal sense of locus of control, a respondent would get +1 for choosing 25a and 0 for choosing 25b. The scale has 29 questions, so by summing responses, scores could range from 0 to 29. Similar to a Likert-type scale with no midpoint, forced-choice questions force respondents to choose a position to take. However, unlike the Likert scales, forced-choice scales do not have scale gaps, making it more reasonable to treat the scores as if they were interval level, especially when several questions are summed.

OSGOOD SEMANTIC DIFFERENTIALS

This technique is designed to assess the connotative meanings of words on 7-point scales that are anchored on either end by opposite adjectives. Figure 7.2 shows how

FIGURE 7.2

For each of the following adjective pairs, place an X in the circle that best represents your perception of the profession of *firefighter.*

Good	O	O	O	O	O	O	O	Bad
Warm	O	O	O	O	O	O	O	Cold
Strong	O	O	O	O	O	O	O	Weak
Active	O	O	O	O	O	O	O	Passive
Decisive	O	O	O	O	O	O	O	Indecisive
Serious	O	O	O	O	O	O	O	Fun loving

Example of a semantic differential scale.

a *semantic differential* might look to collect data on the meaning of the occupation *firefighter*.

Researchers select the adjectives to fit the attitude that is being assessed. You should examine them to see whether they are relevant dimensions, given the object of the attitude, and are anchored by semantic opposites. You should look for indications that all the respondents understood the task and had no difficulty accomplishing it.

Now that you know the array of types of rating scales available to researchers and the potential pitfalls of each, you should examine the scale used in a study to decide whether it was well-chosen; to determine whether the items are worded in a way that is intelligible to respondents; and to determine whether the items are free of difficult vocabulary, compound and multiple thoughts in a single item, and double negatives. You should consider whether the items cover the issues under inquiry and whether an item analysis (if called for) was provided. You should look for documentation regarding the scale's reliability and validity; if the scale is homemade, your attention should be given to how the researchers went about establishing reliability and validity, and how satisfactory the outcome was.

Raters and Judges

Criteria in psychological research often do not lend themselves to measurement by physical apparatus or by psychological tests. Sometimes ratings made by people other than the research participants have to be quantified. Who the raters or judges are (these terms are used interchangeably), how able they are to make valid and reliable judgments, and the way that the researchers provide for them to go about their task are all vital to the construct validity of the study and your assessment of it.

THE RATERS

The raters and judges may be experts or competent observers. Raters are brought in to assess behaviors that are potentially too ambiguous to be counted and call for complex and qualitative assessments. In recognition of the fallibility of human judgment, researchers do not like to rely on a single rater. When multiple raters are used, they must work independently and be free of the experimenters' influence when making their judgments.

Interrater reliability becomes an important issue for you to consider. In some fields of endeavor, particularly when judgments must be made instantly, experts receive extensive training in making rapid judgments. For example, the decisions that baseball umpires make are relatively simple, dichotomous ones—ball or strike, safe or out, fair or foul. There is no "maybe" category and no opportunity to think about the decision. Until recently, there was no video replay to see if the call was made accurately. The rectitude and certitude of the judges prevail.

Umpires, referees, and judges in interactive sports are part of the process and are in the game. Their instant judgments have an immediate effect on the strategy of

the opponents and the course of the game. However, judges in research (with some exceptions) are usually external to the process, and it is hoped that their judgements (or the process of being judged) have no effect on the behavior of participants.

In contrast to interactive sports like baseball, the judging and quantifying of the qualitative merits of performances such as diving, figure skating, and gymnastics are not left to one person. Panels of experts are required to make ratings on numerical scales. In psychological research, panels of raters are similarly preferred over individuals as performance evaluators. The performances are in the form of the behaviors or work products of participants.

Harkening back to the our discussion of the ways of seeking the truth (see Chapter 1), it may have occurred to you that psychological researchers, within the framework of doing scientific experiments, are obliged to place their faith in the judgment of one or more persons who are viewed as competent to make judgments, which is not the best way to acquire knowledge. Researchers try to compensate for this apparent weakness by exercising stringent controls over the entire process of judging and rating to make it as valid and as reliable as possible. There is a tendency to place more faith in the pooled judgment of many than in the judgment of a single expert. However, some would prefer the judgment of the single most knowledgeable expert over the pooled judgments of more than one lesser lights. Others who are more cautious would hold out for a panel of the most knowledgeable experts over any single one.

TRAINING OF RATERS

Because of the lack of numerous outstanding experts available for research purposes, researchers have to enlist competent people and train them for their judging role in a particular study. The raters must learn what to look for, what the criteria are, what the ratings mean, and how to use the rating scales. As part of this training they have to practice on samples of people or materials that are analogous to those of the study proper. The researchers must review the practice ratings with them, make sure that the criteria are clearly understood, and give them feedback. So, if the researchers are studying disruptive behaviors in classrooms, they would train raters by defining disruptions as unambiguously as possible, make sure raters understood its application, play videos of classrooms for the raters, and then ask them to rate the students in the video.

NUMBER OF RATERS

Typically, more than one rater is used. The scores of the raters are summed and averaged to provide a score for each participant. Just as relying on one rater in psychological research can be problematic, you have reason to question the validity of the pooled judgments if only two raters were used in a study. If on a 10-point scale, one rater rates a behavior as 8, whereas the other rates it as a 2, the pooled rating of 5 is of doubtful meaning.

RATING CONDITIONS

For some research, it is necessary for raters to do their work "live" as on-scene observers. This is most difficult because decisions must be made on the spot with no chance for reviewing or second-guessing. The researchers also must arrange the setting in such a way that the raters can work independently of each other.

For other projects, researchers can provide raters with audio, video, or written productions of participants. Here, raters can review materials as carefully as they wish (e.g., they can stop the video and replay it as often as is necessary).

Factors such as the sheer amount of material to be rated, the time that it takes to do a rating, the length of the sessions, and when they are held can affect the accuracy of the ratings. The researchers should be concerned with rater drift, fatigue, boredom, flagging of attention, and loss of interest and motivation. The added experience over a long series of ratings makes some raters better than less experienced ones, but rater drift can bring about a systematic change in their ratings as they proceed. Other raters become slipshod and are in a hurry to finish. The researchers must think of ways to deal with these artifacts, and, when reading the report, you, as a critical reader, must focus on how successful that effort was.

INTERRATER RELIABILITY

Your concern about the consistency of judges' ratings can be allayed if the researchers report some measure of interrater reliability. Once the researchers are reasonably confident that the raters understand all of the rating procedures and issues, it is wise to give a pretest to check interrater reliability. By examining each rater's level of reliability, for example, the researchers can identify any rater who does not reach an acceptable level when compared with the other raters. The decision can then be made to give this individual more training or to substitute a different rater. Careful attention to this type of rater training usually assures that the ratings done in the investigation are reliable.

Several ways of assessing interrater reliability are available. Each rater's ratings can be correlated with those of every other rater to generate a correlation matrix. The researchers can then peruse the matrix to see if there is a rater whose ratings do not correlate with others. Correlation coefficients, *r*s, are converted to z *scores*, the *z* scores are averaged, and the average is reconverted to *r*. This can be used as an overall estimate of the reliability of the team of judges. Rosenthal and Rosnow (2007) presented a table for computing effective reliability of the mean of judges' ratings. Another way to assess interrater reliability is by using the intraclass correlation, which uses analysis of variance. When critiquing the research, you should closely examine the reported training of the raters and the level of interrater reliability.

Categorical Judgments

When the judgments are categorical rather than scaled, a whole different set of problems is presented (Meltzoff & Kornreich, 2007). Categorical judgments can be of the

dichotomous yes–no variety (e.g., did the student answer the question correctly?), may have several alternative choices (did the participant appear happy, sad, or express no emotion?), or may even be in the form of a checklist of things for the judge to identify if seen (which of the following behaviors did the participant engage in?). Judges have a relatively easy job if there is no ambiguity among the categories. If the responses are clear and evident, a count is all that is necessary. The greater the ambiguity, the greater the need for multiple judges, and the greater the need to rule out chance in the agreement among the judges. The type of judgment that a judge is asked to make has a bearing on the number of judges that the researchers should use and on the standards of agreement that the researchers adopt. If the judgments are categorical, there must be a way for the researchers to predetermine how many judges to use and what to do when they disagree.

When categories are used, a certain amount of chance agreement is inevitable. The expected agreement by chance can be calculated as a function of the number of categories (C) and the number of judges (J). The number of possible outcomes can be expressed as CJ. Therefore, if there are two choices, A and B, and two judges, the possible ways of agreement and disagreement can be obtained from the formula $(A+B)^2$. This binomial expansion comes to $A^2+2AB+B^2$, which indicates that there is one way of both judges saying A, one way of both saying B, and two ways of one of them saying A and the other saying B. With only these four possible outcomes, unanimous agreement of the two judges (both saying A or both saying B) would be expected one half of the time by chance even if the judges were making random guesses.

Three judges would agree unanimously by chance one quarter of the time, as can be seen in the expansion of $(A+B)^3$, which comes to $A^3+3A^2B+3AB^2+B^3$. Here, two of the eight possibilities are unanimous agreement (all three judges saying A or all saying B). If the researchers alter the question and inquire about the chances of all three judges saying A without regard to B, there is only one way out of eight possibilities that this level of agreement could happen by chance. This is still far from the conventional 1 in 20 level of confidence, however, and retains more possibility of chance agreement than critical readers might find acceptable.

If the researchers abandon the idea of unanimous agreement and adopt a less rigorous standard, say two out of three judges agreeing is all that is needed, other problems are introduced. A majority decision may be democratic, but you should recognize that agreement by two out of three judges (or better) when there are only two possible categories is inevitable. There simply is no way that three people can all disagree if they are given only two choices. Claims of agreement among judges are not convincing to critical readers if they are made on the basis of elevated levels of agreement that are forced by the limited number of choices and judges.

If additional categories are introduced, the researchers can determine how many judges to use and what level of agreement to require to be confident that the agreement did not come about by chance alone. With three choices and four judges, for example, the polynomial would be $(A+B+C)^4$, and the chances of unanimous agreement would be 3/81, or less than 4%. If much agreement was obtained, you could be quite confident that it did not come about by chance.

Instead of using the Pearson *r* to check reliability, as is the case for scaled ratings, categorical judgments call for a different measure of association. *Kappa* (Cohen, 1960) is widely used because it considers chance agreement and adjusts reliabilities accordingly. Values of kappa range from negative to positive, and a kappa value of +1 means agreement was perfect. Kappa does not assume that all categories will be used equally but rather defines chance agreement based on the marginal totals of each category and therefore takes into consideration the fact that the occurrence of categories may not be evenly divided. The range of values of kappa is restricted when there is a preponderance of instances in some categories, as is often the case when the crucial target behavior happens only occasionally. For example, time sampling of what a child is doing in class might find only infrequent instances of "talking to another student during lesson" compared with "sitting facing the teacher." The size of kappa indicates the extent to which chance agreement is playing a role, but even with an acceptable coefficient, chance agreements still occur in the data; this technique does not remove them.

The binomial or polynomial expansion is an especially rigorous way of dealing with the problem of judges agreeing by chance. Chance agreement among judges can be kept to an acceptably low level for each judgment that is made. If that low level is not reached, the observation or response is discarded as "too ambiguous to call." Therefore, only those data that are protected against chance agreement remain in the data for analysis. This way the researchers can confidently vouch for what is presented to the reader as evidence. Researchers who do not feel that it is necessary to use this method in their particular situation, and those who do not have the resources, may select from among a number of alternative approaches and standards of agreement. However, just as researchers make the decision about which approach to use for interrater reliability, researchers must also convince you that the standards applied were sufficiently rigorous to assure the accuracy of the judgments.

RATER RELIABILITY AND CONSTRUCT VALIDITY

The number of raters and the interrater agreement that you demand are based on the level of confidence that you are willing to use as a guide. In Gortyn on Crete, in the 5th century B.C., the written legal code specified that only one witness was needed to convict a slave, whereas the testimony of five witnesses was required to convict a free man (Fisher & Garvey, 1995). The decision about what is an acceptable level of confidence is based on the value of reaching a correct judgment and the consequences of arriving at an incorrect one. You should consider what the consequences of inaccuracy are to the study. If judgments used to categorize the levels of the independent variable are inaccurate, the construct validity of the entire study is compromised. If individuals are misclassified to start with because of the judging process, the meaning of the entire study is in serious question. Imagine a small-sample study contrasting the behavior of a group of children with hyperactivity and a group of children with no hyperactivity under two different conditions. The children are categorized by judgments made by the teacher and the language specialist. If, because of lax judging standards, several children with hyperactivity are misjudged and placed

in the nonhyperactive group, and several children with no hyperactivity are misjudged and placed in the hyperactive group, the construct validity of the study is in jeopardy. Inaccurate judgments that form the dependent variable measure are also very much a matter of concern because they could lead to erroneous results and distorted conclusions.

Reading critically, you should take into consideration the level of ambiguity of what is being judged, the number of choices available to judges, the number of judges, the obtained agreement, the interrater reliability, the standards of agreement adopted by the researchers, and the rationale that is given for using these standards.

When reviewing the judging process in its entirety, you should be alert to how well all of the following issues were handled:

- Were the judges qualified, and were they able to do the job effectively?
- Were the judges adequately trained and pretested on their ability to do the task?
- Were the judges properly motivated?
- Did the judges' interest and effort sustain throughout the judging period?
- Were the researchers sensitive to the setting and needs of judges?
- What was done to assure the independence of the judges?
- Did the procedures protect against bias on the part of the judges?
- Was interrater reliability satisfactory?
- Was the number of judges large enough to enable the researchers to rule out chance agreements, and were the standards of agreement appropriate?
- How were disagreements among judges handled?

Answers to these questions will give you a good idea about the adequacy of the procedures that the researchers used.

Summary

In this chapter, we have stressed the necessity of constructing manipulated treatment variables that are valid and impactful. We highlighted the importance of using pilot tests and manipulation checks to demonstrate that the operations have these characteristics. The construct validity of studies that use organismic, psychological, and status variables rests on accurate classification of people into static groups. We also discussed the types of dependent variable measures and ways of demonstrating their reliability and validity. Types of tests and measures include counts, behavioral observations, physical measures, and psychological measures (standardized tests and tailor-made measures). The type of data collected by different measurement approaches include rating, scaling, and ranking methods such as the Thurstone scales, Likert scales, forced-choice scales, the Osgood semantic differential, numerical rating scales, visual analogue scales, and pictorial analogue scales. We examined the use of judges and raters to provide ratings on participants, along with issues of reliability for categorical and scaled judgments, the number of judges recommended, and standards of agreement. We also surveyed nonreactive (unobtrusive) measures. Where a manipulation or measure fits into these classifications dictate to you, the critical reader, what to consider when assessing the validity and reliability of a study.

Data Analysis, Discussion, and Conclusions

8

A critical reader of research

- knows the relationship between levels of measurement and the appropriate statistical tests for each level;
- can identify the assumptions needed to apply different statistical tests appropriately;
- can calculate degrees of freedom and check to see if the statistical analyses are consistent with them;
- considers the possibility of an inflated experiment-wise Type I error, if many significance tests have been run;
- can distinguish between planned and post hoc comparisons, and how this distinction affects statistical analysis and interpretation;
- recognizes the difficulty in interpreting measures of change;
- understands the different metrics for calculating effect size; and
- looks for distortions when data are presented in graphs.

The next phases of the research involve the analysis of the data. Some readers of research who are more comfortable with verbal concepts than with mathematical ones give in when it comes to this part of the research report. They accept on faith whatever numbers the researchers present. Needless to say, statistical procedures that are poorly selected, improperly used, and incorrectly interpreted are cause for your concern. This text is not

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a statistics book, and we will not elaborate on the myriad of statistical tests that are available. Nor will we show you how the calculations are carried out. If you are unfamiliar with statistical methods, more detailed information is available in any standard text on statistics.

This chapter addresses a few salient issues with which you, the critical reader, should become familiar. In it, we address the importance of being able to evaluate whether the researchers (a) used statistical tests that were appropriate to answer the research question or test the hypotheses, (b) had data that met the assumptions of the statistical tests, (c) adequately handled the data if there were multiple dependent variables in the same study, (d) controlled properly for *Type I errors* (i.e., rejecting the null hypothesis when it is in fact true), (e) used *planned comparisons* and *post hoc tests*, (f) demonstrated pretest equivalence, and (g) used proper change scores (if they were used at all). We also discuss effect sizes and power to detect significant effects. Finally, we illustrate some distortions in the graphic presentations of data.

Levels of Measurement and Their Appropriate Analysis

As we introduced in Chapter 7, you should be able to identify the level of measurement used by researchers and to determine whether it is consistent with the statistical techniques selected.

NOMINAL DATA

If the data are nominal or categorical, did the researchers use nonparametric statistics? For example, a *binomial test of proportions* could be used if the researchers want to test whether men or women are more likely to sign up for a college workshop on avoiding date rape. Here, there are two static groups (men and women) and two proportions to be compared. A *phi coefficient* can be used to express this relationship in terms similar to a correlation. Men and women are assigned a value of 0 or 1, and sign-up (no or yes) is also assigned 0 or 1. The two sets of 0s and 1s are then correlated using nonparametric assumptions. If the researchers want to compare more than two groups, say students majoring in science, social science, and humanities, a *chi-square* or *Cochrane Q* test (for matched samples) is appropriate.

ORDINAL DATA

For a comparison of two independent groups with ranked data, the *Mann–Whitney U test* can be used analogous to the *t* test to determine whether members of one group or another ranked higher. For example, if a rater ranks the children in a class regarding how well behaved they are, the boys' and girls' rankings could be placed in different groups and their average rank compared. Or, if three or more groups are of interest, the *Kolmogorov–Smirnov test* is appropriate. If the rankings are correlated

(e.g., one rater's ratings of two groups of pictures, perhaps with different backgrounds), then the *Wilcoxon matched-pair signed ranks test* can be used.

If more than two independent groups are used, a *Kruskal–Wallis one-way analysis of variance* (ANOVA) is appropriate, and the *Friedman two-way ANOVA* is appropriate for ordinal data with related samples. If correlating ordinal data, the *Spearman rank-difference correlation* or the *Kendall Rank correlation coefficient (Tau)* is appropriate.

INTERVAL OR RATIO DATA

Interval or ratio data open the way for the full range of parametric statistical techniques, including *t* tests for independent and correlated groups, all types of univariate and multivariate analyses of variance, correlation coefficients (*r*s), multiple regression, discriminant function analysis, factor analysis, and *path analysis*. Some techniques, such as *biserial correlation* (r_{bis}) and *logistic regression*, are capable of handling mixed parametric and nonparametric data.

This is not meant to be an exhaustive listing of all the statistical techniques available. It is merely a sampling of the kinds of statistics that are commonly encountered in research articles. You should wonder whether the researchers used the most appropriate and powerful statistical technique given the available data. For example, sometimes researchers analyze interval data suitable for using a parametric ANOVA with a chi-square statistic. If a valid explanation of the reasons for the choice is not given, you may speculate that the less sensitive test was applied because the data did not meet the assumptions of the more powerful and discriminating test.

Distributed Versus Categorical Variables

As we discussed in Chapter 7, you may come across studies in which the dependent variable was measured on a continuous numerical scale but the data were then collapsed into dichotomies or into a limited number of categories. This procedure is related, but it does not exactly parallel the procedure we discussed in Chapter 3 for establishing levels of the independent variable. Unless the researchers provide good reasons for doing this, much information can be lost this way. Cohen (1990) warned that "when you so mutilate a variable, you typically reduce its squared correlation with other variables by about 36%" (p. 1306). His flat-out dictum, "Don't do it," must be tempered with mercy because at times researchers have no other choice. Sometimes, continuous data do not even vaguely resemble the shape of a bell, the assumption needed for parametric tests; they cannot be transformed no matter how they are stretched, squeezed, pulled, or mashed; and they cannot be made to ring true by calling the statistical technique "robust." In such cases researchers are forced to retreat to categories and to use nonparametric, distribution-free statistics.

Sometimes there are theory-based reasons to use categories even though the data are recorded in discrete integers. For example, in a controlled experiment testing a smoking cessation program, a daily record is kept of the exact number of cigarettes

smoked. The criterion of success is "quitting," defined as zero consumption. Nothing short of that is considered by the researchers as a success. Therefore, even though the participants may have been asked, "How many cigarettes did you smoke last week?" the data at the end are dichotomized appropriately into success (0 cigarettes) and failure (1 or more cigarettes). Another study is designed to test whether meditation is a good cure for high blood pressure. The criterion of success is "return to blood pressure within the normal range." It would be appropriate to determine how many people returned to normal in the treated and untreated groups and how many did not. A comparison of mean blood pressures in the two groups would not be helpful in this situation because everyone's blood pressure could still be high. If the study asked, "Did meditation reduce blood pressure?" and the dependent variable is normally distributed, it should be left continuous and parametric tests used.

Assumptions

In addition to the measurement level of the data, other assumptions should be met for the proper application of various statistical tests. Knowing that an ANOVA was used, for example, you should look for a statement in the text that assumptions about normality, *homogeneity of variance* (the similarity of variance within a condition), and additivity have been met well enough to warrant proceeding with the test. The results of the test for *skewness* and *kurtosis* may even be reported along with these other diagnostic tests of the data. Although older research reports may not contain this information, new reporting standards call for researchers to report these statistics. Lacking any such assurance, you should look for departures from normality in the data that are shown. One way to do this is to examine the standard deviations of the various groups to see whether there are any gross inequalities (i.e., the variances within groups might be heterogeneous), or if the standard deviations are as large or larger than the mean, suggesting skewness.

A favorite defense of researchers who feel justified in violating the assumptions of parametric tests is to adopt the frequently used claim that the statistical test is sufficiently robust to withstand violation of assumptions. As a critical reader, you must judge whether the violation is too serious for comfort and wonder why data transformations or other fallback solutions (e.g., categorization and use of nonparametric statistics) were not applied. Reference material should be consulted for guidance on the assumptions about whatever statistic is used in the study under review, so that you can identify the assumptions. For example, you should identify whether the violation of the assumptions in regression analysis (called *multicolinearity*), *specification errors*, or nonlinearity has occurred, or whether the added assumption of a unidirectional relationship in path analysis has been met. In discussing specification errors in regression analysis, Achen (1982) eloquently, but with tongue in cheek, asserted,

> Significance tests, analysis of variance, R^2 and other statistical calculations are carried out on the assumption that the model in use is the one true specification. Work of this kind curls the lip of a theoretical statistician, and understandably

> so. It looks foolish. If social scientists mean to sally forth under the banners of conventional statistical theory, as they themselves claim to be doing, then they will seem a motley crew. They pledge allegiance to a lord, and then throw off his livery. (p. 11)

You, as a critical reader, become the liege of the theoretical statistician.

Degrees of Freedom

You will find it worthwhile to figure out the correct *degrees of freedom* for each variable and test statistic to see whether it matches what is reported. Having the proper degrees of freedom in mind will help you cross-check the number of participants reportedly used in the statistical analysis with the total number of degrees of freedom in the analysis, which should be equal minus one. Doing so gives you a better understanding of what was done and occasionally turns up inconsistencies.

Statistic Selection and Presentation

Sometimes the researchers' biases become evident when more powerful statistical techniques are used, even though the data are best suited to less powerful ones, or worse yet, when the less favorable results from one set of approaches are rejected, and the more favorable results from the other are adopted.

In other instances, you may sense that the researchers have been selective in analyzing and presenting material. That is, the researchers have not given a full and candid presentation of all the analyses and all the findings of their study. We cover this in Chapter 9. This kind of selective reporting can distort the true results.

Computation

The use of computers reduces the chances of mistakes in computation, but computers are no more accurate than the numbers that are fed to them. Coding and transcription errors are not evident to a reader. Nevertheless, you should look at all the reported statistical values to assess whether they appear to be in line with the data that have been provided. Entries in tables and text should be scanned for consistency and obvious error. For example, if the composition of a sample in a study is described as being 58% male and 32% female, clearly an error has been made. Sometimes (many times, we hope), you can even recalculate statistics. For example, if the means, standard deviations, and cell sample sizes are given, you can recalculate an associated *t* test. You need not do this for every test, but knowing that one was done correctly (and they are mostly done by computer) is evidence that the researchers understood what they were doing.

Some research reports are marked by what appears to be an attempt to make results look more precise than the measurements warrant. Ordinary values are reported to

three, four, and even five decimal places, implying far greater precision than is actually the case. The researchers are not talking about the trajectory of a rocket to Mars, where small imprecisions can lead to big mistakes, but about, for example, an average anxiety scale score of 52.46342, a *t* value of 1.98403, or a *p* value less than .036282. Entries like these give the impression of extreme accuracy (and that is probably why they are reported this way) but in actuality, they contribute no more than calculations that are judiciously rounded down. It will be truly odd to see numbers like these in more recent articles, as reporting standards now recommend describing results to no more than two decimal places, except for *p* levels, which can go to three or four decimal places if all but the last digit is 0.

Multiple Dependent Variables and Type I Error

You should be alert to studies in which many measurements are taken and analyzed with no correction for *experiment-wise Type I error*. Appropriate corrections, such as the *Bonferroni correction*, protect against obtaining "significant" results by chance when many measures are taken and tested on the same set of data.

In some instances, secondary and exploratory forays into the data are launched after the main analysis did not pan out well. As you critically read the report, you get the impression that data are being scrutinized for what little additional value may be derived. In such cases, the possibility that a Type I error has been committed becomes a real consideration. Reporting standards now request that such analyses be labeled as exploratory so that readers can distinguish between them and the hypotheses that originally drove the study.

Control of Type I error when the researchers use multiple dependent variables is a difficult problem. The easiest way for researchers to handle it is to follow Cohen's (1990) dictum "simple is better" and to keep the number of statistical tests down to a meaningful minimum. Of course, this advice does not help a critical reader who has had no part in the decision. One popular approach used by researchers is to do a *multivariate analysis of variance* (MANOVA). The MANOVA merges the multiple dependent variables into a multivariate distribution and tests to see if these distributions differ across conditions. Only if the MANOVA shows this composite distribution to be statistically significant do the researchers proceed with a series of univariate ANOVAs in the belief that this will have controlled Type I errors.

This approach has been vigorously criticized from several directions. Most important, MANOVA should not be used as a substitute for the well-reasoned choice of dependent variables. As a critical reader, you must assess whether the dependent variables were chosen to clearly capture the changes meant to be brought about by the independent or predictor variable. For example, Dar, Serlin, and Omer (1994) cautioned against being driven "into a multivariate frenzy" (p. 81). They observed that "we are rarely interested, however, in arbitrary linear combinations of variables" (p. 81). They thought that before doing tests of hypotheses the researchers should come

up with reasoned combinations of variables. To do this, the researchers can use a *principle components analysis*, to help them find meaningful composite variables. When the main interest is in the individual variables, Dar et al. recommended univariate analyses. They observed that similar considerations apply in the case of regression analysis; the predictor variables should be chosen with care because there is reason to believe they are related to the criterion variable in hypothesized ways.

Planned Comparisons and Post Hoc Tests

Researchers know that it is reasonable to follow up an *omnibus test*, in which the variables have more than one degree of freedom, with planned comparisons for which there are a priori hypotheses. The number of planned comparisons should approximate the degrees of freedom that have been allotted to the independent variable as a source of variance. For example, researchers might test whether religious affiliation relates to people's beliefs about whether life exists on other planets. If there are four religious groups in the study, this predictor variable would have four groups and the omnibus test would have three degrees of freedom associated with it. Finding a significant omnibus test (which simply asks if there are differences somewhere between the religions) and having hypotheses, the researchers might then test the beliefs of religious group A against group B, group C against group D, and the average of groups A and B versus the average of groups C and D. These three comparisons are determined on the basis of a priori reasoning; they are planned comparisons.

As the number of comparisons increases beyond a handful, the chances of making a Type I error increase and should be corrected to minimize the experiment-wise error rate by a procedure. Such corrections include the *Dunnett multiple range test*, or the modified Bonferroni Test. Where there are hypotheses and comparisons that have been planned in advance, the *Fisher exact probability test* can be used with the experiment-wise p value of $\alpha = .05$. It is likely that the researchers do not want to miss out on potentially interesting findings that were not predicted or planned. The number of possible comparisons increases dramatically as the number of variables and levels increase. The researchers must make a decision about whether to explore them all or to restrict the inquiry to the areas that are most likely to yield results.

Post hoc analyses present the same problem, only more seriously so. Here, the researchers do not know what relationships between variables might emerge from the data and are also concerned with the danger of making a Type I error. They are faced with the knowledge that critical values of significance tests are much more difficult to reach with some post hoc tests than with others. For example, in comparing two treatments, the *Newman–Keuls test* might show a significant difference that would not have been regarded as significant had the very conservative *Scheffé test* been used. The researchers must choose among the Scheffé test, the *Dunnett test*, the *Tukey tests*, the Newman–Keuls test, the *Duncan multiple range test*, and the Fisher exact probability test.

If the researchers use a procedure you are unfamiliar with, guides such as Keppel and Wickens (2004) or other statistics texts cover this issue and should be consulted.

As a critical reader, you are faced with the following tasks:

1. Find out which of the comparisons were planned and which were of an exploratory post hoc nature.
2. Determine whether the number of planned comparisons approximates one fewer than the number of levels of the predictor variable and, if an individual comparison approach has been used, determine whether a correction has been made if the number of comparisons is large enough to introduce a Type I error risk.
3. Determine how many post hoc tests have been conducted. You should consider the rationale given by the researchers for selecting the particular test used. If no rationale is furnished, you should try to determine whether it is the most appropriate test to control for Type I error.

Testing for Group Equivalence at Pretest

Some researchers use the *t* test or ANOVA to demonstrate that the groups are equivalent at pretest. They assume equivalence if there are no significant pretest differences. Dar et al. (1994) considered this to be an inappropriate exercise because the null hypothesis (that the groups do not differ at pretest) cannot be proven. Also, large (though not statistically significant) differences could still exist between the groups, and only small changes in scores due to chance could then make the posttest significant. The relieved researchers who declare that the groups are equivalent on the grounds of a nonsignificant (e.g., $p = .08$) difference in group means at pretest have difficulty convincing you, the critical reader, that initial differences were in no way associated with posttest differences. If you read that "group equivalence" has been established by this kind of pretest comparison, you should carefully examine how equivalent the means and standard deviations suggest they really are and should try to weigh what influence pretest differences might have had on posttest scores.

Change Scores

Some investigations with measures that are taken before and after a treatment often are designed to assess change. Two-group, pretreatment–posttreatment designs are interested in the differential change in the groups over time.

One of the problems with the use of raw change scores is that an equal amount of absolute change may not have the same meaning at all levels. In assessing a weight reduction diet, for example, a 320-pound person who loses 20 pounds is compared with a person who begins at 220 pounds. Their raw change scores of 20 are identical, but the meaning of this amount of change is not the same for the two people. This could be corrected by transforming the raw change scores into percentages of change, which would come to 6.25% and 9.09%, respectively.

This is not the only problem with change scores. Raw change scores do not take into account the possibility that the amount of change may be associated with the initial level. There may be ceiling and floor effects. In a program designed to increase SAT math scores, the student who scores 790 out of 800 at pretest cannot gain more than 10 points, whereas a student who begins at 400 has considerable room for improvement. Further complicating the issue, extreme scores will regress toward the mean (see Chapter 6) simply because they were extreme on the pretest because of chance factors.

In other instances, the initial level on a pretest may predict the amount of change across the entire range of scores for experimental and control group participants alike. For example, in a study of the effects of stress on anxiety, people who were anxious to begin with (on the pretest) might react with even greater anxiety to stress than people who are initially less anxious. Unless the pretest level is considered, raw change scores may simply reflect where people were at pretest. This creates a substantial problem if an experimental group and a control group start off at different levels.

When there is a perfect correlation between initial and final scores, the final scores are entirely predictable from knowledge of the initial scores. You should not take this to mean the treatment had no effect, however. If the treatment changed each score the exact same amount, the correlation between pretest and posttest would be $r = 1$, but the mean pretest and posttest scores would differ by the amount of change (and the variance in the pretest and posttest scores would also be identical). At the other extreme, when the pretest level is not at all associated with the final scores ($r = 0$), there is no further need to be concerned with the pretest levels of the groups.

One way for researchers to deal with the effects of the pretest scores in experimental studies in which individuals have been randomly assigned to groups is to treat the pretest level as a covariate in an analysis of covariance (ANCOVA). This statistical procedure adjusts posttest scores for pretest level (by converting posttest scores to their deviation from what the pretest score would predict) and analyzes the differences between the adjusted posttest means. This ANCOVA solution is suggested by many authors (Keppel & Wickens, 2004; Rosenthal & Rosnow, 2007) and should be acceptable to a critical reader.

However, this approach is not considered appropriate when preexisting groups are being compared before and after treatment. It is useful for equalizing the effects of chance differences in randomly assigned groups but not for removing the systematic differences that may be present in preexisting groups. Systematic initial differences between preexisting groups might be central to the study question, and adjusting the final measure of the dependent variable for initial differences might distort the final measure.

Another way for researchers to examine the pretreatment and posttreatment differences is to focus on the Treatment × Time interaction in an ANOVA. This is still a problem in *nonequivalent groups designs*, where one group starts off significantly higher than the other group.

Effect Sizes

As Cohen (1988) defined it, the *effect size* is the discrepancy between the null hypothesis (H_0) and the experimental or *alternative hypothesis* (H_1), or "the degree to which the phenomenon is present in the population, or the degree to which the null hypothesis is false" (pp. 9–10). Reporting effect size is becoming routine in practice.

Cohen (1988) described several metrics in which the size of an effect could be expressed. However, three effect size metrics dominate in the literature. The effect size metric you will encounter as a critical reader is the *r*-index, or the *Pearson product-moment correlation coefficient*, which is used to express the degree of linear relation between two variables. It is most appropriate when describing the relationship between two continuous variables, though we previously mentioned some instances where variations on the *r*-index can be used when at least one of the variables is dichotomous. For example, if researchers were interested in describing the relationship between IQ scores and annual income, they would use a correlation coefficient. Higher absolute values of correlation coefficients denote stronger relationships. The sign of the *r*-index, plus or minus, denotes the direction of the relationships.

A second effect size metric is called the *standardized mean difference*, or *d* index. This is a scale-free measure of the separation between two group means. Calculating the *d* index for any study involves dividing the difference between the two group means by either their average standard deviation or the standard deviation of the control group.

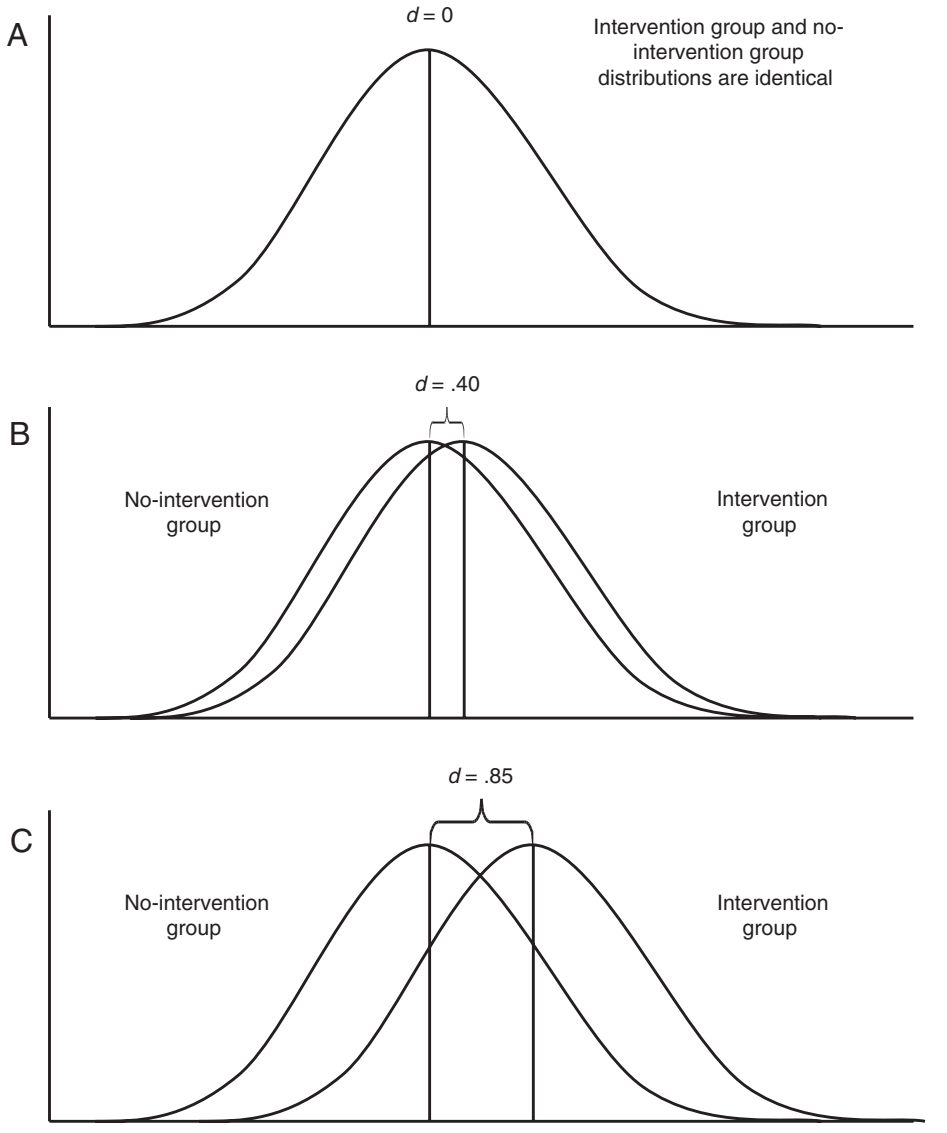
For example, Figure 8.1 presents the *d* indices associated with three hypothetical dependent variables. Assume that these three figures show the effect of three kinds of exercise on a measure of short-term memory. In Figure 8.1A, exercise (e.g., weight training) has no effect on short-term memory. So, the two distributions of scores overlap perfectly and $d = 0$. In Figure 8.1B, the average short-term memory score in the exercise group (e.g., balance exercises) is four 10ths of a standard deviation higher than the average in the no-exercise group. Here, $d = .40$. In Figure 8.1C, $d = .85$, (e.g., aerobics) indicating an even greater separation between the two group means.

The third effect size metric you will encounter is the *odds ratio*. The odds ratio metric is used when both variables are dichotomous and findings are presented as frequencies. The previous example of the proportion of men and women who sign up for a rape prevention workshop would strike you, as a critical reader, as a perfect candidate for use of the odds ratio. Men and women form two dichotomous groups and signing up or not signing up is a dichotomous outcome variable.

Odds ratios appear most frequently in medical studies (e.g., has a patient survived for a certain amount of time after surgery?) but is also used in studies of educational interventions when the outcome of interest is dichotomous (e.g., did the participants graduate high school?) or criminal justice studies (e.g., was a convict charged with another crime after a given amount of time?).

Conceptually, to calculate an odds ratio, the researchers first calculate the odds of the behavior occurring within each group. What are the odds of women attending the workshop on rape? What are the odds of men attending the workshop? Then, the ratio of the odds is calculated, with the numerator chosen depending on how the

FIGURE 8.1



Three examples of the *d* index, standardized mean difference. From *Research Synthesis and Meta-Analysis: A Step-by-Step Approach* (6th ed., p. 212), by H. Cooper, 2017, Thousand Oaks, CA: Sage. Copyright 2017 by Sage. Reprinted with permission.

results will be described. So, if the odds of women attending the workshop is 2:1 and men is 4:1, the odds ratio is 2; women are twice as likely to attend as men.

In many instances, you will still find that researchers do not report effect sizes. If this is the case, there are numerous ways that you can use to calculate them from other pieces of data that are presented in reports (Borenstein, 2009; Fleiss & Berlin, 2009).

Power

The concept of power and its determination was introduced in Chapter 4. Cohen (1988) presented detailed power tables that researchers and readers can use as a guide. Many of these are also available online. The researchers planning the research try to balance power needs, the probability of rejecting a null hypothesis when it is false, and their available resources, in particular, the size of the sample they can collect. With the benefit of hindsight, you can look at these calculations and conclude that the researchers did not plan wisely, used too few participants, and/or included too many groups or variables.

This is different from the situation in which the researchers complete a study with a sample size of, say 26 participants, do not obtain significant results, then go on to do a power analysis of the results. The researchers then say, "All we needed were 367 more participants and we could have gotten $p = .05$ if the mean differences were the same as they are in this study." However, as a critical reader you know that there is no assurance that there would have been the same mean differences with this greatly augmented sample size, nor is there reason to believe that the variability within groups would have remained the same.

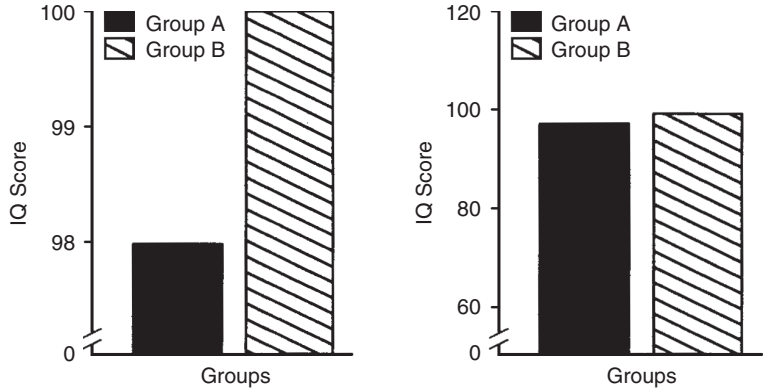
Graphs

Graphic presentations are meant to illuminate, not to obfuscate or distort. In his book, *The Visual Display of Quantitative Information*, Tufte (2001) remarked that "the representation of numbers, as physically measured on the surface of the graph, itself, should be directly proportional to the numerical quantities represented" (p. 56). He defined the "lie factor" as the degree to which the effect that is shown in the graph gives the impression that the effect is larger than its actual size. There are several ways this can happen.

DISTORTION OF SCALE

Distortions of scale can make differences between groups appear to be much larger than they really are. This is done by magnifying a small portion of the scale's ordinate. Suppose you are reading a study that claims a drug can enhance IQ scores. The researchers present the histogram depicted on the left side of Figure 8.2. The double slash lines on the ordinate indicate the line is broken and correctly show that the distance represented from 98 and below is out of scale. The graph, however, makes the mean IQ of

FIGURE 8.2

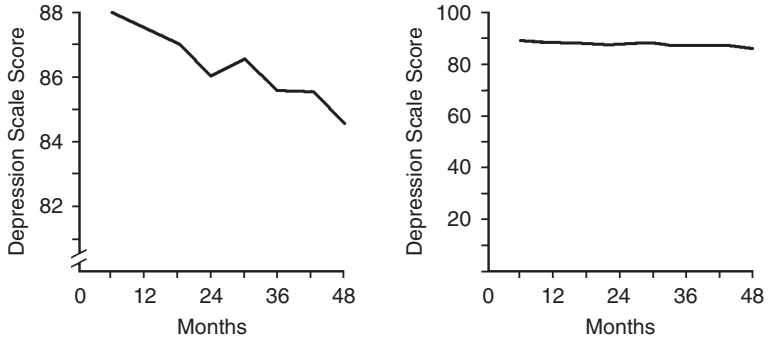


IQ differences between two groups as represented by a distorted scale (left) and a more accurate scale (right).

Group B appear to be triple that of Group A even though the means are only 2 points apart. A more accurate alternative is shown in the right panel of Figure 8.2. Here, the 2-point difference between the group means is portrayed as trivial (which in fact it is).

The same kind of distortion can be made with a line graph instead of a histogram. The left panel of Figure 8.3 illustrates a steady decline in depression scale scores over a 48-month period. For this scale, a score above 70 represents clinical depression. The apparent improvement over time, consisting of a drop of 3.5 scale points over a 4-year period, is inconsequential. The ordinate makes it appear as though the scale begins at 80. In the right panel, the graph is drawn in correct scale.

FIGURE 8.3



Depression scale scores over a 4-year period as represented by a distorted scale (left) and a more accurate alternative scale (right). Scores greater than 70 are considered to represent clinical depression.

DISTORTION OF MEANING

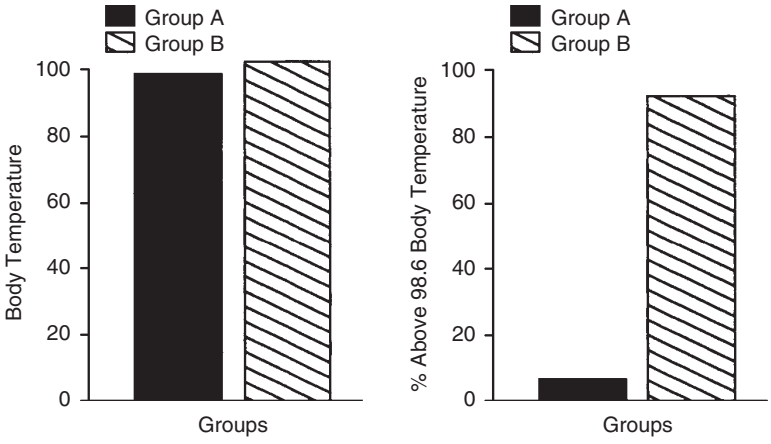
Figure 8.4 accurately portrays the mean body temperature of two groups (Group A, $M = 98.6°$; Group B, $M = 102°$). The graph on the left makes it appear as though there is hardly any meaningful difference between the groups, although 98.6° is a normal threshold value and 102° is distinctly elevated. The graph on the right presents a more meaningful alternative; it shows the percentage of people in each group whose temperature exceeds 98.6°.

ASSUMED LINEARITY DISTORTIONS

When pretest and posttest values are given in a study with a notable time lag between measurements, the tendency to connect two points with a straight line can distort the true picture. The left panel of Figure 8.5 shows a graph purporting to show a doubling of weekly income over a 7-month period. With measures taken only at the beginning and 7 months later, the straight connecting line implies that income increases steadily over the entire period. The right panel shows more comprehensive data, with a distinct curvilinear trend. Income went down for 2 months, returned to where it started after 4 months had elapsed, continued up for the next 2 months, leveled off, and may be starting a downward cycle again.

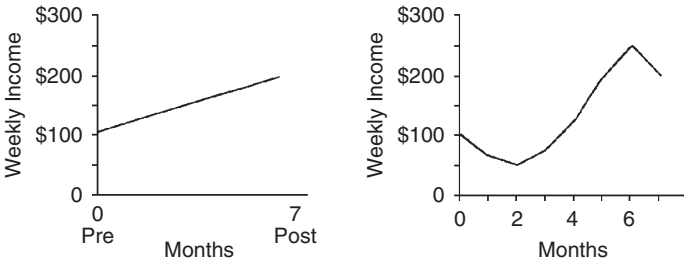
The same type of distortion is shown in Figure 8.6, which presents the results of a two-group design. The left panel shows the pretreatment and posttreatment measures of adjustment of a group of treated and untreated patients with schizophrenia over a 6-year period. The points are connected by straight lines, and the graph sug-

FIGURE 8.4



Body temperature differences between two groups as represented by actual differences in degrees (left) and as percentages of individuals in each group whose temperature exceeded 98.6° (normal threshold; right).

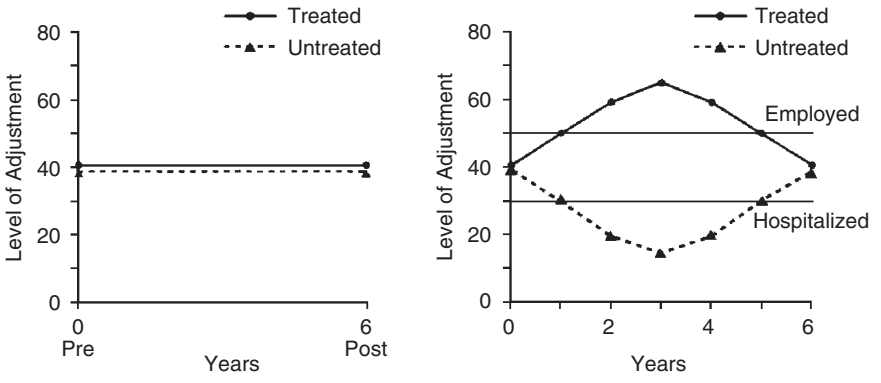
FIGURE 8.5



Changes in weekly income over a 7-month period. The graph on the left shows measures taken only at the start and end dates, which misleadingly suggests a steady rise in income. The graph on the right provides monthly data.

gests that treatment made no difference at all. In the right panel, annual measures are entered. The upper dotted line represents a level of adjustment characterized by an ability to be gainfully employed, to have meaningful family and social relations, and so on. The lower dotted line represents a level that is so impaired that hospitalization is required. From this graph, it can be seen that the treated group members improved enough in a year to enjoy some semblance of a "normal" life for a 4-year period, after which they slipped back to where they were at the start. The untreated group members went downhill and required 4 years of hospitalization, after which

FIGURE 8.6



Results of measures of adjustment of a group of treated and untreated patients with schizophrenia over a 6-year period. The left-hand panel suggests that treatment made no difference; the right-hand panel provides adjustment measures by year for a more accurate representation of the data.

they reverted to their original level. The two panels of this figure give two different trend lines representing two different courses of progress over the 6-year period of the study. The different courses of progress can be analyzed by *trend analysis*, a type of repeated measures ANOVA which can detect significant linear, quadratic, cubic, and other higher order curvilinear trends. Particularly when any appreciable time intervenes between pretreatment and posttreatment measures, you have reason to wonder why no intermediate markers were provided and whether the change over time can justifiably be regarded as linear.

VISUAL INSPECTION OF GRAPHS

Some studies use visual inspection of graphs in place of statistical tests. In psychological research, reliance on visual inspection of graphs is most commonly found in the presentation and interpretation of the results of single-case designs. Kazdin (2011) cautioned, "Studies of how individuals invoke the criteria for visual inspection have shown that judges, even when experts in the field, often disagree about particular data patterns and whether the effects were reliable" (p. 346).

When you encounter a study that uses nonstatistical visual inspection as the primary form of data analysis, you should ask whether other modes of analysis were available. If the data could have been analyzed in other ways, it would be reasonable to wonder why the other methods had not been used. If other methods were used but yielded different impressions, you should consider how the differences were reconciled.

Summary

We have examined how researchers should select the appropriate statistical test for nominal, ordinal, interval, and ratio data and determine whether their data meet the necessary assumptions. As a critical reader, this will be your first consideration when evaluating the statistical conclusions presented in a research report. Next, you will consider whether the researchers' data meet the assumptions of the statistical techniques they used. We called your attention to the need for control of Type I errors when there are multiple dependent variables and put forth reservations about the MANOVA–ANOVA approach. We introduced how important it is for you to consider whether issues relating to planned comparisons and post hoc tests were used properly. We highlighted the importance of critical examining the pretest equivalence of groups, the dangers of using raw change scores, and the use of transformed change scores (or analysis of covariance when there has been random assignment). Pointing out their importance for making inferences or drawing conclusions, we explicated the concepts of statistical significance, effect size, and power. You were alerted to the false implication researchers can make that the data are far more precise than they actually are. We illustrated distortions of scale, meaning, scale and meaning together, and false assumptions about linearity in graphic presentations.

Inferences, Conclusions, and the Research Report

9

A critical reader of research

- knows that statistical significance is not synonymous with practical or clinical significance;
- compares effect sizes in a study to other effects that relate to it, like other treatments of the same disorder or other relevant outcome variables;
- looks to see that the descriptions of methods and results are consistent across the different sections of the report; and
- feels comfortable that the report is providing a complete description of the research that meets reporting standards.

After completing their analyses, the researchers draw inferences from the data and generalize their findings to people, places, times, and variations in procedures not included in their study. They state their conclusions. Here, the researchers accept or reject the hypotheses, compare their findings with those of others, and place results within the context of theory and past research. Their final step is the production of a manuscript that reports their results. As a critical reader, you must not assume that what the researchers concluded was supported by what was found. Instead, you must look to make sure that the findings of the study and what the researchers claim about the findings are consistent with one another.

http://dx.doi.org/10.1037/0000052-010
Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper
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Drawing Inferences From the Data

When reading a research report, you should be able to distinguish circumspection, evenhandedness, and candor from overreaching, rushing to judgment, showing bias, ignoring or dismissing unfavorable results, and misinterpreting data. Painstaking research work can be vitiated by a warped discussion of results.

Examination of the data can tell you if the research question was answered. You can go back and hold the hypotheses and the data side by side to see whether the tests of the hypotheses are definitive and to determine whether those tests have been interpreted properly. You can judge whether the conclusions are justified from the data and whether the generalizations are warranted in the light of the sample, methods, and measures that were used in the study. You should be able to see how the research fits the theoretically derived propositions that have been cited and whether the findings conflict with prior research. The balance and the cogency of the arguments offered to explain any discrepancies can be appraised. You should judge whether there is substantiation for whatever inferences are drawn.

If the study has been well-conceived, well-executed, and correctly interpreted, and if the results come out as predicted, problems with interpretation and conclusions should be minimal. However, when the results are equivocal or run counter to expectations, the researchers' explanations and interpretations are liable to become strained. Researchers are understandably reluctant to relinquish cherished beliefs, even when faced with their own data. This becomes evident if you can identify conclusions that sometimes contradict the data in the study.

In other cases, predicted results are obtained, but the conclusions and generalizations go far beyond the data. Suppose that in a static-group comparison, the researchers make a conclusive generalized statement: "Women are better at abstract reasoning than are men." This conclusion comes from a study comparing 30 male college seniors with 30 female college seniors, all psychology majors in the college where the researchers teach. The means on a test of abstract reasoning were significantly different ($p < .05$). Yet, there is little justification for the general statement about men and women. Strictly speaking, what can be said is that, were the null hypothesis true, a difference of this size or larger would be expected by chance only one in 20 times if successive random samples were drawn from the same population and administered the same test. The significant test statistic would instill confidence that the group difference was not an accident of sampling, given the constraints mentioned.

The researchers, however, would be in error if they interpreted the result to indicate that similar mean differences would be expected 95% of the time if the study were repeated with samples that differed in meaningful ways. At best, the researchers could propose that similar differences might be obtained on similar measures given to other samples drawn from this college and perhaps even from other similar colleges. The reach would be very long indeed, but they might even speculate that the results might also hold for noncollege groups of this age, and perhaps might even be true for other groups of older men and women. If they made such claims, they would be

well-advised to suggest in their discussion why men and women who are college students are unlikely to show differences in abstract reasoning that would not also be evidenced in noncollege populations. Could age matter, with the difference emerging or disappearing over the course of development? In the absence of replications of the study on these other groups, however, these generalizations would have to remain at the level of speculations. On the basis of this study, a critical reader should recognize that a generalized conclusion about differences in abstract reasoning between men and women is not warranted and is overgeneralized.

In another study, the productivity of 30 workers, whose job is to assemble small electronic parts, is compared during mornings and afternoons. Productivity is counted on 25 consecutive work days. The finding is that morning productivity significantly exceeds afternoon productivity ($p < .05$). The researchers conclude, "People are more productive in the morning than they are in the afternoon." One cannot generalize from this sample of "assembly line workers" to "people" or from this kind of assembly work to other kinds of assembly work, or to productivity in other kinds of activities (e.g., clerical work). Perhaps productivity for some things is better in the morning than in the afternoon, and for other activities it is worse. The findings might not hold for other workplaces in other locations. The most that the researchers can conclude is that the morning productivity for the activity in this factory was greater than it would be expected to be by chance were the null hypotheses true. As a critical reader, you must examine with care statements that are offered as conclusions, as distinct from speculations. It is okay for researchers to speculate, to argue for why certain generalizations are plausible. But these speculations must be well-reasoned and clearly labeled as such.

We do not want to give you the impression that a study that rejects the null hypothesis is of little use. But it is one part of a much larger puzzle. Hypothesis tests that support a theory make the theory more credible, and researchers can say this. Replications of results will also extend the inferential reach of findings. And sometimes researchers can argue that some variations in samples and methods should make no difference, given the nature of the phenomenon (remember our example of Alzheimer's patients in New York and New Orleans from Chapter 3?).

Statistical Significance, Generality, and Practical or Clinical Significance

Researchers have been known to conflate the concept of statistical significance with generality and practical or clinical importance. Statistical significance does not relate to generalizability. A finding that is more highly statistically significant tells nothing about how robust a finding is across samples, settings, and measures. It tells how likely it is that the null hypothesis is true, or how likely it is that the null hypothesis will be rejected, in another sample of people like those sampled in this study.

Statistical significance also does not mean the finding is important; more statistically significant is not synonymous with more important; highly significant is not

the same as highly important. When much is made of the statistical significance of a difference or of a relationship, there is still the possibility that the difference is in fact trivial. This is especially the case when studies are conducted with very large samples. For example, a statistically significant 1-point difference in IQ scores using 5,000 men and 5,000 women in the sample would be of no practical importance. Using a standard deviation of 15 for IQ, such a finding would result in $t = 3.33$, $p < .001$.

The importance of a finding is a judgment that is based on sound theoretical and practical considerations and by an analysis of the size of the effect. You should look for guidance in the text in the form of the computation of effect sizes to help gauge the practical or clinical meaning of the obtained results, not just their statistical significance.

The opposite is also true. In studies with small samples, results that are not statistically significant but at first glance look promising may strike you as too readily dismissed. In such cases, you should speculate about what might have happened had the sample size been larger, and you should search the text to determine whether a power analysis was done.

Results that hover around statistical significance are difficult to interpret. If the researchers have prudently selected an alpha level of .05 but obtained $p = .09$, problems arise when this is reported as a "trend." The word *trend* would better be reserved for describing the changing course of data points over time (e.g., a gradual rise in prices). It is more circumspect to refer to p values that fall between .05 and .10 as "equivocal" or "indeterminate." Dar, Serlin, and Omer (1994) surveyed this problem in 30 years of psychotherapy research. They found that 46.3% of the studies discussed effects that were close to $p = .05$ but were not significant according to this standard. They reported that such findings were described as

> marginal, trends, tendencies, borderline significant, approaching significance, near significant and almost significant. . . . We found that p-values of up to .15 were considered a trend. . . . A t-test that resulted in a p-value of .24(!) was interpreted to indicate that one group showed more change although it did not reach statistical significance. (p. 77)

As a critical reader, you must recognize that a difference that reaches a highly significant level can still be practically trivial. The example of a 1-point IQ difference between 5,000 men and 5,000 women is a case in point.

Interpreting Effect Sizes

To help gauge practical or clinical significance, you need to interpret the effect size, the size of the standardized difference between the means (d index), the strength of the correlation (r index), or the ratio of the odds in the two groups. Cohen (1988) subjectively classified effect sizes that can be awarded into the categories of small ("smaller than medium but not so small as to be trivial"), medium ("likely to be visible to the naked eye of a careful observer"), and large ("the same distance above medium

as small was below it"; Cohen, 1992, p. 156). According to Cohen, in correlational terms, small, medium, and large *r*s would be .10, .30, and .50, respectively. For the difference between means of two independent groups, according to Cohen (1992), a *small d* index is .20, a medium effect is .50 (i.e., half of a standard deviation), and a large effect size is .80. In the example of a 1-point IQ difference between 5,000 men and 5,000 women, assuming $\sigma = 15$, a large effect size would be produced by a 12-point difference (.8 of a standard deviation), a medium one by 7.5 points, and a small one by 3 points. The 1-point difference would produce a trivial effect size ($d = .07$) in contrast to .20 for a small effect size. As demonstrated, however, the same 1-point difference is statistically significant at $p < .001$. Significance testing without taking the magnitude of effect into consideration would lead to different conclusions than had effect size been calculated.

But interpretation should not stop there. Cohen (1988) was careful to stress that his conventions were to be used "only when no better basis for estimating the ES [effect size] is available" (p. 25). Today, effect sizes are easy to calculate online (Wilson, 2015) and are contained more frequently in reports. Therefore, the researchers (and you) can find effect sizes that are more closely related to the effect sizes in the report to use as comparators, rather than Cohen's benchmarks. For example, if an evaluation of a new program to help veterans cope with PTSD-related anxiety was found to have an effect of $d = .30$ using Cohen's guide, we would label this effect small. However, what if the effect size of the treatment program currently used most frequently is $d = .15$? Compared with this treatment, the effect of the new treatment is quite large. If the *d* index associated with the current treatment is .50, then the effect of the new treatment is indeed small. Therefore, one way to interpret the effect of a treatment would be to ask whether it revealed smaller or larger effects than other treatments currently in use.

Other contrasting elements might be available as well (e.g., examining the same treatment but using a different outcome variable). For example, does the PTSD treatment have a larger or smaller effect on anxiety than it does on self-confidence? Often, contrasting effect sizes come from meta-analyses, which use effect sizes as their basic unit of data (see Chapter 10).

But other things can be considered when appraising the size of an effect. Cooper (2017) pointed out that effect sizes also need to be interpreted in relation to the methodology used in the primary research. For example, studies conducted in field settings can make it harder for researchers to minimize the variability in participants' responses within conditions, maybe being done in environments with varying ambient noise and other distractions, when compared with studies conducted in the lab. This might make the expected effect sizes smaller. Also, studies that provide an intervention can vary in the frequency and intensity of the treatment. For example, more frequent or more intense aerobic exercise would have a greater likelihood of showing a large effect than another intervention given with less intensity. What would the relative effects be if the two interventions were tested under comparable conditions? Or studies that use measurements of the outcome that are more reliable should produce larger effects than studies that use less reliable measures. Therefore, Cooper wrote, "research

design differences must be considered when drawing a conclusion about the relative size of effects; more sensitive research designs and measures with less random error can be expected to reveal larger effect sizes, all else being equal" (pp. 274–275).

The moral of the story for the critical reader is that almost any effect size can be made to look large or small depending on the choice of the contrasting effect sizes. Your job is to decide whether these contrasts have been chosen well—that they were not meant to give the impression that the effect is too large (or too small)—and allow a sound judgment of the theoretical or practical significance of the effect.

The Research Report

In recent years, efforts have been made to improve the reporting of research in journal articles. It used to be the case that limitations on the number of pages that could be printed in a journal constrained the amount of information that any article could contain. Today, there are journals that appear only online. Printed journals use online supplemental files to provide additional information about the methods and results of the studies they publish. So, limitations on the information researchers can provide about what they did have largely disappeared.

The loosening of length constraints makes it easier for you to critically examine studies and for other researchers to conduct replications. If replications find different results, it makes it easier to discover why this might be the case.

Also, it is not unusual, and it is required in some cases, for researchers to place the raw data they collected into data repositories open to the public (e.g., *Archives of Scientific Psychology*). This way, other researchers can see the actual collected data. This allows you the opportunity to examine the data and determine whether the report is consistent with it and whether the analyses were conducted correctly. Although such close examination is rare, if questions arise about the data or its analysis, it is nice to know this option is becoming increasingly available.

Along with the loosening of constraints on the details in reports, there is a growing interest in the details of what should be contained in a research report. For articles in psychology and other social sciences, the Journal Article Reporting Standards (JARS) have been developed (American Psychological Association Publications and Communication Board, 2008; see also American Psychological Association, 2010).

The rationale for the inclusion of each entry in the JARS can be found elsewhere (Cooper, 2011). To give you an idea of the kinds of things to look for, we present examples of some red flags to look for in manuscripts. In all these instances, you are required to move back and forth in the manuscript to ensure that what the researchers write is consistent between the different sections.

ABSTRACT TO ARTICLE

An abbreviated summary or abstract of the study is presented at the beginning of the journal article or research report. You should not ignore what is written here

because the Abstract is all that many busy readers will review. Occasionally, the resemblance between the Abstract and the study appears to be largely coincidental. The Abstract reads like the researchers' wishes, and the rest of the article like the reality. Sometimes this will happen unintentionally because the Abstract will have a strict word limit and the authors leave out qualifications to their findings. However, the word limit is no excuse for presenting a false, or overly rosy, summary of what the study found.

INTRODUCTION TO DISCUSSION

On reading the Introduction, you may discover no hypotheses are explicitly stated. Sometimes, for example, when an article reports the results of a study evaluating an intervention meant to reduce anxiety in people with PTSD, the hypotheses will be self-evident. At other times you will not be sure what the hypotheses are, assuming the researchers had any in mind when they started the research. In the Discussion, however, the researchers are pleased to report that the results support their hypothesis. The rationale for the originally nonexistent hypotheses is now given and is illuminated by the results. You should assume that these were not prespecified hypotheses but were post hoc—they were adopted after the results were seen. The Discussion should clearly label these as such and make a call for further study.

METHOD TO DISCUSSION

The researchers may report that, during the research, the sampling plan was altered or the procedures had to be modified for the study to be completed. The changes, which clearly compromised the internal validity of the study (e.g., differential attrition, a study starting out with random assignment to conditions but ending up with many subjects choosing their own condition), are not mentioned as limitations when the researchers discuss the results.

Related to this error is a slippage in the use of causal language. For example, the Method section may clearly report that a static-group comparison was conducted. But, when discussing the findings, the authors slip into the use of causal language, using words such as *produced*, *influenced*, or *brought about* to describe the results.

RESULTS TO DISCUSSION

The same result may be referred to several times in a research report. Be careful to check that it is described with similar language each time. Suppose an inconclusive p value of $p = .13$ is obtained in the data analysis. The researchers call it a trend in the Results section. In the Discussion, the word *trend* is dropped and the finding is now referred to as probable. In the Abstract or summary of conclusions at the end of the discussion, the trend has metamorphosed into an established finding, with no qualification. This kind of progressive aggrandizement may not be noticed if you are

a hurried reader who skims the Results section or goes directly to the Discussion or to the Abstract.

METHOD TO RESULTS TO DISCUSSION

Suppose a study used three measures of the dependent variable, all mentioned in the Method section. Only one of them yields results in the predicted direction. There is no advance reason given for you to believe that this measure was more important or more valid than the others. Then, the researchers base the entire discussion and conclusions on the one measure that came out as predicted, while giving short shrift to the other two. This is a post hoc interpretation and should be treated as such. Could the same deficiencies in the two nonsignificant measures noted by the researchers also be leveled against the one that produced significance (e.g., its reliability was hardly different from the measures that were dismissed as unreliable)? Could the one significant finding be due to chance, given that multiple tests for statistical differences were conducted?

Here is a similar example. Researchers use four independent variables and 10 dependent variables, and obtain four statistically significant results out of 40 comparisons tested. In the discussion, the researchers dismiss the 36 that do not come out as expected and make much ado about the four that did. This is something called data mining, fishing for significance, or *p*-hacking. It capitalizes on chance findings.

The matter is even more troubling if measures are mentioned in the Method section but no mention is made of some of them in Results and Discussion. This is called *selective reporting* or *data censoring*, and it is a very serious matter. If results are selectively reported and nonsignificant results are omitted, it is impossible for you to tell how seriously to take the results that are reported and suggests that other researchers trying to replicate the findings may be unlikely to do so.

Also, remember that a null hypothesis cannot be "accepted"; research can only "fail to reject" the null hypothesis. For example, finding no significant difference between study groups A and B (the differences are not large enough to reject the null hypothesis), the researchers conclude that "there is no difference between A and B." A kindred statement, such as, "There is no relationship between A and B," is similarly unwarranted, as affirmative statements claiming "proof" of the null hypothesis are impossible.

RESULTS TO DISCUSSION

Suppose that your reading of the Results section reveals that the data seriously violate the assumptions of the statistical technique used in the study. The researchers proceed without considering any alternatives, arguing that the technique is "robust." Thereafter, the issue is completely disregarded, and there are no reservations expressed in the Discussion section. For a critical reader, this should cast serious doubt on the researchers' interpretation of the results.

Summary

In this chapter, we presented some mistakes researchers can make in interpreting and discussing their results. Special attention was paid to what it means for a result to be found statistically significant and how this does not speak to the generality or practical significance of the findings. We also provided guidance on how to interpret the size of an effect. Finally, we highlighted the need for complete reporting of a study's rationale, method, and results. An example of one set of reporting standards, the JARS, was presented.

Research Syntheses and Meta-Analysis

10

A critical reader of research

- recognizes that syntheses of past research need to meet the same standards of scientific rigor as do new data collections;
- can identify the stages in conducting a research synthesis—problem formulation, literature search, coding of studies by judges, evaluating of study methods, meta-analysis, and interpreting and presenting the results;
- knows that meta-analysts use individual studies' characteristics and effect sizes as data, and effect sizes are averaged to find overall conclusions, as well as used to search for influences on the individual studies' results; and
- recognizes that meta-analysts face many of the same problems faced by collectors of new data (e.g., coder reliability, missing data).

We have now examined the research process from beginning to end, from the formulation of a problem and hypotheses to the statistical analysis of the data collected in a study and its interpretation. We have considered many of the ways that researchers can go wrong. They can define their problem ambiguously. They can choose poor manipulations and measurements. They can use the wrong research design given the question

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they wish to answer. They can leave confounding variables uncontrolled. And they can apply the wrong statistics.

Now we turn from research meant to collect new data to research that summarizes the existing studies on a topic. Rather than do a new study, researchers who conduct *research syntheses* (also called *research reviews* or *systematic reviews*) want to answer the question "What does the research literature already say about the problem that interests us?" In fact, all researchers, including students doing their master's thesis or doctoral dissertation, ask this question before they begin a new investigation. They do so because they do not want to pursue a question others have already answered (unless replication is their goal), nor do they want to repeat mistakes in their study that others have already made and shown them how to avoid. Others do it to bring order to a research area, to identify the most fruitful areas for future research, and the save readers the chore of summarizing and integrating the research themselves. This type of research can be an important contribution to the development of science.

In this chapter, we describe how a rigorous research synthesis is conducted and what you need to look for if the results of the synthesis are to be believed. We focus special attention on *meta-analysis*, a set of procedures for summarizing the quantitative results from multiple studies. We highlight the criteria you will use to evaluate the quality of research syntheses, in general, and meta-analyses in particular. When you read a research synthesis, you need to think about what distinguishes a good synthesis from a bad one and how trustworthy is the summary.

The Stages of Research Synthesis

Just as a new data collection follows the stages of problem formulation, design of research, choice of measurements, analysis of data, and interpretation of results, research synthesis follows a set of sequenced stages that parallel new research, and these are meant to ensure that results meet scientific standards of rigor and systematicity. Cooper (2017) divided the process of research synthesis into seven stages:

- Stage 1: Formulating the problem
- Stage 2: Searching the literature
- Stage 3: Gathering information from studies
- Stage 4: Evaluating the quality of studies
- Stage 5: Analyzing and integrating the outcomes of studies
- Stage 6: Interpreting the evidence
- Stage 7: Presenting the results

Here, we briefly discuss each, paying more attention to those stages that use the procedures that are most unique to summarizing the results of a group of studies.

STAGE 1: FORMULATING THE PROBLEM

The first task in any scientific investigation, whether a new data collection or research synthesis, is to formulate the problem. At this stage, you, the critical reader, ask,

"What are the variables or interventions the researchers wanted to study?" and "What operational definitions have the researchers claimed are measurable expressions of these variables?" In answering these questions, you make a judgment about whether the research synthesis properly fits the concepts and operations together. Typically, you will do this by examining what research evidence the researchers claim was relevant (and irrelevant) to the problem or hypothesis of interest, just as you would do for new data collections. The challenge for you is that in a research synthesis you will likely find many more manipulations and measures claimed to represent the independent and dependent variables than you will find in any individual study. For example, suppose you are examining a synthesis in which the synthesists summarize the research on whether "class size influences academic achievement." The studies include many different measures of achievement, including class grades, cumulative grade averages, standardized tests, and so on. They also include achievement in many different academic domains (e.g., math, English, social studies). But you also find that the researchers included a study of grades in a physical education class. Does this study meet the research synthesis's (and your) definition of "academic achievement" or is physical education not "academic," even though it happens in school? Should the study have been included in the synthesis? Or should the conceptual definition guiding the synthesis have been recast to be more inclusive than academic achievement? As another example, suppose you come across a research synthesis of the effect of violent video games on aggression. Would you consider the game *Super Mario Brothers* to qualify as an operationalization of "violent video games," or is this game too tame? What games do the researchers say qualify as "violent"? Did all the games included in the synthesis meet this qualification? Just as in new data collections, as a critical reader of a research synthesis, you must consider whether the conceptual and operational definitions are carefully defined and fit together well.

STAGE 2: SEARCHING THE LITERATURE

Searching the literature and screening studies for their relevance is how research synthesists define their sampling frame, but in most cases the synthesists want to find the entire population of "all studies relevant to the topic." They cannot find them all, and different sources of studies will reveal different research.

As a critical reader, you first check to see which reference databases the synthesists consulted (e.g., PsycINFO for psychology, Medline for medicine and health, ERIC for education). For example, you would expect that a synthesis about violent video games would use all these resources to find studies, because studies on violence can be found in journals in any of these disciplines. Syntheses about class size and achievement might pass on using Medline but add EconLit (for economics literature). Syntheses of PTSD and anxiety research might pass on ERIC.

Synthesists would also carefully choose the search terms they use in the database to find content related to the topic. For example, starting with "video games" AND "aggression" seems natural, as does "posttraumatic stress disorder" AND

"anxiety." You should assess how well the search terms "capture" the relevant research.

As a critical reader, you should also assess how the synthesists supplemented their reference database search results by examining references mentioned in the relevant articles they find (called a *backward search*), and by searching for articles that mentioned important articles on the topic (called a *citation search* or *forward search*) You should evaluate whether the synthesists made a good effort to find all the past research, or at least an unbiased sample of all the relevant literature.

STAGE 3: GATHERING INFORMATION FROM STUDIES

In research synthesis, data collection involves the information about each study the researchers have decided is relevant to their problem. This information will include not only characteristics of the studies that are relevant to the theoretical or practical questions (i.e., the nature of the independent and dependent variables), but also about how the study was conducted, when and where it was conducted, its research design, its measurements, and its statistical results. This stage also requires that the synthesists train coders so that they can gather information from studies in a reliable and consistent way, just like new data collectors must train coders of observations, videos, or written documents.

As a critical reader, you need to examine the methods the synthesists used to extract information from each study, much like new data collectors using judges (see Chapter 7). Were the definitions and instructions to coders clear and thorough? Were coders well-trained? Did the synthesists present evidence about the reliability of coders?

STAGE 4: EVALUATING THE QUALITY OF STUDIES

After the studies have been gathered, the synthesists should make critical judgments about their *quality*, or how well the study methods allow the inferences the topic calls for. Each study is examined to determine whether it used valid and reliable measures and appropriate research designs and statistical techniques. In other words, the researchers act as critical readers. If the study is found wanting, the synthesists should not use it or give it reduced credibility. For example, if a study of the effectiveness of a new PTSD treatment assigned the first 50 volunteers to the new treatment and the second 50 to the old treatment, the synthesists might discard this study, or they should say why they feel it may still be trustworthy even though the two groups are likely different in more ways than having received the treatment. Sometimes all the research on a topic has flaws. Then, the synthesists may include flawed studies but be careful to temper their conclusions accordingly. What if all or most studies involved self-selection into conditions? The synthesis could still be done, but the careful reader looks for the disclaimer that softens the inferences the synthesists draw on the basis of the design flaw (more on this in the following sections).

STAGE 5: ANALYZING AND INTEGRATING THE OUTCOMES OF STUDIES

During the data analysis stage, the separate research reports collected by the synthesists are integrated into a unified statement about the results of the research. Next, we outline some methods for combining the results of separate studies, or methods of meta-analysis.

STAGE 6: INTERPRETING THE EVIDENCE

After evaluating the studies, the synthesists interpret the cumulative evidence. Just as in reading new data collections, while reading the conclusions of research syntheses, you must ask, "What conclusions are warranted by the set of studies?" and "Are the conclusions drawn by the researchers consistent with their data?" You will also need to consider the generalizability of the findings over different types of units, treatments, outcomes, and situations. You must decide whether the interpretation is consistent with the cumulative evidence from the studies. Assessing the strength and shortcomings of conclusions in a synthesis is part of your job as a critical reader.

STAGE 7: PRESENTING THE RESULTS

Finally, the synthesists prepare their report. Today, most research syntheses, especially meta-analyses, follow a prescription for reporting results that looks a lot like how collections of new data are presented. Guidelines for doing this can be found in numerous places (e.g., American Psychological Association Publications and Communications Board, 2008; Equator Network, 2016).

The Elements of Meta-Analysis

Suppose a team of researchers is interested in whether a new college counseling workshop that includes training in study skills, time management, and avoiding the temptations of life away from home can help freshman complete their degrees (4 years later). After a comprehensive search of the literature, the researchers find 18 studies that present data from separate evaluations of the workshops. Of these, 13 evaluations reported no significant differences in completing college between the workshop and no-workshop groups. Two studies reported a chi-square statistic that was significant at $p < .05$, and two more studies report a statistic at $p < .01$. One study found a significant negative effect ($p < .05$); freshmen were less likely in this study to finish college if they took the workshops. Can the synthesists reject the null hypothesis that the workshops had no positive effect?

There are several ways the research synthesists could answer this question. First, the synthesists could examine the 18 reports, carefully scrutinize those studies that

present results that are not supportive of the workshop, discard those studies because of methodological flaws, and present the four supportive studies as the only ones to be believed. If you read such a synthesis, it is an understatement to say you should view its results with extreme skepticism. It is more of an advocacy piece (and a bad one at that) than a scientific work.

AVERAGING EFFECT SIZES AND MEASURING DISPERSION

The principle results that meta-analyses report are average effect sizes and measures of effect size dispersion or variance. In the workshops example, the meta-analysists would calculate the odds ratio associated with each study, weight each by its sample size (more precisely, the inverse of its variance), and average the weighted effect size across studies. In fact, the original study researchers might have provided tables with the raw number of graduates and nongraduates in each workshop condition. These could be added together directly, giving a raw score estimate of the effects of the workshops on graduation. Confidence intervals would then be calculated to estimate the interval of which 95% of findings should fall within. The confidence interval could then be used to test the null hypothesis that the difference between two conditions is zero (if zero does not fall within the confidence interval).

TESTS OF INFLUENCES ON EFFECTS

Another advantage of performing a meta-analysis is that it allows synthesists to formally test hypotheses about why the outcomes of studies differ, about why the effect sizes vary from one study to another. For example, suppose the meta-analysis in the workshop example finds more variation in odds ratios from study to study than expected by chance (called a *homogeneity analysis*). Synthesists also note that eight of the studies used only participants who were first-generation college students, whereas 10 studies did not restrict participants in this way. They could then test whether the average odds ratios in these two groups of studies differed from one another (put more formally, whether this characteristic of studies is associated with variance in effect sizes). Researchers could also test, for example, whether those workshops with a component that encouraged responsible drinking were more effective than those without.

If this reminds you of an *analysis of variance*, it should. Analyses looking for influences on effect sizes in meta-analysis are analogous to analyses of variance using new data. As a critical reader, you should look for whether the meta-analysis includes analyses of well-chosen possible influences on the study outcomes.

In sum, a typical meta-analysis should contain (a) estimates of average effect sizes with confidence intervals, (b) homogeneity analyses to assess whether the variance in effect sizes is greater than would be expected by sampling error, and (c) moderator analyses that examine study features that might influence study outcomes. There are other procedures and statistics that you will find reported in meta-analyses (Borenstein, 2009) but these are the basic elements. If you do not find them in the report, something may be amiss.

Summary

There are many other important decision points in conducting a research synthesis that you need to consider as you read a report. Cooper (2007, 2017) listed 20 questions (see Exhibit 10.1) that critical readers of research syntheses might ask that relate to the validity of conclusions. Each question was posed so that an affirmative answer by you would mean you can place more confidence in the conclusions of the synthesis. Although the 20 questions are not an exhaustive list of those that might be asked, most of the concerns you would have about the validity of the synthesis's conclusions are addressed in the questions.

EXHIBIT 10.1

A Checklist of Questions Concerning the Validity of Research Synthesis Conclusions

Step 1: Formulating the problem

1. Are the variables of interest given clear conceptual definitions?
2. Do the operations that empirically define each variable of interest correspond to the variable's conceptual definition?
3. Is the problem stated so that the research designs and evidence needed to address it can be specified clearly?
4. Is the problem placed in a meaningful theoretical, historical, and/or practical context?

Step 2: Searching the literature

5. Were proper and exhaustive terms used in searches and queries of reference databases and research registries?
6. Were complementary searching strategies used to find relevant studies?

Step 3: Gathering information from studies

7. Were procedures used to ensure the unbiased and reliable (a) application of criteria to determine the substantive relevance of studies and (b) retrieval of information from study reports?

Step 4: Evaluating the quality of studies

8. If studies were excluded from the synthesis because of design and implementation considerations, were these considerations (a) explicitly and operationally defined and (b) consistently applied to all studies?
9. Were studies categorized so that important distinctions could be made among them regarding their research design and implementation?

Step 5: Analyzing and integrating the outcomes of studies

10. Was an appropriate method used to combine and compare results across studies?
11. If a meta-analysis was performed, was an appropriate effect size metric used?
12. If a meta-analysis was performed, (a) were average effect sizes and confidence intervals reported and (b) was an appropriate model used to estimate the independent effects and the error in effect sizes?

(*continues*)

EXHIBIT 10.1 (*Continued*)

A Checklist of Questions Concerning the Validity of Research Synthesis Conclusions

13. If a meta-analysis was performed, was the homogeneity of effect sizes tested?
14. Were (a) study design and implementation features (as suggested by Question 8) along with (b) other critical features of studies, including historical, theoretical, and practical variables (as suggested by Question 4) tested as potential moderators of study outcomes?

Step 6: Interpreting the evidence

15. Were analyses carried out that tested whether results were sensitive to statistical assumptions and, if so, were these analyses used to help interpret the evidence?
16. Did the researchers (a) discuss the extent of missing data in the evidence base and (b) examine its potential impact on the findings of the synthesis?
17. Did the researchers discuss the generality and limitations of the synthesis's findings?
18. Did the researchers make the appropriate distinction between study-generated and synthesis-generated evidence when interpreting the synthesis's results?
19. If a meta-analysis was performed, did the researchers (a) contrast the magnitude of effects with other related effect sizes and/or (b) present a practical interpretation of the significance of the effects?

Step 7: Presenting the results

20. Were the procedures and results of the research synthesis clearly and completely documented?

Note. Adapted from *Evaluating and Interpreting Research Syntheses in Adult Learning and Literacy* (pp. 65–66), by H. Cooper, 2007, Cambridge, MA: National Center for the Study of Adult Learning and Literacy. Retrieved from http://www.worlded.org/WEIInternet/inc/common/_download_pub.cfm?id=16699&lid=3
In the public domain.

11 Research Ethics

A critical reader of research

- adheres to the ethical principles that govern the discipline in which he or she works;
- looks for an indication that if human subjects were used in a study they were treated in compliance with their Institutional Review Board approval; and
- keeps an eye out for indication that the study's methods, data, and/or analyses might have been manipulated in a way that biased the conclusions, or that the data and writing might not be original.

At the very beginning of the planning process, the researchers must decide whether the benefits of the study outweigh the risks. The reader of a research report is not able to second-guess a decision that was made to go forward with the research long before any of the results were known. However, you can be attuned to questionable ethical decisions about who was selected to participate, how they were assigned to treatments, the kinds of inducements that they were offered, whether they were properly informed about the study and freely gave their consent, and whether any harm was done to them that could have been avoided or ameliorated.

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Ethical standards apply not only to the treatment of participants but at every point throughout the research process, including the choice of criteria measures and how they were applied, how the data were analyzed, how the data were reported, what generalizations were made, and what conclusions and inferences were drawn. This chapter deals with some of the key ethical issues in research.

Adherence to ethical standards in conducting and reporting research has attracted attention and debate for well over a half-century. There are two major aspects to the matter. One involves the way individuals are treated before, during, and following their participation in the research; the other has to do with the scientific integrity of the research and the way that the work product is presented.

Protection of Human Participants

The Nuremberg Code (Annis & Grodin, 1992), the Declaration of Helsinki (Schmidt & Frewer, 2007), the Belmont Report (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1978), and the American Psychological Association's (APA, 2017) *Ethical Principles of Psychologists and Code of Conduct* (hereinafter, Ethics Code) all provide principles and standards for the ethical treatment of participants. Researchers are expected to master these principles and standards and to weigh them heavily when deciding that the risk–benefit ratio of a study justifies going forward with the planned research. And they cannot make this decision alone. All research conducted at major research facilities must undergo external review by an institutional review board (IRB). As a critical reader, you should have a good understanding of the APA's Ethics Code; they relate to many matters (e.g., treatment of clients in therapy, teaching), but there is an entire section on research. APA publishing requires researchers to certify that they have followed the Ethics Code when they submit their article and are encouraged to mention this certification again when they describe their procedures or in the Author Note. You should look for this in the journal article.

As a critical reader, you should have reasonable assurance from the researchers that people participated in the research willingly and without coercion of any kind. While obtaining approval from an IRB should ensure this, you can look for assurances that conscientious researchers will provide for readers. Therefore, the researchers might mention that (a) participants took part only after being fully informed about their role in the study and any risks that might be involved, (b) inducements to participate must not be so excessive as to be seen as undue pressure, and (c) signed informed consent was obtained in all except minimal-risk studies in which the identity of individual participants is not disclosed.

VULNERABLE POPULATIONS

Did the researchers heed the special precautions that are recommended in the ethical code for dealing with vulnerable participants (e.g., children, mental patients, prisoners,

or others who may be involuntarily institutionalized)? When children are participants, signed consent is required of parents or guardians, as well as the assent of participants who are under their jurisdiction. When participants want to withdraw from a study, they must not be pressured to continue against their wishes. The authors of a research report have the obligation to describe the methods that were used to comply with the ethical principle, with particular emphasis on any special safeguards that they used. Also, researchers who work with animals have unique obligations to protect the animals' welfare. Careful researchers use a few sentences in their report to inform readers about how animals were handled.

USE OF DECEPTION IN RESEARCH

Some degree of *deception* may be a necessary part of the research design. When deception is involved, diligent researchers tell readers when the debriefing was done and whether the participants were permitted to leave with false beliefs about themselves. For example, in some experiments participants may be deceived into having failure experiences (e.g., they are given a puzzle to solve that is unsolvable), some are falsely led to believe that they lack an ability (e.g., they take a bogus test), and some are given the impression that they have done something reprehensible that reveals a moral deficiency (e.g., they refuse a request for help from a research confederate). These kinds of deceptions may be necessary for a study that has great potential benefit, but the IRB will be particularly concerned that exceptional care is taken to debrief the participants as soon as possible.

ADVERSE REACTIONS

Once again, decisions regarding the justification for the study have been made before the research begins and without knowledge of the outcome or how the participants would react. However, critical readers are interested in what the researchers did to determine whether participants had any untoward reactions and if so, what was done about this. For example, did some veterans with posttraumatic stress disorder experience extreme anxiety caused simply by answering questions that had them focus on stressors in their lives?

Details about the handling of any problems with participants are reported by ethical researchers, along with ways that the procedures may have been modified to prevent further incidents. If no problems arose, a negative report is expected stating, in effect, that there were no signs of any adverse reactions as determined either by observations during the experiment or by inquiry in the postexperimental debriefing session. Enough information should be provided so that you can judge compliance with this ethical principle.

Related, the researchers must respect the individual's freedom to decline to participate in or to withdraw from the research at any time (APA Ethics Code, Section 8.2(2), p. 10). The number of times this happens should be revealed in the Results section when the researchers discuss attrition. A good report of a study will tell you if participants who began it did not complete it and the reason for this attrition.

Conducting and Reporting Research

There is more to conducting ethical research than treating participants well. Other deviations from accepted ethical standards include how the research is conducted (e.g., making rules that allow certain participants to be dropped after their results are known) and how it is reported (e.g., conducting statistical analyses that come out differently than the researchers would like, but which are never mentioned). These ethically questionable behaviors occur after the study has gained IRB approval. Such practices are not easy to detect, but things do appear in manuscripts that will alert you to their possibility.

Gross ethical violations, such as the *fabrication* or *falsification* of data, are usually uncovered by complaints of associates who become aware of them and by scholars who become suspicious when reviewing data that contain unexplained inconsistencies or when examining data that look "too good to be true." If you believe such a violation has occurred, ethics committees of professional organizations are in place to deal with such matters when allegations are made. The Office of Research Integrity of the Department of Health and Human Services (https://ori.hhs.gov/) also receives and investigates complaints. Summaries and outcomes of these investigations are reported regularly on the Office of Research Integrity website. If ethical violations were uncovered by others, you will see that the article you are examining has been retracted, and a notice will be printed to this effect with the paper online. Obviously, this article is not to be believed. If a less severe error has been spotted, the article will appear with a statement of correction or *erratum*.

Other Ways to Manipulate Results

Researchers can also mislead readers by introducing other ways of treating data that are more in the service of the hypothesized outcome than they are in the service of the truth. Even when there is no hard evidence of outright falsification or fabrication of data, your doubts can be raised by procedures that give the impression of being deliberate attempts to influence results in a particular direction. It is no easy matter to differentiate between deliberate acts and acts that are accidental, inadvertent, or simply the result of an unwise decision. You, the critical reader, do not possess all the facts and do not have to make that differentiation. It is worthwhile, however, to be able to recognize some of the most common paths to biasing results.

EXPERIMENTAL CONFOUNDS

One way researchers can manipulate results is to not hold things constant when "keeping other things equal" is central to the design of the study. For example, researchers studying reaction times may always schedule subjects in the condition they expect to react faster earlier in the day, in a fresh, cool, and comfortable environment.

Subjects hypothesized to react slower are scheduled in the late afternoon in hot and uncomfortable surroundings. In this instance, the different environmental conditions and times were not randomly assigned to experimental conditions. Subjects may have been randomly chosen to show up early or late but the experimental conditions are confounded with the time of day. This type of confound will be difficult for you to discern because the (unscrupulous) researchers are unlikely to report it. The result, however, may be difficult for others to replicate, and the confound may be uncovered when the researchers are asked to share their data.

TAMPERING WITH RANDOM ASSIGNMENT

Tampering with the random assignment of subjects so that the hypothesis is favored is another way researchers can bias the study and mislead the reader. Researchers may succumb to the temptation to place a few individuals in the experimental or control group even though the random assignment procedure has them designated for the control group. The researchers may even try to justify this on the grounds of clinical expediency or clinical ethics. However, the process of picking conditions on the basis of participant characteristics defeats the purpose of random assignment. Citing clinical ethics as justification for placing especially deserving participants in the experimental treatment group is irrelevant, for if the researchers had been certain about the effects of the experimental treatment, there would have been no point in doing the study in the first place.

POST HOC DETERMINATION OF CUTOFF SCORES

The decision about cutoff scores should be made in advance of the data analysis. If they are decided on when the data are in hand, you may wonder whether the grouping was made to suit the results. Might a grouping with three categories have led to a null result so the researchers used a median split which led to a statistically significant difference? If gerrymandering placed some of the "best" participants in the favored group and some of the "worst" into the not favored group, the results will be misleading. As a critical reader, you should look for a statement by the researchers that the cutoff scores were determined before data analysis and were based on theoretical or other meaningful considerations.

BIAS IN SELECTION OF DEPENDENT VARIABLES

Dependent measure selection can reflect a bias. For example, in a cross-cultural study of IQ or learning ability, researchers who are out to prove a point might select highly verbal tests, printed in English, that will clearly disadvantage the group that is not as fluent in this language. You must always examine the dependent measures to determine whether they are valid, reliable, and equally appropriate for all participants.

ALTERING SAMPLING PROCEDURES

The procedures of some studies call for a fixed number of participants, trials, or sessions. This is decided on rational grounds (e.g., power analysis) in advance. During the study, however, the researchers run a statistical analysis, find significance, decide to stop running subjects or trials within the subject, and "quit while ahead." This will, of course, give the predicted results, but they will be misleading. If the series is not shaping up as hoped, unethical researchers may extend it until it is going in the predicted direction. Taking advantage of normal fluctuations of chance events is a gambling strategy, not an ethical research practice. Although researchers who would do this are unlikely to say so in their report, you should be on the lookout for odd numbers of participants (e.g., very small or very large, decidedly different numbers in different conditions) or for numbers of trials that vary from participant to participant.

CHANGING STATISTICAL ANALYSIS

Short of outright falsification of data, you may see signs of shifts of mode of analysis in different parts of the article, even though the type of data has not changed. For example, the researchers might not explain why one test of group differences is made by a parametric test and another by a nonparametric test, and there seems to be no statistical rationale evident. The least you can expect from the researchers is a statement explaining why different analyses were used. If multiple types of analysis were done, all results also should be reported. If perfectly good, normally distributed scores were obtained in the study, why did the researchers shift to using nonparametric statistics? Do you get the impression that they were fishing for the best result instead of using the most appropriate and most powerful statistics for data of this type?

GRAPHIC DISTORTIONS

Graphic distortions (see Chapter 8) when introduced with the intent of misleading readers, are of considerable ethical concern. Pictorial illustrations and representations can make an even greater impression than technical text materials and numerical tables. Everyone who has studied psychology remembers Harlow's picture of a little, sad-eyed, orphaned monkey snuggling up to its "cloth mother" and rejecting its "wire mother." This picture, backed up by data, was not a distortion, but no matter what other data were included, or in what form they were presented, this vivid pictorial representation dominated and shaped people's impressions of the results.

Pictorial representations that illuminate solid data and bring them to life are an important feature of the presentation. They provide excellent highlights when they emphasize a valid point. However, they can be deceptive when isolated and when nonrepresentative cases that run counter to the overall results are selected as illustrations. Graphs, too, although nowhere near as powerful as expressive pictures, can create more of an impression than dense tables of numbers. As a critical reader, you should try to determine whether the graphs and the illustrations are accurate representations and not intentionally selected to exaggerate or distort a result.

USE OF ANECDOTAL EVIDENCE

Researchers can use postexperimental inquiries to find out how the participants experienced the research and how the experimental manipulation, if any, appeared to them. Because they are subjective qualitative data, it is easy for researchers to give a highly selective and biased reporting of this supplementary information. When the quantitative results are not as favorable as anticipated, and biased supplementary anecdotal material is used to nullify or to replace them, your confidence in the scientific integrity of the authors should be strained.

Plagiarism

Occasionally you will question the language used to describe research. You may encounter something that sounds familiar or may even recognize the original source (particularly if you happen to be the original author!). Acts of plagiarism can range from stealing an entire work, by simply changing the name of the author, to paraphrasing someone's work and not attributing the ideas to the original written document (Turnitin, 2012). If you think you have discovered an instance of plagiarism, you can use a plagiarism detection service, such as iThenticate (http://www.ithenticate.com/) or HelioBLAST (http://helioblast.heliotext.com/), to check it out. These programs compare the target paper with millions of other documents found online and in scholarly reference databases.

Duplicate Publication

You may see a publication that also looks suspicious not because the words are the same but because the data are familiar. As a critical reader, you should attempt to find the original publication. If you discover that the same data have been published twice, this is called *duplicate publication*. It distorts the scientific record, perhaps making the results look like they have been replicated when they have not.

There are instances in which reproducing already-published data can be legitimate (e.g., a study involves a follow-up and the original data are described along with the new data). In these cases, the researchers are obligated to let readers know where the original report can be found. If they do not include the reference to earlier work, this is an ethical violation.

Summary

In this chapter, we emphasized the importance of adherence to ethical standards when conducting and reporting research. Ethical violations may take the form of mistreating participants, not controlling for biases in research design, fabricating

or falsifying data, distorting or exaggerating data analysis, plagiarizing the work of another, or presenting results that have already been published as original.

These and other related ethical concerns are elaborated in detail in the APA (2017) Ethics Code, and a good critical reader should be well-versed in the principles it contains. These are the provisions currently enforceable by APA regarding its members (you can get tossed out of APA if you are found to violate them), and they are also required of authors who are not APA members but wish to publish in APA journals. You may refer to Cooper (2017) or Meltzoff (2011) for more in-depth discussion of ethics in research and publication. These materials can serve as background for your critical examination of these aspects of the research process.

CRITICAL THINKING USING PRACTICE ARTICLES

II

Introduction: Part II

Over time, the reporting of research in science has become highly stylized. The basic structure of research reports is very consistent across science disciplines, although there are some variations from one discipline to another. As we reviewed in Chapter 9, in psychology and many other social sciences the standard (and efficient) format consists of the Introduction (including a statement of the problem and hypotheses and a review of closely related literature), and the sections for Method (including a presentation of the procedures and the instruments), Results, Discussion, and References. The research report is usually preceded by the Abstract.

This familiar format takes readers through the investigation in an orderly, sequential fashion. Readers know where to find essential details. Dressed in its formal regalia, the research report is ready to be taken seriously. Unwary readers lower their guard because form appears to legitimize content. Many social and behavioral science journals use the *Publication Manual of the American Psychological Association* (APA, 2010) as a requirement for manuscript format and style. This manual contains a set of recommendations regarding the information that should be contained in each section of a research report. More detail on reporting standards can be found in Cooper (2011) and on the APA Style website (http://www.apastyle.org/).

As part of the process of science, the manuscripts of researchers wishing to publish their work is reviewed by a group of peers (see Chapter 1) and editors. Most journals have manuscript coordinators who examine manuscripts before they are published for adherence to formatting guidelines, but not for content. All this would not be necessary if reputable and capable researchers never made errors. But we are all human. The scientific process

is a complicated one, and mistakes and misjudgments are inevitable. Despite all the checks that are designed to catch them, errors of commission and of omission, of logic and of procedure, nonetheless appear in print.

After reading a published article that describes original research, researchers in the same field are in a position to improve on the work of their predecessors by correcting errors (if present), modifying procedures, closing loopholes, and testing rival hypotheses in their own work which could explain the results. This process of replication with planned variations in methods is how science progresses.

Contents of Part II

Part II presents a series of short practice articles that have been written in journal article format. Each article is followed by a critique that highlights its main flaws. These flaws were intentionally built into the articles. Neither the articles as a whole nor any of their contents are real. They were written for the express purpose of giving you practice in the critical evaluation of psychological research without giving offense to real authors of real articles by selecting their work as exemplars of how research ought *not* to be done or interpreted.[1] All of the authors' names are made up; any resemblance between the names used in these articles and the names of real persons is accidental and coincidental.[2] The affiliations of the fictitious authors are (we hope) purely imaginary and nonexistent. The names of the authors of citations in the texts and references, the titles of the articles and books cited, the names of the journals, and the names of the publishers are also all fictional. None of the literature reviewed has any basis in fact. The data, tables, figures, reported results, and the conclusions drawn from them are all contrived.

In the interest of realism, however, some of the articles use or refer to well-established psychological, educational, or research methods, procedures, programs, and tests. All the statistical procedures in the articles are recognized research tools, and the names of the statisticians attached to them are genuine. Further information about any of the statistical procedures that are cited can be found in standard statistical texts. Some prominent psychological tests of IQ, personality, or special abilities are featured in some of the articles. Further information about these established tests can be found in texts on tests and measurements or in sources such as the regularly updated PsycTESTS (APA, 2016). To avoid any confusion, we have attached to this introduction to Part II a list of nonfictional tests, procedures, and programs that are in common use and are mentioned in one or more of the articles (see Exhibit 1). All other tests, procedures, and programs that are referred to in the articles but do not appear on this list are imaginary.

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[1]These articles represent varied published research and do not in all cases conform to APA Style, but they are close.

[2]Critiquing articles is, in a sense, a kind of puzzle solving. All the invented authors' names have a meaning that, in most instances, is related to the topic of the article. An explanatory note to the authors' names is provided in the Author Note of each article.

EXHIBIT 1

Nonfictional Tests and Procedures Mentioned in Fictional Articles

Tests and Manuals

Beck Depression Inventory
California Achievement Test
Diagnostic and Statistical Manual of Mental Disorders (DSM)
Hamilton Rating Scale
Harter Self-Perception Profile for Children
Hollingshead Index of Social Position
Minnesota Multiphasic Personality Inventory (MMPI)
Necker Cube
Porteus Maze
Rotter Locus of Control Scale
Stroop Color–Word Test
Tennessee Self–Concept Scale
Wechsler Intelligence Scale for Children—Revised (WISC–R)

Procedures or Programs

Contingency Reinforcement
Day Treatment Center
English as a Second Language
Health Maintenance Organization (HMO)
Implosion Therapy
Individualized Education Program (IEP)
Systematic Desensitization
Talented and Gifted Program (TAG)

Flaws are built in for you to find. None of the types of flaws are original; we have seen them all in published articles and unpublished manuscripts. After reading a real article, you, the critical reader, could rightfully criticize it because the literature review is incomplete (i.e., it is missing important reference to relevant scholarship or research) or is interpreted improperly. However, because our examples are made up, your interest is exclusively on what follows the introductory material. You should assume that the "theoretical foundations" that are furnished and the previous "research findings" that are presented are correct. Even if there is good reason to believe that they are not true or accurate, you should accept them in the spirit of the exercise and begin by focusing on whether the hypotheses flow logically from whatever theoretical foundations are stated. You should suspend judgment about the introductory materials in each article, assume that they are accurate, and begin the critique with the research that is reported.

The content of the articles should not enter your knowledge bank; some of them may appear to be realistic, but they are all, in fact, contrived. These articles are analogous to the "war games" played by an army unit: The "enemy threat" appears realistic, but their guns fire blanks and nobody is injured in the exercise; the threat feels somewhat like the real thing and gives everybody a chance to practice tactics, to learn, and to prepare for the real encounter. When it is over, the maneuvers are critiqued.

So it is with the simulated articles. They can be used for practice in critiquing research without fear of hurting the feelings of actual researchers.

When reading a real journal article, you may be intimidated by the knowledge that the authors are accomplished scientists, that the article was peer reviewed and selected by an editor based on its quality, and that the researchers may even have succeeded in passing through grant review teams before it was begun. Still, some articles survive this selection process but are faulted by readers in subsequent publications. We hope that knowing this is sufficient to encourage you to critically approach research published in the most selective journals. With the articles in this section, however, you should not feel inhibited about critiquing them; they are intentionally flawed.

We strongly recommend that you read each article and assess it on its own before consulting the critique that accompanies it. To do otherwise would be like looking at the completed crossword puzzle at the back of the newspaper before attempting to solve it. You may find shortcomings that are not mentioned in the critique. If so, all to the good. We do not claim to list all the weaknesses in each article. This is especially true because our practice articles are shorter than those you will read in journals. Therefore, there are many errors of omission that we do not point out. Conversely, if you do not notice all the flaws that are listed in the critique, you should not be discouraged. Some flaws have been purposefully obscured. They were included to encourage you to look at everything carefully. As in reading a real article, it is best to skip nothing along the way.

Most of the articles are empirical and contain the typical Introduction, Method, Procedures, Results, Discussion, and References components. The articles are not presented in any order, and they do not have to be read in sequence. They should be approached as you would approach a journal, by looking through the titles of the various articles and beginning with something that looks interesting (or your instructor may assign them out of order for your class). Once you have selected an article to read, most people start with the Abstract to get an overview. Next, some people find it helpful to read through the article quickly to get the flavor of what was done. In this first go-round, it is worthwhile to highlight parts that require scrutiny in subsequent line-by-line readings. Other readers prefer to read carefully and deeply right from the beginning. This approach likely still requires that the reader backtrack to previously read portions of the article. As we noted in Chapter 9, you will want to make sure that the authors are being consistent in how they fit together their hypotheses, method, results, and inferences. The section that follows provides a checklist of questions that can help to guide and systematize your review of the individual articles.

Systematic Approach to the Critique of Research Reports

When you intend to read an article critically, it is useful to know what to look for. Experienced critical readers carry the outline in their heads, but individuals who are still trying to develop and sharpen their abilities to review scientific articles can benefit from using a written checklist (even experts can, really). For instructional purposes, the checklist provided in Exhibit 2 may be used as a form that lists the key

EXHIBIT 2

Checklist of Questions to Ask to Evaluate the Trustworthiness of Research

Introduction

1. Is the research question clearly stated?
2. Do the introduction, statement of the problem, and overview of the literature adequately set the background for the reader?
   2a. Is the material that is cited (e.g., theories, social issues) related to the research question?
3. Given the research question and the background material, are the hypotheses appropriate and clearly stated?
4. Are the key terms clearly defined?
5. Are the independent variables (manipulated and measured) appropriate, given the question of the study?
   5a. Are the levels of the independent variables appropriate?
6. Are the dependent variables appropriate for the study?

Method

Participants

7. Are the participants properly selected?
   7a. Is the sample representative of its intended population and not biased?
   7b. Do the procedures adhere to the guidelines for the protection of experiment participants?
   7c. Is the size of the sample appropriate?

Manipulations and Measures

8. Are the operations used to realize the independent variable appropriate, valid?
9. Are the operational measures of the dependent variables appropriate, valid, and reliable?
   9a. Is the scoring, rating, or judging procedures valid and reliable?
   9b. Is the apparatus (if any) accurate and reliable?

Procedures

10. Are the methods and procedures clearly described in sufficient detail to be understood and replicated?
11. Is the research design suitable to test the hypotheses and answer the research question?
12. Are the experimental and/or statistical controls appropriate?
    12a. Can the results be affected by variables that have not been controlled?
    12b. Are control or contrast groups (if present) properly selected?
13. Are the participants properly oriented and motivated?
    13a. What is their understanding of the task?
    13b. Were there checks on the manipulations, if any?
    13c. Are the instructions to participants sufficiently clear and precise?
    13d. Is communication among the participants a factor?
14. Are there signs of experimenter bias in the design, data collection, assessment, analysis, or reporting?
    14a. Are appropriate procedures used to assign participants to groups, treatments, or conditions?
    14b. Are suitable techniques used to establish group equivalence, such as matching, equating, or randomizing? Does participant attrition occur, and if so, does it bias the sample?

(*continues*)

EXHIBIT 2 (*Continued*)

Checklist of Questions to Ask to Evaluate the Trustworthiness of Research

Results

15. Are the statistical tests appropriate, and are the assumptions for their use met?
    15a. Are the degrees of freedom correct and the error measures valid?
    15b. Are there errors in calculation or presentation of statistical results?
16. Are the tables and figures clearly labeled and accurately presented?

Discussion

17. Are results correctly interpreted, properly reported, given meaning, and placed in context?
18. Is the discussion reasonable in view of the data?
19. Are the conclusions valid and justified by the data?
20. Are the generalizations valid?
21. Do the references match the citations in the text?
22. Are ethical standards adhered to in all phases of the research?
23. What are the strengths of the study?
24. What would you do to improve or redesign the study?

elements to be looked for in a report. These can be used as the basis for a systematic narrative critique should one be needed. For even more detail, you can use the Journal Article Reporting Standards mentioned in Chapter 9, although this is meant more for judging the completeness of reporting than for research evaluation. Also, we provided a similar checklist in Chapter 10 regarding the key elements of a research synthesis and meta-analysis that you can use for Practice Article 17.

The research critique questions in the checklists are provided only as a guide to issues on which you should focus your attention. Your comprehensive assessments of the merits and shortcomings of a study should be based on a firm understanding of the principles of research design, logical analysis, and common sense.

The checklist is not transformed into a critique until all the judgments about the items in the form have been made in a reasoned manner. For example, the checklist reminds you to determine whether the experimental and statistical controls are appropriate. The uninformed reader might look intently but still not be able to determine whether they are appropriate. Part I explicates the meaning and the import of the various items in the checklist. It is meant to help you acquire the understanding that is necessary to make informed critical judgments.

This checklist was influenced by Davitz and Davitz (1967). Other approaches to critiquing research can be found in Tuckman and Harper (2012); Kantowitz, Roediger, and Elmes (2015); Katzer, Cook, and Crouch (1998); and Stern and Kalof (1996). Especially worth consulting is Maher's (1978) *A Reader's, Writer's, and Reviewer's Guide to Assessing Research Reports in Clinical Psychology*. Maher's guide, originally formulated for reviewers of articles submitted to the *Journal of Consulting and Clinical Psychology*, was the product of experience gleaned from about 3,500 manuscripts that were submitted to editors of the journal.

Practice Article 1

Use of Counselor "Street Talk" to Stimulate Self-Disclosure of Inner-City Youths

Consuela Eusis-Lang and Anita Reveles
Garden State Community College

Counseling interviews were conducted with 60 male and female inner-city youths. Half were interviewed in standard English, and half were interviewed in a vernacular commonly used in urban neighborhoods, or "street talk." Self-disclosure of females significantly exceeded that of males under both conditions. Overall, there was a greater amount of self-disclosure when the counselor spoke in the vernacular of the client. A significant interaction showed the least amount of self-disclosure by males but the most self-disclosure by females in the standard English condition.

The rate of emotional disturbances and behavioral problems among inner-city youths is disproportionately high (Battersby & Landsman, 2012; Green, 2009). These individuals are exposed to far more than the ordinary stresses and problems associated with growing up. As documented by Schoenwald (2015), many of them also face problems associated with being

> members of ethnic minorities, low on the ladder of socioeconomic status, economically disadvantaged, in lack of socially acceptable role models, frequently raised in single-parent households, exposed to gangs and peer pressures, menaced by actual and threatened dangers, and presented a short or dim future. (p. 96)

Perceiving themselves as outsiders to mainstream society (Bickel, 2008), inner-city youths, in turn, trust only others from their own social group (Rodriguez & Hernandez, 2013).

Graber (2012) argued these circumstances pose a special problem when counselors try to provide psychological interventions intended for the youths' benefit. As Hernandez (2012) pointed out,

> Professionals who have not shared the experiences of their clients, who do not intimately understand their milieu and way of life, and

Project COPE, which provided the setting and support for this research, is sponsored by the Melton Foundation and matching Title XI funds. Correspondence concerning this article should be addressed to Consuela Eusis-Lang, PhD, Psychology Department, Garden State Community College, Newark, NJ, 07102 or clang@gsc.edu

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Consuela Eusis-Lang = counselor uses slang; Anita Reveles = Anita Reveals.

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> who do not even speak the same "language" are greatly handicapped when it comes to establishing a trusting relationship. (p. 118)

Aside from personal appearance and attire, one of the first things evident about mental health professionals is their manner of talking. Here, we refer not only to the level of abstraction and the nature and complexity of the concepts being discussed, but to the actual phrases and verbal expressions that are employed. According to McNulty, Spires, and Antonovich (2009), it does not take a prospective client long to "size up" an interviewer and to decide, on the basis of the things that are said and the way in which they are said, that the interviewer, an outsider, is determined not to be trusted.

The judgment about whether someone is "one of us" or "one of them" influences what will happen in the dyadic interaction of therapy. As Kelly (1998) showed in his study with 80 outpatients in a mental health clinic, clients will remain guarded and withholding unless there is some degree of interpersonal trust. Under these circumstances, self-disclosure, a *sine qua non* for successful intervention (Pemberton, 2012), will be minimal.

This study is designed to determine whether the use of "street talk" (i.e., the vernacular used in clients' inner-city neighborhoods) by an interviewer in a counseling situation will lead inner-city youths to self-disclose more than they would if spoken to in standard English. It is predicted that the use of street talk will stimulate positive transference, make clients feel more at ease and more trusting, and (therefore) pave the way for more self-disclosure.

Method

Participants

Participants for this study included 60 youths from inner-city high schools in Newark, NJ. There were 30 male and 30 female teenagers (ages 15–18 years) of diverse ethnic backgrounds. The ethnic distribution included African Americans (48%), Hispanics (32%), Caucasians (10%), and others (Southeast Asians, Middle Easterners, etc., 10%). All youths were born and raised in the United States, and are primarily English speaking, although some are bilingual. Classification by the Hollingshead Index placed all youths at the lower socioeconomic status levels (IV or V). The participants were referred by principals, teachers, and guidance counselors for special help under the auspices of Project COPE. Participants had been singled out for attention for reasons that primarily included disciplinary or truancy problems, a history of aggressive or antisocial behavior, depression, or substance abuse. All agreed to participate in counseling. In addition, permission was obtained from parents or guardians, and the youths assented to participate in a study designed to "improve communication between youths and their adult counselors." Project COPE is headquartered in a downtown office building, and the counseling sessions were conducted in this facility.

The first 15 participants of each sex to enter counseling were interviewed in standard English, whereas the second group of 15 males and 15 females were interviewed in street talk.

Counselors

Two counselors were used for this study. Counselor A is a 42-year-old Caucasian woman with a master's degree in special education and experience in guidance counseling. Counselor B is a 26-year-old, Caucasian woman with a master's degree in guidance counseling. Both are regularly employed in the public schools. Counselor A was asked to conduct all interviews in standard English, whereas Counselor B, who was raised in the Newark area, was instructed to use vernacular English similar to that used in the surrounding inner-city neighborhoods.

Interviews

The study interviews were the initial sessions of an indeterminate series to be held with each participant. The purpose and goals of the sessions were explained initially, and participants were encouraged to discuss how they were getting along in school, at home, and in the neighborhood, and to focus on any special problems that they were currently having and on any future plans. Sessions were open-ended and were not preplanned for content. Rather than being counselor centered, directive, and advice-giving, counselors strongly encouraged participants to talk about themselves. Interviews were 45-min long. Aside from the fact that two counselors were used, the only way that the interviews differed was that half were conducted in standard English and half in street talk.

Independent Variable Conditions

Chisholm and Grady's (2014) standards for street talk were adopted for this study. The standards include the frequency of use of colloquialisms, slang, local pronunciations and syntax, vulgarisms and obscenities, interjections (e.g., "like," "bro," "you know"), and assorted nonverbal gestures that accompany, accentuate, and highlight vocal expressions. Counselor B, who was not unfamiliar with street talk, was nevertheless given orientation and instruction in its use. A 2-hr practice session was held, featuring recording and instructional feedback. Not until she was judged to have met the standards was she permitted to proceed with actual study participants. The principal investigator used Chisholm and Grady's checklist to determine proficiency.

As a manipulation check, 5-min time samples of the counselor's talk from early, middle, and late portions of recordings of actual experimental interviews were subjected to the same checklist appraisal. The speech of Counselor A, who was to speak standard English, was similarly checked to make sure that this condition was correctly applied.

Dependent Variable Criterion

The dependent variable, self-disclosure, was rated by Hannigan's (2002) method. This method asks raters to focus on 10 aspects of speech pattern, including use of first-person pronouns and self-referential themes, and it involves talk about one's feelings, attitudes, hopes, wishes, fears, relationships, and personal experiences. Self-disclosure during the interview was rated on a 4-point Likert scale (4 = *a great deal*, 3 = *considerable*, 2 = *a little*, and 1 = *not at all*).

Interviews were video recorded and were rated by two research assistants. As described previously, early, middle, and late time-sample segments were rated. When the two raters disagreed on a rating, they reviewed the tape segment together and reevaluated the rating until they reached agreement. If they could not reach agreement, the decision was made by the principal investigator.

The three segment ratings were averaged for each of the 10 items, and then the average scores for each were summed. This provided a score for each participant with a possible range of 10 to 40.

Results

As can be seen in Table 1, the self-disclosure score for the females ($M = 22.57$) across the two conditions exceeds that for the males ($M = 17.13$), and the sex variable is significant ($F(1, 56) = 9.67$, $p = .003$), as shown in Table 2. As predicted, self-disclosure to

Table 1
Means and Standard Deviations of Self-Disclosure

	Street talk			Standard English			
Sex	*M*	*SD*	*n*	*M*	*SD*	*n*	Total
Boys	20.4	3.80	15	13.87	4.24	15	17.13
Girls	22.33	11.37	15	22.80	5.53	15	22.57
Total	21.37			18.33			

Table 2
Analysis of Variance of Self-Disclosure by Type of Talk and Sex Group

Source	Sum of squares	*df*	Mean square	*F*
Sex	442.817	1	442.817	9.67***
Type of talk	138.017	1	138.017	3.01*
Sex × talk	183.750	1	183.750	4.01**
Error	2,565.067	56	45.805	
Total	3,329.651	59		

*p = .088, ** p = .05, *** p = .003

street talk (M = 21.37) is greater than to standard English (M = 18.33). The F-test for this comparison shows a strong trend in support of the hypothesis ($F(1, 56) = 3.01$, p = .088). There is a significant Sex × Type of Talk interaction ($F(1, 56) = 4.01$, p = .05). This interaction is portrayed in Figure 1. The least amount of self-disclosure is shown by males in interviews that were conducted in standard English (M = 13.87), and the most by females in the standard English condition (M = 22.80).

Discussion

As the data demonstrates, there is a strong trend to support the proposition that the use of street talk by counselors leads to increased self-disclosure, at least in the initial interview with male inner-city youths. The first interview is critical in determining the future path of the intervention, particularly for disaffected youths who are not strongly committed to the idea of counseling in the first place. If resistance to counseling is reinforced instead of diminished during the initial interview, the outlook for future sessions is less optimistic.

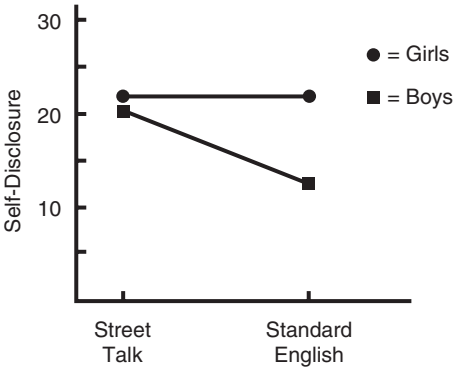


Figure 1. Sex × Type of Talk interaction.

The use of street talk by counselors who are faced with this type of client has proven to be a promising way of reducing initial resistance, increasing comfort, decreasing negative attitudes toward the counselor, and thereby providing an atmosphere that facilitates self-disclosure.

It is interesting to note that the data reveal a significant effect of the clients' sex. Females displayed significantly more self-disclosure than did males. Further, it made no difference for females whether they were interviewed in street talk or standard English.

This sex difference in self-disclosure is shown in related contexts. Females, in general, are known to be more expressive of feelings (Elkin, 1999), more verbally communicative (Schwartz & Sopwith, 2011), quicker to relate interpersonally (Feldman, Crouch, & Rodriguez, 2014), and more apt to seek out psychotherapy (Feodor, 2009). The finding that they self-disclose more readily is consistent with all of these prior findings.

This study has implications for counseling and therapy with various other groups. It shows that it is important to "talk the talk" of the group with which a client is principally identified. Counselors and psychotherapists are advised to speak using the vernacular of the client to maximize the chances of early self-disclosure.

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Critique

PARTICIPANT GROUPS

Sex has been made into a static-groups predictor variable (see Chapter 2), but no attention has been paid to the equivalence of these two groups in other regards. The researchers should have determined and reported whether the two groups were comparable in age, ethnic composition, other demographic characteristics, the percentage of habitual users of street talk in the group, and the types of behavioral problems and reasons for referral. If these two static groups are not equivalent on these other variables, you, the critical reader, cannot assume that differences in tendency to self-disclose are due to the participants' sex (see Chapters 5 and 6).

Participants were assigned to the street talk and standard English conditions on the basis of when they came into the project. The first 15 in each group were interviewed in one manner, and the second 15 in each group in the other manner. This predetermined systematic assignment could unwittingly introduce confounding variables (see Chapter 4). It is possible, for example, that the first 15 participants were the first to be nominated to take part and had more severe problems. This might have made them less willing to self-disclose than those who came in later. They also were referred at a different time of the school year, meaning their problems might have been more or less severe. From a design point of view, it is better to assign participants randomly to the two type-of-talk conditions, so as to give each participant an equal opportunity to be in either of the two conditions as the research progresses.

INTERVIEWERS

Different interviewers were used for the standard English and street talk conditions. Therefore, results may be associated with the interviewer rather than with the type

of talk. It is possible that if the two interviewers were to switch roles, the one who elicited more self-disclosure with street talk might elicit more self-disclosure with standard English as well (see Chapter 5). The interviewer effect is a confounding variable that should have been controlled by having both counselors equally fluent in the street talk and standard English and randomly assigned to participants.

Other possible confounds include the age, sex, socioeconomic status, and ethnicity of the interviewers. The younger interviewer used street talk, the older one used standard English. Did the age disparity contribute to or account for any obtained differences? Both were Caucasian, whereas the youths were mostly other racial/ethnic identities. Both interviewers were females. What was the effect on the behavior of these youths when they found themselves in an encounter with a White, middle class, professional woman speaking street talk? You do not know and cannot tell whether there is a same-sex/opposite-sex effect. Had there been male interviewers as well as female interviewers, perhaps males would have disclosed as much or more than females, and maybe females would have disclosed less.

INDEPENDENT VARIABLE

The use of street talk by the interviewer (the experimental manipulation) was judged using a checklist by Chisholm and Grady (2014). However, as a critical reader, you are at a disadvantage because you cannot check the checklist; the citation is missing from the References. Readers could look it up using a reference database but this does not speak well for the carefulness of the authors. You are left to assume that it was validly and reliably applied (see Chapter 6).

Even if the counselors were able to follow a checklist of street-talk usage during training, a question would remain about the credibility of the performance in the actual interview. A manipulation check during data collection would have helped in this regard. Also, a savvy interviewee could detect whether correctly used expressions had a false ring to them. Further control of this variable is necessary.

DEPENDENT VARIABLE CRITERION MEASURE

A 4-point numerical rating scale was used, which is not a true Likert scale. The scale is deficient in that it does not contain equal-appearing intervals. The distance between a rating of *a great deal* (4) and *considerable* (3) does not appear to be equal to the distance between *considerable* (3) and *a little* (2). A midpoint value would have been helpful (see Chapter 7). No information is given about the reliability or validity of the scale.

The judging method leaves much to be desired. The two judges were "research assistants" and they most likely were aware of the study's hypothesis and the condition of each recording they rated (because this would have been evident if the video recording captured the client and the counselor). Certainly, the principal investigator, who cast the deciding vote when an agreement couldn't be reached, was aware of the hypothesis. A better approach would be to use at least three judges working independently of each other, to average their ratings, and to present evidence that the interjudge reliability met acceptable standards (see Chapter 7). The tapes should

have been edited so that only the voices of the participants could be heard, not the counselors.

STATISTICS AND DATA ANALYSIS

Examination of Table 1 reveals that the standard deviation of the females in the street-talk condition was much larger than that of any other subgroup. The variability in responses of the females in this group is not addressed in the text. This disparity in variability suggests the possibility of a violation of the analysis of variance assumption of homogeneity of variance that should have been checked. There is no indication that it was either tested for or dealt with. There is also no interpretation of the wide range of outcomes that is suggested by the variability in this group, nor is there appraisal of the normality of the distributions (see Chapter 8).

The authors interpret a $p = .088$ as a "strong trend" and view it as evidence of support for the hypothesis, even though the conventional level of significance, $p = .05$, has not been reached. It would have been better for the researchers to "suspend judgment" or to call the results "equivocal" or "inconclusive" (see Chapter 9). The word *trend* implies that things are moving in the right direction well enough for critical readers to reject the null hypothesis. The addition of 10 or 20 more cases might have led to more confidence about that conclusion. Then again it might have suggested the trend was not sustained by the addition of more participants. In this article, once having adopted the notion of a trend, the authors proceed to interpret the results as though they are established facts.

The statement that females in the standard English condition self-disclosed the most is deceptive in view of the fact that the mean self-disclosure score for the females interviewed in standard English was 22.80, compared with 22.33 for street talk. The Sex × Type of Talk interaction suggests that it would have been helpful to do tests of simple effects (i.e., males vs. females for standard English, standard English vs. street talk for males). These would have been post hoc tests (no hypotheses about a Sex × Type of Talk interaction was offered), but it is clear that the males who were spoken to in standard English had the lowest rate of self-disclosure.

INTERPRETATION AND DISCUSSION

Beyond the establishment of statistical significance lies the task of giving meaning to the group averages that were obtained (see Chapter 9). In this study, scores ranged from a possible minimum of 10 to a maximum value of 40. This came from a summation of 10 separate ratings, each made on a scale that ranged from 1 to 4. A score of 10 indicates no self-disclosure at all. A score of 20 reflects "a little" self-disclosure. This provides a context for the males' mean self-disclosure score of 17.13. As a group, they did not disclose very much at all. The females, with a mean of 22.57, can in no way be considered as high in self-disclosure. The difference, although statistically significant, may be of little substantive practical import. When interviewed in street talk, the self-disclosure mean for all participants combined was 21.37, just 3 points higher than the mean of 18.33 for standard English. Because Table 1 does not

provide the standard deviations for the males and females combined in each Type of Talk condition, an overall effect size must be estimated from the results of the ANOVA (Cooper, 2017). The effect size is approximately $d = .23$, which Cohen (1988) would categorize as small, and the article provides no other contrasting elements. Again, the psychological difference appears to be trivial. Regardless of sex or type of talk, there was not much self-disclosure in any of the groups.

The discussion continues to expand on the implications of an inconclusive result that was promoted to a trend and then graduated into a significant finding. The study provides no evidence to support the claim that use of neighborhood vernacular by the counselor reduced resistance to self-disclosure, or increased comfort (participants could simply have been asked "How comfortable were you talking to this counselor?"). Statements about how readily females self-disclose pale when the meaning of the females' average self-disclosure rating is evaluated properly (see Chapter 9). There is no evidence that the findings of this study generalize to other types of client groups and to other types of psychological interventions (see Chapter 3). What happens in the way of self-disclosure in an initial interview may indeed be important, but this investigation has not established a link between this particular first-session behavior and the ultimate outcome or benefit of the treatment. There certainly is not enough research-based knowledge in this study to warrant recommending clinical applications (see Chapter 9).

ABSTRACT

The Abstract perpetuates the flaw in the text and implies that an inconclusive result is a firm finding (see Chapter 9).

Practice Article 2

Treatment of Flying Phobia: Comparative Efficacy of Two Behavioral Methods

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In this naturalistic study, 50 men and women with flying phobia were matched in pairs for pretest flying anxiety level and randomly assigned to systematic desensitization or implosion therapy. After 5 min of exposure to the most realistic of preflight conditions, they completed a state anxiety questionnaire. This experience was repeated after 4 therapy sessions. The anxiety of both groups declined, but the decline was steeper for the implosion therapy group. Widespread application of this rapid and cost-effective treatment is recommended.

According to Woodward (2012), approximately 32.5 million Americans are afraid of flying to some degree. This commonplace fear is sufficiently strong to rule out the option of air travel for close to 15 million people (Abramson, 2013). Aside from the effects that this has on the lifestyle, recreational opportunities, and job requirements of these individuals, it represents a loss to airlines and their affiliates of $13.5 billion annually (Abramson, 2013).

It is not our goal here to speculate about the ultimate causes and origins of flying phobia. A comprehensive summary of the various causal factors has been provided by deSousa and Loring (2007). Just as there are multiple causes, there is no single preferred treatment modality. Psychoanalytic (Pryczk, 2008), interpersonal (Ellsworth, 2014), rational emotive (Megwarthy, 2010), multimodal (Glauber, 2013), existential (Eulith, 2012), behavioral therapy (Logan, 2010), and hypnosis (Reznikoff & Belitsis, 2008) are among the approaches that have been tried with varying degrees of success. The purpose of the present study is to compare the effectiveness of two promising behavioral approaches: systematic desensitization and implosion therapy.

This research was supported by Midflight Airlines and by a grant from the Field Foundation. Requests for reprints should be sent to Marcel Beauchamp, PhD, Psychology Department, 115 College Hall, Great Lakes State University, Waukegan, IL, 60085 or MBeauchamp@glsu.edu

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Beauchamp = beautiful field (French); Campobello = beautiful field (Italian).

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Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper
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Systematic desensitization features the pairing of a hierarchy of images of the feared object or activity with relaxation techniques. When systematic desensitization is applied to fear of flying, clients are gradually advanced in imagery from visualizing going to an airport, entering the aircraft, strapping in, taking off, gaining altitude, cruising above the clouds, descending, and landing. At each stage, relaxation is induced until that particular activity can be contemplated in imagery without anxiety. When successful, clients are able to imagine flying without experiencing fear (Nagamo, 2008).

In the *implosion therapy* approach, clients are directly confronted with images of their fears unaccompanied by positive or negative reinforcement. A series of the most frightening consequences are sequentially presented, as described by Eaglesberg (2007). They include such scenes as crashing on take-off, plummeting in a steep descent into the ground, looking out of the window and seeing an engine fall off, a midair collision with another airplane, fire and smoke on board, and spinning out of control into water.

Method

Participants

Fifty participants, all of whom professed a fear of flying on commercial airlines, were recruited for this study. All participants were volunteers who responded to newspaper advertisements offering free treatment for their fear of flying in return for their participation in a research project. Participants were 18 men and 32 women between the ages of 21 and 65 years who lived in or around Chicago, Illinois.

Each volunteer was seen individually for an initial interview to determine his or her suitability for the study. The research was explained in general terms, time requirements and scheduling were discussed, and those who were ready to make a commitment signed a consent form.

Assignment to Groups

Participants were assigned either to the systematic desensitization or implosion therapy condition following the formation of matched pairs based on pretest flying anxiety scores. A distribution of anxiety scores was prepared for men and for women. Two women who had approximately the same pretest scores were considered to be a matched pair. A coin flip decided which one was to receive systematic desensitization and which one implosion therapy. This was repeated until 16 pairs had been assigned. An identical procedure was followed with the men, yielding nine pairs for assignment. Therefore, each treatment group had 25 participants matched in anxiety score to the 25 in the other group.

Criterion Measure

Flying anxiety was measured by the Mulhausen State Anxiety Scale (SAS; Mulhausen, 2009). This is a 10-item scale on which respondents are asked to tell how they feel at the moment. Each item is a 5-point Likert scale ranging from *not at all anxious* to *extremely anxious*, and the scores from the 10 items are summated to yield a total score. The Mulhausen SAS was standardized on 280 college students, ages 18 to 23 years. Convergent validity with the Chicago SAS was $r = .65$, and test–retest reliability after a 1-week interval was $r = .75$.

Procedure

A Boeing 717 was made available by Midflight Airlines for research purposes. The plane was berthed at O'Hare Airport, Chicago, Illinois. The aircraft was in its usual preflight mode in preparation for a scheduled flight. The airline dedicated 20 min for research purposes; the study did not interfere with its regular flight schedule.

The preflight experience and preflight instructions when boarding a plane can arouse extreme stress in phobic flyers. This is advantageous in a study on fear of flying. At the initial security check, pockets must be emp-

tied, shoes removed, and passengers, along with their handbags and their carry-ons, are scanned in a search for concealed weapons or suspicious objects. In the terminal, passengers are repeatedly admonished over the public address system not to leave luggage unattended and not to accept gifts from strangers. Once on board, flight attendants tell passengers to sit upright and to fasten their seat-belts so as to prevent injury from any air turbulence. Flight attendants demonstrate how oxygen masks are used, and they assure passengers that these masks will drop down in front of their faces if there is a sudden loss of cabin pressure. Passengers are warned not to smoke. Flight attendants describe how to inflate a life jacket should the pilot have to make a landing on water and point out the location of the nearest emergency exit.

Participants in the study were ushered on the airplane to assigned seats and were given the standard preflight instructions. As is customary, a check was made to make sure that seats were in an upright position and that seat belts were fastened. Shades had been drawn in all windows so that motion could not be visually detected. Engines were started and accelerated.

After 5 min, participants were asked to lower their trays and to fill out the Mulhausen SAS, which had been placed in the flap of the seat in front of them. Following this, everyone deplaned and assembled in a lounge area that had been set aside. Refreshments were served, and individuals were told when and where to report for their treatment. One month later, following treatment, a second session that duplicated the first was held aboard the aircraft. The SAS was completed in the same manner as it had been on the first occasion.

Treatment

Four half-hour weekly sessions of therapy were scheduled for each participant. Six therapists assisted in this study; all were advanced psychological interns who were trained and experienced in both therapeutic approaches. Half of each therapist's assigned clients were given systematic desensitization, and half received implosion therapy. Therapists closely followed procedures outlined in a prepared procedural manual for both types of intervention. All sessions were held at the Chicago Center for Behavioral Intervention (CCBI). Sessions were video recorded and spot checked by supervisory personnel to ascertain compliance with the procedural manual. Participants who did not attend at least three of the four sessions and the final posttest were dropped from the study.

Results

At the first airport session, two people panicked while aboard the aircraft. They left their seats and asked to be let off the plane. They both withdrew from the study. Treatment was nevertheless offered to them, but neither accepted. One of them was scheduled to be in the implosion group, and one had been assigned to receive systematic desensitization. Six more participants did not attend the required number of treatment sessions and did not come for the postsession meeting. All eight individuals had to be dropped from the study, bringing the final sample to 42.

Data from the 42 participants who met the attendance requirements were analyzed in a 2×2 analysis of variance with one between-subjects and one within-subjects factor. There were two orthogonal levels of treatment, systematic desensitization and implosion, and a repeated measure for time, with pretest and posttest levels. The dependent variable was the participants' score on the Mulhausen SAS. Obtained means are presented in Table 1.

The analysis of variance (Table 2) reveals a significant reduction in reported state anxiety across time for both treatment groups combined ($F(1, 40) = 86.57$, $p < .0001$, $d = 2.94$). The mean for the combined groups declined from 39.62 at pretest to 27.07 at posttest, as shown in Table 1. Although both groups experienced a diminution of anxiety, there was a

Table 1
Pretest and Posttest Means of Two Treatment Groups

Group	Pretest	Posttest	Total
Desensitization ($n = 24$)	43.70	33.54	38.63
Implosion ($n = 18$)	34.17	18.44	26.31
Totals	39.62	27.07	

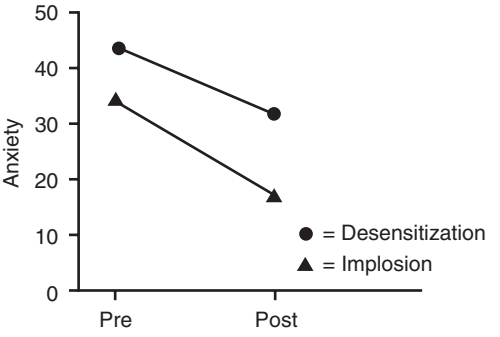


Figure 1. Mean anxiety scores by time and treatment.

significant Treatment × Time interaction ($F(1, 40) = 4.17$, $p < .05$). As illustrated in Figure 1, there was a steeper reduction in anxiety for the implosion therapy group than for the systematic desensitization group. The former dropped about 16 points in contrast to a 10-point reduction for the latter.

Discussion

The significant reduction in anxiety as a consequence of both treatments demonstrates the effectiveness of behavior therapy regardless of the specific form of the intervention. The results clearly show, however, that the two approaches were differentially effective, with a marked decline favoring implosion therapy.

This study was not done in the artificial setting of a laboratory or in an analogue situation. People who were troubled by their flying phobia and who were motivated to seek help in gaining relief were exposed to a real-life situation that duplicated the feared situation in all respects save one. All elements of the prelude to flight were reproduced in their actual settings. Even though the aircraft never took flight, the realism of the experimental arrangement was sufficient to stimulate a high level of anxiety.

The fact that two participants had overt panic reactions in the first session on board attests to this realism. The finding that implosion therapy in particular was successful in reducing flying anxiety to a very manageable level after only four half-hour sessions is highly encouraging. In view of these findings, this rapid and cost-effective treatment merits widespread application.

Table 2
Analysis of Variance Results of Flying Anxiety for Two Groups, Pretreatment and Posttreatment

Source	*df*	Sum of squares	Mean square	*F*
Treatment	1	3,122.009	3,122.099	30.749**
Subjects within groups	40	4,061.389	101.535	
Time	1	3,306.298	3,306.298	86.573**
Treatment × time	1	158.730	158.730	4.165*
Time × subjects within groups	40	1,524.472	38.112	

*$p < .05$, ** $p < .0001$

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Critique

CONTROL GROUP

Because of the omission of an untreated control group, it is impossible to tell whether either treatment had any effect (see Chapter 6), that is, whether the pretest–posttest change in scores can be attributed to the treatments. Many factors other than the treatment could have accounted for change in anxiety over the two occasions:

1. Lower posttest scores might be attributed to familiarity with the experimental situation. Having survived the first session in the airplane, participants might not be as anxious on the second occasion even if they had no intervening treatment (see Chapter 9).
2. Participants may have been less frightened on the second occasion because they had learned that the plane was not really going to take off.
3. Group support of shared anxiety, and perhaps reassuring remarks made by other participants, might have been more helpful in controlling anxiety on the second occasion.
4. People who had a recent stressful experience with flying might have been more likely to respond to the ad and their anxiety might have naturally diminished over time (regression to the mean).
5. A placebo effect was possible: Participants might have believed, hoped, or expected that the treatment (whatever it was) would be beneficial (see Chapter 5).
6. A desire to please the therapist or the researcher could have lead respondents to underestimate their state anxiety on the final session aboard the airplane.
7. Some participants might have underestimated their anxiety at the final session because they perceived their lack of improvement as a personal failure that they are reluctant to admit.

8. As the first factor suggested, familiarity with the experimental situation might have reduced anxiety. We must also consider that there could be a testing effect; familiarity with the test, independent of the context, could be partially responsible for lower anxiety scores on the second administration.

One, all, or a combination of these variables could account for lower posttest scores. Use of a control group of randomly assigned participants who received no treatment would rule out most of these possibilities (see Chapter 6). Had the researchers used an additional placebo control group consisting of participants who received some inert (i.e., not intended or believed to be therapeutic) intervention that took the same amount of time as the real sessions, they would rule out most of the remaining possibilities.

MATCHING, SELECTIVE ATTRITION, AND SAMPLE BIAS

Initially, there were 25 pairs, matched for state anxiety level on the basis of their anxiety scores obtained in the introductory session. It is very unusual to be able to come out with 25 well-matched pairs when starting with 50 people. Pairing does not work out that evenly; typically, researchers must have a larger initial pool of participants to locate 25 matched pairs. Therefore, it is likely that some of the "pairs" did not have identical scores on the pretest anxiety measure, though random assignment of the pairs should have minimized differences between the groups getting the two types of therapy (more on this next).

More troubling, the final sample was composed of only 24 participants in the systematic desensitization group and 18 in the implosion therapy group. One person withdrew from the systematic desensitization group but this occurred before therapy began, so his or her departure could not have been related to the therapy. Seven were eliminated from the implosion therapy group. Of these seven, one left the experiment before the pretest because they could not cope with being on the airplane. Six, or 24% of the sample, in the implosion condition completed the pretest and began therapy but did not complete it. This is concerning because their withdrawal might have been related to the treatment itself. They might have left the study because the therapy was not working for them and/or it was too aversive.

The pretest means were nearly 10 points apart on the anxiety scores obtained at the first experience on board the airplane ($M = 43.70$ for systematic desensitization and $M = 34.17$ for implosion therapy). This suggests that the average anxiety score of the seven people who were excluded from the implosion therapy group may have been considerably higher than the mean of those who remained in this group and also much higher than the mean of the systematic desensitization group. Therefore, a subgroup with a high level of pretest anxiety may have been eliminated from one of the treatment groups. The two samples are probably not equivalent, because of differential attrition, and may bias the results (see Chapter 4) because the most severe cases dropped out of one group but remained in the other group. The researchers provide no test of the pretreatment equivalence of the two groups and do not provide standard deviations, so you cannot test this yourself.

From pretest to posttest, participants in the desensitization condition dropped about 10 points and in the implosion condition over 15 points. At first glance, this speaks well of the implosion condition (and explains the statistical interaction between Time and Treatment). However, because of the pretest differences it could be due to initial level of anxiety influencing the success of the treatment, with lower levels of anxiety at pretest allowing for more change (see Chapter 8).

HARMFUL EFFECTS

Potentially harmful effects of implosion therapy are not mentioned; the dropouts are simply no longer a part of the study sample. This led to the distorted conclusion that "implosion therapy, in particular, was successful in bringing down flying anxiety to a very manageable level" and that it "merits widespread application" (see Chapter 8). You do not know if the participants who withdrew became even more anxious about flying because of the treatment. The researchers should have addressed this in the Discussion.

SAMPLE COMPOSITION

Although more women than men participated in the study, the authors do not indicate whether the ratio of women to men was equivalent in the two treatment groups when the posttest was completed. You are not told about the sex distribution of the dropouts and remaining participants. This is a possible confounding variable, and it requires attention (see Chapter 4).

THERAPISTS

You are not told whether the therapists were equally skilled and equally committed to both forms of treatment, though the attempt by the researchers to make them equivalent is commendable. Results could easily be affected if they were more adept at or favored one approach over the other (see Chapter 5).

FOLLOW-UP

It is a good idea to have a follow-up assessment sometime after the end of treatment in studies of therapies. Gains that are reported immediately at the end of the treatment may disappear in a short while. Advocates of a specific treatment should state in advance how long a remission is expected to last to justify their claim that the treatment can be deemed worthwhile (see Chapter 6). You can then decide whether to accept this judgment.

ETHICS

Participants were not informed that the airplane would remain on the ground. The researchers withheld this information because they wanted to keep the conditions of the research realistic, but by so doing, they perpetrated a deception. The investigators

did not explain the conditions fully when obtaining consent. The intense reaction of some of the participants is a matter of concern. One might argue that a worse deception could have been practiced (e.g., if the airplane had actually taken off despite assurances that it would not), but this does not mitigate the potentially damaging effects of the study conditions on the participants (see Chapter 11). It would have been good for the researchers to state in the Author Note or Method that Institutional Review Board approval had been obtained.

ABSTRACT

The Abstract is misleading. This becomes apparent when you examine the research and discover its flaws that make the stated conclusions untenable.

Practice Article 3

The Effect of Divorce on Sons' Aggression

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Forty-eight boys (ages 13–15 years) with divorced parents were given paper-and-pencil mazes to solve. Mazes in two pretest trials were readily solvable, whereas mazes in a frustration condition were impossible to solve. Ratings of the anger displayed by the boys were made by the researchers on a 10-point scale, and self-ratings were obtained as well. The observed anger ratings were significantly greater when attempting the unsolvable mazes. The research clearly demonstrates that divorce increases boys' aggression.

The divorce rate has dramatically and progressively increased in recent years (Morton & Blatchford, 2013; Twilley, 2014). Many authorities have cited the varied effects of divorce on offspring. Of particular import are affective (Israel, 2007) and behavioral (Canasta & Bridges, 2001) disturbances. Males, especially, have been observed to be prone to anger (Affertide & Stone, 2009; Miller & Metaluma, 2010). Albright and Tinderly (2008) have posited that the increase of conduct disorders and juvenile delinquency can be attributed, in part, to the increasing divorce rate. In their study of 185 juvenile delinquent boys, King and Arthur (1995) noted a divorce rate among parents that was significantly higher than in a group of comparable age and socioeconomic status who were not delinquent. Truancy and referrals for disciplinary action were more frequent in a sample of Chicago junior high school youths whose parents were divorced than in a sample of students from intact families.

Clinical theorists have presented a compelling rationale for these findings. Waverly (2007) pointed out that the breakdown of the nuclear family and, as is typical, the departure of the father from the household, leads to anger stemming from the change in family status, disruption of secure routines, economic

This research was supported by a grant from the Eldridge G. Thomson Foundation. Requests for reprints should be sent to Prof. Bernard D. Partes, Psychology Department, Fairweather University, Billings, Montana, 59105 or PartesB@bigsky.edu

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Bernard D. Partes = Bernard departs; Junior Fites = Junior fights.

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stresses, and blame casting. Boys, who are apt to be more closely identified with the father, are more likely to display anger in the form of aggressive acting-out behavior (Bingham & Newman, 2006; Ellsworth & Kripke, 2005).

It is important that these clinical observations and theoretical formulations be subjected to more rigorous scientific appraisal. The purpose of this study is to test the hypothesis that boys of divorced parents are prone to anger, thereby providing research support for existing theoretical formulations. This investigation aims to offer experimental confirmation of the idea that anger is a distinguishing characteristic of boys of divorced parents.

Method

Participants

Forty-five male, middle class, Caucasian, junior high school students, ages 13 to 15 years, participated in the study. They were recruited from three junior high schools in neighboring suburban communities. All of the participants were children of divorces in which the father had been out of the home for 1 to 5 years. None of the mothers had remarried, and no adult males were living in the household. All of the boys were volunteers, and their mothers signed an informed consent form.

Procedure

Participants were seen individually, and each of the authors served as the experimenter for approximately one-half of the boys. The boys were given two paper-and-pencil mazes of moderate difficulty. The mazes were selected to be comparable in difficulty to mazes for that age level in the Heathrow (2010) revision of the Porteus Maze series. The boys were told that most boys in their grade could solve the maze puzzles without too much difficulty.

Boys were then given a second set of two mazes that had been altered to seal off the one escape path, thereby turning it into a cul-de-sac. They were told that they would be given two mazes for practice and that they would receive a $10 prize if they finished the mazes more quickly than the average participant. The aim, unbeknownst to the participants, was to frustrate them and to observe consequent signs of aggressive behavior.

All of the boys successfully completed the first two mazes and were praised for their effort. As arranged, they were unable to find their way out of the second set of mazes. When they indicated that they were ready to give up on the first unsolvable maze, they were encouraged to try the next one. They were told that even though they did not complete the maze, they were still in the running for a prize.

On completion, each boy was debriefed about the purpose of the study and the reason for the deception created by unsolvable mazes. They were reassured about their true ability on the basis of their performance on the two solvable mazes. All were given the $10 reward and were thanked for their participation. Nobody left the session in an angry mood.

Dependent Measures

Aggressive behavior was assessed on a 10-point visual analog scale completed by the experimenters. The raters took into account facial expressions, direct verbal expressions of anger, vocalization of expletives, broken pencils, hitting of the table, and other signs of emotional distress. The anchor values of the scale were 1 (*no apparent signs of anger*) and 10 (*extremely angry*). Ratings were done for boys' performance on each of the four mazes. Therefore, there were two pretest (no frustration condition) ratings and two posttest (frustration condition) ratings to compare. Scores of the two pretest ratings were averaged for comparison with the averages of the two posttest ratings.

At the end of the last maze, the subjects were asked to make ratings of their own subjective feelings of anger while doing each of the four mazes. This was done from recollection instead of at the time of the task because it was thought unwise to place the focus on anger when the participants were led to believe that the study was about problem solving.

Results

There were no differences in the performance of the boys' ratings by the two different experimenters. Their results were therefore combined for purposes of analysis.

As shown in Table 1, the pretest mean of the objective rating was 2.39, in contrast to a mean of 7.12 at posttest. These two means were compared by a *t* test for independent groups yielding, $t(88) = 4.76$, $p < .001$, $d = 1.01$. The comparable means of the self-ratings were 2.00 and 2.15, respectively, $t(88) = .33$, *n*s, $d = .07$ (see Table 2).

Table 2
Means and Standard Deviations of Self-Ratings of Anger at Pretest and Posttest

Trial no.	Pretest (nonfrustration)	Posttest (frustration)
1	1.98	
2	2.02	
3		2.13
4		2.17
M	2.00	2.15
SD	2.11	2.21

Discussion

The finding that objectively observable signs of anger and overt aggressive behavior nearly doubled under frustration clearly supports the hypothesis that anger is a distinguishing characteristic of boys of divorced parents. Even such a minor frustration as inability to solve a maze puzzle was sufficient to evoke clearly observable anger. Divorce apparently takes its toll on adolescent and preadolescent boys who are deprived of the presence of a father in the home at this critical time in their development.

The fact that self-ratings did not show a significant difference under the two conditions, even though there was more self-reported anger under frustration, suggests either that the defense of denial is strongly operating or that there is considerable unawareness of their own affective inner states. Each of these explanations has important clinical implications.

Because anger is such a prominent feature of the psychological makeup of sons of divorced parents, thought should be given to the application of early interventions to help manage and control it. More research is needed to help pinpoint those aspects of development that are most severely affected and to determine which therapeutic approaches are most effective in helping boys of divorced parents cope with their anger.

Table 1
Means and Standard Deviations of Observer Ratings of Aggressive Behavior at Pretest and Posttest

Trial no.	Pretest (nonfrustration)	Posttest (frustration)
1	2.25	
2	2.53	
3		6.56
4		7.28
M	2.39	7.12
SD	2.60	6.13

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Critique

CAUSATION

The title indicates that this is meant to be a study of cause and effect, with divorce as the cause. This was not the case, however; the study included only one preexisting group (consisting of sons of divorced parents) and lacked any measure of the aggressiveness of these boys prior to their parents' divorce. Even if all the contrast groups discussed had been used, this study still would not be able to determine a cause–effect relationship (see Chapter 3).

CONTRAST GROUPS

Two essential contrast groups for the static group "sons of divorce" are missing. Most important, a contrast group composed of sons of intact families is missing. This is necessary if one is to rule out the possibility that sons of intact families may become just as, or even more, angry after frustration than sons of divorced families. Moreover, unless sons of divorced parents are contrasted with daughters of divorced parents, no statements can be made about anger as a distinguishing characteristic of sons. The introduction to the article implies that males "in particular" become angry as a consequence of divorce. A test of this hypothesis demands a contrast with daughters (see Chapters 3 and 6). For all we know, daughters may become just as angry or even angrier than sons. To support this statement, the design should have had two static-group variables, each with two levels: divorced–not divorced and sons–daughters, all equated on other important variables.

According to the frustration–aggression hypothesis, aggression is a natural consequence of frustration. To the extent that this is true, people in general (not just the subset "sons of divorce") tend to respond to frustration with aggression. To show that sons of divorce share this tendency with people in general shows nothing distinctive or unique about them (see Chapters 3 and 6). It merely demonstrates what may be a typical transitory state in contrast to a trait that characterizes their usual demeanor. The pretest does provide an estimate of a baseline state, but it is unclear whether the amount of anger that they feel in this experimental situation is typical for them in other situations (see Chapter 3).

DEPENDENT MEASURES

There are two criteria: observed behavior and inner feeling state. The measure for observed behavior is the experimenters' ratings on a visual analog scale, and the measure for the second is participant self-ratings. No attempt was made to establish the validity or reliability of either (see Chapter 7).

RATERS

The experimenters are the sole judges of aggressive behavior displayed by the participants. The potential for bias is substantial, because the experimenters know the hypotheses and the experimental conditions under which the males are performing. In this study, it is essential that independent raters be unaware of both the hypotheses and the conditions (see Chapter 7), and it would be preferable to use the average rating of at least three independent raters.

Self-ratings must always be received with some skepticism and interpreted with caution. Here, the shortcomings of self-ratings are compounded by the procedure of using ratings of recollections of prior affective states after other experiences have intervened. Self-ratings are subject to inaccuracies because of the social desirability factor in self-revelation, and they may be influenced by distortions of self-perception. Nevertheless, once having decided to use self-ratings, results obtained with them ought not be dismissed out of hand because they do not support the hypothesis (see Chapter 8). No data in this study buttresses the authors' claim that lack of awareness or the mechanism of denial account for failure to obtain predicted results with self-ratings.

COMMUNICATION

Clusters of participants came from the same schools and some possibly from the same classes. This introduces the possibility of communication among participants (see Chapter 5). This is a potential threat to internal validity, especially in a study that involves the use of deception. It would be exceedingly damaging for the integrity of the study if any males came to the session with grapevine foreknowledge that some of the mazes were unsolvable and were there for the express purpose of making them angry. No mention is made of whether participants were asked to refrain from talking about the experiment—not that this would have sufficed to address the problem.

GENERALIZABILITY

Even if the study had appropriate contrast groups, its generalizability would be limited to middle-class, Caucasian suburbanites, ages 13–15 (see Chapter 3). It is not established that the correlates of divorce would be comparable at other socioeconomic levels, at other locales, and with other ethnic groups. This limitation should at least have been acknowledged in the article.

DATA ANALYSIS

There is an error in Table 1: Posttest means of 6.56 and 7.28 average to 6.92, not to 7.12. You should routinely scan and check all tables for obvious errors in entries (see Chapter 8). Data were analyzed by a *t* test to assess the significance of the difference between the means of two independent groups. There is, however, only one group in this study. Measurements were taken on this group under two different conditions. This, then, is a repeated-measure design calling for a dependent *t* test for correlated samples, not for independent groups. The calculations, on the basis of the differences within each participant's performance under two conditions, would be different than those in an independent groups design. With 45 participants, the correct number of degrees of freedom would be 44, not the 88 used here (see Chapter 8). The result is likely not affected in this instance because the difference is so large, but in cases where the mean difference is smaller, an error in designating the proper number of degrees of freedom could change the conclusions.

Practice Article 4

Dyslexia in Fifth-Grade Girls: Personality and Perceptual Factors

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Personality and perceptual factors in dyslexia were studied in 60 readers with dyslexia and 60 readers without dyslexia. There were no significant differences between the two groups of readers or between boys and girls on the Minnesota Self-Esteem Scale or the Luft Locus of Control Scale. In contrast to readers without dyslexia, children with dyslexia displayed significantly fewer perceptual shifts on the Necker Cube, significantly more distractibility by extraneous cues on the Stroop Color–Word test, and more difficulty identifying words from minimal cues using the Fuzzy Copy Technique. There were no sex differences on the perceptual measures.

Dyslexia is a common learning disability that has a profound effect on an estimated 7 million children in the United States (Blither, 2014). Many children grow to adulthood with it and cannot overcome it regardless of what they do. Some continue to further their education despite this obvious disadvantage. As many as 3% of college students are dyslexic (Lauber, 2012). Dyslexia has been linked to personality factors and to perceptual deficits (Shasta & Kovich, 2012). Some of these are of particular interest in the present investigation.

Personality

Self-Esteem

Krantz and Trilby (2011) asserted that children with dyslexia often have one or more overpowering parents whom they can never please enough. Their lowered self-esteem leads them to believe that they will not be able to learn or to keep up with other children. This becomes a self-fulfilling prophecy when they get to the point where the mastery of reading becomes essential for further progress. This peaks in the fifth grade (Ekkles, 1995) and appears to be a problem particularly for young girls. Narwhal (2012) speculated that boys have more diversified avenues for establishing a positive self-image than do girls.

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Jane Kent-Reid = Jane can't read; John E. Duzzent = Johnny doesn't.

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Locus of Control

According to Mason and Mitchell (2009), individuals with dyslexia typically have an external locus of control. Their domineering parent(s) make them feel that they are powerless to determine their own destiny, including the mastery of the meaning of printed symbols.

Perceptual Factors

Shifting Mental Set

Mastery of reading requires an assortment of perceptual skills. One of them is the ability to shift mental set when encountering letters and letter combinations in one word that are pronounced differently when seen in other words. For example, *most–lost* and *rough–dough* are combinations that look alike but sound different. *Fluff–tough*, *graph–laugh*, and *ghost–toast* are examples of pairs that look different but sound alike. To add to the confusion, the letters *b* and *d* and the words *look* and *took* not only resemble each other, but sound similar as well. The fluent reader has to shift set rapidly to be able to make out the meaning of visually perceived words that are not familiar to the ear (*couch* read as *cutch* to be consistent with the way the word *touch* is pronounced) or to accommodate words that are familiar but which do not fit the context of the sentence (*couch* read as *coach*).

Extraneous Cues

Another perceptual factor that is important for reading is the ability to shut out extraneous cues so as to keep attention focused exclusively on the specific task at hand. Delft (2006) has shown how easy it is to be influenced, misled, or distracted by irrelevancies. Distracters can take many forms. Some, such as competing auditory or visual stimuli, are external but can have an effect, as demonstrated by Perls and Nicholas (2009). Others, of especial interest here, are intrinsic to the words being read. An example of this is the visual conflict created when one is asked to read the word *red* printed in green ink (as in the Stroop effect).

Minimal Cues

Another perceptual skill that is necessary for rapid, fluent reading, according to Hofheimer and Schmidt (2013), is the ability to perceive a meaningful whole from minimal cues. "Good" readers can take in at a glance a combination of letters that make up a word, or a combination of words that comprise a sentence, without having to pore over each letter and decipher the sound for which it stands. They make rapid saccadic eye movements as they pass back and forth over the text (Benjamin, 1996), but they do not have to examine each letter combination and word as "poor" readers are obliged to do. Good readers are able to "get" a word with minimal cues. As Gruno (2012) showed, good readers can make out words that baffle individuals with dyslexia when the words are presented tachistoscopically.

To summarize, the present study aims to demonstrate that two personality measures, self-esteem and locus of control, are associated with dyslexia in fifth-grade students, as are three different perceptual skills. It is hypothesized that readers without dyslexia do significantly better than readers with dyslexia on all five of the measures that have been selected to tap these variables and that these differences are exaggerated in girls.

Method

Participants

Participants in this study included 30 girls and 30 boys with dyslexia and 30 girls and 30 boys with no history of dyslexia. All participants were in the fifth grade. Students with dyslexia were identified as having reading problems by their teachers and had been referred to the remedial reading program in the school. The judgment of the classroom teacher was

confirmed by reading specialists who conducted individual assessments. All participants with dyslexia were classified as having moderate to severe dyslexia.

The students came from six elementary schools in an urban school system, and the research was carried out over a 3-week period. The study was conducted with the approval and cooperation of the school board, the superintendent, the six principals, and the teachers who released children from class for study purposes. The parents of all participants signed an informed consent form, and no child who did not wish to participate was included. Children in the study signed an assent form. All participants were given an attractive box of mixed candies as a reward for participating.

Students were matched in quadruplets for age, grade, nonverbal IQ, and ethnicity. A quadruplet consisted of two girls, one of whom was dyslexic, and two boys, one of whom was dyslexic. The matching procedure assured equivalence on these critical variables. All schools were in comparable neighborhoods, so equivalence of socioeconomic status for the four groups was assumed. Only native-born children for whom English was the first language were included.

A total of 16 children had to be excluded from the study sample because they did not fall into one of the 30 matched quadruplets. The final sample was 48% Caucasian, 24% African American, 21% Hispanic, and 7% other.

Procedure

All children had completed a nonverbal IQ test in the second grade. These scores were used for matching purposes. The following measures were completed by participants in three individual meetings over a 3-week period.

Minnesota Self-Esteem Scale

This scale is a paper-and-pencil questionnaire that was standardized on a broad sample of preteenagers. It has good convergent validity, with established correlation coefficients of .78 with the Tennessee Self-Concept Scale and 74 with the Harter Self-Perception Profile for Children. Correlation with teacher judgments is .61. Its test–retest reliability is reported as .79. Scores are scaled from 0 to 100.

Luft Locus of Control Scale

This questionnaire was devised for children and was standardized on a large sample of elementary-school children. Its correlation with the Rotter Locus of Control Scale is .75. The scale's scores range from 0 to 50, with scores at the lower end representing internal locus of control and scores at the higher end signifying external locus of control.

Necker Cube

The Necker Cube has a long history of use in perceptual research. It is a line drawing of an open-faced cube. Lines that would ordinarily be hidden are drawn in the same way as the ordinarily visible lines. When people fixate their gaze on it, the cube flips in appearance, and the inner lines appear to become the outer borders. It shifts back and forth as one continues to look at it. In this study, the instructions were to look at the cube and to signal every time that it changed form by pressing a button on a hand-held recorder. The recorder counted presses over a 3-min period. This procedure was used to measure the children's ability to shift mental set.

Stroop Color–Word Test (Modified)

The Stroop Color–Word Test (Stroop, 1935) consists of a list of color names printed in colors that do not correspond to the words. This conflict interferes with efficient naming of the colors. A number of different versions of this procedure have been used. For this study, 50 words were flashed on a computer screen for 1 sec duration. The words *black*, *red*, and *green* were spread throughout the presentation. The distribution was random with the exception that no word was duplicated by the

word that immediately preceded it. The words were presented either in black, red, or green, and in no instance did the presentation color match the name of the word. Therefore, *red* was presented in green or black, *black* in red or green, and *green* in black or red.

In the first presentation, the children were instructed to respond as quickly as possible, ignoring the word and just naming the color. In the second presentation, they were instructed to ignore the color and read the words. Their score was the number of errors made. In a pretest of this procedure, with 50 children not in the experiment, a test–retest reliability of $r = .76$ was obtained after a 2-week interval. This test was used to test children's ability to shut out extraneous stimuli.

Fuzzy Copy Technique

The Fuzzy Copy Technique was used to assess the ability of the students to identify words from minimal cues. Twenty 4-letter words were typed in a Microsoft Word document using 12-point Times New Roman font. The single page was printed on standard grade white paper. A fuzzy photocopy was then made by placing a translucent sheet between the clear copy and the copier's window. The resulting copy is barely legible to some people, but others are able to make use of the minimal cues available and can identify some or all of the words. The procedure has been used for various purposes in a number of research projects (Nicholson, 2005), and it is reported to be of satisfactory reliability. The translucent paper was obtained from the test developers.

Results

Personality and perceptual measures data were analyzed in five 2×2 analyses of variance, one for each variable. Then, a logistic regression analysis was conducted to predict the students' group membership, both for boys versus girls who were either dyslexic or not dyslexic. The means for each group on each outcome variable are presented in Table 1.

Table 1
Scores of Dyslexic and Nondyslexic Students on Five Variables

	Dyslexia		Nondyslexia		
Variable and sex	*M*	*SD*	*M*	*SD*	Total
Self-esteem					
Girls	47.0	20.3	51.0	10.1	49.0
Boys	49.0	27.6	53.0	11.7	51.0
M	48.0		52.0		
Locus of Control					
Girls	34.0	18.6	32.0	9.2	33.0
Boys	38.0	24.3	38.0	8.9	38.0
M	36.0		35.0		
Necker Cube					
Girls	10.0	4.1	18.0	3.9	14.0
Boys	8.0	3.7	16.0	3.8	12.0
M	9.0		17.0		
Stroop Color–Word					
Girls	12.0	3.1	6.0	2.2	9.0
Boys	16.0	3.4	8.0	2.9	12.0
M	14.0		7.0		

Note. All means have been rounded.

Table 2
Logistic Regression Summary Table of p *Values of Discriminators Between Groups*

Variable	Dyslexic vs. nondyslexic	Girls vs. boys
Self-esteem	.52	.75
Locus of Control	.49	.73
Necker Cube	.01	.71
Stroop Color–Word	.05	.71

The *p* levels associated with each variable in the logistic regression are presented in Table 2.

Self-Esteem

Contrary to the hypotheses, there was no significant difference between the students with dyslexia ($M = 48$) and the students without dyslexia ($M = 52$) on the Minnesota Self-Esteem Scale ($F(1, 116) = 1.21$, *n*s). The self-esteem of the girls ($M = 49$) did not differ from that of the boys ($M = 51$; $F(1, 116) = 1.32$, *n*s) and there was no interaction ($F(1, 116) = 2.01$, *n*s). Logistic regression also showed that self-esteem did not discriminate between the groups with and without dyslexia or between the sexes.

Luft Locus of Control

The Luft Locus of Control score for the students with dyslexia ($M = 36$) was essentially the same as that of the students without dyslexia ($M = 35$; $F(1, 116) < 1$, *n*s). The means for boys and girls were 38 and 33, respectively ($F(1, 116) = 1.94$, *n*s), and no interaction appeared ($F(1,116) = 1.50$, *n*s). Therefore, locus of control did not discriminate between groups with and without dyslexia or between the sexes.

Necker Cube

As predicted, the children with dyslexia displayed a significantly smaller ($M = 9$) number of perceptual shifts than did the children without dyslexia ($M = 17$; $F(1, 116) = 4.12$, $p < .05$). There were no sex differences ($F(1, 116) = 1.03$, *n*s) or interactions ($F(1, 116) < 1$, *n*s). The logistic regression showed performance on the Necker Cube discriminated significantly between the groups with and without dyslexia.

Stroop Color–Word Test

The mean for the students with dyslexia ($M = 14$) significantly exceeded that of the students without dyslexia ($M = 7$; $F(1,116) = 4.11$, $p < .05$) on the Stroop Color–Word Test, indicating a greater degree of distractibility by extraneous cues for the students with dyslexia. There was no significant sex effect ($F(1, 116) < 1$, *n*s) or interaction ($F(1, 116) < 1$, *n*s). The color–word procedure was also a significant discriminator between the two reading groups on the logistical regression.

Fuzzy Copy Technique

In support of the hypothesis, the children with dyslexia had more difficulty identifying words from minimal cues ($M = 4$), in contrast to the children without dyslexia ($M = 11$; $F(1, 116) = 4.15$, $p < .05$). As with the other variables, there was no difference between girls and boys ($F(1, 116) < 1$, *n*s) and there was no interaction ($F(1, 116) < 1$, *n*s). Logistic regression confirmed the finding that ability to identify words from minimal cues significantly discriminates between the two reading groups.

Discussion

It is apparent from the results of this study that the personality variables of self-esteem and locus of control did not discriminate between any of the groups, whereas the three perceptual measures significantly discriminated between children with dyslexia and those with not history of dyslexia. It seems likely that the personality measures did not discriminate because of some weakness in the measuring devices that were used. Although the Minnesota Self-Esteem Scale and the Luft Locus of Control Scale are supposed to be valid with preteen children, it is possible that

the children in this particular research sample were too young and too homogeneous for the scales to show the kind of discriminatory power that was anticipated. The possibility also must be entertained that personality factors do not, in fact, loom as large in reading disabilities as was heretofore believed. Further research with alternative measures on children of different ages is certainly warranted.

In sharp contrast to the failure of personality factors to discriminate among reading groups, the consistent results on the three perceptual variables places renewed emphasis on the importance to reading skills of basic perceptual processes. The three perceptual processes that were the focus of this study had to do with the abilities to shift mental set, to shut out extraneous visual cues, and to recognize visual stimuli on the basis of minimal visual cues. This does not imply that the perceptual deficiencies of children with dyslexia are limited to these three, but we have presented solid evidence that at least those that we studied are involved.

The findings have interesting implications for remedial work. Although it is premature to report any details, we are already striving to develop remedial procedures for improving the ability of children to shift mental set, to devise training methods for enhancing the ability to shut out extraneous cues, and to discover ways of teaching children how to identify stimuli that have reduced cues. Once these techniques have been fully developed and field tested, they will be subjected to rigorous research appraisal and reported.

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Critique

CAUSALITY

The authors confuse the independent variable and dependent variable by being unclear about whether low self-esteem is a cause or effect of dyslexia (see Chapters 2 and 3). In fact, they state that "lowered self-esteem leads them to believe that they will not be able to learn or to keep up with other children", thus implying low self-esteem may cause dyslexia. Equally likely (and perhaps even more compelling) is the possibility that low self-esteem is a consequence of having a learning disability. The same could be true of locus of control.

Even if this issue is disregarded and the predictor variables are considered to be independent variables, they could lead to overly specific conclusions had the expected results been obtained (see Chapter 9). Self-esteem and locus of control have been associated with so many disorders that it is inappropriate to think of them as having a specific causal link that is central to dyslexia in particular.

GROUPS

The study makes specific reference in the title and in the text to fifth-grade girls with dyslexia. Contrast groups of students without dyslexia and the two sexes have been correctly used. But, no mention is made in the discussion of the failure to find the predicted interaction between the two predictor variables.

Suggestions in the text that the anticipated effects peak at about the fifth grade beg for contrast with other grades. The fifth grade should be bracketed, preferably with lower and higher grades (see Chapters 3, 5, and 6). This would enable the authors to determine whether the results are specific to fifth graders and would have resolved the speculation in the Discussion about the participants being too young for the tests to be valid.

DEPENDENT MEASURES

Both of the personality outcome measures (see Chapter 7) are inappropriate because they require reading on the part of participants with dyslexia. Items on the Minnesota Self-Esteem Scale could easily be misread or answered inaccurately by respondents with dyslexia who are guessing at meanings. The same is true of the Luft Locus of Control Scale. The fact that the variability of responses for children who had dyslexia is much greater on these scales than it is for the readers without dyslexia suggests that the children with dyslexia may have had trouble reading the questions and were randomly responding. The similarity of the means of the two reading groups could well be accidental and not reflective of a true similarity between the groups.

The Stroop Color–Word Test also requires reading. This could at least partially account for the greater number of errors by the children with dyslexia. The result may have more to do with dyslexia itself than with the diminished ability to shut out extraneous cues. In addition, the color–word conflict featured the colors red and green. The likelihood of red–green color blindness being an unanticipated confounding variable for some of the children could have been prevented by changing the colors to blue and yellow or by testing for color blindness and restricting the sample to children with normal color vision.

Inferior performance on the Fuzzy Copy Technique may also reflect poor reading skills as much, or more than, inability to identify words from minimal cues. The children with dyslexia likely would have performed more poorly than the readers without dyslexia even if they had been shown the original copy instead of the fuzzy copy.

In trying to explain the results, the authors cast doubt on the validity of the personality measures that they used. However, even if the personality measures were

highly valid for students who could read and comprehend the items, the validity of their use in this study with individuals who did not have the requisite reading skills is in serious question (see Chapter 7).

DATA ANALYSIS

As has been already noted, on several of the measures there is a considerable difference in variability in scores within the group of the readers who are dyslexic compared with the variability in scores of readers who are not dyslexic. This violation of the analysis of variance assumption of homogeneity of variance was not addressed (see Chapter 8).

With five dependent variables, each treated separately, there is a heightened opportunity for making an experiment-wise Type I error. No attempt was made (e.g., using a modified Bonferroni correction or other method) to reduce the alpha level required to reject the null hypothesis (see Chapter 8). Had it not been for this problem with heterogeneity of variance, an argument could be made in favor of pooling the three perceptual measures and perhaps the two personality measures into synthetic composites, or using two multivariate analyses of variance.

In fact, the description of the logistical regression is incomplete. A logistical regression uses a binary criterion variable and predicts it from multiple predictor variables. You might assume that the researchers predicted group membership for the readers with and without dyslexia and the boys–girls groups separately but this is never mentioned.

Insufficient tabulated material does not allow you to study any of the details of the analyses of variance or the logistic regression analysis (see Chapter 8). Effect sizes are not reported, though you could estimate these from the results of the analyses of variance.

REFERENCES

References to the criteria measures are missing. This is a disservice to readers who are not familiar with the measures and who wish to look them up. Two citations in the text, Nicholson (2005) and Stroop (1935), are missing from the reference list, and two authors (Blither or Bilther and Gruno or Gurno) are spelled inconsistently in the References and in the text citations.

Practice Article 5

Time Estimation: Effect of Depression and Pleasantness or Unpleasantness of an Experience

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Sixty college students were classified as depressed or not depressed on the basis of scores on a depression inventory and were asked to write short essays about pleasant, unpleasant, or neutral experiences. They then were instructed to estimate how much time had elapsed while writing the essays. No support was found for hypotheses that depressed individuals give longer estimates than not depressed individuals and that time spent writing about unpleasant experiences appears to be longer than when describing pleasant experiences. The findings prove that time estimation is unaffected by affective states.

Psychologists have long been interested in how people experience the passage of time (Kluger, 1999). Unlike physical objects, time is an intangible that is not perceived through the usual senses. In ancient cultures, time was associated with observable events such as the rising and setting of the sun, the phases of the moon, and the seasons of the year. Recognizing the advantage of fashioning their own markers, ingenious ancestors invented sand clocks, sun dials, and water clocks. Arriving much later in the history of civilization, mechanical clocks were devised with energy stored in a wind-up spring. An escapement device released the energy, bit by bit, so as to move the hands in measurable units. This enabled people to "tell" time accurately in small units. Until clocks were miniaturized and made portable, town dwellers were dependent on the tower clock, church bells, and town criers who periodically announced the time. Pocket watches and wristwatches enabled people to make appointments, to follow schedules, to be more acutely aware of the passage of time, and to be more accurate in their ability to estimate it.

Psychologists have been intrigued by time perception because it is something that is sensed and experienced even though it cannot

The authors wish to acknowledge the assistance of Mary Brunell, Carl Isherwood, Henri Aronson, and Meryl Twill. Requests for reprints should be sent to Jean Malheure, PhD, Psychology Department, Louisiana Wesleyan University, 325 S. Cedric Ave., Shreveport, LA, 71119 or MBrunell@lwu.edu.

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Malheure = unhappiness (literally, bad hour in French); Bontiempo = good time (Italian).

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be touched, lifted, seen, heard, or smelled. According to Kolodny (1994), the realization that this perception could be subject to the influence of the state of the perceiver is something that has sparked inquiry in many quarters.

The observation that "Time flies when you are having fun" has been argued persuasively by Folsom (1979), Yardley and Toll (1982), and Caldarone and Bliss (2008). They maintained that time does seem to pass more rapidly during pleasant experiences and appears to be interminable during unpleasant ones. Experimental studies by Cargill (1989), Ripley and Winkle (1992), and Zimmer (2001) tend to support this view. On the other hand, some have asserted that people defensively block out unpleasant experiences, so that they therefore seem to have passed rapidly (at least in memory). They maintain that individuals tend to linger and to savor pleasant experiences, which consequently seem prolonged. Research by Gladbach (1987) and Latke (2009) provided some support for this point of view.

According to Wallis (1997), time sense seems to be affected by depression. Time seems to "drag on" and to "hang heavy" for individuals with depression (Toliver, 2002). The mechanism for this alteration in time perception is not fully known but has been the subject of much speculation (Archer & Horowitz, 2011). It is consistent, however, with such standard characteristics as exhibiting a low energy level, decreased concentration and attention, diminished involvement in pleasurable activities, and feelings of being slowed down.

Based on the belief that the person as well as the event are part of the equation, the aim of this research is to determine whether pleasant, unpleasant, and neutral experiences yield different time estimations and whether they are related to depression in the perceiver. The following hypotheses were tested:

1. There will be a significant variation in time estimation of pleasant, unpleasant, and neutral time experiences. The shortest estimates are predicted for pleasant experiences, followed by neutral ones, and the longest estimates are for unpleasant experiences.
2. Individuals who are depressed will perceive time as passing more slowly.
3. Individuals who are depressed will give longer time estimates than will individuals not depressed.
4. The longest estimates will be given for unpleasant experiences by individuals who are depressed and the shortest by individuals not depressed to pleasant experiences.

Method

Participants

Sixty college students volunteered to participate in return for free movie tickets. Half were males and half were females. They were given information about the study and then were asked to sign a consent form. Those who agreed to participate completed the Hilger Depression Inventory. They were then given an appointment for the experimental session.

Procedure

Participants were first divided into groups of participants who were depressed and not depressed. Scores on the Hilger Depression Inventory were used for classification. All those above the median of the total sample (20.63) were placed in the depressed group and those who scored below the median were assigned to the not depressed group.

All participants in both groups were first exposed to a neutral experience as a baseline condition. Half of the participants, randomly assigned from each of the depressed and not depressed groups then received an unpleasant condition, whereas the remaining half were exposed to a pleasant condition. These conditions are described next.

Stimulus Conditions and Time Estimations

Neutral. Participants were seen individually and were asked to remove their watches and to place them inside a brown envelope that was

on the table. There were no clocks in the room. They were then given a pencil and lined paper and were asked to write a description of their living room at home. They were instructed to continue writing until told to stop. When 5 min and 30 sec had elapsed, they were asked to stop and to estimate how much time had passed. They had known in advance that they were going to be asked to make a time estimation; that had been the reason given for the removal of their watches.

Pleasant. After completing the neutral task and time estimation, participants assigned to the pleasant condition were asked to write a description of a pleasant experience that they had had last summer. This was followed by an estimation of the time spent writing that description.

Unpleasant. After completing the neutral task, participants assigned to the unpleasant condition were asked to write a description of an unpleasant experience that they had had last summer and then to estimate the time spent on that task.

Measure of Depression

The Hilger Depression Inventory (Hilger, 2009) is a measure of depression that is specifically normed for young adults and was therefore thought to be most appropriate for use in this research. It has good convergent validity, correlating $r = .75$ with the Beck Depression Inventory, $r = .69$ with the Hamilton Depression Sale, and $r = .62$ with the Depression Scale of the MMPI. It has a split-half reliability of $r = .85$. Scores on the inventory can range up to 100, and individuals who score 50 or more are considered to be clinically depressed.

Results

Data were tested for normality of distributions and homogeneity of variance. On both of these counts, the data were determined to meet the assumptions for use of analysis of variance. Means of time estimates under the three conditions by the two groups are shown in Table 1. The obtained mean time estimation for the group with depression was 20.53 sec longer than the mean of the not depressed group. This difference, however, was not statistically significant (see Table 2), $F(1, 84)=1.287$, ns. The respective means (in seconds) for the pleasant, neutral, and unpleasant conditions were 330.67, 342.3, and 377.43. These, too, although in the predicted direction, did not vary significantly, $F(2, 84)=2.413$, ns. There was no significant interaction between the two independent variables, $F(2, 84)=1.166$ ns.

Table 1
Means (in Seconds) of Time Estimations for Depressed and Nondepressed Participants Under Neutral, Pleasant, and Unpleasant Conditions

Condition	Depressed ($n = 15$)	Nondepressed ($n = 15$)	Total
Neutral	370.33	314.27	342.30
Pleasant	325.00	336.33	360.67
Unpleasant	385.87	369.00	377.43
Totals	360.40	339.87	350.13
	($n = 45$)	($n = 45$)	($n = 90$)

Discussion

None of the hypotheses was supported. Time estimates were not shown to be a function of depression, and they were not affected

Table 2
Analysis of Variance of Time Estimates for Two Groups Under Three Conditions

Source	Sum of squares	*df*	Mean square	*F*
Condition	35,568.067	2	17,784.033	2,413
Diagnosis	9,486.4	1	9,486.4	1,287
Condition × diagnosis	17,186.6	2	8,593.3	1,166
Error	619,047.333	84	7,369.611	

by the pleasantness, unpleasantness, or neutrality of the experience. These findings challenge earlier theoretical formulations and research findings that suggested that the perception of the passage of time is dependent on what is going on in the phenomenal world of the perceiver, particularly in terms of its pleasantness or unpleasantness. These data prove that one's intangible time sense, like the physical senses, is relatively unaffected by internal states. It places time as a true "sense" that is right in line with the tangible physical senses.

To solidify this conclusion, further research is suggested with other affective states and with other stimulus conditions. In the interest of greater generalizability, it would also be illuminating to replicate the study on broader samples with a greater range of age, education, IQ, and other demographic variables than were present in our relatively restricted and homogeneous college student sample.

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Critique

INDEPENDENT VARIABLES CONDITIONS

The validity of the research hinges on exposing participants to experiences that are truly pleasant, unpleasant, or neutral. It is quite possible that some people would find the experience of having to sit down and produce a written description or personal story to be intrinsically unpleasant, regardless of whether they had to write about pleasant or unpleasant events in their past. Others might take pleasure in writing, even if it were a description of an event that was unpleasant at the time that it happened. There is also no assurance that thinking about one's living room at home would be neutral for everybody. Thoughts of home and its living room might arouse very pleasant feelings in some participants and markedly unpleasant feelings in others. Neutrality cannot simply be assumed. No manipulation check was used, not even as simple as a self-rating of how pleasant or unpleasant each participant actually found the writing experience to be (Chapter 7).

DEPRESSION

Whereas people could be randomly assigned to conditions, which were manipulated by the experimenters, diagnosis is formed by static-group memberships to which participants obviously cannot be randomly assigned. The possibility that static groups may not be equivalent in other respects is always a possibility that requires attention (Chapter 3); it was not addressed in this study.

More crucial, however, is the issue of the accuracy of classifying participants into groups. The researchers used a median split of the scores on a depression inventory, classifying all of those who fell above the median as depressed and those who fell below the median as not depressed. This is a risky procedure (Chapter 3).

Whenever a trait or characteristic is normally distributed, people who closely resemble each other are more likely to congregate near the middle of the scale than anywhere else. Participants who are 1-point apart in a 100-point inventory (undoubtedly closer than the standard error of the measure) are placed in contrasting groups despite their similarity. There is no good rationale for calling one depressed and the other not depressed. A median split contributes to the possibility of making the Type II error of failing to reject the null hypothesis when in fact it is false. In this study, participants were placed in categories on the basis of continuous inventory scores.

Some would argue in favor of retaining the scores as continuous and doing a multiple regression analysis to obtain F-values (Chapters 3 and 8). The choice of method depends on the theoretical rationale of the hypotheses. In this study, there is no prediction that time estimates would vary across the entire range of inventory scores. The intent of the researchers was apparently to find out if people who are truly depressed would perform differently than people who are clearly not depressed. They therefore transformed scores into the depressed and not depressed categories. The median split, unfortunately, did not allow them to meet their categorization objective. Furthermore, the normative cutoff to qualify for classification as depressed is 50 (the higher the score, the greater the depression). The median score of the study sample was 20.63. Obviously, then, people with scores well below 50 were being erroneously categorized as depressed (Chapter 3). It is possible that none or very few of the participants could have been legitimately classified as depressed.

Neither of the two independent variables, condition and diagnosis, instills confidence. The study has highly questionable internal validity.

STATISTICS

The researchers used a peculiar design in which all of the subjects were exposed to the neutral condition. In addition, half were exposed to the pleasant condition and half to the unpleasant condition. Within-subject and between-subject error variance are not separated in the analysis. The actual design is summarized in Table 5.1.

Half (1–15) of the 30 depressed subjects are measured again under "pleasant" conditions and the other half (16–30) are measured a second time under "unpleasant" conditions. Half of the 30 not depressed subjects are repeated under pleasant (31–45), and half (46–60) are repeated under unpleasant (46–60) conditions. Therefore, there

TABLE 5.1
Actual Study Design

Group	Neutral	Pleasant	Unpleasant
Depressed			
S	1–30	1–15	16–30
n	30	15	15
Nondepressed			
S	31–60	31–45	46–60
n	30	15	15

were 60 subjects, each measured twice for 120 total measurements. This makes this a repeated-measures design. The analysis of variance was (incorrectly) calculated as if it had been a completely randomized independent group design, with a total sample size of 90, instead of the actual sample size of 60, divided into six different groups (Chapter 7). The degrees of freedom for error are necessarily wrong, and an invalid estimate of error has been used. No mention is made of the missing subjects under the neutral condition (Chapter 8): In Table 1 and in the data analysis, 15 participants who are depressed and 15 not depressed subjects disappear from the neutral condition.

The total seconds in the pleasant condition in Table 1 (fourth column) is incorrectly reported as 360.67 instead of 330.67 (the average of the two depression conditions). No effect sizes are reported.

ORDER

The neutral condition is given first to all participants, creating a possible order confound. This problem could have been avoided, along with the statistical problems cited previously, had there been 45 participants who are depressed and 45 not depressed randomly assigned to the neutral, pleasant, or unpleasant conditions.

DISCUSSION

The authors' claim that they have found proof that internal states have no effect on time estimation is incorrect because the null hypothesis cannot be proven. Failure to find support for the hypotheses may represent a Type II error brought on mainly by the lack of internal validity (because of the study's two flawed independent variables) and faulty statistical analysis.

ABSTRACT

The Abstract summarizes the flawed results and falsely restates "proof" of the null hypothesis. Anyone who relies on this abstract without reading the article would be misled.

Practice Article 6

Bossiness in Firstborn Girls

Beverly Capofiglia
Southern California Child Study Center

The social interactions of 40 firstborn girls and 40 firstborn boys was observed in 20-min play periods when they were paired with younger children. Bossiness was rated by two independent judges using video recordings of the sessions. Girls were significantly more bossy than were boys, and the highest mean was observed when firstborn girls were playing with younger girls.

Birth order of children is alleged to have considerable influence on their personality development (Schwartz & McNair, 2002). Firstborn children, particularly sons, have traditionally occupied a privileged place in families and in society. Historical laws of primogeniture favored the firstborn in the inheritance of properties and titles. In families, the firstborn becomes the centerpiece whose every move and developmental advance is celebrated. Firstborn children are doted on and are given as many advantages as the family can afford. The place or rank of those who are first born is guaranteed by the fact of their being, whereas later-born children have to "try harder" to establish their place. For many crucial years, the firstborn is bigger, knows more, and possesses more than younger siblings. Younger children have to make do with hand-me-downs from older siblings who have outgrown clothes and toys. Studies by Polard and McQuirty (1996), Engles (2000), and Thurbridge and Caruthers (2005) have all found differences in temperament and personality in firstborns compared with later-born children.

Of special interest in the present study are differences in the ways that firstborn children relate to other children. As stated by Flynn (1985), "Firstborns who have younger siblings are used to dominating and having their way. They jealously guard their rank as numero uno" (p. 84).

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Capofiglia = Boss-girl (Italian).

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The form that this domination takes appears to differ in boys and girls. Boys tend to dominate physically, to "throw their weight around" literally and figuratively (Milsap, 1989, p. 18). By contrast, girls tend to dominate verbally by ordering their younger siblings around. According to Ebel (1983), they become bossy as a way of guarding and enforcing their dominant position. As Ebel (1983) described it, "Males stand up and challenge all-comers with sharp-swords as king of the hill, whereas females use sharp tongues to organize things to their own advantage as queen of the castle" (p. 99). These views may appear sexist by today's standards, yet they probably still contain instances of verifiable differences in how boys and girls are treated differently as they develop.

The word *bossy* has several meanings. As a noun, it is an endearing term for a cow. There is no doubt, however, of its pejorative meaning when used in its adjectival form to describe the behavior of people. This distinction is epitomized in the old Scottish saying, "Better a lazy bossy than a bossy lassie." People are regarded as bossy when they adopt the worst qualities of a boss by being dictatorial, high-handed, arrogant, overbearing, domineering, and officious in their use of power.

The purpose of this research is to determine whether firstborn girls are indeed bossy and, if so, whether they are bossier than firstborn boys. Cognizant of the possibility that the behavior of girls might vary as a function of the sex of the person with whom they are interacting (the protagonist), sex is made into an independent variable in this research. The prediction is that firstborn girls are bossier than firstborn boys, and particularly so when the protagonist is of the same sex.

Method

Participants

A total of 40 girls and 40 boys, ages 4 to 5 years, participated in this research; all of them were Caucasian, firstborn, and had at least one younger sibling. The children were recruited from five private preschools in Orange County, California. Parents of these children were mostly affluent, college graduates who occupied professional or managerial positions. Parents were approached through school meetings, and all volunteered the participation of their children. Parents signed a consent form and were promised a full report of the findings at a future school meeting. Parents were assured that the performance of individual children would remain confidential. All children were number coded for purposes of data analysis.

Protagonists were 3- and 4-year-old younger siblings of the study participants. They were not placed in interaction with their own siblings to see whether behavior patterns generalized outside of the family.

Setting and Play Sessions

Attractively decorated playrooms, approximately $10' \times 10'$, were equipped with assorted toys and were set aside in each school for study purposes. Two strategically placed video cameras attached to computers unobtrusively recorded the interactions within the room. Each firstborn child was paired with another younger and unrelated child. The two were introduced to each other and were instructed to play while their mothers were "right outside talking to the teacher." Half ($n = 20$) of the firstborn girls were paired with other girls, and half ($n = 20$) were paired with boys. Half of the firstborn boys ($n = 20$) were paired with boys, and half ($n = 20$) were paired with girls. Sessions lasted 20 min.

Ratings and Raters

Ratings of bossiness were made from the video recordings. A Checklist of Bossy Behaviors was constructed for this research. It included such behaviors as (a) tells other what to do, (b) tells other what not to do, (c) tells other how to do something, (d) insists on having own way, (e) claims possession of toy or game,

(f) makes demands on other, (g) verbally intimidates other, (h) puts other down, (i) behaves in overbearing, dictatorial manner, and so on. Specific examples were given to differentiate *telling* or *commanding* from *suggesting*. An example of suggesting was, "Would you like to play house with me?" An example of telling/commanding was, "Let's play house. I'll be the mother; you'll be the baby." Parallel examples were furnished for other bossy behaviors.

One male and one female psychology major served as raters in return for course credit for assisting with the research. They were given orientation about how to rate and were asked to study the manual before beginning. Ratings were made independently by the two raters at different times. They had access to stop–start replay apparatus so that they could back up to take another look at a scene if they needed to make sure of what had transpired.

The older child was identified by a head-on camera shot that appeared at the beginning of the recording. The recordings (all of which were 20 min in length) were divided by markers into 20 sec segments. There were 60 segments in each recording. Raters were asked to circle yes or no to the question, "Did the older child initiate bossy behavior in this segment?" If there were more than one incident per segment, it was counted as only one. An incident that was continued from the preceding segment was not counted again. When the data were collated, behavior during a segment was considered as bossy only when both raters agreed. Total scores could range from 0 (no bossy behavior) to 60 (bossy behavior initiated in all segments).

Table 1
Mean Bossiness Ratings of Participants When Playing With Boys and Girls

	Protagonists		
Participants	Girls	Boys	*M*
Girls	30.50	24.50	27.50
Boys	20.50	19.20	19.63
M	25.28	21.85	23.56

Results

Data were treated in a 2 × 2 analysis of variance (ANOVA). Tests of skew, kurtosis, and homogeneity of variance met ANOVA assumptions. The respective means for girls and boys were 27.5 and 19.63 (see Table 1). As predicted, the highest mean, 30.5, was obtained by girls when playing with younger girls. This is in contrast to their mean of 24.5 when playing with boys. A Tukey test revealed this difference to be statistically significant ($p < .01$). The means for boys were 20.5 when interacting with girls and 19.2 when interacting with other boys. As can be seen in Table 2, girls were significantly more bossy than were boys ($F(1, 76) = 10.694$, $p < .01$, $d = .75$). Table 2 and Figure 1 indicate that the Participant Sex × Protagonist Sex interaction was significant ($F(1, 76) = 4.131$, $p < .05$).

Table 2
Two-Factor Analysis of Variance of Participants' and Protagonists' Gender

Source of variation	*df*	Sum of squares	Mean square	*F*
Participant gender	1	990.312	990.312	10.694**
Protagonist gender	1	234.613	234.613	2.533
Participant gender × protagonist gender	1	382.613	382.613	4.131*
Error	76	7,038.150	92.607	

*$p < .05$, **$p < .01$

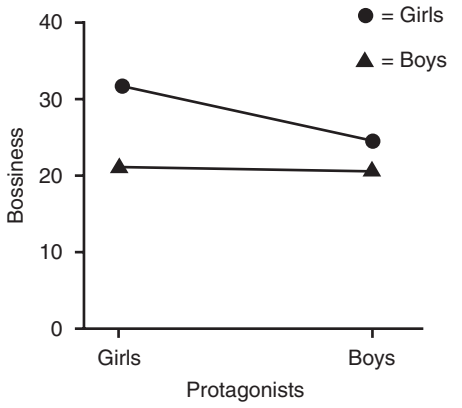


Figure 1. Mean bossiness scores by subject gender and protagonist gender.

Discussion

The aim of this research was to test the hypothesis that firstborn girls, as a group, have a distinctive character trait of bossiness. When contrasted to firstborn boys, this proved to be the case: The data clearly show that firstborn girls are significantly more bossy than firstborn boys. The bossiness ratings of the girls were higher when they were interacting with other girls than they were when interacting with boys. These data support the experimental hypotheses and lend credence to the formulations of Ebel (1983) and Milsap (1989). Not only does birth order make a difference in the trait under inquiry, but it has a differential effect on boys and girls.

We must keep in mind that we were dealing here with preschool children. It would be of interest in future research to extend the finding to other age groups. Only-children, although firstborn, were intentionally excluded from the study, but it would be illuminating to see whether they share the same behavioral characteristic of bossiness shown by firstborn girls who have younger siblings. It would also be worthwhile to develop a program of intervention that could ameliorate this type of behavior in firstborn girls.

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Critique

CONTRAST GROUPS

Essential contrast groups consisting of later-born female and male children are missing. In the absence of data about the level of bossiness of later-born children, no statements can be made pertaining to the distinctive characteristics of firstborns. The conclusions could be different if later-born children turned out to be even more bossy than firstborns (see Chapters 3 and 6, this volume).

SAMPLE

The sample is too homogeneous in several regards for generalizations to be made about people who are first born (see Chapter 4). The children are from affluent Caucasian

families. There is the possibility that children from affluent families, who may be more used to getting their way, are more bossy than children from families with more modest means. If this is true, the results could be at least partly attributed to the economic status of the children. This could be determined or ruled out with a broader sample.

Furthermore, all the children in this study are 4 to 5 years old. Perhaps children are in a particularly bossy stage at this age and would not show the same behavior at age 8 or 12. To the extent that these speculations are true, age and socioeconomic status rather than birth order might account for most of the variance in bossiness.

All the protagonists were younger than the participants. Therefore, it cannot be determined whether bossy behavior is restricted to interactions with a younger protagonist. A more representative sample (see Chapter 4) and appropriate contrast groups are needed if generalizations are to be made. The question about whether family patterns generalize outside of the family cannot be answered because family patterns are never observed: All of the pairs are unrelated. Data obtained by observing children with their own siblings are required to address this question.

Do you remember the conditions described in Chapter 2 that had to be met for a statement to be taken as causal? Condition #2 said there needed to be few side conditions. In this study, the defensible statement would be "Girls are bossier than boys but we know this only for affluent, Caucasian, 4- to 5-year-old girls interacting with a younger, nonfamily member in a controlled setting." Does this evidence convince you of a causal link between sex and bossiness? It might be a piece of the puzzle but there are many pieces missing.

DEPENDENT MEASURE

The dependent variable measure is a rating from a checklist that has not been validated or standardized. Without norms, it is impossible to know what score level is indicative of bossiness. The obtained means could represent high, average, or low levels of bossiness; they cannot be interpreted without norms. You would have more confidence in the measure had the authors demonstrated criterion-related validity by showing that the checklist discriminates successfully between groups of children who are known to be of different levels of bossiness (see Chapter 7).

RATERS AND RATINGS

Two undergraduates are used as raters after having received "orientation." Whenever behaviors may be ambiguous, as they certainly are here, raters should be given extensive training, not merely orientation. Raters should be trained until there is reasonable certainty that they fully understand the rating categories and are able to identify the pertinent behaviors accurately. Their ability to do this should be pretested before they are exposed to the research rating task. There are 20 minutes of recordings from each of 80 participants, for a total of 1,600 minutes (26.7 hours) of viewing. If this is not done by well-trained individuals under conditions that maximize motivation and alertness, inaccuracies can be expected (see Chapter 7). Also, you are not told whether raters were unaware of the hypotheses of the study.

Unanimous agreement of two judges who are asked to make a dichotomous judgment could occur by chance. If the two judges were blindfolded and wore earplugs when shown the tape, they still would agree on 50% of the judgments (assuming bossy behavior occurred half the time). No interrater reliability data are provided. Lacking any information to the contrary, and knowing that the odds on agreeing by chance are great, you have to consider the possibility that much of the data consisted of chance agreements about ambiguous behavior (see Chapter 4).

DISCUSSION

The authors refer to the "effect" of birth order in this study. This formulation is inappropriate because causation cannot be demonstrated in a static-group design (see Chapter 3). In this study, the homogeneity of the samples in the four different groups rules out many confounding variables. However, there are still clear concerns about making causal statements (e.g., missing control groups, homogeneous sample, unknown reliability of measures) from this one study.

ABSTRACT

The Abstract depicts what was done with reasonable accuracy, but you still must read the entire article to evaluate the validity of the results that are summarized.

Practice Article 7

Effect of Context Upon Accuracy of Recall of Affective Experiences

Rhea Kahl and I. K. Ferguson
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The authors examined the effect of context on recall of neutral stimuli that had been embedded in a series of either funny, sad, or fearful stimuli. The stimuli consisted of cartoons and captions. Neutral cartoons were recalled as being funny when they had been embedded in a series of funny cartoons more often than funny cartoons were recalled as being neutral. Fearful cartoons were remembered as being neutral more often than neutral ones were recalled as being fearful. Context had no effect on recall of neutral stimuli that had been embedded within a sad series. The role of defensive mechanisms in neutralizing fearful images in memory is discussed.

The process of memory has long been of central interest for cognitive psychologists, neuropsychologists, personality psychologists, and clinical psychologists alike. This includes how people remember, what they remember, and how memories can be distorted over time. The research reported in this article focuses on a particular aspect of memory: distortion that is based on the category of events with which the remembered event is associated.

Some theorists assert that memories of unhappy events that were embedded within a contextual class of happy events will be distorted so that they are recalled as having been happy (Martin & Abelson, 1989). This putatively defensive reaction is akin to the defense of denial according to Herschel and Wall (1992). This formulation, however, may be too simplistic, as Gerber (2000) has pointed out, because it does not explain why pleasant memories that are placed within a context of unhappy events tend to be recalled as unhappy. This, too, could not be considered as denial. Some theorists have tried to explain the phenomenon in terms of schema (Trowbridge, 2100). Memories are fitted into some overall schema. If they are incongruent

This study was supported by a Northeastern Alabama University faculty grant. We are deeply indebted to Marco Sherman for his outstanding artistic endeavors in preparing the stimuli for this research. Correspondence about this research should be sent to Rhea Kahl, PhD, Psychology Dept., Northeastern Alabama University, 15 Marcus Road, Anniston, AL, 36201 or kahlr@NAU.edu.

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Rhea Kahl = Recall; I. K. Ferguson = I forget (in German and Yiddish).

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with the schema, they are altered in a way that makes them fit. Cognitive dissonance theory has been invoked as an explanation by Parador (2007). Dissonant experiences are corrected in memory and the dissonance neutralized.

As an illustration of the phenomenon, Baldwin (2008) wrote, "People remember their years in high school as the best time of their lives. In the bloom of their youth they had no real responsibilities, lots of friends, good times, dances, games, and long vacations. At class reunions, remembrances of happy times dominate the joyful reminiscences of the good old days. Forgotten are the adolescent glooms, the tensions, the awkwardness, the failures, the zits, the rejections, the tests, the powerlessness" (p. 21). In his critique of Baldwin, Farber (2009) wryly observed that those who had a miserable time in high school probably did not attend the reunion. In his rejoinder to Farber, Baldwin argued that Farber not only missed the point but unwittingly lent support to it. Classmates not attending classified that entire period of their lives as unpleasant. Their unpleasant memories predominate, and even the few pleasant events that happened are distorted in recall.

There is much anecdotal clinical evidence of these kinds of memory distortions (Feldman, 1980; Port & Neufchattel, 1989; Williams, 1983). When it comes from a clinical population, the memories are usually sad or fearful. Clients obviously do not go to therapists to report happy experiences. Memory distortions and memory selectivity are to be expected. The schema that govern their selection and representation of past events are dictated by the setting and the purpose (Mossy, 1995).

Experimental evidence is scarce. What does exist is flawed, as Seltzer (2010) concluded in his astute research synthesis and critique of the literature. The purpose of this research was to gain further experimental evidence of this phenomenon. The plan was to present a series of different cartoons with neutral captions that were embedded in a larger series of cartoons (a) with funny captions, (b) with sad captions, and (c) with fearful captions. These were presented on three different occasions. In three subsequent sessions 1 month later, participants were given a recall task to determine whether they correctly remembered the cartoons that were neutral as neutral or whether they distorted them in memory and recalled them to match the particular series within which they were first embedded. The hypotheses of this study are as follows:

1. There will be more funny choices than neutral choices in recall of neutral captions of cartoons that were originally presented within the context of a series of funny cartoons.
2. There will be more sad choices than neutral choices in recall of neutral captions of cartoons that were originally presented within the context of a series of sad cartoons.
3. There will be more fearful choices than neutral choices in recall of neutral captions of cartoons that were originally presented within the context of a series of fearful cartoons.[1]

Method

Participants

The participants in this experiment were 48 college juniors (48% females and 52% males) who were enrolled in a class on Communication. (Initially, there were 59 participants, but 11 had to be excluded because they were absent for at least one of the six experimental sessions.) The class met Mondays, Wednesdays, and Fridays during the spring semester. All participants were advised that their participation was voluntary and were asked to sign a consent form in which they agreed to participate in a study on cartoons as a communication medium. They were not told

[1]The full set of all cartoons and captions is available for research purposes. Details about how to obtain them will be furnished by contacting the author.

that it was a study of memory because conscious effort to remember the cartoons would have confounded the study. The researchers gave them a full explanation of the study at its conclusion and explained the reasons for the deception. The investigators reported the findings to them after the data had been analyzed and led a discussion on the psychology of communication.

Procedure

Funny Stimuli. A series of 21 cartoons with captions were projected on a screen to the assembled class with an exposure time of 30 sec each. The investigator introduced them as a series of funny cartoons. Fifteen of them had humorous captions, and six had neutral captions. One of the latter, for example, portrayed a man dozing off in front of a TV set. The caption had the announcer saying, "Now for the late-night news." The six neutral stimuli were randomly spread throughout the series. One month later, the 21 cartoons were again shown to the class, but without captions. The task of each participant was to recall the original caption. A multiple choice of two captions (one neutral, one humorous) was displayed on a video screen and responses were recorded by a clicker provided to each student. For the cartoon in the previous example the two choices were "Now for the late-night news" or "Now for a late-night snooze."

The cartoons that were originally humorous were presented at the recall session with the funny caption that appeared in the initial viewing along with a neutral caption as the alternative choice. For example, two choices accompanied a *New Yorker*-based cartoon in which a doctor said to a patient "I'm sorry I can't help you. You are too sick for managed care" or "I'm sorry I can't help you. You are too rich for Medicaid."

Sad Stimuli. Two days after the first exposure to the set of humorous cartoons, participants were presented with a set of 15 sad cartoons intermingled with 6 neutral ones. They were introduced to the class as a series of sad cartoons. For example, one of the neutral cartoons was a drawing of a small man holding out his shaggy-haired poodle to a veterinarian and saying, "Please shave my dog." In the recall exposure 1 month later, participants were asked to select the original caption from among the following two captions: "Please shave my dog" or "Please save my dog."

The 15 sad cartoons were presented with the original sad caption and an added neutral caption. In one of these, a man is lying in bed and a doctor is taking his pulse but is glancing toward the window. The captions options were "It looks like he's going" and "It looks like it's snowing."

Fearful Stimuli. Two days after the second series, the group was shown a third series presented as a collection of fearful cartoons. Fifteen had fearful captions, and 6 had neutral ones. By way of example, one of the neutral cartoons portrayed the back of a woman whose figure partly obscured a man who was facing her. He was apparently holding something in his hand, but only his raised forearm could be seen. The caption was, "I'm on a diet. I beg you to put that bun away." One month later, when the cartoons were shown again, the students were asked to recall the original and were given the following two captions: "I'm on a diet. I beg you to put that bun away" and "I don't want to die. I beg you to put that gun away."

An example of an originally fearful cartoon, with a neutral caption added for the recall session, was of a dimly lit scene of a large man confronting a woman. The two choices included "Let go or I'll scream" and "Let's go for ice cream."

For the funny, sad, and fearful stimuli alike, the two captions that were presented for recall were counterbalanced. Half presented the original neutral caption first, and half led off with the affective caption that they had not seen previously.

Results

The mean percentages of distortions (incorrect choice on recall) were compared in three separate analyses: (a) for the funny series, (b) for the sad series, and (c) for the fearful series. Because there were 15 cartoons used in each of these series to establish the predominant context, but only six neutral cartoons embedded in each series, the numbers of incorrect choices were transformed into percentages. These percentages are referred to as the distortion percentage, and it is used as the dependent variable measure.

Funny Series

The mean distortion percentage (neutral pictures recalled as funny) was significantly greater than funny cartoons that were recalled as neutral ($t(46) = 2.13$, $p < .05$). The means were 33.4% for the former and 15.2% for the latter.

Sad Series

A mean distortion score of 20.5% was obtained for cartoons that were originally neutral but which were recalled as sad when intermingled in the sad series. This mean was not significantly different than the mean of 18.1% for cartoons that were originally sad being recalled as neutral ($t(46) = .84$, *n*s).

Fearful Series

Contrary to expectations, and in distinction to the funny and sad series, the fearful cartoons were recalled as neutral (28.2%) more often than neutral cartoons were recalled as fearful (17.9%; $t(46) = 2.41$, $p < .02$).

The three comparisons of funny versus neutral cartoons, sad versus neutral cartoons, and fearful versus neutral cartoons are illustrated in Figure 1.

Discussion

Partial support was obtained for the hypotheses that predicted that neutral stimuli, after having been seen within the context of affective stimuli, tend to be distorted in recall to match the predominant context of the series. This notion held up admirably for a series that was composed of funny cartoons. Context, however, had no effect on recall of neutral stimuli that were originally embedded within a sad series. A contradictory result was obtained for fearful stimuli. Defensive mechanisms apparently operate to neutralize powerful fearful images, or at least to wall them off and prevent their spread. They become, in effect, "benign," as Antonio (1996) suggested. Funny events become "more so" in recollection and generalize to their surroundings, fearful ones become "less so" and are contained, and sad events are not changed.

There is no evidence, however, that these relationships remain fixed. In this study, the time interval was 1 month. It is quite possible that shorter or longer intervals might have yielded different results. Further research is needed to determine whether fearful events become progressively weaker in memory over longer periods of time. A second variable that may be a factor is the intensity of

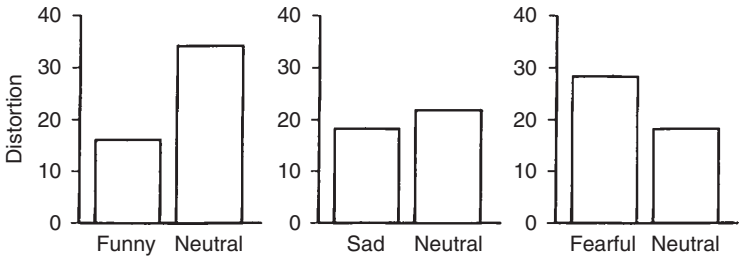


Figure 1. Distortion percentage in funny, sad, and fearful series.

the stimulus or event. Different results could be obtained if the sad and fearful events were real, personal, and more intense.

Another variable that could well be involved is the personality makeup of the respondent. Some people may have a proclivity to place a sad slant on the recall of events, of whatever kind. Others may slant memories in a fearful direction. We need to be cognizant of the possibility that these four variables—context of the event, intensity of the event, personality of the respondent, and amount of time lag—may be operating singly and in interaction with each other to affect recall. There would be considerable value in future research that controlled or accounted for all these variables.

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Critique

HYPOTHESES AND DESIGN

The wording of the hypotheses is inconsistent with the analyses that were done (see Chapter 2). For example, the first hypothesis predicts that "there will be more funny choices than neutral choices in recall of neutral captions of cartoons that were originally presented within the context of a series of funny cartoons." A test of this hypothesis would call for a comparison of the number of funny choices with the number of neutral choices to the six cartoons that were originally neutral. What was done instead was a *t*-test between the mean percentage of incorrect recalls for the neutral cartoons versus the mean percentage of incorrect recalls for the funny cartoons. This approach is more informative, looks at all of the 21 stimuli (not just the six neutral ones), and is closer to the theoretical rationale of the study.

Nevertheless, the design is deficient because respondents were given only two choices in recall of the funny series: funny and neutral. Had they been given the opportunity they might have recalled sadder choices and more fearful choices than either neutral choices or funny ones. This may be unlikely, but it has not been ruled

out. The investigators should have added a neutral series, to address the possibility that people give structure and meaning to neutral events by recalling them as funny, sad, or fearful. Even if the investigators did not consider projection in this form to be likely enough to warrant a prediction, a neutral series could have served as a useful control condition (see Chapter 5).

COMMUNICATION AMONG PARTICIPANTS

All the participants were in the same class. They had an opportunity to discuss the cartoons among themselves, to repeat the jokes, and to share their curiosity about the meaning of the neutral captions (see Chapter 5).

STIMULI CONTENT

The authors state that the captions were funny, sad, fearful, or neutral but they show only a few samples. The cartoons should have been submitted to a panel of judges whose agreement would establish content validity (see Chapter 7). The investigators also should have done a manipulation check (see Chapter 7) in the form of systematic inquiry of the participants to ascertain that they truly perceived the cartoons as intended.

ORDER OF PRESENTATION

The series of cartoons were presented in a fixed order: funny, sad, fearful. An order effect is a possibility, and the researchers might have gotten different results had they changed the order. For example, having seen the funny cartoons first, the participants might have been set to see more funny ones when the sad series was presented. Conceivably, this could have altered the perception of some of the sad cartoons. Because order of presentation may be a variable, the investigators should have made efforts to control it (see Chapter 5).

MULTIPLE TREATMENT INTERFERENCES

The three series of cartoons were presented sequentially with a 2-day interval. The three sets could interfere with each other to affect recall (see Chapter 6). Also, after the first series, the subjects were asked to recall them. This might have cued them to memorize the cartoon and captions in later trials. Multiple treatment interferences should have been controlled by assigning different groups of participants to each series of stimuli.

STATISTICS

The degrees of freedom used for the three linked-pair *t* tests was 46. The correct degrees of freedom is 47, or 1 less than the number of linked pairs (see Chapter 8). This error would not make much difference in this instance, but it is a sign of carelessness that you should note.

DISCUSSION AND INTERPRETATION

The authors claim that the results support the idea that neutral stimuli seen within the context of a funny series tend to be recalled as funny. In actuality, a mean of 33.4% distorted recalls of the neutral cartoons was obtained. Thus, about two-thirds were correctly recalled as neutral. This hardly represents a large effect toward distort (see Chapter 9). By the same token, the mean of 28.2% distortions of neutral stimuli that had been embedded in the fearful series does not reflect a clear tendency to defuse fearful events. Evidently, a mean of 71.2% of the recalls were accurate. The three disparate results—one positive, one neutral, and one negative—may simply reflect chance variation among three analyses of data coming from a study that contained a number of uncontrolled variables.

Practice Article 8

The Effects of Medication and Cognitive Behavior Therapy on Insomnia

Eleuthra Gaye-Schloffen and Sam Nussbaum
Bainbridge Sleep Center

This study on the treatment of insomnia compares medication (Flummoxadon), cognitive behavior therapy, and a wait-list control group. The patients in each group kept a sleep diary for 4 weeks of treatment and 4 weeks of follow-up. Results showed that the medication group (89%) and the psychotherapy group (75%) had higher rates of improvement than the wait-list controls (47%).

In one form or another, insomnia, a sleep disorder, affects virtually everyone at some time. More often than not, difficulty falling asleep or maintaining sleep is a transient episode associated with particular life events or physical states (Kittle, 2006). Even when it is chronic, some people are able to function effectively and appear to get along on less sleep than most people require. For others, insomnia is a persistent pattern that leaves them fatigued the following day and interferes with work and other activities. Insomnia can take several forms: difficulty in falling asleep, awaking during the night and having trouble getting back to sleep, or awaking prematurely in the morning and not being able to doze off again. These are not necessarily exclusive categories—they may be found in combination in the same individual.

The particular form of the disorder that is the focus of this research is sleep onset. The first resource for individuals with this disorder is usually a pharmacy that carries any number of patent medicines that claim to promote sleep or the health food store that sells herbal and other substances meant to aid sleep onset. Some people with insomnia are attracted to the sedative use of alcohol.

This research was supported by a grant from Archer Pharmaceuticals, Inc. Requests for reprints should be sent to E. Gaye-Schloffen, MD, Bainbridge Sleep Center, 352 So. Meil Ave., Boston, MA, 02123 or ESG@BSC.com.

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Eleuthra = peaceful or free (Greek); Gaye-Schloffen = go to sleep (German, Yiddish); Sam Nussbaum = Somnus (sleep, in Latin); and (in the address) So. Meil = Someil (sleep, in French).

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If none of these remedies brings relief, insomnia sufferers usually go to a general practitioner, who usually suggests that they cut down on the use of stimulants or relinquish them entirely, exercise more, avoid dietary indiscretion, and, if all else fails, take a prescription drug that is a more potent sedative than anything that can be bought off the shelf. If this regimen fails, individuals may be referred to a mental health professional for appraisal and amelioration of psychological factors or to a sleep clinic that specializes in sleep disorders. If nothing works, people with insomnia have no choice but to be stoic and to spend their days fighting off the onset of somnolence that remains out of reach when they are tossing and turning in bed at night.

Insomnia, or dissatisfaction with sleep, is classified in the *Diagnostic and Statistical Manual of Mental Disorders* (fifth ed.; American Psychiatric Association, 2013) as difficulty initiating and/or maintaining sleep on at least three nights per week for at least 3 months. It leads to fatigue, sleepiness, poor concentration, poor mood, or the impaired social activities during the day.

The present investigation compares and contrasts the relative efficacy of a newly available prescription drug and a relatively new form of short-term cognitive–behavioral approach. Both are used with individuals having a sleep disorder of the sleep-onset variety. The medication, Flummoxadon, is orally administered in the form of a tablet; is alleged to act rapidly; does not cause stomach upset; can be safely taken by pregnant women, people with respiratory disorders, and men with benign enlargement of the prostate; and is not followed by fatigue and lethargy the following morning. It is meant to be used exclusively at bedtime, and it is not to be used before operating machinery or driving a motor vehicle. Developed and used with reported success in the Czech Republic (Švejk, 2010), it has not yet been widely used in the United States.

The Seldrick (2012) manual for the behavioral treatment of insomnia is reported to have been used effectively at the Garbrand Sleep Clinic, but the data are largely anecdotal in nature (Ingraham, 2013). We have used it along with other treatments at our clinic since 2011. Clinical results appear to be favorable, but there remains a need to establish its value scientifically and to determine how it compares with medication. Our intention, then, is to appraise these two contemporary approaches side-by-side in a naturalistic field setting and to compare participants with an untreated control group.

Method

Participants

Participants were all those who responded to a newspaper advertisement for people who were interested in receiving free treatment for difficulty in falling asleep. A total of 135 individuals applied. Each applicant filled out a detailed health history form and was interviewed by a specialist at the Bainbridge Sleep Center to screen for any coexisting physical or mental disorder, and to judge whether the claimed insomnia fell within the established criteria for the disorder. Anyone who was currently being treated for insomnia elsewhere, taking medications that might contaminate the results, or admitted to having a substance abuse or alcohol problem was eliminated. Where there were any doubts, the applicant was interviewed by an internist, a neurologist, or a psychologist as appropriate. All participants were at least 21 years old.

To keep the research as naturalistic as possible, all of the participants were asked to choose whether they would receive medication or psychotherapy. In real-life situations the patient has that choice. Giving participants that choice in research equalizes the expectancy factor, which can be crucial in the outcome of treatment. In addition, the medication was not administered in a double-masked design because it is not possible for the patients receiving psychotherapy (or for their therapists) to be unaware of what they were doing. Just as there was no placebo therapy, there was no placebo drug.

As soon as 25 participants had elected one of these treatments, the quota for that treatment was considered to be filled. After 25 people had been placed in each of the two treatment groups, all others who expressed a preference for these groups were placed on a waiting list and were promised that they would receive the treatment of their choice in 8 to 10 weeks. Assignment continued in this fashion until there were 25 people in the waiting list group. They were told that they would be expected to keep a record of their sleep experiences while waiting. Demographic characteristics of the three groups of 25 participants each are shown in Table 1.

Treatment Conditions

Medication. Participants who were assigned to the medication group were given a 1-week supply of Flummoxadon on their second visit, with instructions to swallow one 5-mg tablet each night 30 min before bedtime. They were required to come to the Sleep Center once a week to check for any side effects and to receive a renewal of their medication. They were also to bring in their sleep diary (described next). This regimen was continued for the 4 weeks of the treatment phase and 4 weeks of a follow-up phase, participants who failed to come in on a scheduled date and who could not be reached by phone to schedule a make-up visit were dropped from the program.

Psychotherapy. Patients who were assigned to the psychotherapy group were required to come to the Sleep Center twice a week for 4 weeks beginning with their second visit. At this time they participated in individual 45-min sessions. Sessions were conducted by three experienced Sleep Center professionals who were instructed to follow Seldrick's (2012) manual. Sessions were spot-checked by a supervisor who sat in on random sessions. Participants were expected to bring in their updated sleep diary at every

Table 1
Demographic Summary of the Three Groups

Characteristic	Medication	Psychotherapy	Wait list
Mean age (years)	28.6	52.8	41.4
Gender			
Male	62%	32%	44%
Female	38%	58%	56%
Marital status			
Single	28%	30%	29%
Married	44%	42%	40%
Divorced/widowed	13%	16%	15%
Cohabitating	15%	12%	16%
Occupation			
Professional/management	25%	48%	26%
Clerical/sales	36%	25%	32%
Skilled	28%	15%	29%
Semiskilled	10%	11%	10%
Unskilled	1%	1%	2%
Education			
High school graduate	37%	22%	38%
Some college	28%	32%	28%
College graduate	29%	35%	26%
Graduate school	6%	11%	8%

session. Attendance of at least six out of the eight sessions was required for continuance in the program.

Waiting List. Patients on the waiting list were supplied with prepaid envelopes and were asked to mail in their sleep diaries on a specific day once a week.

Sleep Diary

All participants were given printed forms on which to keep a daily log of their sleep patterns. They were told to enter their most accurate estimate of the time it took for them to fall asleep the night before; this was to be done each morning, as soon as they woke up. The estimate was to be from the time that they set out to initiate sleep to the time that they actually fell asleep. Time estimates were to be made in minutes. They were also to note whether they felt rested or overly fatigued on awakening. The pretest was the recording of the first night before any medication had been taken or other treatment received. The estimate on the first night for those on the waiting list was also used as the pretest.

Results

Of the 75 original participants, 18 either dropped out, had to be excluded because of incomplete data (failure to properly maintain sleep diary), or failed to meet the attendance requirements. These losses were spread fairly evenly among the three groups, so that 18 remained in the group that received medication, 20 in the psychotherapy group, and 19 in the wait-list group.

Participants were rated as either improved or unimproved on the basis of a comparison of their pretest and posttest sleep-onset estimates. The posttest estimate was derived from the average of estimates made over the 4-week follow-up period. The data were analyzed in a 2×3 chi-square analysis. The result of this overall analysis yielded ($\chi^2(2, N = 57) = 9.70$, $p < .01$). Frequencies for this analysis are shown in Table 2. Here it can be seen that 89% of the medication group improved in comparison to 75% of the psychotherapy group and 47% of the wait-list group. A second analysis omitted the wait-list group and isolated the medication and psychotherapy groups for comparison. The advantage of the medication group is not statistically significant ($\chi^2(1, N = 38) = 1.22$, $p > .28$, $OR = 2.66$).

Table 2
Chi-Square Analysis of Improvement for the Three Groups

Status	Medication	Psychotherapy	Wait list
Improved	16	15	9
Unimproved	2	5	10
Total	18	20	19

Discussion

The results of this research clearly demonstrate the benefits of Flummoxadon and the Seldrick approach to the behavioral treatment of sleep-onset disorder. Both of these treatments resulted in significantly more improvement than was shown by a wait-list control group. Although the percentage of those who showed improvement was slightly higher for the medicated patients than for those who received psychotherapy, the difference was not statistically significant. These results should encourage more widespread clinical application.

It would be of interest to determine whether a group that receives the medication and the psychotherapy combined would show even better results than those receiving either treatment alone. Furthermore, a longer follow-up period would help to determine the durability of these favorable results. It would also be of value to see whether results would be comparable with other forms of sleep disorder.

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Critique

GROUPS

Participants were not randomly assigned to groups. Giving them a choice of treatments in an attempt to make the study naturalistic is described as a virtue of the design. Unfortunately, it adds to the likelihood of obtaining nonequivalent groups and nonequivalent expectancies (see Chapter 4). It is also possible that some of the late applicants who were assigned to the waiting list (because the group of their choice was filled) did not have the same level of motivation as that of earlier applicants. In addition, people assigned to the waiting list had no choice about when their treatment was to begin. This may be clinically realistic, but it imposes a confound when it is done for research.

Inspection of Table 1 reveals nonequivalence in several respects. The group that chose psychotherapy is considerably older than the group that chose drug treatment (52.8 years vs. 28.6 years), with the wait-list group in between. No effort was made to determine the statistical significance of this variation, but there is no doubt that it is a factor that has to be seriously considered. To the extent that sleep patterns and treatment outcomes may be associated with age, this age difference could be an important factor.

Similarly, the male to female ratio varies among all three groups. Men predominate in the choice of medication, women in the choice of psychotherapy. Note the arithmetic error in computing the percentages of men and women in the psychotherapy group; the total is 90% (see Chapter 8). There may also be an imbalance in the occupation and education of the groups. The psychotherapy group appears to contain more individuals who are better educated and have higher occupational status, although the authors did not note or analyze the apparent imbalance in the occupation and education of the groups. As far as expectancies are concerned, there is no assurance that people who opt for medication and those who select psychotherapy have comparable expectations of success or that those administering these treatments convey equal optimism about the outcome (see Chapter 5).

The 18 participants who were dropped from the original 75 appear to be spread fairly evenly among groups. There is, however, no information about whether there was equivalence in groups on the number who were eliminated because of irregular attendance, dropping out, or noncompliance with record-keeping requirements (see Chapter 4). For example, if some of the most difficult and least responsive cases

dropped out of one group, their data would be dropped from the analysis. This would make the outcome for that group look more favorable than it actually was.

TREATMENT

The authors correctly argue that psychotherapy cannot be given in a double-masked manner, meaning neither the client nor therapist would know the condition the client was in. It would have been possible, however, to test the specific effects of the medicine against a placebo substance and the psychotherapy against an "expectancy control" treatment. The latter consists of the same number of sessions as the psychotherapy, but it includes no known therapeutic components other than the expectancy effect that comes from feeling that somebody who cares is trying to help and that benefits are expected. It is similar in many respects to an "attention placebo" (see Chapter 5), but the central intent is to control expectancies.

There is no assurance that patients took the drug as prescribed. Some may have skipped some nights, doubled up on others, or combined the prescribed medication with other drugs. There was no check on this (see Chapter 5).

CRITERION MEASURES

Participants were asked to keep a sleep diary estimating the time it took to fall asleep each night. There was no way of determining the accuracy of these estimates. People cannot note what time they lose consciousness. Self-estimates can be valuable if they are accompanied by a more objective check. For example, these could have included a device worn on the wrist. It would be interesting to know whether the accuracy of self-estimates varies as a function of the kind of treatment given.

COMPLIANCE

Researchers can ask participants to fill out a diary as soon as they wake up each morning, but they cannot assume that participants complied. Some people may remember to make an entry later in the day or may even try to reconstruct several of the memories at a later date. The researcher has no control over the accuracy of these data (see Chapters 5 and 7).

TIME OF MEASURES

The pretest consisted of data from a single night, which were to be compared with an average over multiple nights. The single night could have been atypical. Estimates were obtained throughout the treatment phase, but these measures were not put to any apparent use. The posttest was an average over the 4 weeks following treatment for the two treatment groups, but the medication group continued to take medication throughout this period. The period is erroneously referred to as "follow up" for the medication groups even though it was an active treatment period (see Chapter 6).

DATA ANALYSIS

After requiring participants to go to the trouble of keeping a daily log of sleep onset, the authors collapsed the data into the categories *improved* and *unimproved*. An individual could be classified as improved if the sleep onset estimate changed from 40 minutes to 39 minutes. Detailed (albeit probably inaccurate) data were available that would have given the researchers distributions of sleep onset over time for all groups. This would have enabled the researchers to calculate mean sleep onset for the three groups at various time points. They could then have done a Group × Time analysis of variance to see whether groups differed in sleep onset estimates as they progressed over time.

The chi-square analysis on continuous data that were needlessly converted into categorical data had the effect of wasting data and causing a loss of potentially important information (see Chapter 8). Also, the analysis does not appear to meet the requirement for valid chi-square statistics because the frequency of occurrence of one cell ($n = 2$, for unimproved in the medication treatment) was too low. The odds ratio ($OR = 2.66$) for the comparison of treatments is quite high, suggesting the study may have lacked power to detect a meaningful difference (see Chapter 9).

When you, as a critical reader, see what looks like a missed opportunity for appropriate data analysis, you cannot help but wonder whether the opportunity was really missed or whether the choice was informed by a desire to present the results in the most favorable light (see Chapter 9). It is possible, for example, that the differences between the treated groups and the wait-list group in mean time to fall asleep were not statistically significant, but that the number of people who improved (on the basis of very small changes) favored the treated groups.

RESEARCH FUNDING

A critical reader will note that the study was funded by a pharmaceutical company. Although there is no evidence that this influenced the researchers to present the medication result in a favorable light, the potential for bias should be kept in mind (see Chapter 9).

Practice Article 9

Midlife Crises of Men at Age 50

Richard Malesnatt, Morris Tuskor, and Jonathan Anten
Franklin Health Center

Thirty 50-year-old men were interviewed to elicit their feelings about reaching midlife. The top 10 themes that emerged from the interviews were ranked and discussed. The main issues were concern with physical health and well-being, mental health problems, concern about the health and dependency needs of elderly parents, problems with children, job and career issues, financial worries, issues of goal achievement in life, and apprehension about future security.

Much has been written about midlife crisis and its association with depression, the leading form of severe mental illness (Jamison, 2015; McNulty, 2015; Stritch, 2005). Physical, social, economic, sexual, and interpersonal factors have all been cited (Silverman & Green, 2010) as central to crises of midlife. Whereas the onset of midlife may occur earlier in some individuals and later in others, age 50 is a recognized turning point for most people. As 50 is the halfway point to the usually unattainable age of 100, people's lives begin a long downhill slope at this time. Age 50 is a decade marker that has come to symbolize a new level of maturity. As Seldrake (2002) observed, it elevates a person to the category of "older."

On gaining this status, one's existence is suddenly discovered by The American Association of Retired Persons. All kinds of catalogs advertising health aids ranging from jar openers to bedpans begin to clutter one's mailbox. Suddenly at age 50, people realize that TV commercials about hair dye, baldness, arthritis, dentures, urinary problems, laxatives, and energy restorers are pitched to them, and they begin to take them personally. Health insurers are more likely to reject 50-year-olds, and life insurers raise their premiums as one's life expectancy grows shorter. Medical problems abound as the body begins to complain about its half-century of use and abuse (Aurelio, 2007).

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Malesnatt = males at; Tuskor = two score; Anten = and ten.

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The opportunity to be considered as a "wunderkind" in one's chosen field of endeavor is lost forever. In fact, employers begin to think of a 50-year-old as perhaps "a bit too long in the tooth for the job" (Carney, 2013, p. 74) and too old for retraining for some other occupation. Marital problems come to a head, once children have left home (Sanders, 2013). Problems with elderly parents, difficult children, large mortgages, and growing credit card debt all take their toll. As Liebowitz (2015) observed, anxiety and depression are exacerbated by the realization that, at age 50, the time left to prepare for the uncertainties of the future is not as limitless as it may have seemed a decade earlier.

In the effort to anticipate problems and to provide professional guidance to clients who are approaching midlife, it is important to grasp the principal themes that are of greatest concern to them. This qualitative research aims to go directly to the source and to learn, first hand, what are those concerns.

Method

Participants

The participants in this study were 30 men who had attained the age of 50. Men who were included had celebrated their 50th birthday no more than 11 months before their participation in the project. The participants consisted of a networking sample that was organized by the three investigators. Each investigator began with a few acquaintances who expressed willingness to participate. They asked these men to refer them to other men who were within the required age range. Each co-investigator recruited 10 men in this fashion.

In addition to being 50 years of age, the participants had to be sufficiently articulate, introspective, and willing to disclose themselves to professional interviewers for research purposes. Confidentiality was assured to the participants, and all of them were offered a free counseling session in return for their participation. Demographic characteristics of the final study sample are shown in Table 1.

Table 1
Sample Characteristics

Characteristic	Percentage
Education	
Graduate school	58
College graduate	32
Some college	10
Occupation	
Professional/management	32
Business owner	33
Technical	21
Sales	4
Marital status	
Single	11
Married	58
Divorced/separated	31

Note. Participants' income range was $39,000–$150,000; the median was $69,385.

This background information was obtained by means of a questionnaire that was filled out by the participants before the interviews.

Interviews

The three investigators conducted open-ended interviews with the volunteers that lasted at least 1 hr. None of them saw volunteers that they had recruited. The volunteers who had been recruited by Investigator A were interviewed by Investigator B, those recruited by Investigator B were interviewed by Investigator C, and those recruited by Investigator C were interviewed by Investigator A. The interviews were open-ended, and the interviewers followed the responses that were generated as long as they were fruitful before moving to another topic. All of the interviewers asked the following standard questions when they could be smoothly inserted in the dialogue:

1. What does it feel like to turn 50?
2. Did turning 50 make you reflect on your life? If so, how?
3. In your experience, what major problems come up at age 50?
4. What are the major issues in your life now?

5. What is your outlook on the future now that you are 50?
6. What advice would you give to other men who are going to turn 50?

Each investigator conducted interviews with 10 volunteers. With the permission of the participants, all interviews were recorded. At the end of each interview, the investigator replayed the tape for the participant and asked him to identify what he thought were the major themes that surfaced during the interview. At the end of the session, anyone who felt a need to continue the discussion was offered an appointment. Fourteen participants decided to pursue further the issues that had been raised.

Judging of Themes

When all of the interviews had been completed, the recordings were transcribed verbatim. A 3-inch margin was retained on the right-hand side of each page. Copies of the transcriptions were given to each investigator in advance of the judging conferences. Prior to each conference, everyone studied the transcripts of the cases to be discussed and recorded the themes in the document. The investigator who conducted the interview led that case discussion at the conference. The participant's own designation of themes was added to the mix, so that there were four sets of judgments involved. After the investigators reached agreement about the themes that predominated, they rank-ordered them. When they did this for all participants, they averaged the major themes to attain an overall ranking of themes for the group.

Results

The 10 most prominent themes that emerged are listed in rank order:

1. Physical health: Concern with physical health and well-being, including a wide variety of existing ailments or ones that they were apprehensive about acquiring.
2. Mental health: Concern about symptoms of anxiety and depression predominated, with some mention of a considerable range of disorders.
3. Aging: Concern with getting older and losing one's youth, alteration of appearance, diminished physical prowess, and awareness of issues related to death and dying.
4. Sex and marriage: Problems with spousal, extramarital, or dating relationships, and sexual concerns.
5. Parents: Concern with health and/or dependency needs of elderly parents.
6. Children: Problems associated with children growing up and separating from parents.
7. Job and career: Concern with job productivity and job stability, and misgivings about continued interest and motivation for the work.
8. Financial: Concern about finances and money problems.
9. Achievement: The extent to which the respondent feels that his life goals and status in life have been achieved or were no longer likely to be achieved.
10. Economic security: Concern with future economic security in retirement years.

A variety of other themes were mentioned by individuals but did not come up frequently enough to merit ranking among the top 10.

Discussion

As shown by the ranking of these themes, the primary concern of people turning 50 is their physical health. The aggregation of small but nagging ills, and the presence of larger threats to well-being, apparently come to the fore at the age of 50.

A surprising second concern is with mental health. The stresses and strains of living, and the cumulative problems that are encountered, become an issue at age 50. Judging from the number who sought further discussion of their individual problems, there is a growing recognition of the need for help at

this stage of life. This theme is probably associated with all of the other themes on the list.

As expected, awareness of aging is one of the major concerns. Awareness of getting older, of losing one's youthful appearance, seeing one's hair start to turn gray, one's waistline expand, and one's strength and agility decline, is one of the foremost characteristics of those who have turned 50.

Problems of love, sex, and marriage, too, surface at age 50. Married participants averaged 20 years of marriage. About 24% were already separated or divorced, and some were on their second or third marriages. Many of those who were not married were encountering relationship problems or difficulty finding suitable partners at this age. Sexual problems were associated with physical conditions and with marital relationships that were strained in other ways.

An unanticipated fifth-ranking theme was concern about elderly parents. It takes the form of concern regarding the illness or approaching death of parents and the responsibilities for satisfying the dependency needs of elderly parents. Some reported a strain on their marriage as a consequence. Some felt sandwiched between the dependency needs of their parents and their children. Many had children of college age, and concern with college tuition overlapped with the financial theme. Others were concerned with separation issues. Several men expressed concern with their relationships with their children, with their children's behavior, their children's drug use, or their lack of achievement.

We thought that job and career would have occupied a more prominent place than it did, but it is nonetheless a theme of importance for those reaching 50. The primacy of this theme probably fluctuates with the occupations of the individuals and the economy of the times. Related to job and career, themes of financial well-being are present, but are not as predominant as we expected. Actually, the participants interviewed were fairly well-off financially.

Achievement of life goals is one of the lesser of the top-10 themes. Some thought that life had already passed them by, but the feeling was not widespread. Perhaps they felt that there was still time to get where they want to get in life after age 50.

Lastly, security is a theme that we expect will increase decade by decade. Concern with security and retirement will move to the forefront at ages 60, 65, and 70.

We have found that themes that bother the participants in this study are commonly shared by other people of that age. Awareness of the top three concerns—physical health, mental health, and aging—highlights the need to begin to plan a comprehensive program for people in their 40s to prepare them to deal with the issues that they are likely to encounter in a few years. A coordinated approach to the physical and psychological aspects of aging is recommended.

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Critique

THE SAMPLE

The sample is not remotely representative of 50-year-old individuals or (specifically) 50-year-old men (see Chapter 4). All volunteers were obtained through a networking procedure which attracted a narrow group of friends, acquaintances, and perhaps relatives starting with the researchers. Participants were affluent, educated, and articulate achievers. One cannot generalize from this sample to men or to people generally, as the authors did in interpreting their results.

Very different concerns might have been expressed by less affluent and less successful men whose life experiences and priorities are different. The fact that the participants were burdened with these kinds of problems may have induced them to volunteer for the interview. The offer of further help may have been added inducement. This is an instance in which nonvolunteers might be quite different in outlook (see Chapter 4).

The authors should have made reference to the ethnic composition of the sample.

CONTRAST GROUPS

In this qualitative descriptive study, the researchers attempt to establish commonalities within a sample of 50-year-old men, but they do not differentiate them from anybody else so as to establish their distinctiveness. To state anything distinctive about men, it would be necessary to compare them with a group of women. By the same token, 50-year-old men would have to be compared with men of other ages to determine whether the themes that were elicited differed from the themes of men who are 45, 40, or even 35 years old (see Chapters 3 and 6).

INTERVIEW PROCESS

In light of the introductory remarks to the study, the kinds of questions that were routinely asked, the setting, and the orientation of the interviewers, it is very likely that interviewer preconceptions served as demand characteristics that led participants to emphasize their problems and concerns (see Chapter 5). There was no reference to anybody reaping the rewards of a life well-lived, of a happy marriage, loving children, satisfying career, good friends, and comfortable lifestyle that they had earned by age 50.

THEME JUDGMENTS

The conferencing procedure was potentially contaminated by interpersonal dynamics within the small group of investigators. Independence of judgment is easily lost as the judgment of dominant member(s) prevail. The judges were the authors and the interviewers. Any biases that they had could be carried through all phases of the inquiry

(see Chapters 6 and 7). Many of the people who were being judged were friends and acquaintances of the judges. This arrangement is not conducive to impartiality and objectivity.

It is unclear what rank was given for a theme that did not appear on the list of a given participant but was mentioned by others. If a theme were absent from the list of several participants, the average rank of that theme would be distorted. If these individuals were omitted from the total for that theme, the rank of the theme would be artificially elevated.

DATA

The list of demographic characteristics does not add up to 100% for occupation. Either an occupational category was omitted, or the figures are wrong (see Chapter 8).

INFERENCES

In the absence of appropriate contrast groups, statements like those about (a) themes that "come to the fore" at 50 and (b) recognition of the need for help is age-based are not warranted from the data in this study (see Chapter 9). Even within the population of 50-year-old men, the rank order of themes may be sample specific and therefore incapable of providing the researchers with a basis for general statements.

ETHICS

Participants were promised confidentiality, but the investigators discussed each case and examined transcripts of the interviews. The authors do not indicate that the participants had consented to the contents of their interview being discussed (see Chapter 9). Privacy may have been invaded.

A networking sample, beginning with friends and acquaintances of the investigators, was used. This places each investigator in the dual role of being a participant's friend and being a researcher who is studying the participant.

Practice Article 10

Effect of Job Stress and Stress Management on Low Back Pain

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The effect of job stress and stress management on low back pain was studied in a two-part experiment. In Experiment 1, 60 hospital employees with chronic low back pain were shown to experience more job stress than those without back pain. In Experiment 2, four sessions of stress management training resulted in a greater reduction in job stress for the low-back-pain group than for the control group.

Pain in the lower back is one of the principal reasons that employees use sick leave in commerce and industry (Oglethorpe, 2006). The symptoms may range from mild discomfort to acute and incapacitating distress. The pain may be short lasting, intermittent, or chronic. It attacks people in all walks of life—all ages, sexes, ethnic groups, and occupations. Treatment may include analgesics, muscle relaxants, bed rest, special exercises, spinal manipulation, massage, ultrasound, biofeedback, acupuncture, and surgical intervention. In view of the central location of the disorder, almost all activities, even sitting and sleeping, are affected.

There is little doubt that there are multiple causes of low back pain—structural, mechanical, postural, and trauma-related (Wrigley & Overstreet, 1998). More attention is now being paid to linkages between pain and stress (Shorter & Longstreet, 2013). According to Baumann's (2013) stress-diathesis model of low back pain, the predisposition that some individuals have to low back pain manifests itself when they are under stress. If the truth of this assertion can be demonstrated, it would follow that stress management techniques should be helpful in warding off the onset of low back pain and in alleviating the pain when it occurs.

Job stress is one of the primary sources of stress in adults (Zweig, 2010). People spend a major portion of their waking hours at work and in commuting to their places of employment. Jobs vary in the amount of stress that

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Jeb Strauss = job stress; Laura Backe = lower back.

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they place on employees, but it is difficult to imagine a job that is stress free. As Erlanger (2012) stated, "Stress can be identified in all occupations, from the lowest paid to the highest paid, from the physically demanding to the sedentary, from the skilled to the unskilled" (p. 76). Low back pain is common across the same range of occupations.

The purpose of this study is to test the hypotheses that low back pain may be brought on by job stress and that stress management is an effective treatment. The research consists of two parts. Experiment 1 aims to show that low back pain is a consequence of job stress. Experiment 2 represents an effort to determine the effects of stress management on low back pain.

Experiment 1

Method

Participants. Sixty employees at Metropolitan Hospital in Santa Monica, California, volunteered to participate in this research, which had the full support of the hospital administration. A total of 30 employees who had a record of using sick leave because of low back pain volunteered. The hospital has employees at all levels (e.g., physicians, dentists, nurses, medical technicians, executives, clerical workers) and skilled craftsmen (e.g., plumbers, electricians, painters, maintenance workers, food service personnel, elevator operators, guards). The low-back-pain sample was representative of the work force. There were 18 men and 12 women, ages 22 to 65 ($M = 39.2$), of varying occupations, and they had all tried a variety of treatments for their condition.

A control group of 30 employees was selected from among the 60 original volunteers. The group consisted of 18 men and 12 women, ages 21 to 66 ($M = 40.2$), of varying occupations (comparable to the low-back-pain sample). None of the members of the control group had ever experienced low back pain.

Procedure. Participants from both groups were assembled in a large meeting room to fill out a consent form and to complete the Paulson Job Stress Inventory (Paulson, 2006). Six different time slots (day and evening) were available for the convenience of the employees.

The Paulson Job Stress Inventory is a summative rating scale with fifty 5-point Likert scale items that have to do with the experience of job stress. Total scores can range from 50 to 250, with high scores indicative of higher job stress. Reliability was assessed by $\alpha = .82$. Criterion-related validity coefficient was reported as .65. The inventory has been used successfully in a number of previous studies (Meredith, 2013).

Results and Discussion

The low-back-pain group had a mean score of 187.4 on the Paulson Job Stress Inventory, in contrast to a mean of 168.7 for the control group. In a t test of the difference between these two means, $t(58) = 2.04$, $p < .05$. This supported the hypothesis that the job stress mean would be significantly higher for the low-back-pain group.

Experiment 2

Method

Participants. The same two groups of participants in Experiment 1 were the participants in Experiment 2.

Procedure. At the time of the introductory session, after the completion of the Paulson Job Stress Inventory, participants were given a packet of instructions for time and place of stress management training sessions. These sessions were held in the hospital, and employees were excused from their duties so that they could attend. Sessions were conducted for small groups of six employees at convenient times.

To assure that they received equal training, each group was made up of three participants from the low-back-pain group and three from

the control group. Closely following Perconi and Groteer's (2011) stress management protocol, each session contained didactic material and guided practice in the application of relaxation techniques, self-messages, and ways of handling stress. Participants were strongly encouraged to attend all four sessions. Attendance of at least three of the four sessions was required for inclusion; no participant had to be dropped from either group. Sessions were conducted by the authors, both of whom have had extensive experience in applying the Perconi and Groteer protocol with pain patients. At the end of the final session, participants again filled out the Paulson Job Stress Inventory.

Results

Mean scores and pretest–posttest difference scores are shown in Table 1. The low-back-pain group exhibited a decrease in job stress score from 187.4 at pretest to 165.4 at posttest, with an average difference score of –22.0. In contrast, the control group had a mean difference score of –8.1, with a pretest mean of 168.7 and a posttest mean of 160.6. In a *t* test of the significance of the difference between the mean difference scores of the two groups, $t(58) = 3.14$, $p < .01$.

Table 1
Job Scores for the Two Groups

Back pain	Pretest	Posttest	Mean difference
Low	187.4	165.4	–22.0
None	168.7	160.6	–8.1

General Discussion

This two-part research demonstrates that, as hypothesized, low back pain is at least one of the consequences of job stress. The second experiment clearly showed a significant reduction in job stress as a result of brief stress management training for employees who suffered from low back pain.

This finding holds great promise for alleviating this pervasive symptom, which interferes so drastically with worker productivity. Employers are well-advised to introduce this cost-effective stress management program for the benefit of their employees and for the welfare of the organization. Health maintenance organizations should widely endorse and support these kinds of interventions for low back pain and for other stress-related conditions that we infer would respond just as favorably and with great savings.

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Critique

INDEPENDENT AND DEPENDENT VARIABLES AND CAUSALITY

Judging from the words "effect of" in the title of the article and from the statement of the problem, the investigators intended to design a study that would show a causal link between job stress and low back pain. The design, however, is set up backward.

Back pain (with the two levels, low-back-pain and no-back-pain) is incorrectly conceptualized to be the independent variable, and job stress is made into the dependent variable. To be consistent with the intent of the research, and with the hypothesis of Experiment 1, levels of job stress should have been varied as the independent variable, and the incidence of back pain measured as the consequence, or dependent variable.

As designed, this was not an "experiment." The two groups are static groups, no variable was manipulated, and no causal statements about effect can be made (see Chapters 2 and 3).

The results showed that the employees who had low back pain scored higher on the job stress inventory than those who did not have back pain. This does not enable one to conclude that job stress caused low back pain. It would be just as plausible to speculate that people who have the misfortune of having to work with low back pain experience more job stress as a consequence of the discomfort and interference of their affliction. A static group design of this sort can at best indicate an association or two-way relationship (each can cause the other) between two variables; it does not permit causal inferences to be drawn.

CONTROL GROUP

In Experiment 2, the group that is free of back pain is incorrectly called a control group. To test the effects of the treatment, a control group is indeed required. The outcome of the treated group should be compared with the outcome of an untreated control group or a placebo control group (see Chapter 3).

The independent variable in this part of the research should have been treatment, not the presence or absence of back pain. If one wishes to study the effects of treatment, one must vary the treatment. The aim of Experiment 2 is to study the effects of stress management on low back pain. There is no reason to include and treat a control group of people who do not have low back pain. Doing so is analogous to putting a "control group" of nonsmokers through a smoking cessation regimen for comparison with a group of smokers on their success in quitting.

DATA ANALYSIS

Change is measured by using difference scores. These scores may be associated with pretest level (see Chapter 8). In this study, the marked difference in the pretest stress levels between the two groups (187.4 vs. 168.7) could have been a function of all participants signing up at a particularly stressful time, one that would have diminished over time whether or not the treatment had been given (see Chapter 8). The change from pretest to posttest is impressive, but again it cannot be unequivocally attributed to the treatment. Effect sizes are not reported.

CONCLUSIONS

The study lacks internal validity, and it is conceptually muddled because of the mix-up of independent and dependent variables in the first phase and the selection

of the wrong outcome measure and the wrong control group in the second phase. As a result, none of the conclusions are warranted. Generalizations and recommendations go well beyond the obtained data.

REFERENCES

Erlanger (2012) and Paulson (the Job Stress Inventory, 2006) are cited in the text but are missing from the list of references.

ABSTRACT

The Abstract refers to the "no-back-pain" group in Experiment 2 as a "control group." Unless one reads the entire article, one might think that the control group was a no-treatment group (as it should have been). The fact that the "control group" in this treatment outcome phase of the study was an inappropriate no-low-back-pain group changes the interpretation of the results and makes the Abstract misleading.

AUTHOR NOTE

No information is given on how to contact the authors.

Practice Article 11

Evaluation of Health Maintenance Organizations

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Health maintenance organizations (HMOs) are the predominant form of health care in the United States. Along with cost consciousness and increased emphasis on efficiency in the delivery of health services is an inherent commitment to accountability. As Shawcross (2012) suggested, an HMO is a collective or syndicate of medical professionals. When a number of qualified individuals pool their skills, their accountability is multiplied rather than divided (Lassiter, 2013). There have been more calls for the evaluation of HMO services than there have been efforts to accomplish such evaluations (Arlington, 2010; Hollander & McKeachie, 2013; Meany & Nottingham, 2008; Parsighian, 2012).

Two principal strategies may be used to evaluate any service program. One way, advocated by Klipstein in his summary of the Joint Commission Report (2013), is to (a) identify and define the services offered, (b) develop standards for the delivery of these services, and (c) measure the extent to which the HMO meets or exceeds these standards. This is particularly difficult when the HMO features a wide range of diverse services. The task of developing performance standards in and of itself is exceptionally demanding (Vaslek & Harodny, 2013).

Another approach is to ask the consumers of the services how they feel about the outcome. Consumer attitudes are the principal source of survey material in market research (Albright, 2008). Although an HMO's very

NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Harlan M. Oberlin = HMO; Pauline P. Olmstead = PPO.

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existence depends on clients using their services, the goodwill of the clients is central to the continued success of the syndicate. Client attitudes are a key to appraisal of services rendered, and by extension, a vital link to their continued membership in the program.

Honesty and candor on the part of the respondents is a sine qua non of attitude surveys (Brady, 2005). Health service providers cannot simply ask their clients how they feel about the services that they have been receiving. Such an approach would be far too subject to slanting of the expressed attitudes in a direction favorable to the provider-inquirer. Some people would be reluctant to express their true attitudes for fear of hurting the feelings of the provider and of appearing ungrateful. This phenomenon, known as *bias of the auspices*, is well-known in the attitude survey literature (Carlsson, 1994).

The danger of this bias can be minimized by using printed questionnaires that are filled out anonymously. If the survey that is developed is quantitative rather than just qualitative, it can be used as a means of longitudinal comparison to assess progress and change over time. In addition, once such a quantifiable survey form has been developed, it can be used in other similar settings. Ultimately this provides an additional advantage of establishing norms for service delivery programs. These would enable any program to compare itself with others and would offer valuable information to insurers and to prospective consumers.

Method

Survey Development

The researchers interviewed 20 patients who were receiving a variety of different kinds of medical services that could affect their attitude toward the HMO. The interviewers made it clear that they were seeking positive as well as negative experiences and that the information was being gathered for the sole purpose of improving services. Interviews were also held with representatives from all of the sections whose work brought them in contact with patients in person, by mail, or by telephone. The purpose of these interviews was to gather information about the kinds of things that they thought might have a bearing on patients' attitudes.

Information from these disparate sources was categorized, and the most salient issues were converted into questionnaire items. Redundancies were then eliminated and the final pool of items assembled. Each item was placed in the form of a Likert scale with the following format:

1. *strongly disagree*
2. *disagree*
3. *disagree as much as agree*
4. *agree*
5. *strongly agree*

Respondents were asked to place a check or "X" next to the statement that most closely described their attitude to that item. For scoring purposes, the scale values ranged from 1 to 5 and corresponded to the numbers in front of each choice. High scale values that represented negative attitudes, and low scale values that represented positive attitudes, were reverse keyed. For example, when someone strongly agreed to an item that described an unfavorable attitude, the score value of 5 was scored as 1. The converse was true when someone disagreed with a negative item. When this happened, a score of 1 was converted to a 5. The range of total scores across 24 items was therefore 24 to 120. The higher the score, the more favorable the summated attitude.

The items were placed as much as possible in a hierarchical temporal order, beginning with making an appointment, and covering all aspects of the visit and beyond. Item 25 was a global item that was analyzed separately. The scale items are presented next.

1. When I call for an appointment, the appointment clerk answers promptly and is courteous and efficient.
2. Appointments are arranged expeditiously, and I have no difficulty changing appointments if the need arises.

3. When I have an emergency and have to be seen as soon as possible, special attention is usually given to my needs by the appointment personnel when I call.
4. I receive courteous treatment by the receptionist when I arrive for an appointment.
5. I am not kept waiting too long when I arrive for a scheduled appointment.
6. The furnishings in the waiting room are comfortable, and there is plenty of interesting reading material.
7. The examining rooms are not warm, comfortable, or clean, and there is no place to sit while you wait.
8. They usually keep you waiting too long in the examining room.
9. The doctors I have seen are skilled at diagnosis and treatment.
10. The doctors I have seen are courteous and treat me with dignity.
11. The doctors I have seen often do not listen to what I have to say or understand how I feel.
12. The doctors do not order unnecessary tests, nor do they fail to order necessary ones.
13. My doctors are usually available to talk with me on the phone if I have questions about my condition and/or my medications, or if not available, they call me back in a timely manner.
14. There is excessive variability in the abilities of the doctors in this HMO.
15. My doctors do not prescribe unnecessary drugs.
16. After a visit I receive follow-up calls to find out how I am doing.
17. The nurses are skilled, courteous, and show compassion.
18. The medical technicians (e.g., x-ray, EKG, EEG) are competent and courteous.
19. The medical equipment does not seem to be first rate or up to date.
20. The pharmacy is accurate in filling prescriptions and does not keep me waiting too long.
21. Special requests that I have, such as a note verifying my visit or a letter about my condition, are cheerfully received and are handled promptly.
22. I can expect a call to notify me when a follow-up visit is needed.
23. Business accounts are handled efficiently by the HMO.
24. The office is courteous and quick to supply me with any medical records or reports that I need.
25. Overall, I am satisfied with the services that I receive from this HMO, and I am confident that my health needs are well attended.

Survey Sample and Method

The survey form was sent to the last 500 patients who had visited the HMO. A cover letter explained that the purpose of the survey was to provide information that would be used to improve services. Respondents were urged to be as candid as possible in their replies. There was no space for a name on the form, and respondents were informed that they should respond anonymously. A stamped and return-addressed envelope was enclosed. An unobtrusive code was placed on each form so that, for ongoing research purposes, respondents could be categorized on demographic variables and types of services used.

Results

Three hundred and ninety questionnaires were returned. The return rate of 78% is considered high for a mail survey (Feldman, 2010). The average rating for Items 1–24 was 3.12. Only five of these items yielded mean scores of less than 3, reflecting negative attitudes. The remainder were all favorable. Responses to Items 1 and 2 revealed a problem with the appointment system. Unfavorable rating means to Items 7 and 8 revealed some dissatisfaction with the examining room arrangements. The mean response to Item 24 showed concern with the production of reports and records.

A global evaluation of the HMO was obtained from Item 25: "Overall I am satisfied with the services received from this HMO, and I am confident that my health needs are well-attended to." Patients were resoundingly affirmative in their agreement with this statement. The mean scale value was 4.17.

Discussion

The HMO Patient Attitude Survey was constructed and applied to a random sample of 500 patients. It successfully reflected the favorable attitudes of the patients and proved itself to be an effective method of program evaluation. Analysis of clusters of related items enabled us to determine which aspects of the program needed special attention and remediation.

In this regard, a follow-up study of the reasons for poor ratings of the appointment process led to a revision of some procedures and the retraining and reassignment of some personnel. We anticipate that a future attitude survey will reveal that this problem has been corrected. Similarly, we believe that some procedural changes introduced as a consequence of this survey will improve the problem that patients experienced with the receipt of medical records and reports. As a result of responses to items about the examining rooms, we have placed a comfortable chair for patients to sit in, and reading material, in each examining room. Nurses have been instructed to be particularly vigilant to the issue of timing and coordination of the process so that patients are never kept waiting too long.

The benefits of the survey are evident. Not only does it fulfill the need for accountability through objective evaluation, but the ability of the survey to identify particular areas of discontent enables it to serve as a guide for remedial action. Now that the survey form has been developed and its successful application demonstrated, we strongly recommend its adoption by other HMOs.

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Critique

SAMPLE

Selecting the last 500 patients to visit opens the study to bias. The last 500 probably have an ongoing health problem or have just resolved one. Their present condition and status may color their responses. Random selection from the patient roster would have drawn a more representative sample (see Chapter 4, this volume). The systematic sample that they used is erroneously described as "random" in the Discussion.

SCALE CONSTRUCTION ITEMS

The value of a survey depends on the quality of its items. Certain simple principles apply when it comes to constructing items (see Chapter 7). The following categories of item deficiencies can be observed in this survey.

DOUBLE NEGATIVES

Items with double negatives can confuse respondents. Items 2, 5, 12, and 15 have this problem (e.g., Item 15: "My doctors do not prescribe unnecessary drugs").

MORE THAN ONE THOUGHT PER ITEM

Multiple thoughts in the same item can be seen in Items 1, 6, 7, 9–12, 17–21, 24, and 25. For example, Item 20, "The pharmacy is accurate in filling prescriptions and does not keep me waiting too long," presents two issues, accuracy and timeliness. If the pharmacy is accurate but slow, or inaccurate but fast, the respondent is faced with a dilemma.

VOCABULARY AND WORDING

Items 2 and 14 have words and concepts that are too difficult to permit the assumption that they will be understood by all. Item 2 uses the word "expeditiously," and Item 14 assumes that everybody will know what is meant by "excessive variability in the abilities."

ITEM LENGTH

Items 13 and 21 are too long and too unwieldy to be grasped easily.

ITEM ORDER

As a means of avoiding response set, it probably would have been better to randomize the order of the items instead of having them arranged sequentially (see Chapter 6, this volume).

RELIABILITY

No effort has been made to establish the test–retest reliability or the internal consistency of the survey scale. The researchers should demonstrate that the scale is reliable before they recommend its widespread use (see Chapter 7).

VALIDITY

The researchers made no effort to validate a scale that is being suggested for general use (see Chapter 7). As a first step, they should have checked the content validity

of the scale and eliminated and added items as appropriate. The content validation process probably would have suggested inclusion of items about inpatient facilities and services, surgical facilities and outcomes, emergency care, extended care, and mental health treatment.

In addition, the establishment of some kind of construct or criterion-related validity is desirable, especially if other HMOs are expected to adopt the scale. For example, an auxiliary validation study could have been undertaken to show that the scale successfully discriminated between a group that was known by other criteria (e.g., frequent criticisms and grievances or letters of complaint) to be dissatisfied and a group that was known, without doubt, to be satisfied.

RESULTS

The presentation of results is sketchy. The reader is told only a few of the item means and is given no information about the variability of response to the various items. The average item mean of 3.12 is only modestly favorable, and the authors place more emphasis on the success that is reflected in the global evaluation in Item 25 ($M = 4.17$). They do not dwell on or try to explain this discrepancy; instead, they put more weight on this single item than they do on the aggregate of the other 24 items.

DISCUSSION

In light of the poorly constructed items and the lack of established reliability or validity, the Discussion statements claiming a successful demonstration of accountability are overblown (see Chapter 9). Recommendations for the adoption of the survey by other HMOs are unwarranted.

ETHICS

Even though respondents were assured anonymity, the investigators used a code to identify them. This is a clear violation of existing ethical principles. A pledge assuring anonymity should be scrupulously honored (see Chapter 10).

REFERENCES

Feldman (2010) and Lassiter (2013) were cited in the text but were omitted from the References.

ABSTRACT

There is no Abstract.

AUTHOR NOTE

The author note gives no information about how to contact the authors.

Practice Article 12

Effects of Jury Selection Consultation on Trial Outcome

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Jury consultants were randomly assigned to the prosecution teams in the trials of 13 violent crimes and 13 nonviolent crimes. An equivalent number of cases were tried without benefit of jury consultants. Although consultants and attorneys thought that the consultation process was helpful, a comparison of conviction rates showed no significant differences for either type of crime when consultants were employed.

The use of forensic psychologists who specialize in consulting on jury selection has been burgeoning in recent years as prosecution and defense lawyers attempt to gain any advantage that they can (McElheny, 2012). Jury consultants are of two main types: (a) idiographic (i.e., those whose judgment is informed by individual clinical impressions, taking into consideration the nature of the offense, characteristics of the defendant, and the demographic characteristics and demeanor of the prospective jurors) or (b) nomothetic (i.e., consultants who rely heavily on profiles developed from predictive regression formulae; Bligh, 2010). There is some dispute over which of these two approaches is most efficacious (Alper & Ipswich, 2009), and some have expressed doubt that either approach contributes to the successful prosecution or defense of the case (Schwinger, 2009).

During the *voir dire* process (i.e., the process of determining prospective jurors' suitability to serve), the prosecution and the defense are each allowed a certain number of peremptory challenges in which prospective jurors can be dismissed without cause. Many experienced attorneys believe that they are perfectly well-equipped to make informed judgments (Alonzo, 2006), and they resist bringing others into the case. Others are open to any edge that they can gain and are receptive to

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advice on any factor that they might have overlooked (Carmichael, 2013).

It is important to recognize that the attorneys in charge of the case are not under any obligation to follow the recommendations, and they bear the final responsibility for accepting or dismissing a prospective juror and for picking the kind of jury mix that they believe will best serve their client's interests.

Researchers on jury consultation have done analog studies using mock trials (Emerson & Holmes, 2003; Greentree & Oliver, 2000; Pemberton, 2000) or naturalistic field studies (Gable & Griffith, 2010; Herflinger, 2004). As Douglas (2011) pointed out, studies use either a micro or macro approach. In the former, the success or failure of the consultant is gauged by the performance of the individual juror. In the latter, the performance of the jury as a whole, which is determined by the favorableness or unfavorableness of the verdict, is the outcome criterion. In the present investigation, conducted on actual trials, we chose the macro approach because of inherent difficulties in the accurate determination of individual juror performance and because the sponsoring district attorney's office had an overriding interest in case outcomes.[1] Most of the cases are settled by plea bargaining before reaching a trial or before the trial is completed. In fact, less than 1% of the people who are charged ever receive a jury trial. Of these, 85% to 90% of the verdicts that come down favor the prosecution.

According to Asherman and Bilitnikoff (2007), the composition of the jury is more critical in criminal cases because of the feelings that such cases arouse and because of the responsibility entailed in making a judgment that will set the defendant free or send that person to jail. Furthermore, within the realm of criminal cases, Asherman and Bilitnikoff asserted that people who serve on juries are more apt to let their feelings sway their judgment when hearing about the details of such crimes. Jurors bear the added burden of having to make decisions that could lead to a severe penalty (e.g., long-term incarceration or even death). The present research is restricted to the study of criminal cases of a violent and nonviolent nature so as to maximize any of the possible advantages of jury consultation that are anticipated.

Method

Defendants and Charged Offenses

In the jurisdiction in which this research took place, 75% of the jury pool is Caucasian, 12% African American, 8% Hispanic, and 5% other. The defendants are 41% African American, 31% Caucasian, 23% Hispanic, and 5% other. The jurors and defendants in this study closely approximated these ethnic breakdowns.

Offenses with which the defendants were charged can be classified into two categories: violent and nonviolent. Violent crimes included murder in the first degree, 8%; murder in the second degree, 8%; manslaughter, 12%; rape, 12%; assault on police officer, 32%; and assault with a deadly weapon, 20%. Nonviolent crimes included controlled substance violations 36%; burglary, 20%; petty theft, 12%; grand auto theft, 12%; driving under the influence, 8%; forgery, 8%; and embezzlement, 4%.

Case Assignments

The district attorney was extremely cooperative in this research and made all of the resources of that office available in the interest of finding out whether expert jury consultation actually affects trial outcome. The study did not violate any defendant's civil rights, because no defense attorneys were deprived of the opportunity to use consultants if they wished. A pool of three trained jury consultants was available.

A total of 52 criminal cases served as the source of data for this investigation. By design, half the cases were violent crimes and half were nonviolent. The specific nature of these crimes and the percentages of each are as described previously.

[1]This research was conducted under the auspices of the district attorney, who wishes that the jurisdiction be unnamed for reasons of confidentiality.

One of three participating jury consultants was randomly assigned to work with the prosecuting attorney on each of 26 randomly selected cases (13 violent and 13 nonviolent). The remaining 26 cases (also 13 violent and 13 nonviolent) had no consultant.

For a case to be included, it had to be heard in its entirety by jurors who then had to go through the deliberation process. If a case was settled before going to the jury, or if no verdict was reached because of a hung jury, the case was removed from the sample and was replaced by a new case in the same crime category.

Attorney Ratings

In addition to the case outcomes, prosecution attorneys who had been assigned jury consultants were asked to fill out a rating scale after the final conclusion of the case. The 5-point scale consisted of the following:

Please circle the number that precedes the following statement which best describes your attitude about the helpfulness of the jury consultation that you received in this case:

1. *not at all helpful*
2. *somewhat helpful*
3. *helpful*
4. *very helpful*
5. *extremely helpful*

Results

Violent Crimes

As shown in Table 1, defendants in cases of violent crime to which a consultant was assigned were found guilty in 11 out of 13 instances (85%). Without a jury consultant, there were 10 convictions (77%). This difference is not statistically significant, $\chi^2(1, N = 26) = .24$, *n*s. For all cases combined, a total of 81% were declared guilty.

Table 1
Outcomes With Violent Crimes

Outcome	Consultant	No consultant	Total
Guilty	11	10	21
Not guilty	2	3	5
Total	13	13	26

Nonviolent Crimes

Table 2 shows that 12 (92%) of the 13 nonviolent cases that had jury consultants were declared guilty. For similar cases, 11 (85%) were convicted when there was no consultant. Combined, the conviction rate for nonviolent crimes was 88.45%. The difference between the outcome of cases with and without consultants was not statistically significant, $\chi^2(1, N = 26) = .37$, *n*s.

Table 2
Outcomes With Nonviolent Crimes

Outcome	Consultant	No consultant	Total
Guilty	12	11	23
Not guilty	1	2	3
Total	13	13	26

All Crimes Combined

When the violent and nonviolent crimes were combined, 23 of 26 (88.5%) cases were successfully prosecuted when a consultant was used, compared with 21 of 26 (80.8%) without a consultant, $\chi^2(1, N = 52) = .59$, ns. These results are shown in Table 3.

Attorney Judgments

The 11 prosecuting attorneys who participated in this study were asked to rate on a

Table 3
Outcomes for All Crimes Combined

Outcome	Consultant	No consultant	Total
Guilty	23	21	44
Not guilty	3	5	8
Total	26	26	52

5-point scale the helpfulness of jury consultants for each case that they tried. The average rating was 3.96, which is close to the adjective description *very helpful*. Point-biserial correlation between the ratings and the case outcomes was $r = .25$.

Survey of Consultants

There was no difference in the case outcomes of the three consultants. All consultants appeared to have comparable results. No statistical test was applied because two of them participated in only nine cases each, and one of them in eight cases. At the conclusion of the study the consultants were asked for their impressions of how useful they had been. All of them were firmly convinced that they had made valid recommendations and that their input had been important in influencing the outcome of the cases.

Discussion

Case outcomes when jury consultants were furnished were slightly enhanced for violent and nonviolent crimes, but the difference was not statistically significant. Attorneys who had consultants, however, were favorably disposed, rating the consultants as very helpful. It made no appreciable difference whether the trials were for violent or nonviolent crimes. Given that the attorneys found the process to be helpful and that the consultants were convinced of the validity of their own judgments, it is difficult to explain why stronger differences were not obtained. In view of the findings, however, only a qualified endorsement can be given to the use of consultants to assist in the process of jury selection. Because this study was intentionally restricted to criminal cases, no generalizations can be made to civil cases.

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Critique

BASE RATE

The study is handicapped by a high base rate of convictions. The authors state that 85% to 90% of verdicts are for conviction. Given their ability to settle cases without trial, it would stand to reason that prudent district attorneys do not squander their strained resources by trying cases that they do not expect to win. With such a high

base rate of success, it is difficult to demonstrate that any added procedure would produce a statistically significant improvement.

The accuracy of chi-square results is in doubt when there are expected cell frequencies of less than 5, as in this study. If the Fisher Exact Probability Test had been used, or had the Yates correction for continuity been applied, however, the result would not be significantly different; use of either approach would have reduced an already nonsignificant chi-square. A *z* test of the significance of the difference between proportions would give the same result as chi-square. Here, too, there are problems interpreting the *z* test as a deviate of the unit normal curve when p (proportion guilty) and q (proportion not guilty) are extreme values, as they are here, rather than in the .40 to .60 range (see Chapter 8).

PARTICIPANTS

The sum of the percentages in the listing of charged violent offenses is only 92%. Either the arithmetic was incorrect, or one or more categories were omitted.

CASE (GROUP) VERSUS INDIVIDUAL OUTCOME

The authors gave a reasonable explanation for using case outcome data instead of the performance of individual jurors as the criterion measure. What remains unknown, however, is how individual potential jurors who were rejected would have performed had they been chosen. Those who were rejected would fall into the categories of true negatives (those who would have voted against the prosecution, and thus were rejected correctly) and false negatives (those who would have supported the prosecution, and thus were wrongly rejected). We do not know if the record of convictions would have been better, the same, or worse if the jurors who were excused had been seated. One cumbersome way of finding out would be to have a shadow jury of rejects sitting on a case and rendering a separate verdict, not knowing that they were rejected and believing that their verdict was going to count.

In this study, the only cases that counted were those in which a verdict was reached. Cases with hung juries were omitted. To all intents and purposes, such cases are losses for the prosecution. Because unanimous agreement of 12 jurors is required, no verdict would be reached if anywhere between one and 11 jurors voted against the prosecution. Cases that resulted in hung juries should have been counted as losses for the consultants in this study, instead of being discarded.

DEFENSE CONSULTANTS

The authors point out that defense attorneys cannot ethically or legally be deprived of jury consultants, but they give no information about how many, if any, were used by the defense. This introduces a confounding variable (see Chapter 5) that was neither studied nor accounted for. When both sides use consultants, it becomes a zero-sum game with one consultant winning and one losing. If the success of the winning case could be attributed to their consultant, then the failure of the losing side could

just as well be blamed on the other consultant. The mere presence or absence of a consultant (without regard to differences in skill) is neutralized when both sides use them. Nothing is said in the article about the qualifications of the jury consultants to do their work.

ATTORNEY AND CONSULTANT RATINGS

The global 5-point rating scale does not provide enough data points to permit fine discriminations to be made (see Chapter 7). The centrist tendency of raters to stay away from extreme scores (1 and 5) leaves them with only three other scale points among which to choose.

Perhaps most critically, the scale has no true midpoint. The middle response should have been *neither helpful nor unhelpful*. Scale point 4 should have been *somewhat helpful*.

Raters also may have some reluctance to down-rate someone who is earnestly trying to help them. The average rating of 3.96 may reflect this positive bias. Furthermore, they won 88.5% of their cases and knew the outcome of the case at the time that they made the ratings, so magnanimity toward the consultants would be expected.

As far as the consultants themselves are concerned, a success rate of 88.5% provides sufficient reinforcement for them to be convinced of the validity of their judgments, whether or not their judgments had any bearing on the outcome of the case.

A point biserial correlation of $r = .25$ is not statistically significant with a sample of this size (see Chapter 8). In addition, 26 ratings were made by 11 attorneys. Therefore, at least some of the attorneys contributed multiple ratings.

Practice Article 13

Contingency Reinforcement in the Treatment of Talking Aloud to Self: A Single-Case Study

Earl Singleton, Janet Kase, Paul Enoff, and Enrico Juan
Day Treatment Center, Memorial Hospital

In this single-case experimental design, a 34-year-old ambulatory patient with schizophrenia who attended the Day Treatment Center was treated by contingency reinforcement for 1 hr a day in an ABAB design to reduce his talking out loud to himself in public places. Following a 7-day baseline period, he was given points that could be redeemed for goods in the hospital store whenever he refrained from talking aloud. This 7-day contingency reinforcement phase was followed by a return to 7 days of nonreinforcement. Lastly, contingency reinforcement was reinstituted for 7 more days. The target behavior was seen to decline during the periods of contingency reinforcement. Contingency reinforcement is an effective way of modifying the behavior of self-talk in patients with schizophrenia.

The research reported in this article uses a single-case ($N = 1$) experimental design to assess the effectiveness of contingency reinforcement in modifying a socially disruptive behavior that is commonly displayed by individuals with schizophrenia. Talking out loud, in the absence of any evident social interaction, has been interpreted by Galworthy (1947), Furth and Glabner (1959), and Ashencroft (1974) as an external manifestation of hallucinatory experiences. The patient is engaged in illusory conversations with fantasized persons. Einhoven (2009) called it "the externalization of the inner voice" (p. 173). Greentree (2010) believed that it reflects an obliviousness to external surroundings and not merely a lack of control. Such "communications" can be bland and seemingly matter-of-fact, but are more often accompanied by strong affect whereby the patient appears to be engaged in an angry confrontation with inner adversaries or may giggle and laugh outright in response to some imaginary encounter that is perceived to be witty or amusing (Svendsen, 2012).

This behavior is a matter of concern because it stamps a person as obviously mentally ill. When people talk to themselves in public, strangers give them a wide berth. People who talk to themselves are particularly avoided

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Singleton, Kase, Enoff, Juan = single case, N of 1.

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when their demeanor and the content of their verbalizations are hostile in nature. Bystanders are well able to distinguish the meaning of talking out loud in private as opposed to public spaces. A man who is moving his mouth while driving alone in his own car with the windows raised is looked on as likely speaking on a cell phone with a hands-free device or singing along in accompaniment to the car radio. No one looks askance at young children who do it, but when it is done by an adult on the street, on a bus, in a restaurant or other public place, it is seen as socially noxious behavior. If done habitually, even if harmlessly, it can seriously impair an individual's adjustment in the community.

Contingency reinforcement has a history of successful application in eliminating or modifying a variety of socially unacceptable behaviors. Bernardi's (2012) review gives a comprehensive historical accounting of its use. The purpose of this study is to test the efficacy of contingency reinforcement specifically in an effort to modify the inappropriate talking-out-loud behavior of a patient with schizophrenia.

Method

The Patient

The patient in this study is a 34-year-old Caucasian, ambulatory man with schizophrenia. He was first diagnosed with schizophrenia at the age of 21, and he has been in and out of mental hospitals on numerous occasions. The earliest clinical reports about him indicate that he has persistently displayed the behavior in question.

All previous efforts to get him to desist, including the use of antipsychotic medication, were of no avail. The behavior has been a consistent manifestation of his illness for at least the past 13 years. For the past 9 months the patient has been out of the hospital, has been living with his elderly parents, and has been coming for outpatient treatment in the Day Treatment Center.

Setting

The setting for the treatment and the study was the psychiatric Day Treatment Center, located in the outpatient department of a large mental hospital. The center is primarily for ambulatory individuals with schizophrenia who are invited to attend daily and to spend much of the day engaged in supervised therapeutic activities. It is staffed by a clinical psychologist, a clinical social worker, an occupational therapist, trainees in psychology and in social work, and a receptionist. There is also a part-time psychiatrist and many volunteers.

All personnel are engaged in milieu therapy with all of the patients, meant to assist the patient in arranging their day in a way that promotes healthy living. A wide range of activities are featured, such as group discussions, patient government, arts and crafts, music, games, and other recreational activities. Social skills activities include luncheons, special events, dances, and escorted excursions into the community to enhance the use of community facilities. Trained professionals mingle with the patients and talk with them about their personal problems, their feelings, and their reactions to incidents that occurred during the day and the way that they handled them.

The interactions take place informally within the context of ongoing activities—during a game that is being played, while a picture is being drawn, or while sitting in the lounge—in distinction to fixed and formal individual office sessions. Social interactions among patients, who are typically very supportive of each other, are an important part of the treatment regimen. In addition to being engaged in the ongoing treatment program, the clinical social worker is involved with patients and families on issues associated with daily outside living. Antipsychotic medication is prescribed by the attending psychiatrist who is in daily contact with the staff about patient status and behavior.

In addition to this general milieu therapy program, behavioral techniques are applied according to the needs of individual patients to modify particularly troublesome behavior that is impeding their overall progress. The case in hand is one such instance in which a single case experimental design was implemented

in an effort to disrupt a disturbing behavior pattern and simultaneously to evaluate the progress made.

Procedure

The patient was observed for 1 hr every day for 7 consecutive weekdays. An observer unobtrusively noted whether talking-out-loud behavior occurred in each of 12 consecutive 5-min segments. The purpose of this series of observations was to establish a baseline for the target behavior. The next phase, contingency reinforcement, was introduced for the following 7 consecutive weekdays.

For every 5 min interval in which the patient had not talked out loud to himself, he was rewarded with a chit worth 1 point. Accumulated points were exchangeable at the hospital store for gum, candy, notions, magazines, or toiletries. The observer signaled an associate to hand out the chit. If the patient had talked inappropriately, he received no chit for that 5-min period. The maximum number of points that could be accumulated in the hour were 12.

The observer usually stayed across the room from the patient and other people sometimes were in the way, so it was not always clear whether the patient was talking to himself or to other persons. As a check on reliability, a second observer watched along with the primary observer 25% of the time. Points were still authorized exclusively by the primary observer.

At the end of the 7 days of contingency reinforcement, there was a return to the baseline condition with no points being awarded. Observations continued, however, in the same manner. This phase was succeeded by a return to the contingency reinforcement condition. The procedural sequence is as follows: Days 1–7, baseline; Days 8–14, contingency reinforcement; Days 15–21, baseline; and Days 22–28, contingency reinforcement.

It should be noted that the patient was absent on 6 days. Whenever that happened, the trials were extended so that there were the same number of days (7) in each of the four phases.

Results

Reliability Check

The 25% sample of observations judged by two observers was analyzed by Cohen's kappa. The resulting $\kappa = .75$ suggests a satisfactory degree of agreement between the two observers and encouraged us to accept as valid the observations of the primary observer.

Change in Target Behavior

The number of episodes of talking out loud could range from 0 to 12, because a judgment was made about whether the patient had talked in each of twelve 5-min periods during the hour of observation. The way that this varied across the four phases can be seen in Figure 1.

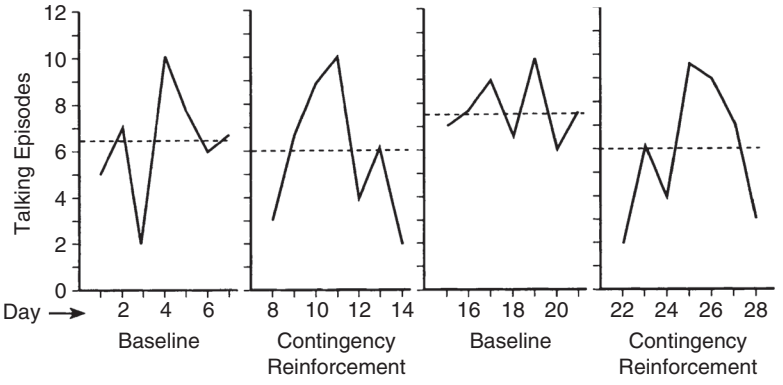


Figure 1. Talking to self during baseline and contingency reinforcement. Means are represented by dotted lines.

The dotted line in each phase indicates the average number of talking episodes during that phase. Visual inspection shows a decrease in talking aloud during the first 7 days of contingency reinforcement. The undesirable behavior increased in the third phase when reinforcement was withdrawn, but again decreased when it was reintroduced.

Discussion

This study demonstrated that talking-aloud behavior could be decreased by contingency reinforcement in a patient with schizophrenia who had consistently displayed this disruptive behavior for at least 13 years. It is evident that contingency reinforcement is an effective way of modifying the disruptive behavior that some individuals with schizophrenia display when they talk out loud to themselves. Other socially unacceptable behaviors are probably equally susceptible to this form of behavior therapy, and we recommend the further application of this treatment approach in clinical settings. We intend to apply it in future studies with such behaviors as exaggerated facial grimacing, repetitious patting of various body parts, constant dry-spitting gestures, inappropriate affective reactions, extreme social withdrawal, reluctance to engage in any activities, and lapses in personal hygiene.

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Critique

PROCEDURES

The procedures did not control some potential confounds in the study.

1. The researchers do not indicate when the six missed sessions occurred. If the missed sessions are related to treatment, this could indicate a problem. For example, if the patient absented himself on six occasions but attended unfailingly during contingency reinforcement sessions, the interpretation would be quite different than if a converse pattern had been shown, or if the absences were spread evenly throughout the four conditions. It is impossible to gauge the significance of this in the absence of information.
2. There was no control in the study of the patient's interaction with other people during the periods of observation. When the patient was not alone but was talking with other people, there would be reduced opportunity to talk to himself. If interaction with others differed in the various phases of the study, this could be a confound; no information is given about this (see Chapter 5).
3. There is no mention of any effort to hold medication constant during the four phases of the study. Differences in medication levels taken could be a confound (see Chapter 5).

4. The procedures for assuring accurate observations and for establishing interobserver reliability are questionable (see Chapter 7). You are told that the view of the observer was sometimes obscured and that judgments were not always easy to make. Under these circumstances, reliance exclusively on the judgments of the one primary observer is a risky procedure. It is not helpful to say that when a 25% sample was checked against a second observer an acceptable level of interobserver reliability was obtained. Incorrect judgments on the part of the single primary observer are not eliminated by this procedure. Any incorrect judgments (and surely there were some) would either provide unintended reinforcement when the patient was actually talking to himself, or no reinforcement in some instances when he was engaged in a social interaction, was not talking to himself, and should have been rewarded. The presence of any such misclassifications would jeopardize the internal validity of the study. A more rigorous judging procedure to minimize such observer errors should have been used.
5. Because the primary observer and the reliability observer knew which reinforcement phase the patient was in, there is a strong possibility of coding bias.

DATA ANALYSIS

In Figure 1, the initial baseline was not at all stable. You cannot determine the base rate of the behavior because it fluctuated excessively. It is difficult to assess change without a clear starting point.

The data analysis is based exclusively on visual inspection of graphs and of mean levels portrayed graphically (see Chapters 6 and 9). The means look a bit higher under reinforcement conditions, but it is impossible to know whether the differences are reliable or are merely chance fluctuations. If the differences are reliable, it is unclear whether the effect is of sufficient magnitude to make any real difference. A statistical analysis would help to answer these questions.

CONCLUSIONS AND INFERENCES

The authors overreach when they generalize about the effectiveness of contingency reinforcement in modifying this particular behavior in individuals with schizophrenia. Unless they replicate the study, they can only talk about what happened in this individual case. They stretch their conclusions even further when they infer that they have grounds for recommending the clinical application of the technique not only to other individuals with schizophrenia who exhibit the target behavior, but to those who exhibit other disturbing behaviors as well. There is no basis for such generalizations (see Chapters 3 and 9).

Moreover, any claim of clinical value is premature, if not ill-advised, if the patient reverts quickly in Phase 3 to talking aloud to himself when the rewards are withdrawn. To recommend clinical application, one must demonstrate that the behavior was reduced and remained at a lower level, or better yet, that it was permanently eliminated. The issue of external validity (see Chapter 3) leaves one wondering whether the targeted

behavior decreased on contingency reinforcement days during the patient's hours of not being observed in the center and whether he decreased the amount of self-talk when he was outside the center. This issue was not addressed. It is possible that there was a dramatic increase in the undesirable behavior as soon as the experimental hour was over.

ETHICAL CONCERNS

You might wonder whether conducting this research without first obtaining informed consent poses ethical problems (see Chapter 10). Ethical standards suggest that it is acceptable to conduct research without special informed consent if the procedures are part of an ongoing treatment effort. This research could be regarded in that light and thought of as the internal evaluation of one stage of a treatment program.

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ABSTRACT

The Abstract perpetuates the overgeneralization in the Discussion that claims, on the strength of this single case without replication, that contingency reinforcement is an effective way of reducing talking aloud behavior in individuals with schizophrenia (see Chapter 8).

Practice Article 14

The Effect of Race of Examiner on IQ Scores of American Indians

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A total of 40 American Indian middle-school students were administered the WISC–R. Students were randomly assigned to either an American Indian or a White examiner. The verbal half of the test was administered to half of the students in a first session, and the performance half was administered on the following day. The order of the tests was reversed for a random half of the students. Results showed that the mean IQ of American Indian students was significantly higher when examined by the American Indian examiner. Differences are accounted for by diminished anxiety and increased comfort of examinees when tested by a member of their own racial group.

IQ test scores are generally considered to be relatively stable and robust measures that are not materially affected by environmental conditions of administration or by differences among trained examiners (Haverstraw & Kolodny, 1988). In cross-cultural studies, however, the race of the examiner is a variable that has been singled out as possibly influencing results. Charles (1980) and Mercer and Glover (1990) found that African American youths scored higher when examined by a member of their own racial group than when tested by a White examiner. La Farge and Lascot (2011) obtained similar results with a sample of White French adolescents compared with a group of Black French residents from North Africa. A comparable finding was presented by Osuna (2012) in a study with Hispanics. In all of these investigations, the authors theorized that respondents performed better because they were more comfortable and felt less threatened by examiners who shared their racial group membership.

Another series of studies have produced results that are in direct contradiction to this by showing that higher IQ scores are generated when the examiner is White and the examinee is not White than when the tester and person being tested are of the same racial group. This result was reported in two studies with African American examinees by Schultz (1991) and by Washington and Marshall (1988). The third

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Whyte = white; Cheryl Key = Cherokee.

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study was with Mexican American examinees (Muñoz & Kirby, 2006). The authors of these studies reasoned that examinees are not accustomed to seeing members of their own racial group in the role of tester, experience cognitive dissonance, and become unsure of how to react.

A third series of investigations are in opposition to the other two. They report that the racial group membership of the examiner and the examinee are irrelevant. The presence of cross-racial pairings did not have any effect in studies with African Americans (Cernoff, 1990; Lerner & Diaz, 1991) or with Asian American examinees (Chen, 2012).

In summary, the results are variable, with some finding for higher IQs with pairings of testers and test takers of the same racial group, some finding for lower IQs, and the remainder claiming that racial differences do not affect results. None of these studies was carried out with American Indians, but all of them have implications for other cross-racial pairings. The present study aims to help resolve this conflict in the literature and to contribute findings about American Indians to the topic. Our hypothesis is that American Indian students score higher when the examiner of an IQ test is of the same racial background.

Method

Participants

The participants were 40 American Indian 7th and 8th grade students in a middle school on the Cherokee Reservation. Psychological testing is a normal part of the school procedures, and no parental consent was required. The students were about evenly matched for sex (21 females and 19 males). There were 20 each from the 7th and 8th grades.

Examiners and Test

Both examiners were women in their mid-30s. Both were well-trained and experienced psychometrists. One of them was White, and the other was American Indian (Cherokee).

The IQ test selected was the Wechsler Intelligence Scale for Children–Revised (WISC–R). It is considered to be one of the most reliable, valid, and culture-fair measures of IQ available (Musgrave, 2012). The WISC–R consists of verbal and performance subtests. Examiners were given special orientation to adhere closely to the instructions about test administration in the manual. Each was observed as she examined two sample cases; both proved to be competent examiners, and they did not noticeably differ in the way that they administered the test. Their scoring of these sample cases was checked by supervisory psychologists, and was found to be accurate.

Procedure

Children were released from class two at a time for about 30 min per examining session. One was escorted by the White psychometrist to an examining room, and the other was shown to a different room by the American Indian psychometrist. A random half of the students were given the verbal subtest at the first meeting, and half were given the performance subtest. The order was counterbalanced to enable a check on the possibility that students who start off with performance tests might feel more at ease and score higher than if they begin with verbal tests. Each student was brought back for the other half of the test on the following day. Testing was done on every day except Friday. Participants who were absent for the second session were seen on the next day that they were present in school.

Grade level of the participants (7th or 8th) was also counterbalanced by randomly assigning an equal number to Order 1 and Order 2. Grade was not treated as an independent variable. Counterbalancing of grade levels was introduced to rule them out as factors that could affect the results.

Scoring

Each psychometrist scored her own tests. Scoring was checked for discrepancies and accuracy by supervisory psychologists. Only

occasional minor changes were found to be necessary.

Results

Mean IQs obtained by the two examiners under the two orders are shown in Table 1. The scores were normally distributed, and skew and kurtosis were within acceptable limits. Bartlett's test of homogeneity of variance revealed no violation of this assumption for the analysis of variance (ANOVA).

As can be seen in Table 1, the total IQ for the students when the test was administered by an American Indian examiner was 101.45, in contrast to 99.35 when the examiner was White. This difference is statistically significant. A two-factor ANOVA with repeated measures on one factor yielded $F(1, 72) = 4.01$, $p = .05$, for race of the examiner (see Table 2). Neither the order of administering the subtests nor the Order × Race interaction was significant.

Discussion

The hypothesis of the study was supported by the finding that this sample of American Indians in the 7th and 8th grades scored, on average, significantly higher when the examiner was an American Indian than when the examiner was White. Precautions were taken to rule out examiner preferences and biases as factors. Both of the psychometrists were given special instructions to follow the test manual closely and to administer the test in a consistent and standard manner. Preexperimental observations of their performance by expert supervisory psychologists showed no detectable tendencies on the part of either examiner to favor or give advantage to the sample cases. We have no reason to believe that this was a factor in the higher scores attained by children when examined by the American Indian examiner. Following this reasoning, the difference must lie in the children who were being examined. Evidently, the added comfort in the presence of an American Indian examiner (or added anxiety in the presence of a White examiner) is conducive of better (or worse) demonstration of their abilities.

There is one added issue that needs to be addressed. We have been referring to American Indians in a way that might lead people to think that the findings were generalizable to all American Indians. In this study the examiner and the children were Cherokee. We have not explored the possibility that an examiner who was a member of another tribe might obtain different results. Future research would also have to determine whether the findings apply to American Indians examinees of different ages and different stages of acculturation outside of their racial group.

Table 1
Student Scores by Race of Examiner and Order of Test

Test order	Examiner		Total IQ
	American Indian	White	
Order 1			
M	102.50	101.00	101.75
SD	14.85	15.03	
Order 2			
M	100.40	97.70	99.05
SD	15.15	14.98	
M	101.45	99.35	

Note. Order 1 = verbal first; Order 2 = performance first.

Table 2
Analysis of Variance of Race of Examiner and Order of Test

Source of variation	Sum of squares	*df*	Mean square	*F*
Race of examiner	82.83	1	82.83	4.01*
Order of test	42.48	1	42.48	2.05
Race × order	48.76	1	48.76	2.36
Error	1,488.55	72	20.67	

*$p = .05$.

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Critique

DATA ANALYSIS

It is necessary to call attention to an error in data analysis first because it affects the results and their interpretation (see Chapter 8). There are 40 students with a total of 39 degrees of freedom. With 1 degree of freedom each for the two races of the examiners, the two test orders, and the Race × Order interaction, and assuming that the two subtests were added together and not analyzed separately, there should be a residual of 36 degrees of freedom for error, not 72. The error term should consist of the variation between the subjects within each of the four groups. Ten students were examined by the White psychometrist with Order 1 and another 10 with Order 2. Two more groups of 10 were tested by the American Indian examiner with the two orders again varied. The 9 degrees of freedom for each of these four groups contribute to the total of 36 degrees of freedom.

It appears the researchers left the verbal and performance subscales separate in their analyses but did not include it as a factor in the analysis of variance. Had they, it would have been a within-subjects factor. The consequence of having used a larger (but incorrect) number of degrees of freedom is to reduce the error mean square from 41.35 to 20.67. This, in turn, incorrectly raises the obtained *F*-value from 2.00 (*n*s) to 4.01 ($p = .05$).

Claims that the authors make about the statistical significance of the 2-point difference no longer hold when the correct degrees of freedom are applied. A reinterpretation of the results would find no support for the hypothesis. The small difference between the means (the *d* index is approximately .14) is not a reliable one and probably can best be regarded as a chance fluctuation.

In light of the statistical error, remarks in the Discussion that are based on the erroneous results are no longer appropriate. That said, it is interesting to consider whether you would describe a "less-than-small" effect of this size to be clinically significant if it were found to be reliable, especially because the contrasted means straddle the normed average IQ.

COMMUNICATION

Some of the children might have been sharing their experiences. Tales about how difficult the experience was or how unfriendly the examiner was could influence a child to be anxious or resistant when coming in to be tested. The children also had the opportunity to give each other information about the test and the answers. They all came from the same school, and the testing apparently went on for some time, so it is likely that some interparticipant communication took place. Nothing was done to protect against it or to assess whether communication may have affected the results (see Chapter 5).

EXAMINERS

The investigators were sensitive to the possibility of systematic examiner influences on the test results, and they took steps to try to rule it out. It would have been useful to use several examiners to represent each racial group, so as to diminish the possibility of personal (nonracial) characteristics of a single examiner having excessive weight on the outcome (see Chapter 5).

CONTRAST GROUP

It would have been helpful to have a contrast group of White students to determine whether any influences were indeed systematic. In such a design there would be groups of American Indian children and White children examined by American Indian and White examiners. This would enable the critical reader to determine the differences in IQs obtained by American Indian and White examiners regardless of the group membership of the examinees. In other words, did either examiner consistently obtain higher or lower scores from those whom they tested? The design would also enable the critical reader to determine whether there were any differences in IQ scores obtained by American Indian and White examiners regardless of the race of the persons whom they were testing. Moreover, any Race of Examiner × Race of Examinee interaction would demonstrate a selective pattern of higher or lower scores associated with race (see Chapter 6).

ABSTRACT

The Abstract repeats the incorrect finding that resulted from incorrectly determining the degrees of freedom. People who limit their reading to the Abstract and do not read the entire study will be badly misled. The final sentence of the Abstract explains the (incorrect) result and presents it as a confirmed conclusion. The comfort or the anxiety level of the children while they were being tested was not assessed in any way. The authors present as facts their speculations about intervening variables (see Chapter 3).

Practice Article 15

Social Effects of Tax Deadline

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Some of the social effects of the April 15th IRS tax deadline are examined in this study. Archival statistics from 15 regionally diverse states provided comparative data for April 15–20 and March 15–20 on mental health (depression), monetary and property crimes, domestic violence, and birth rate 9 months later. There was a significant increase in hospital admissions for major depression during the April period over the March period and a significant increase in money and property crimes and in domestic violence. There was a corresponding decline in birth rate 9 months later.

Taxes in one form or another have been levied to support regimes since the government of ancient Rome began to collect a poll tax from its citizens. About 2,000 years ago, Augustus introduced property taxes and inheritance taxes (Ernestine, 1997). Taxes on all kinds of products soon ensued. In the United States, resistance to taxation was one of the causes of the American Revolution, and taxation has been met with distress and opposition ever since (Schucker, 2011).

Direct taxation in the form of the income tax is a relatively new idea; it was first introduced in Great Britain in 1842 (Farrell, 1986). The United States government, adopting the ways of its former rulers, began taxing people's income in 1861 to help pay for the Civil War, but the tax was repealed 10 years later (Anderson & Oliphant, 2003). Efforts to restore it were declared unconstitutional by the Supreme Court in 1895. The 16th Amendment to the Constitution, authorizing the collection of income taxes passed in 1913. The first $3,000 of income were exempt, which meant that less than 1% of the people qualified for paying taxes. Three thousand dollars represented a substantial income in the early twentieth century. Tax rates have been subject to marked fluctuation over the years. Top rates have ranged from the original 7% up to 94% during World War II.

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Levy = (tax) levy; Hertz = hurts.

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The historical and contemporary resistance to paying income taxes has become the centerpiece of numerous campaigns for public office. Commodity taxes (e.g., sales taxes) do not have the same effect on people's psyches as do personal income taxes. As Wellington (2009) has observed about sales taxes, "They may nickel and dime you to death, but they don't cause people to jump from the windows of high buildings" (p. 85). April 15th is probably the only date that most people instantly recognize but do not celebrate. It is a date that is associated with anxiety among the populace. The spin-off of this national anxiety is something that has not been investigated.

Failure to pay income taxes is a crime that is very likely to be detected. Financial insolvency can stimulate some to commit monetary crimes such as forgery, counterfeiting, and embezzlement, which may place them at less risk for being apprehended than does the federal crime of tax evasion. Another likely consequence is the threat to domestic tranquility. Money problems are a major source of domestic fights (Abernathy, 2009). Squabbles over family finances escalate into episodes of domestic violence in some cases, according to Dalrymple and Kashigian (2010). The most common reaction, however, is profound depression, in which people feel dismayed, discouraged, dispirited, and helpless (Cureton & Massingal, 2004).

Hypotheses

The research plan was to compare the rates of certain key variables in two time spans: April 15–20 and March 15–20 in the year 2015. Research hypotheses were as follows:

1. Rates for diagnosis of major depression will be higher for the mid-April dates than for the mid-March dates.
2. Rates of monetary crimes will be higher in April than in March.
3. Rates of incidents of domestic violence will be higher in April than in March.
4. Birth rates will be lower 9 months after the April dates (i.e., January 15–20) than they will be 9 months after the March dates (i.e., December 15–20).

Method

A broad sample of archival data were used to test the hypotheses.

Depression

We used admission data from 15 large state hospitals in three western states (California, Colorado, Arizona), two midwestern states (Ohio, Illinois), two southern states (North Carolina, Alabama), and three eastern states (New York, Pennsylvania, Massachusetts). Only admissions during the specified dates for a primary diagnosis of depression were included.

Crimes

Recorded incidence of crimes of burglary, embezzlement, forgery, counterfeiting, auto theft, and shoplifting during the target dates were obtained from 15 urban communities in these same states.

Domestic Violence

Complaints of domestic violence during the time periods were furnished by the same police departments that supplied the crime rates.

Birth Rates

Birth rates were national figures of births that took place in the specified period 9 months after the original April and March dates.

Results

Depression

The 15 hospitals had 110 cumulative admissions for depression during the mid-April dates, in contrast to only 80 for the mid-March dates. This represents a 37.5% increase in April. In a one-sample test, $\chi^2(1, N = 190) = 5.26, p < .05$. The contrast in admission rates is shown in Figure 1.

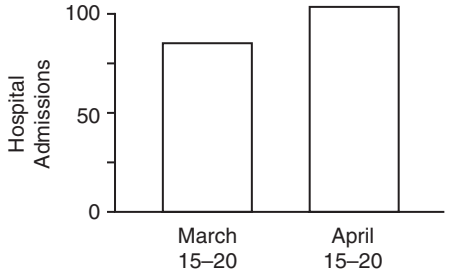


Figure 1. Hospital admissions for depression for March 15–20 and April 15–20.

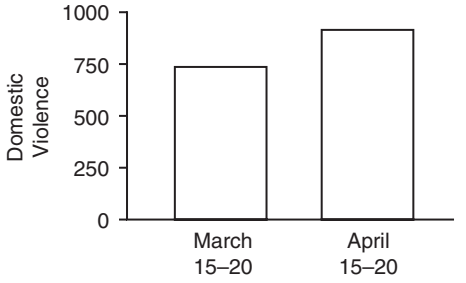


Figure 3. Domestic violence for March 15–20 and April 15–20.

Crimes

The cumulative crime rates reported by the cities studied were 1,326 in April, as compared with 1,130 in March ($\chi 2(1, N = 2{,}456) = 18.64$, $p < .01$). This represents a 17.3% increase. Figure 2 illustrates the difference.

Domestic Violence

During the period in March, there were 746 incidents of domestic violence as opposed to 882 for the corresponding period in April (Figure 3). This is an increase of 18.2%. The one-sample test was significant ($\chi^2(1, N = 1{,}628) = 11.36$, $p < .01$).

Birth Rate

As shown in Figure 4, there was a 5.6% fall in birthrate 9 months after the critical 6 days in April 2015 from the birthrate in a comparable time following the comparative dates in March ($\chi^2(1, N = 147{,}944) = 122.78$, $p < .01$). In January 2016, there were 71,841 births in comparison with 76,103 in December 2015, a decrease of 4,262.

Discussion

Some of the criteria measures that we selected for this study are controversial and need some explanation. The measure for depression consisted of the number of admissions to state mental hospitals for major depression. By far the majority of people who seek treatment for depression first go to their private general physicians or health maintenance organizations for diagnosis, medication, or referral to mental health professionals for psychotherapy (Larchfort & Calvino, 2012). Those whose condition is so acute that they present a danger to themselves are hospitalized in private or public institutions. It was

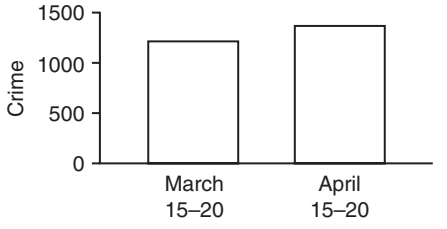


Figure 2. Crime rates for March 15–20 and April 15–20.

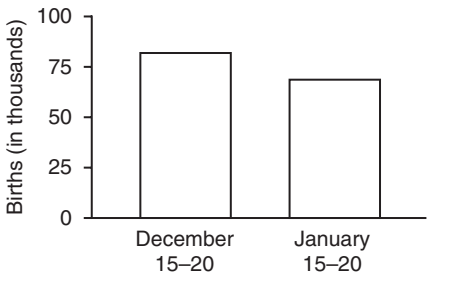


Figure 4. Birth rates for December 15–20 and January 15–20.

beyond our scope to obtain comprehensive data on the true incidence of new depressive episodes. The choice was made to use data that were available but limited to admissions to state mental hospitals for severe depression. We are guided by Stengle's (2011) assertion that the incidence of the more serious forms of depression is a correlate of the overall rate of depression in the general population.

Crime rate is another potentially controversial choice of measure. Obtaining records of incidents, complaints, and arrests is a way to gauge crime (Perlmutter & Jacoby, 2007). We can only guess at the actual number of incidents—they are not a matter of record. Many events are simply not reported by victims (Lefkowitz, 2009). Complaints are not always accurately recorded, and until they have been investigated they cannot be considered as verified incidents. Excellent records of arrests are available, but until convictions have been obtained, there is uncertainty about their accuracy. In addition, there are no arrests for large numbers of incidents. Arrests during any given time period always include crimes that took place sometime earlier. Because there is no easy solution to this dilemma we decided to use recorded complaints plus arrests of individuals who were apprehended at the scene of the crime. Incidents of domestic violence came from recorded complaints that were verified by police investigation, whether or not any arrest was made.

The findings of this study support the hypotheses that the severe stresses imposed by the April 15th tax deadline are manifested in the populace by at least three social indicators—incidence of severe depression, money and property crimes, and domestic violence—as measured by the previously mentioned criteria. Interpersonal stresses are further reflected in lowered birthrates 9 months later. Earlier studies have shown increased birthrates 9 months after major snowstorms or electrical power blackouts (Richman & Milbank, 2010). Now we see the converse, a decrease following tax stress.

These findings do not furnish a base for recommending changes in public policy regarding taxation, but they do show the sensitivity of personal and social phenomena to governmental requirements. The findings emphasize the need to recognize the profound consequences of the tax deadline and to be prepared to adjust social, medical, and law enforcement systems to deal with strains that are placed on the social system.

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Critique

SAMPLE

An effort was made to obtain a large and reasonably broad geographical sample of people, but the sample was not stratified (see Chapter 4). No rationale is given for selecting the particular locations that were chosen. There is no evidence that the perpetrators of the offenses, the patients who were admitted to the hospitals, or the people who did not conceive offspring but who otherwise might have, had any tax problems, were concerned about taxation, or were even under obligation to pay taxes.

DATA COLLECTION

The authors do not describe who and how data were extracted from the records. There is no evidence presented about the reliability with which this was done.

CONTRAST GROUPS

Rates for a week in April are contrasted to a week in March. Without data from the rest of the year, rates for these periods are not informative. It is entirely possible that the rates of depression, crime, and domestic violence may have been lower in all of March than in April, and the conception rate higher, than in most other months. Without looking at the year as well, it remains plausible that the difference between these weeks were part of a longer temporal trend. Rates for a single month do not constitute a baseline (see Chapters 5 and 6).

CRITERIA MEASURES

The authors acknowledged that their measures were controversial. Candid recognition of limitations is praiseworthy, but to plead mea culpa is not enough. Sometimes researchers do not have many alternatives and must use less than ideal measures. Here, the decision to limit the measure of depression to admissions to state mental hospitals remains questionable.

CAUSATION

The article title, "Social Effects of Tax Deadline" indicates that this is meant to be a cause and effect study. The introductory material and the discussion of the results are couched in causal terms. Just because these events occurred right after the tax deadline does not prove that the deadline was their cause. No causal relationship has been demonstrated. This is an instance of post hoc ergo propter hoc (see Chapter 2).

RESEARCH ETHICS

In this study, the hypotheses are stated as specific predictions, and one must assume that they were formulated in advance of searching the preexisting archival data. If

that were not the case, and if the researchers were of doubtful probity, the findings could have emerged from a selective search. The investigators would examine reports of the frequency of a long list of social phenomena and would select several negative phenomena that occurred at a higher rate during the April dates than during the March dates. The search might reveal dozens of frequencies that did not support the general April 15th trauma hypothesis, but these would be disregarded. Especially ignored would be indices that were counter to the general hypothesis. There is no indication that the researchers did anything of the sort in this case; this scenario is provided simply as an example of how someone who had an intent to distort to prove a point might approach this problem (see Chapter 11).

ABSTRACT

The summarized conclusions are misleading. You have no basis for evaluating their validity and must take them on faith unless the study is read in full (see Chapter 9). Although this is true of all studies, it becomes a particular problem when the inquiry is flawed.

Practice Article 16

Comparative Effectiveness of Teacher Management Styles in a Fifth-Grade Classroom

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This study compared the effectiveness of self-regulatory and teacher-directed classroom management styles in the fifth grade. Two teachers were selected who exemplified the two different styles. Children in their classrooms were compared at the end of the year on standardized achievement tests of math and reading. The class means of the tightly managed classroom were significantly higher than those of the other class. The number of disciplinary referrals were significantly greater in the classroom led by the teacher with the tighter management style.

The management style of teachers is believed to be related to the effectiveness of teaching. In the history of education, one style has been variously referred to as *teacher-directed* or *autocratic* (Hubble, 1992). Its opposite has been called *open*, *permissive*, or *self-regulatory*. Although teachers are expected to be fair, firm, and friendly to their pupils, there is a distinct difference in atmosphere under these two classroom management styles. The difference stems from how much control is exercised by the teacher over every aspect of the classroom.

Where the control is tight, the teacher oversees the classroom space and how the students move within it. As has been described in detail by Overmaier (1991), the teacher decides where, when, and how the pupils sit and must give permission for them to move about or to leave the room. The teacher's desk is usually at the front of the room and cannot be approached without permission. Children must raise their hands to speak, and the teacher decides who speaks and for how long. Time is controlled in other ways, as the teacher designates how long pupils are to spend on each activity and when to move on to the next. The organization of the classroom

The authors express their gratitude to Principal Annabelle Grinch and to Mary Tydor and Lena Yent for their outstanding cooperation with this project. Correspondence about this article should be sent to M. Furman, PhD, Oregon State Teachers College, 15 Porter Rd., Portland, OR, 97210 or Annabelle@OSTC.edu.

NOTE: This is a fictional article by fictitious authors to be used for the purposes of research education only. Furman, Loos = firm and loose; Tydor, Lena Yent = tight or lenient.

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is controlled; the teacher has the power, sets the rules, makes the expectations clear, consistently follows up, enforces the rules, and establishes the consequences for violations of the rules. The focus is on behavior, and teachers do not hesitate to discourage violations of prohibitions by enforcing time-out, speaking harshly, and meting out penalties. Pedagogical information is transmitted more directly from teacher to pupil than it is using cooperative learning subgroups.

By contrast, Nance (1991) described the much looser control of the opposite management style. The classroom is conceptualized as an environmental system. Teachers examine the way that their own behavior influences the classroom ecology and use problem-solving techniques to work out issues that arise. The teacher's desk or table is usually at the side or the rear of the room, and it is freely approached by pupils who need supplies. The children control the space, in that they decide on the seating arrangement and choose whether to sit at an individual desk or at a table with other students. They feel free to walk around without permission. They can talk to one another and speak out in class without raising their hand if they have something to say. They help set the expectations and the rules, and the rules can be bent and are subject to negotiation. Teachers are sensitive to the feelings of the children and deal in the realm of affect, as well as behavior. Cooperative learning subgroups are used as pedagogical devices. It is considered important for teachers of both styles to be consistent in their style of classroom management (Martin & Longworth, 2004).

The aim of this study is to determine whether these two different management styles do, in fact, have an effect on the achievement of educational goals. Specifically, we are interested in finding out whether one has the advantage over the other in promoting higher academic achievement and in diminishing behavior that interferes with the educational process. As a convenient shorthand, in this research we refer to the tightly controlled management style as *teacher-directed*. The term *self-regulatory* refers to a style of classroom management that is characterized by the teacher being part of the whole environment and exercising influence (as opposed to control) on the children from a position within the system.

In a study by Adams (2010), the teacher management styles were not clearly distinguished, and the results were somewhat ambiguous. Kimber's (2011) investigation, which found in favor of a self-regulatory management style, was carried out in an exclusive private school for girls. It is doubtful that the results can be generalized to the typical urban public school. A related study by Jesse (2012) was conducted in a school in a severely economically disadvantaged neighborhood of Detroit. The findings favored very tight teacher controls, but here again the results may not generalize to less difficult classroom environments.

Method

Setting

The study took place in an urban public elementary school in Portland, Oregon. The school is attended mostly by children of working-class families. Many of the pupils are children of single mothers. Transient families use the school, and it is not unusual for children to enter or to leave the school at any time during the school year. Ethnic group membership is diverse, and a subgroup of children speak English as a second language. In the fifth grade, children still spend most of their time with one teacher instead of rotating to different classes as they do in middle school and high school.

Class Assignments

Under the direction of the principal, a committee of teachers formed matched pairs of fifth-grade pupils. Pairs were matched on sex, past academic performance, behavior problems, and language proficiency. In addition, they were matched as to whether they had an Individualized Education Plan (IEP), thus

receiving remedial services, or had been identified as talented and gifted (TAG), thus receiving accelerated or enrichment services.

Two comparable classes, each with 28 children, were formed from these matched pairs. The classes were also balanced so that they each contained a similar ethnic mix and a similar number of children who were known to have behavior problems. Each of the two groups contained 10 students served by IEPs, two students identified as TAG, and two students served in the English as a Second Language program. The composition of the classes was considered fixed, and the school administration agreed not to add any new enrollees during the year.

Teachers

Two teachers were selected by the principal. Each was known to have adopted one of the two management styles in question. Other teachers, and the ones selected, confirmed the accuracy of the choices. Each of the designated teachers had at least 9 years of classroom teaching experience. A coin flip decided the assignment of teachers to classes.

Criteria Measures

Academic Achievement. The nationally normed California Achievement Test served as the standardized measure of achievement in reading and math. Percentile scores were obtained for all pupils at the end of the school year.

Behavior. There is a schoolwide policy of "assertive discipline." The pupil's name is placed on a board on receiving the first warning for unacceptable behavior that day. At the second violation, the student receives a second warning, and a check mark is placed alongside the name. Any student who receives three checks in one day is sent to the principal, who determines appropriate administrative action and makes parental contact. For this study, records were kept of all incidents and consequences.

Results

During the school year, 11 children dropped out of the two classes combined. Eight of them were from the teacher-directed class, and three from the self-regulatory class. As far as can be determined, the families moved out of the district. This is not uncommon in this school in view of the number of transient and migrant residents in the community. The children who left the school did not provide sufficient data to be counted, and all were therefore excluded from the analysis of the data. The children who were dropped brought the sizes of the two classes to 20 and 25, respectively.

Achievement

Achievement test scores in reading and math are given in percentile standing in comparison with national norms. As can be seen from Table 1, both classes were below the 50th percentile in reading and math for fifth grade students nationwide. The obtained percentiles of 47.3 and 40.4 for reading and of 49.6 and 42.3 for math are above the level of academic achievement that is typically attained in this school. The reading mean for the teacher-directed classroom is significantly higher than that of the other class ($t(43) = 2.03$, $p < .05$, $d = .62$). This was also true of differences between the math achievement means ($t(43) = 2.51$, $p < .01$, $d = .77$).

Behavior

The number of behavioral incidents that merited administrative action were tabulated

Table 1
Mean Percentiles on Achievement Tests for the Tightly and Loosely Managed Classes

Test	Tight ($n = 20$)	Loose ($n = 25$)	t
Reading	47.3	40.4	2.03*
Math	49.6	42.3	2.51**

*$p < .05$, **$p < .01$.

Table 2
Disciplinary Referrals in the Tightly and Loosely Managed Classes

	Disciplinary referrals			
Style	≤ 10	11–20	21–30	> 30
Tight	2	3	5	10
Loose	16	2	4	3

Note. $\chi^2(3, N = 45) = 11.63, p < .01$.

and divided into four categories: ≤ 10, 11–20, 21–30, >30. Results are shown in Table 2. The results vary significantly ($\chi^2(3, N = 45) = 11.63, p < .01$). By far the most frequent referrals to the principal for administrative action occurred in the class with a teacher-directed style. A total of 75% of the students in this class had more than 20 referrals, in contrast to 28% in the self-regulatory class. The greatest number of incidents occurred during the last marking period in the spring, when pupils characteristically become restless.

Discussion

The results of this study revealed that a teacher-directed classroom management style was more conducive to academic achievement as reflected by nationally-normed math and reading tests. It would not be wise to attempt to generalize these findings to other school grades or to schools whose background characteristics differ from our sample.

If these findings hold up with other children in other schools, and with children of other ages, it would suggest that a more organized classroom, more under the direction and the control of the teacher who sets and enforces rules and expectations, is a better atmosphere for encouraging academic achievement. Of course, we do not know whether more advantaged children from more affluent schools would respond in the same way, or whether they would do better in classrooms with a looser management style. It will take future research to answer this question.

Problem behavior is especially disruptive in the classroom because it diverts the teacher's attention away from the demanding task of attending to the learning needs of an entire group, because it interferes with the learning of the other children, and because it is inimical with that child's own educational experience. There were clearly more reported incidents of problem behavior in the more teacher-directed class. There are several possible explanations for this. The controlling management style may stimulate resistant and challenging behavior. On the other hand, closer attention to rule infractions and the resultant maintenance of order in the classroom may facilitate higher achievement. There is also the possibility that the behavior problems in the two classes are comparable, but that the teacher in the more self-regulatory classroom may not respond to them or be as quick to act. Whatever the explanation of the phenomenon, the two groups clearly differed in this regard.

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Critique

SAMPLE ATTRITION

The authors began the study with 28 students in each group and lost eight in one group and three in the other. Attrition is a factor that looms large in this study, and the fact that it was differential could be even a greater problem in biasing the sample (see Chapter 4). The equivalence of the groups that was established by using matched pairs is threatened. An analysis of who left and how that changed the composition of the groups would have been helpful.

Differential attrition also resulted in classes that were different in size. It is easier to give individual attention to the needs of 20 pupils rather than 25. The difference in class size may have contributed to the difference in achievement test performance.

A final consideration is that the classes do not represent the school. Both classes had fewer students than are typical for that school because they agreed not to accept transfers during the school year.

TEACHER CONFOUNDS

Classroom management styles were set up to differ, and both teachers had over 9 years of experience. However, one of them may have been a better teacher than the other, irrespective of management style. The researchers should have found a way to make sure that the two teachers were equal in teaching ability. Because that was not done, teaching ability and management style are potentially confounded (see Chapter 5).

Other personal characteristics of the teacher that had nothing to do with management style may have confounded the results by affecting the outcome. Had the researchers used several teachers for each style, the influence of the unique characteristics of any single teacher would have been diminished.

MANIPULATION CHECK

The authors state that the classroom management styles of the two teachers exemplify what was intended, but they do not provide any check on how the teachers actually performed in the classroom during the year. Teacher performance should have been checked to make sure that the classrooms were indeed managed in the ways that were specified (see Chapter 7).

CRITERION MEASURE (BEHAVIOR INFRACTIONS)

Giving children warnings and sending them to the principal for behavioral infractions was a poor choice as a criterion measure, because it is confounded with teacher management style. The authors were understandably puzzled when it came to interpreting the findings. The teacher who runs a teacher-directed classroom is probably more inclined to hand out warnings and send children for administrative action than is the

teacher in a self-regulatory classroom. The teacher with the more loosely managed classroom is more likely to overlook infractions or to deal with the problem in other ways. If the misbehavior of children in the classroom is an important criterion measure (and it probably is), it would have been better to have it independently observed and judged by a more objective standard (see Chapter 7).

LEVELS OF VARIABLES

For reasons that remain unstated, the dependent variable of disciplinary referral was divided into four categories instead of leaving it as a continuous variable. This has implications for the statistical analysis.

DATA ANALYSIS

Had the number of disciplinary referrals not been converted into categories, the researchers could have done a *t* test of the difference in the mean disciplinary referrals of the two classes (using the pupil as the unit of analysis) instead of a chi-square. This would have been consistent with the mode of analysis selected for the other dependent variables. As a critical reader, you do not know if the results would have differed, but use of the appropriate test would have removed any doubt that the researchers were not shopping for the type of analysis that favored their preconceptions (see Chapter 11).

Reading and math achievement test results are reported in percentiles. Percentiles are ordinal data. A *t* test requires interval data (e.g., the raw scores obtained on the achievement tests); it is not an appropriate statistical test for percentiles. You cannot tell whether the results were statistically significant as reported (see Chapter 9).

Practice Article 17

Age Differences in Gullibility: A Meta-Analysis

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With the increase in use of the Internet by unsupervised children, concern has been raised about the gullibility of young children, that is, their uncritical acceptance of information that contains clear errors of fact or logical inconsistencies. Research has focused attention on whether young children are more gullible than older children. A meta-analysis was conducted to determine whether gullibility, as measured by the Childhood Gullibility Scale, was related to the child's age. The results of 8 studies revealed that younger children are more gullible than older ones (d = .15) and the relationship varies no more than predicted by chance. It is concluded that young children may have difficulty telling truth from fact and that their Internet use should be monitored.

The advent of the Internet has put an enormous amount of information at the fingertips of users. One consequence of this has been a renewed interest in how easily people can be persuaded by information of questionable veracity and arguments with logical fallacies. This interest is especially keen regarding children and adolescents, who may spend hours reading Internet content without an adult present to assist them in interpreting what they read.

In reaction to this concern, researchers have turned their attention to using psychological measures to distinguish children and adolescents who may be more or less susceptible to influence by Internet content. One construct of interest has been children's individual differences in *gullibility*, or their "general susceptibility to believe and be influenced by others or the things they read that may contain dubious facts or flaws in logic" (Wayne, 2006, p. 814).

A measure of gullibility that has been found to have acceptable reliability and validity is the Childhood Gullibility Scale (CGS; Ford & O'Brien, 2008). The CGS presents children with a series of questions meant to measure their gullibility. For example, one question asks, "I believe the things my friends tell me," to which the responder either agrees or disagrees. Items are counterbalanced to

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NOTE: This is a fictional article by fictional authors to be used for purposes of research education only. Ira M. Green = I am green; E. Stato Ingannati = He was duped (Italian).

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remove acquiescence bias (e.g., this statement is also rephrased as "Sometimes, my friends say things that are hard to believe."). The circumstances depicted in the questions involve other people (friends, teachers, parents), written material, online material, and movies. There are 20 items that are summed, so that higher scores mean greater gullibility.

As part of a large-scale evaluation of measures related to children's openness to persuasion, Townsend, Collins, and Stevens (2010) found the CGS to have high (for children) test–retest reliability ($r = .65$) and to be only weakly related to intelligence ($r = -.19$) and self-esteem ($r = -.24$). Hull (2011) found responses on the CGS also could be distinguished from measures of trust in others and openness to experience.

Among the first research questions developmental psychologists asked that related to gullibility in children was "Are there age differences in gullibility?" For example, Herman (1989) hypothesized that younger children would be more susceptible to influence in general than older children. This proposal was based on findings that as children get older they become less trusting of others and begin to develop critical thinking skills. Researchers who followed Herman proposed that what was true of openness to influence in general, ought also to be true of gullibility in particular.

This study involves a meta-analysis of previous research on the relationship between children's ages and their responses on the CGS. It was our intention to summarize the finding of research rather than conduct a new investigation. In this way, we hoped to discover the accumulated support for the age difference in gullibility hypothesis and point to the most fruitful direction for future research.

Method

Literature Search

Studies for the meta-analysis were located through a search of the PsycINFO reference database and through the use of a reference list provided by the first author of the CGS instrument. Our search terms were "gullibility" crossed with "children" or "age." Document titles and abstracts were searched.

Criteria for Study Inclusion

For a study to be included in this meta-analysis it had to (a) use children and adolescents between the ages of 9 and 15 years as participants, (b) use the CGS as the instrument to measure gullibility, (c) examine age as a moderating variable of responses, and (d) provide an effect size estimate or enough information to allow us to calculate an effect size.

Coding of Studies

Coding Scheme. Twenty different characteristics of each study were included in the database. These characteristics encompassed five broad categories: (a) research report (e.g., published versus unpublished); (b) respondents (age and socioeconomic range), (c) type of stimuli to be read, (d) setting, and (e) effect size information.

Effect Size Estimation. We used the standardized mean difference, or *d* index (Cohen, 1988), to estimate the relationship of age and scores on the CGS. In the synthesis, we subtracted the older children's mean from the younger children's mean and divided the difference by their averaged standard deviation. Because higher scores mean greater gullibility, positive *d* indexes indicate that the younger children were more gullible.

Coder Reliability. Information from complete reports selected for inclusion was extracted independently by two of the authors. Discrepancies were first noted and discussed by the coders and if agreement was not reached a third coder, not involved in the meta-analysis and unaware of its hypotheses, was used to decide the final code.

Methods of Data Integration

Calculating Average Effect Sizes. We used a weighting procedure to calculate average effect sizes across all comparisons, that is, each

independent effect size was first multiplied by the inverse of its variance. The sum of these products was then divided by the sum of the inverses. Also, 95% confidence intervals were calculated for weighted average *d* indexes. If the confidence interval did not contain $d = 0$, then the null hypothesis of no relationship between age and gullibility can be rejected.

Heterogeneity of Effect Sizes. Finally, we tested for whether the variability in effect sizes was significantly greater than that predicted by sample error. To do this, we calculated a Q statistic, which is tested against a chi-square distribution. We used a fixed-effect model of error.

Results

Literature Search

The search of the literature found 8 studies that met the criteria for inclusion. Four studies reported that age had been tested as a moderator but did not provide enough information to calculate an effect size. All these studies simply stated that no significant relationship between age and gullibility was found, but because we could not estimate an effect size they were excluded from the meta-analysis.

Combined Effect Sizes

Table 1 contains information on the eight included studies. Six of the effect sizes were in the positive direction, indicating greater gullibility for younger children. The weighted average *d* index for all eight studies was $d = .15$. The 95% confidence interval was .03 to .26. This interval does not contain $d = 0$, indicating that the null hypothesis of no relationship could be rejected. The test for heterogeneity ($Q(7) = 5.50$, *n*s) indicated that we could not reject the hypothesis that the variability in effect size was due to sampling error alone.

Discussion

The results of this meta-analysis reveal that younger children scored higher on the scale measuring gullibility than older children. The ages of children included in studies ranged from 8 to 15 years old. Although children younger than 8 years old could not be expected to read the scale because of a lack of reading ability (and this was the youngest age used to validate the CGS), it is clear that children older than 15 years of age are missing from the samples. Future research should include older children.

On the basis of these results, it is concluded that younger children are not as able to distinguish truth from facts or to discover errors in logic than are older children. This suggests that their exposure to the Internet should be monitored by adults and programs to enhance critical thinking skills should be developed for children at young ages.

Table 1
Studies of the Relationship Between Children's Age and Gullibility

First author	Year	*Young* age	*Old* age	*N*	*N*	*d*-index
Townsend	2010	9 to 11	12 to 14	108	108	0.12
Jackson	2011	8 to 9	11 to 12	40	40	0.25
Abbot	2009	8	9	27	26	0.09
Swift	2016	9 to 10	11 to 13	40	40	0.41
Hopkins	2012	9 to 11	12 to 14	126	125	0.3
Belmar	2011	10 to 11	12 to 13	32	32	0.01
Ford	2015	11 to 12	14 to 15	60	60	–0.03
Chilton	2013	10 to 11	12 to 14	64	64	–0.04

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NOTE: References preceded by an asterisk were included in the meta-analysis.

Critique

GULLIBILITY MEASURE

The CGS measure is a trait measure of gullibility. Although this is fine, it would have been nice if a measure directly related to the Internet were used (if one existed) or the items on the scale related directly to reading or the Internet were examined separately from each study. This would have required the studies to report Internet gullibility separately and this probably would have been unlikely. The meta-analysts are constrained by what was presented in the individual studies.

It is not clear from the Introduction that other measures exist of the individual difference "susceptibility to believe and be influence by others or the things they read that may contain dubious facts or flaws in logic" (Wayne, 2006, p. 814). If they do, they should have been included in the meta-analysis. If these other measures were constructed in different formats, (e.g., children are presented with different scenarios with clearly erroneous facts or flaws in logic and asked to find them), this would have gone a long way to ruling out possible confounds embedded in the CGS (e.g., social desirability; see Chapter 7).

SEARCH STRATEGY

The search strategy leaves much to be desired (see Chapter 10). The PsycINFO reference database will favor published research and be missing research that is most

recent (a real problem when a topic is new and hot). Contacting the CGS developer was a good idea as that person likely receives many preprints, reprints, and inquiries about it use. The location of unpublished doctoral dissertation suggests this happened. However, researchers who found null results and results counter to the hypothesis (i.e., young children were less gullible), which might cast doubt on the construct validity of the measure (see Chapter 7), might be reluctant to share these with the measurement developer. This suggests the retrieved studies might be a biased sample. The researchers should have (a) conducted a citation (forward) search looking for studies that cited Wayne (2006), and (b) contacted groups of researchers with interests in young children's personalities.

CHILD CLASSIFICATION

The authors are correct to point out that the range of children in studies is restricted at the upper end; the oldest children were 15 years of age and there is good reason for studies of this sort to go up to 18 years of age.

More critically, the authors of the articles included in the meta-analysis artificially dichotomized children into younger and older groups. This was unnecessary and reduces the chances that a significant relationship would be found. Equally, troubling is that it forced the meta-analysts to classify some groups with 11- and 12-year-olds in the *young* group and others in the *old* group. This would clearly have an effect on their ability to find an overall difference between young and old, if one existed. All this could have been avoided had the original researchers left the age variable continuous and reported correlation coefficients.

COMBINED EFFECT SIZES

It is problematic that four studies with null effects had to be eliminated from inclusion in the meta-analysis because an effect size could not be calculated. It suggests that the average *d* index could have been smaller had they been available. The researchers might have contacted these authors and asked for the means, standard deviations, and sample sizes underlying the null statistics. There are also tests for bias against the null hypothesis that they could have employed (see Rothstein, Sutton, & Borenstein, 2005).

TESTS FOR MODERATING VARIABLES

The hypothesis for the meta-analysis is that as children get older they become less gullible. The meta-analysts missed a chance to shed light on the hypothesis, light that was not contained in the original studies. If you look at the Table 1, there appears to be a relationship between the overall range of participant ages and the effect size, the difference between the groups. The first five studies include children ranging in age from 8 to 14, whereas the last three include children ranging from 10 to 15. The *d* indexes range from .09 to .41 and −.04 to .01, respectively. Could this suggest that any age difference disappears around adolescence? The researchers could have tested whether this difference in average effect sizes was larger than that predicted by sample error alone (see Chapter 10).

DISCUSSION

The opening sentence of the Discussion is far too great a generalization. The call for sampling older children is good, but no mention is made of the use in the studies of (a) a single measure of gullibility, (b) the questionable dichotomy between young and old children, and (c) the overlap in these category assignments.

The opening sentence of the second paragraph is also a gross over-generalization, on the basis of these results. Not only are the samples and measurement devices restricted, but there may even be evidence here that the age difference disappears as children great older, without a critical thinking intervention. There is no interpretation of whether a *d* index of .15 is small, medium, or large in this context.

ABSTRACT

The Abstract contains none of the cautions about generalizing the research.

Epilogue

Now that you have seen the flaws in a series of studies, you may get the mistaken impression that research is inherently defective, that it is going to be picked apart no matter what researchers do, and therefore, it is too daunting and rarely results in believable findings. Bear in mind that the practice articles that were generated for this book were intentionally flawed in design and in execution to give you material on which to practice your research critique skills.

Some published research studies do resemble these articles, and examples of each of the errors cited can be found in published articles, despite efforts to weed them out by the process of peer review. However, they do not represent (and were not intended to represent) all articles that are published in quality journals.

It is not necessarily more difficult to do good research than it is to produce flawed research. It just requires more knowledge, more careful planning, and more exacting execution. Many published studies are marvels of rigor and ingenuity. As a critical reader, you will be delighted to find that every possible confound in a study has been anticipated and controlled, that a tight and powerful design has been selected, that pains have been taken to obtain a representative sample of ample size, and that the dependent measures are valid and reliable. Finding all elements in order, you can legitimately admire the work and accept its conclusions.

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Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper
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Even though trained scientists are expected to produce research work of high quality, all studies are potentially vulnerable, and all contain some flaws. This is inevitable; the perfect study is yet to be conducted. It is this potential for vulnerability that researchers protect themselves against by building their knowledge and skills in research design and by rigorously applying the principles that they have mastered. In recognition of this vulnerability, investigators typically critique their own work in the Discussion section and make suggestions for future research. They welcome attempts by others to reproduce their work or to repeat it with improvements.

Results of a single study are generally not enough to establish facts. The findings of a study become more credible when the study can be successfully replicated in different places with different participants and different investigators. Partial replications often correct minor flaws in the original work. Public review, critical evaluation, and repetition with built-in improvements form an orderly progression in scientific fields. The original work, the critique of that work, and the improved partial replications all contribute to our knowledge base in the field of inquiry. Research synthesis is used to evaluate the credibility of the cumulative work on a topic.

There are lessons in this for the producers and the consumers of research. You, the critical reader of research reports, should take nothing on faith alone. You should engage in interactive reading with the research report, do your best to understand all of it, and question everything that was done in the spirit of trying to further your understanding of it, not merely to satisfy an urge to decimate it. You should think about why things were done in this particular way, what was not done, and what could have been added or done differently. The assumptions, hypotheses, design, procedures, measures, and findings should be challenged. The numbers and tables should not be allowed to intimidate you; you should look them over carefully and understand what they mean. You should challenge the interpretation of the data, the inferences that have been made, and the conclusions.

If the research report answers all of the challenges, you will have the satisfaction of having read and learned from a solid, effective, and compelling research report. This will be the case more often than not when reading publications that have been filtered through a diligent review process, but it is something that cannot be taken for granted. If the study is well done, critical reading will increase your understanding of the research and will heighten your appreciation and enjoyment. If the study is poorly done, interactive, critical reading will prevent you from being misled.

Critical reading and the feedback that comes to researchers from it are central factors in improving future research. The ongoing dialogue between the researcher and the reader is rewarding to both and promotes the advancement toward truth that research promises.

Good researchers will not be intimidated by the thought that a portion of readers will be casting a critical eye on the work. As a critical reader, it is important for you to be tactful in your criticisms. But it is also important for you to use a skeptical mind-set when you read and voice your concerns. This is how science progresses.

Glossary

ABSTRACT: A brief summary of a scientific research article that appears at the beginning of the article.

ACCESSIBLE POPULATION: The population to which a researcher can legitimately draw conclusions based on the characteristics of people or units who are available to be part of the sample.

ALPHA (α): The chosen level of statistical significance. The probability, which the researchers have set, that the results could have occurred by chance before the researchers can reject the hypothesis that no relationship exists.

ALTERNATIVE HYPOTHESIS: See *experimental hypothesis*.

ANALYSIS OF COVARIANCE (ANCOVA): An analysis of variance that removes the relationship of variables that correlate with the dependent variable. It is used to help reduce confounding, as well as error variance.

ANALYSIS OF VARIANCE (ANOVA): A statistical procedure used to determine whether the variation among the means of several levels of an independent (or predictor) variable vary more than would be expected by chance. ANOVAs with more than one independent variable (two-way ANOVA, etc.) test for the variables' main effects and interaction effects among the main effects on the dependent variable.

A PRIORI REASONING: Based on logical deductive reasoning before observation or experience. Deductive reasoning from general statements of truth to the specific instance, and from cause to effect.

ARCHIVAL SAMPLING: A sampling procedure that collects data from records or files, rather than directly from the participants themselves.

Artifact: An unanticipated factor that is extraneous to the independent variable but is confounded with it and contributes to systematic variation in the dependent variable.

Assumptions: The acceptance of a proposition as true in the absence of proof. In research, the validity of some statistical tests is based on assumptions about the distribution of the data. If these are violated, the test is inappropriate.

Attribute variable: A static group variable that represents some psychosocial characteristic of the person (e.g., personality, attitudes, beliefs, behavior).

Attrition: The loss of participants during a study.

Backward search: In literature searching, the use of the references in a report, journal article, or bibliography that are examined to see if they are relevant to the topic of the literature search. Sometimes called *footnote chasing*.

Baseline: A starting point for measurements, which is used for comparison with measurements under a different set of conditions.

Beta (β) error: The probability of making a Type II error (i.e., of accepting the null hypothesis when it is in fact false).

Bidirectional relationship: An association between two variables in which the direction of causality can flow either way.

Binomial test of proportion: Nonparametric test to determine whether two proportions are equal.

Biosocial experimenter effects: Extraneous effects brought about by the experimenter's physical characteristics (e.g., age, sex, height, appearance).

Biserial correlation (r_{bis}): A test of association between one dichotomous variable and one continuous variable.

Block and blocking variable: The clustering of participants together into the same group (e.g., sex, ethnicity) before randomly assigning them to treatment groups. The blocking variable can then be used as a predictor variable and its main effect and interactions with other predictor variables can then be examined.

Bonferroni correction: A statistical procedure used to reduce the chances of Type I error in post hoc significance testing.

Categorical variable: An independent variable whose levels are categories (e.g., boys and girls, religious affiliation); a dependent variable whose values are in categories (e.g., survived/did not survive, graduated/did not graduate) instead of having numerical values.

Cause: An antecedent that produces an effect or consequence.

Changing criterion design: A single-case design in which the criteria for performance are changed throughout the experiment. This is often used as subgoals are met by the participants (e.g., reducing the number of cigarettes smoked gradually until zero is reached).

Chi-square (χ^2): A statistical procedure that tests the obtained frequencies of categorical data against what would be expected by chance or by some hypothesis.

Citation search: See *forward search*.

Cluster sampling: A sampling procedure in which subgroups of elements (e.g., men/women) are first identified within a larger population, and then sampled within these groups. Typically, this is meant to ensure that the numbers of participants in each subgroup are properly represented in the final sample.

Cochran Q: A nonparametric test used to determine whether a dichotomous measure (e.g., pass–fail) reveals differences between three or more related groups.

Coder drift: Change that takes place in how coders use a system while the system is being applied.

Coefficient alpha (α): A method for assessing the internal consistency of items on a test using the individual correlations between all items.

Conceptual definition: A description of an abstract variable that is independent of time and space (e.g., aggression and depression are concepts that have not physical manifestation in time and space).

Conceptual replication: An attempt to reproduce a study and its findings using the same conceptual variables but different operational variables.

Concurrent validity: A form of criterion-related validity that measures the extent to which test scores correlate with another variable measured at the same time.

Confounded variable: A correlate of the variable of interest that may contribute to or account for the observed relationship.

Construct validity: How well the manipulation or measure captures the intended construct. It is the degree to which a manipulation or the results of a test or measure correlate with other measures of the same construct and do not correlate with measures of supposedly independent constructs. It is also the degree to which manipulations of measures reveal results consistent with predictions derived from the theory that underlies the construct being measured.

Content validity: The extent to which a test measures the knowledge, skills, behaviors, etc., of the domain that is being tested.

Continuous variable: A variable represented by scores that form an entire distribution of values without interruption, which are distinct from levels or categories.

Contrast group: A group of people that is compared with a target group.

Control group: A group of randomly assigned people who are given no treatment or an alternative treatment, and who then are compared with those people given the experimental treatment.

Convenience sample: A sample that is chosen because the elements are easy to recruit.

Convergent validity: The extent to which a test correlates with other tests of the same construct (e.g., correlation of scores on a new IQ test with scores obtained on established IQ tests).

Correlation coefficient (*r*): A statistic for measuring the linear relationship or association between two variables. Values can range from –1.0 (maximum inverse or negative relationship) to +1.0 (maximum direct or positive relationship), with 0 representing an absence of association.

Counterbalance: A technique used to rule out the effects of the order in which manipulations are presented. Each experimental group is exposed to multiple conditions in a systematically rotated order.

Covariates: Variables, other than the independent variable, that correlate with the dependent variable.

Criterion-related validity: The degree to which the results of a test or measure agree with those of an independent standard. The independent standard can be in the present, future, or past.

Criterion variable: The standard against which a measured variable can be judged. Often used in clinical or intervention research meant to improve responses of participants.

Cross-sectional: A research design that collects all data at onetime but may divide the sample into time-based sections for analysis (e.g., measuring some characteristic in a sample of 3-year-olds, 4-year-olds, and 5-year-olds).

Curvilinear correlation: Correlation that tests for a nonlinear pattern in the data.

Debriefing: The process at the end of a study for explaining to participants the purpose, procedures, and expected outcomes of the study, as well as ensuring participants are returned to a state as good or better than when they arrived for the study.

Deception: The deliberate lying to subjects by researchers or the omission of important information to lead the thinking of subjects away from guessing the actual purpose or hypotheses of a study.

Degrees of freedom (*df*): The number of observations in a study minus the number of restrictions that each observation has to vary when a statistical analysis is conducted.

Demand characteristics: Behaviors or qualities of the experimenter or the setting that appear to place an implicit request on research participants to respond in a particular way.

Demographic characteristics: Life-status statistics (e.g., marital status, occupation, education, age).

Dependent variable: A variable influenced or affected by another variable. The variable whose values depend on variation in the independent variable; the outcome or criterion variable.

Descriptive research: Qualitative research whose purpose is to present the details or characteristics of a group or phenomenon.

Dichotomous variable: A variable having two categories (e.g., yes/no, present/absent).

Differential attrition: The loss of subjects from a study that occurs in different frequencies or proportions for different conditions of the study.

Direct replication: A study that attempts to reproduce an earlier study exactly as it was first conducted (i.e., using the same manipulations, measures, settings, etc.).

Direct sampling: When data for a sample are provided by the elements that make up the population.

Discriminant analysis: A statistical procedure that generates the best regression equation for assigning people (or other units) to one of two or more groups.

Discriminant validity: The degree to which a measure of a concept that is not supposed to be related to another concept, is, in fact, unrelated.

Distortions of scale: The use of scale to give the appearance that the results from one group differ more noticeably from another group than the data warrant; this is usually done by magnifying a small portion of a scale's ordinate.

Distribution: An array of scores or values arranged in order of the frequency of their occurrence.

Duncan multiple range test: A post hoc procedure that tests the significance of the difference between all possible pairs of treatment means.

Dunnett test: A post hoc multiple comparisons test for comparing all treatment means with the mean of a control group.

Duplicate publication: When the same data have been published twice, with no new information or analyses.

Ecological validity: The degree to which the conditions of a study can be generalized to real-life settings.

Effectiveness research: Research designed to evaluate a treatment under conditions as close as possible to those routinely seen in clinical practice.

Effect size: The magnitude of the relationship between two variables, or the magnitude of the effect of the independent variable. The degree to which the null hypothesis is false.

Efficacy research: Research conducted under highly controlled conditions. For example, participants may be screened to maximize homogeneity and the treatment setting may be standardized to provide a strong experimental test.

Empirical: Studies that collect data based on experience or the senses.

Enabling conditions: Conditions or circumstances that allow or facilitate something else to happen.

Epidemiological studies: The study of patterns (associations) in health and disease data in defined populations (e.g., a study of diet and mortality rates in a country).

Equal-appearing intervals: Intervals between scale values that have been judged to have subjective equality (e.g., Thurstone method for measuring attitudes).

Equated groups: Groups whose membership has been adjusted so that their means on key variables are equivalent.

Erratum: A statement accompanying a publish article that corrects an error in the version originally published.

Experiment: A type of empirical investigation in which the conditions are prearranged by the experimenters to rule out alternative explanations or rival hypotheses, and to discount the effects of confounding variables.

Experiment-wise type I error: The probability that at least one of the tests in an analysis will be significant due to chance. This is a concern when, in the study, the investigators conduct more than one test of statistical significance.

Experimental group: In an experiment, the group that receives the manipulation, treatment, or intervention hypothesized to lead to an effect.

Experimental hypothesis (H1): The prediction about the cause–effect relationship an experiment is meant to test.

Experimenter bias: Prejudgments or attitudinal leanings on the part of the experimenter that influence the research procedures, the interpretation of the data, or the report in the direction of the hypothesis.

Exploratory research: Research with imprecise or general hypotheses (e.g., religious affiliation will relate to attitudes), meant to uncover interesting relationships about which little was previously understood.

Ex post facto: Retroactively; in research, attempting to explain something after it has happened.

External validity: The degree to which the results of a study can be generalized to persons, procedures, outcomes, and settings, other than those included in the study.

Fabrication: The data reported in a research report that have been made up.

Face validity: The extent to which a measure appears to be a measure of what it purports to measure, based on the simple examination of the measure's content.

Factor analysis: A statistical procedure for identifying the broader dimensions (fewer in number) that underlie the variables measured in a study or testing whether the pattern of correlations between items conform to a prespecified prediction.

Falsification: When research materials or processes are inaccurately reported or when data or results are changed or omitted so that the research is deceptively presented in the research report.

Fisher exact probability test: A statistical technique for analyzing comparisons between discrete classes of scores of two independent groups that are composed of small samples in 2×2 tables (e.g., number of Republicans and Democrats in a group of men as compared with a group of women). Provides the probability that the table's values were generated by chance and corrects for Type I error.

Forward search: In literature searching, the search for documents that cite a past report or article relevant to the topic the searcher is looking for.

Friedman two-way analysis of variance: A two-factor analysis of variance that uses rankings for matched samples.

Generalizability: The degree to which the findings of a study can be used to make inferences about people, places, or circumstances not included in the study.

Haphazard sample: A nonprobability sample in which participants are selected in an arbitrary, unsystematic way.

History: A change in participants that is not attributable to the independent variable, but instead is a consequence of changes caused by external events and experiences that took place between pretest and posttest. *Local history* refers to a historical event that happens to one group in a study but not to other groups. *General history* refers to an event that happens to all groups.

Homogeneity analysis: In meta-analysis, an analysis that compares the amount of variance in an observed set of effect sizes with the amount of variance that would be expected if only sampling error were making the effect sizes different.

Homogeneity of variance: The expectation that scores of all groups or conditions within a study should have the same variation about their mean within groups or conditions.

Homogeneous: Uniform in composition; showing little variation.

Hypothesis: A deduction from a theory or observations predicting a proposed relationship; a logical conclusion of propositions that is based on theory or limited observations.

Independent variable: The antecedent manipulation or treatment that causes or leads to a consequence or outcome that is dependent on it.

Inductive reasoning: Reasoning in which the premises of the argument are believed true, typically through experience, and are combined to obtain a specific conclusion.

Instrumentation: Apparatus, psychological tests, rating scales, or other measuring devices that are used in research.

Interaction: A condition that exists when groups perform differently at one level of an independent variable than at another level. For example, a high-IQ group might persist more than a low-IQ group on intellectually challenging tasks, whereas a low-IQ group might persist more on simple repetitive tasks more than a high-IQ group.

If true, this would reflect an interaction between level of IQ and type of task on the dependent variable (persistence).

Interactional experimenter effects: The ways in which the experimenter and participants may influence each other's behavior during research.

Internal consistency: A measure of how well the different items on a test measure the same underlying conceptual variable.

Internal validity: The extent to which the outcome of an experiment truly depends on the manipulated variable as opposed to other variables confounded with the independent variables.

Interrater reliability: The degree of agreement among raters.

Interval level of measurement: Measurements in which the size of the difference between participants' scores corresponds to their difference in the level of the variable.

Intervening variable: A variable that is assumed to operate between the independent and dependent variables and whose presence is invoked to explain the way that the former affects the latter.

Intervention: A treatment or action that is meant to improve the situation of those exposed to it (e.g., an intervention to reduce alcohol dependence).

In vitro studies: In social science research, studies taking place in natural environments.

In vivo studies: Studies that involve processes that occur within an organism, instead of in an artificially maintained organism.

Kappa (κ): A statistic that measures the agreement beyond chance between judgments of categorical data.

Kendall rank correlation coefficient tau (τ): A nonparametric test of the association between the ranks of participants on two variables.

Kolmogorov–smirnov test: A nonparametric test for comparing frequency distributions of participants using ordinal data.

Kruskal–wallis one-way analysis of variance: A nonparametric analysis of variance to test the significance of the variation in ranks of independent samples on one factor.

Kurtosis: The degree to which a distribution has a long tail relative to a normal curve, often signaling the presences of statistical outliers.

Likert scale: A rating scale that uses adjectives given numerical values that reflect the respondent's degree of agreement or disagreement, or other dimension (e.g., liking, belief about truthfulness) with a statement.

Logistic regression: A statistical procedure that determines the degree to which several predictor variables differentiate between participants who are in discrete groups.

Longitudinal study: Research that follows the progress of the same group of people by taking successive measurements on some variable(s) over an extended time (e.g., motor coordination in children at ages 1, 2, 3, 4, and 5).

Manipulated variable: A variable whose conditions are established by the experimenter (as opposed to occurring naturally).

Manipulation check: An effort to determine whether the experimental conditions in a study were experienced and perceived by participants in the manner intended.

Mann–whitney u-test: A nonparametric test equivalent of the *t*-test. A test of the significance of the difference between two independent groups.

Masking: A procedure that keeps the experimenters (i.e., the persons interacting with the subjects, recording of judging the data, etc.) unaware of the condition that the subjects are in.

Matching: Selecting participants for assignment to groups on the basis of person-for-person equivalence on one or more variables.

Maturation: The change over time of any internal state or personal characteristic that comes about independent of the experimental treatment (e.g., fatigue, hunger, puberty).

Measured variables: Variables that are not manipulated by researchers, or are naturally occurring events, and measured with minimal interference by the researcher.

Mediator variables: A variable that helps explain a cause–effect relationship that temporally comes between the cause and effect.

Meta-analysis: A statistical procedure for combining the results of any number of studies meant to examine the same research problem.

Moderating variables: A variable that influences the direction or strength of a relationship. Often a contextual variable (e.g., a setting) or a status of participants (e.g., sex, age).

Multicolinearity: When two or more predictor variables in a multiple regression model are highly correlated, meaning their predictive relationship with the criterion variable cannot be separated because the two variables contain highly redundant information.

Multiple baseline design: A single-case design in which baselines are established for several different target behaviors that the treatment is either meant or not meant to change.

Multiple operationism: In a study, the use of more than one measure of the same construct that use different methods of data collection.

Multiple regression: A correlational procedure for determining the combined association between several predictor variables with a dependent or outcome variable and their combined ability to predict values of the dependent variable.

Multiple treatment interferences: The effect of participants' exposure to a treatment on their response to subsequent treatment(s).

Multivariate analysis of variance (MANOVA): Analysis of variance in which two or more dependent variables are analyzed in a single procedure, meant to reduce the chances of committing a Type I error.

Naturally occurring experimental manipulation: A manipulation of an independent variable in an experiment that was not introduced by the experimenter but rather occurs naturally (e.g., a natural disaster, the introduction of a new reading program at a school).

Network sampling: A sampling procedure in which participants who are already in a sample are used to identify and recruit additional sample members.

Newman–keuls test: A post hoc multiple comparison test used to compare all possible pairs of treatment means.

Nominal level of measurement: Measurements in which variables are divided into two or more categories.

Nonequivalent groups design: A research design in which the experimental and control or contrast groups are not equivalent at the outset on variables that could have an association with the outcome variable.

Noninteractional experimenter effects: The ways in which the outcome of an experiment may be effected by either an observer or interpreter.

Nonparametric statistics: Statistical tests that do not require specifications about the shape of the distribution of the data in the population from which the sample is taken. Nonparametric ("distribution-free") statistics are used to analyze categorical (nominal) or ranked (ordinal) data.

Nonprobability sampling: The process of selecting research participants without regard to any specified probability that the population members had an equal chance of being selected into the sample.

Nonreactive measures: Measures or observations that are made in such a way that they have no effect on participant behavior, and that do not disrupt the flow of natural behavior.

Normative rating scale: A scale that uses the responses of other members of a population as a comparison for the rating the responses of the test taker.

Null hypothesis (H0): The prediction that there will be no difference among the groups in an experiment or that there is no relationship between the variables.

Null hypothesis significance test: The statistical test used to determine whether a null hypothesis can be rejected and with what probability that it is false.

Numerical rating scale: A rating scale that has values expressed in numbers.

Odds ratio: An effect size metric used when both variables are dichotomous. It expresses the effect as the ratio of one set of odds (e.g., the odds of success in a treatment group) relative to the odds of another set (e.g., the control group).

Omnibus test: An overall test of a hypothesis that has more than one degree of freedom.

One-tailed tests: Significance tests in which the null hypothesis is rejected only if the obtained value exceeds a critical value for one tail of the distribution (upper or lower) as predicted by the researchers. Also called one-sided or directional tests. A two-tailed test is nondirectional; the null hypothesis is rejected if the obtained value exceeds a critical value at either tail of the distribution.

Operational definition: A definition of a variable in terms of the specific observable and reproducible procedures that are used to measure a variable (e.g., the number of smiles is used as the operational definition of approval).

Ordinal level of measurement: Measurements expressed in the form of ranks.

Organismic variable: A static group variable that represents some biological characteristic of the participant (e.g., age, gender, height, etc.).

Outcome variable: See *dependent variable*.

Overall attrition: The proportion of participants in a study who withdraw before the study is complete, regardless of group membership.

Parametric statistics: Statistical analysis procedures that assume the sampled data come from a population of data whose values follow a probability distribution based on a fixed set of parameters. Most typically, parametric tests make the assumption that the underlying population data exhibits a normal distribution.

Path analysis: A set of correlational analyses that incorporates the theorized order of how variables influence one another to enable the researchers to make informed speculations about causal relationships.

Pearson product-moment correlation coefficient: A metric used to express the degree of linear relation between two variables; produced as an *r* index.

Peer review: The process by which scientific work is evaluated by other knowledgeable scientists before it is published in a scientific journal, accepted for presentation at a scientific meeting, or funded by a research sponsor.

Phi coefficient (Φ): The correlation coefficient between two dichotomous variables.

Pilot test: A preliminary, small-scale study conducted to test whether the manipulations and measurements meant for use in a larger study are feasible and interpreted by participants the way the researchers intend.

Placebo: Any drug or procedure that the recipient benefits from even though it contains no active ingredients.

Placebo effect: An improvement in symptoms experienced by participants taking a placebo (i.e., an inert treatment).

Planned comparison: A test of hypothesized differences between the means of any two subgroups of a multilevel variable that was planned before the experiment began.

Plausible rival hypotheses: An alternative explanation for a claimed causal relationship that remains reasonable because it has not been explicitly ruled out.

Population: The people or other units from which a smaller research sample is drawn but about whom the researchers want to draw inferences.

Post hoc ergo propter hoc: The erroneous belief that an event that preceded another must have caused it.

Post hoc tests: A posteriori (not planned in advance) tests of the significance of the difference between means of subgroups of a multilevel variable.

Postdictive reasoning: An argument for an explanation proposed after the fact.

Postdictive validity: The extent to which a test is shown to be able to accurately reflect a person's status at some time in the past.

Power ($1 - \beta$): The probability of rejecting the null hypothesis when it is false. A power analysis is used to estimate the number of participants needed to give the null hypothesis a good chance of being rejected.

Predictive validity: The extent to which a test is shown to be able to predict future behavior or future status.

Predictor variable: A variable used to forecast something that will happen in the future.

Principle components analysis: A type of factor analysis meant to identify a smaller number of variables from a combination of a larger number of variables by analyzing the correlations between them and weighting them optimally.

Probability sampling: The selection of research participants in a way that assures that participants are chosen so that each member of the population has a known chance of being selected into the sample.

Proof of concept: A demonstration that something is feasible, or can appear under at least one set of circumstances.

Prospective study: A study in which the researchers predict the consequences or effects from known antecedents or causes.

Psychosocial experimenter effects: The ways in which an experiment may be affected by the experimenter's psychosocial characteristics (e.g., personality, demeanor, behavior, dress, regional accent).

Purposive sampling: A sampling procedure that chooses elements because the researchers think they are particularly suited to be in the study.

***p* value**: Level of significance. Probability of differences or relationships of the obtained size or larger occurring by chance.

Quasi-experimental designs: Designs that resemble true experiments because conditions are manipulated but the design does not include random assignment of participants to conditions.

Random allocation: See *random sampling*.

Random assignment: A method of assigning participants to treatments whereby each unit has an equal chance of placement in any condition in the study.

Random sampling: A method of selecting people at random from the larger population, whereby each individual has an equal, nonzero, chance of being selected.

Randomized block design: Design in which subjects are first matched based on static group characteristics (e.g., sex) and then randomly assigned to conditions.

Ratio level measurements: Measurements that use a scale that includes an absolute zero value.

Regression toward the mean: A statistical phenomenon whereby, on repeated testing, individuals with extreme high or low scores will move in the direction of the population mean. The mean of an extreme group on the second occasion will fall closer to the population mean.

Reliability: The consistent accuracy of a test or measure. Stability can be assessed by test–retest correlations, and internal consistency by split-half and related techniques.

Repeated measures analysis: A within-subject design in which measures are taken on the same people when they are exposed to different treatment conditions, or the same treatment but after multiple time lapses.

Replication: The redoing of a study by following the same procedures as in the original study (direct replication). Some aspects may be altered in a conceptual replication.

Representative sample: A sample whose characteristics correspond to or mirror those of the population from which it is drawn.

Research design: The blueprint for the collection, measurement, and analysis of data. The approach that is taken to collect data that integrates the different components of a study, so that it addresses the research problem.

Research synthesis: A study that summarizes and integrates the findings of a set of related studies.

Response set: The propensity to respond to questions or tests in specific ways (e.g., saying yes, giving a socially desirable answer).

Retrospective study: A study that infers from the present what happened in the past; it involves postdiction in contrast to prediction.

Role selection: Participants' decisions about what role to play, what persona to adopt, how to present themselves to make a desired impression, and what to reveal about themselves.

Sample: A portion of a population that is selected to be in a study.

Sample of opportunity: See *convenience sample*.

Sampling frame: A list of all the elements from which a sample can be drawn, or the procedures that can be used to define these elements.

Scheffé test: A post hoc multiple comparison test of the significance between all possible treatment means. The procedure controls Type I error.

Selective deposit: When archival records are used to collect data, some important aspects of sample members' records may be missing because they were never recorded.

Selective survival: When archival records are used to collect data, some important aspects of sample members' records may be missing because their records have disappeared or have been discarded.

Self-fulfilling prophecy: When the expectations of the data collectors influence how they behave toward a research participant to make it more probable the participant will respond in the expected manner.

Self-report: Data acquired from the participant's own ratings or reports about the self, in contrast to archival records or observations by impartial observers.

Self-selected sample: A sample in which participants have chosen to take part in the sample.

Semantic differential: Attitudes toward an idea or object based on a scale of the perceived psychological distance between adjectives (e.g., good–bad) when applied to the construct under evaluation by the respondents.

Significance level: The level of confidence that a difference or relationship of the same size or larger could have occurred by chance; expressed as a probability.

Single-case design ($N = 1$): Experiment with several controlled observations conducted on a single individual or other unit.

Skewness: Departure of a distribution from normality in the form of excessive accumulation of score at the low or high end of the curve.

Solomon four-group design: An experimental design with four groups that allows the assessment of pretest effects, as well as experimental effects. Two groups are given both a pretest and posttest, whereas two other groups are given only the posttest.

Spearman rank-difference correlation coefficient (ρ): A method of determining the relationship between two variables when the data are ranks.

Specification error: The omission of important variables from a regression equation or the inclusion of unimportant ones.

Split-half: Randomly dividing item scores on a test into halves and summing them, then correlating the halves with each other to obtain a coefficient of reliability.

Standard deviation (*SD*, s, σ): A measure of the variation of a set of scores around the mean.

Standardized mean difference (*d*-index): An effect size metric that expresses the difference between two group means in terms of the separation of the terms in standard deviation units.

Static group variable: An organismic or psychological characteristic of people (e.g., age, sex, occupation); these are typically variables that cannot be manipulated (i.e., people's group identity).

Statistical conclusion validity: The extent to which the statistical outcomes of a study permit valid inferences about a study's results. Typically based on whether (a) the statistical procedures are appropriate to the question, and (b) the data meet the procedure's assumptions (e.g., for parametric statistics that the data are normally distributed).

Status variable: A static group variable that represents some aspect of life status (e.g., marital status, educational attainment, occupation).

Strata: See *subpopulations*.

Stratified sampling: Selecting participants from subpopulations (strata) of units so that the proportions of group memberships (e.g., sex, age, educational level, education) in the sample are the same as those found in the general population.

Structural equation modeling: Statistical methods that fit multiple networks of relationships between variable and observed data. Similar to path analysis, it is used to examine support for a causal model of temporally sequenced relationships.

Subjects: The participants in a research study; typically participants in studies that use experimental manipulations.

Subpopulations: Smaller groups of participants categorized from the larger population from which participants in a study were samples.

Survey research: A method for collecting information about a population of people. Questions are asked of a sample of people drawn from the population, and then inferences are made about its characteristics.

Syllogism: A three-line form of argument in which a conclusion is drawn from two premises that are taken as given or assumed.

Systematic review: See *research synthesis*.

Systematic sampling: A procedure that involves choosing every *k*th element from a list of the sampling frame members.

Target population: In a study, the people or other units about which researchers hope to make inferences.

Test–retest reliability: Repeating a test to obtain a second set of scores for correlation with the original set to measure the test's stability over time.

Theory: An overarching, general explanation for a set of verifiable hypotheses.

Third factors: See *plausible rival hypothesis*.

Thurstone scale: See *normative rating scales*.

Time series design: A design that studies behavior of the same intact group over successive samples of time, typically alternating experimental and control treatments.

Treatment variable: See *independent variable*.

Trend analysis: A repeated measures analysis of variance that examines successive mean values over time and assesses linear or curvilinear components.

***t* TEST**: Statistical procedures that assess the significance of the difference between the means of two independent or correlated groups.

TUKEY TEST: A post hoc multiple comparison test of the significance between all treatment means. Also known as the Honestly Significant Difference procedure.

TYPE I ERROR: The error of rejecting the null hypothesis when it is true.

TYPE II ERROR: The error of accepting the null hypothesis when it is false.

UNIDIRECTIONAL RELATIONSHIP: A time sequence of events that logically flows in only one direction when assessing antecedents and consequences.

UNOBTRUSIVE MEASURES: See *nonreactive measures*.

VARIABLE: A quantity that takes on different values.

VARIANCE: The discrepancy between measures of a set of variable values.

VISUAL ANALOGUE SCALE (graphic rating scale): A scale in which respondents make a mark on a horizontal line between two polar adjectives (e.g., strongly agree/strongly disagree). No scale values are printed on the line. Scale values are obtained by measuring the distance of the respondent's mark along the line from left to right.

WILCOXON MATCHED-PAIRS SIGNED RANKS TEST: A nonparametric test of the significance of the difference between ranks of matched pairs of observations.

WITHIN-SUBJECT DESIGN: A repeated-measures design, in which measures are taken on the same individuals under different conditions or different times.

***z* SCORES**: Scores that are expressed in standard deviational units.

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